

Hang Lee

Foundations of Applied Statistical Methods

Second Edition



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Preface

Researchers who design and conduct experiments or sample surveys, perform data analysis and statistical inference, and write scientific reports need adequate knowledge of applied statistics. To build adequate and sturdy knowledge of applied statistical methods, firm foundation is essential. I have come across many researchers who had studied statistics in the past but are still far from being ready to apply the learned knowledge to their problem solving, and else who have forgotten what they had learned. This could be partly because the mathematical technicality dealt with their past study material was above their mathematics proficiency, or otherwise the studied worked examples often lacked addressing essential fundamentals of the applied methods. This book is written to fill gaps between the traditional textbooks involving ample amount of technically challenging mathematical derivations and/or the worked examples of data analyses that often underemphasize fundamentals. The chapters of this book are dedicated to spell out and demonstrate, not to merely explain, necessary foundational ideas so that the motivated readers can learn to fully appreciate the fundamentals of the commonly applied methods and revivify the forgotten knowledge of the methods without having to deal with complex mathematical derivations or attempt to generalize oversimplified worked examples of plug-and-play techniques. Detailed mathematical expressions are exhibited only if they are definitional or intuitively comprehensible. Data-oriented examples are illustrated only to aid the demonstration of fundamentals. This book can be used as a guidebook for applied researchers or as an introductory statistical methods course textbook for the graduate students not majoring in statistics.

Boston, MA, USA

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Chapter 1

Description of Data and Essential Probability Models



This chapter portrays how to make sense of gathered data before performing formal statistical inference. The topics covered are types of data, how to visualize data, how to summarize data into a few descriptive statistics (i.e., condensed numerical indices), and introduction to some useful probability models.

1.1 Types of Data

Typical types of data arising from most studies fall into one of the following categories.

Nominal categorical data contain qualitative information and appear to discrete values that are codified into numbers or characters (e.g., 1 = case with a disease diagnosis, 0 = control; M = male, F = female, etc.).

Ordinal categorical data are *semi*-quantitative and discrete, and the numeric coding scheme is to order the values such as 1 = mild, 2 = moderate, and 3 = severe. Note that the value of 3 (severe) does not necessarily be three times more severe than 1 (mild).

Count (number of events) data are quantitative and discrete (i.e., 0, 1, 2 ...).

Interval scale data are quantitative and continuous. There is no absolute 0, and the reference value is arbitrary. Examples of such data are temperature values in °C and °F.

Ratio scale data are quantitative and continuous, and there is absolute 0; e.g., body weight and height.

In most cases, the types of data usually fall into the above classification scheme shown in Table 1.1 in that the types of data can be classified into either quantitative or qualitative, and discrete or continuous.

Nonetheless, some definition of the data type may not be clear, among which the similarity and dissimilarity between the ratio scale and interval scale may be the ones that need further clarification.

Table 1.1 Classification of data types

	Qualitative	Quantitative
Discrete	Nominal categorical (e.g., M = male, F = female)	Ordinal categorical (e.g., 1 = mild, 2 = moderate, 3 = severe) Count (e.g., number of incidences 0, 1, 2, 3, ...)
Continuous	N/A	Interval scale (e.g., temperature) Ratio scale (e.g., weight)

Ratio scale: If two distinct values of quantitative data were able to be represented by a ratio of two numerical values, then such data are ratio scale data. For example, two observations $x_i = 200$ and $x_j = 100$, for $i \neq j$; the ratio $x_i/x_j = 2$ shows that x_i is twice of x_j , for example, lung volume, age, disease duration, etc.

Interval scale: If two distinct values of quantitative data were not ratio-able, then such data are interval scale data. Temperature is a good example as it has three temperature systems, i.e., Fahrenheit, Celsius, and Kelvin. Kelvin system also has its absolute 0 (there is no negative temperature in Kelvin system). For example, 200 °F is not a temperature that is twice higher than 100 °F. We can only say that 200 °F is higher by 100 degrees (i.e., the displacement between 200 and 100 is 100 degrees in Fahrenheit measurement scale).

1.2 Description of Data

1.2.1 Distribution

A distribution is a complete description of how high the occurring chance (i.e., probability) of a unique datum or certain range of data values is. The following two explanations will help you grasp the concept. If you keep on rolling a die, you will expect to observe 1, 2, 3, 4, 5, or 6 equally likely, i.e., a probability for each unique outcome value is 1/6. We say, probability of 1/6 is distributed to 1, 1/6 is to 2, 1/6 to 3, 1/6 to 4, 1/6 to 5, and 1/6 to 6. Another example is that if you keep on rolling a die many times, and each time you say a success if the observed outcome is 5 or 6 and say a failure otherwise, then your expected chance to observe a success will be 1/3 and that of a failure will be 2/3. We say, a probability of 1/3 is distributed to the success, and 2/3 is distributed to the failure. There are many distributions that cannot be described as simply as these two examples, which require descriptions using sophisticated mathematical functions.

Let us discuss how to describe the distributions arising from various types of data. One way to describe a set of collected data is to describe the distribution of relative frequency for the observed individual values (e.g., what values are very common and what values are how less common). Graphs, simple tables, or a few summary numbers are commonly used.

1.2.2 Description of Categorical Data

A simple tabulation (frequency table) is to list the observed count (and proportion in percentage value) for each category. A bar chart (see Figs. 1.1 and 1.2) can be used for a visual summary of nominal and ordinal outcome distributions. The size of each bar in Figs. 1.1 and 1.2 reveals the actual counts. It is also common to present it as the relative frequency (i.e., proportion of each category in percentage of the total).

1.2.3 Description of Continuous Data

Figure 1.3 is a list of white blood cell (WBC) counts of 31 patients diagnosed with a certain illness listed by the patient identification number. Does this listing itself tell us the group characteristics such as the average and the variability among patients?

How can we describe the distribution of these data, i.e., how much of the occurring chance is distributed to $WBC = 5200$, how much to $WBC = 3100$, ...,

Example of Frequency Table

sex	Frequency	Percent	Cumulative Frequency	Cumulative Percent
Female	21	53.85	21	53.85
Male	18	46.15	39	100.00

Example of Bar Chart

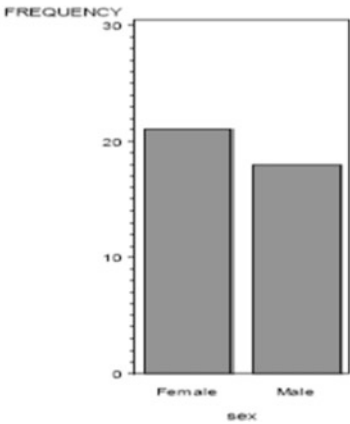


Fig. 1.1 Frequency table and bar chart for describing nominal categorical data

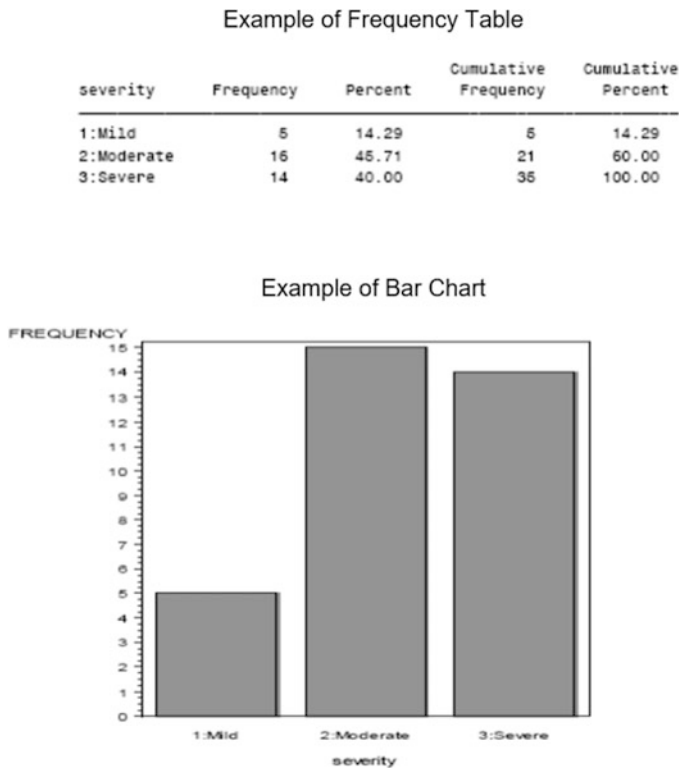


Fig. 1.2 Frequency table and bar chart for describing ordinal data

etc.? Such a description may be very cumbersome. As depicted in Fig. 1.4, the listed full data in ascending order can be a primitive way to describe the distribution, but it does not still describe the distribution. An option is to visualize the relative frequencies for grouped intervals of the observed data. Such a presentation is called histogram. To create a histogram, one will first need to create equally spaced WBC categories and count how many observations fall into each category. Then the bar graph can be drawn where each bar size indicates the relative frequency of that specific WBC interval. The process of drawing bar graphs manually seems cumbersome. Next section introduces a much less cumbersome manual technique to visualize continuous outcomes.

1.2.4 Stem-and-Leaf Plot

The stem-and-leaf plot requires much less work than creating the conventional histogram while providing the same information as what the histogram does. This is a quick and easy option to sketch a continuous data distribution.

Fig. 1.3 List of WBC raw data of 31 subjects

ID	WBC
1	5200
2	3100
3	3000
4	3700
5	4000
6	3700
7	4100
8	8100
9	3500
10	3300
11	3400
12	9300
13	3800
14	2600
15	4800
16	5800
17	4300
18	6100
19	6100
20	1800
21	7500
22	4100
23	11200
24	2800
25	3900
26	3400
27	8900
28	5900
29	4500
30	3800
31	6500

Let us use a small data set for illustration, and then revisit our WBC data example for more discussion after this method becomes familiar to you. The following nine data points 12, 32, 22, 28, 26, 45, 32, 21, and 85 are ages (ratio scale) of a small group. Figures 1.5, 1.6, 1.7, 1.8 and 1.9 demonstrates how to create the stem-and-leaf plot of these data.

The main idea of this technique is a quick sketch of the distribution of an observed data set without computational burden. Let us just take each datum in the order that it is recorded (i.e., the data are not preprocessed by other techniques such as sorting by ascending/descending order) and plot one value at a time (see Fig. 1.5). Note that the oldest observed age is 85 years, which is much greater than the next oldest age 45 years, and the unobserved stem interval values (i.e., 50s, 60s, and 70s) are placed. The determination of the number of equally spaced major intervals (i.e., number of stems) can be subjective and data range dependent.

Figure 1.10 depicts the distribution of our WBC data set by the stem-and-leaf plot. Most values lie between 3000 and 4000 (i.e., mode); the contour of the frequency distribution is skewed to the right, and the mean value did not describe the central location well; the smallest and the largest observations were 1800 and 11,200, respectively, and there are no observed values lying between 1000 and 1100.

Fig. 1.4 List of 31 individual WBC values in ascending order

Order	ID	WBC	
1	20	1800	← Minimum Value
2	14	2600	
3	24	2800	
4	3	3000	
5	2	3100	
6	10	3300	
7	11	3400	
8	26	3400	
9	9	3500	
10	4	3700	
11	6	3700	
12	13	3800	
13	30	3800	
14	25	3900	
15	5	4000	
16	7	4100	← Median Value
17	22	4100	
18	17	4300	
19	29	4500	
20	15	4800	
21	1	5200	
22	16	5800	
23	28	5900	
24	18	6100	
25	19	6100	
26	31	6500	
27	21	7500	
28	8	8100	
29	27	8900	
30	12	9300	
31	23	11200	← Maximum Value

1.2.5 Box-and-Whisker Plot

Unlike the stem-and-leaf plot, this plot does not show the individual data values explicitly. This can describe the data sets whose sample sizes are larger than what can usually be illustrated manually by the stem-and-leaf plot. If the stem-and-leaf plot is seen from a bird-eye point of view (Fig. 1.11), then the resulting description can be made as depicted in the right-hand side panels of Figs. 1.12 and 1.13.

The unique feature of this technique is to identify and visualize where the middle half of the data exist (i.e., the interquartile range) by the box and the interval where the rest of the data exist by the whiskers.

If there are two or more modes, the box-and-whisker plot cannot fully characterize such a phenomenon, but the stem-and-leaf can (see Fig. 1.14).

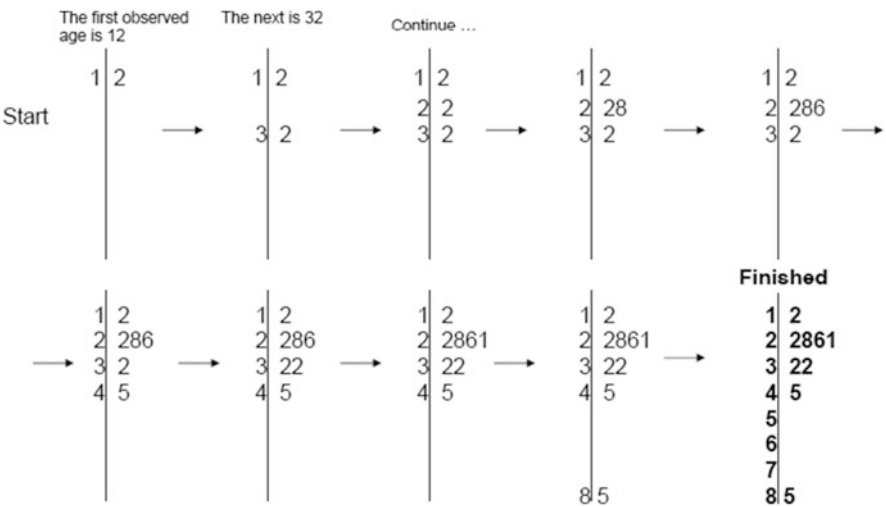


Fig. 1.5 Step-by-step illustration of creating a stem-and-leaf plot

Fig. 1.6 Illustration of creating a stem-and-leaf plot

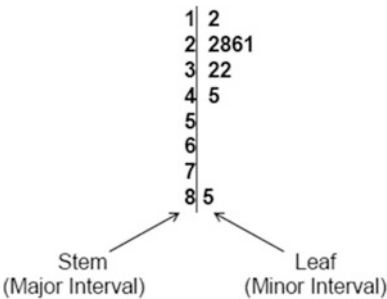
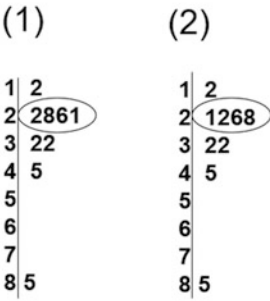


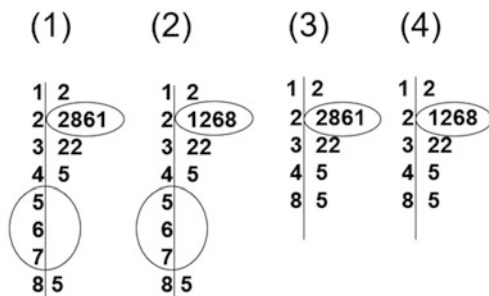
Fig. 1.7 Two stem-and-leaf plots describing two same data sets

Do (1) and (2) describe the same data sets?



Yes.

Do (3) and (4) describe the observed distribution correctly?

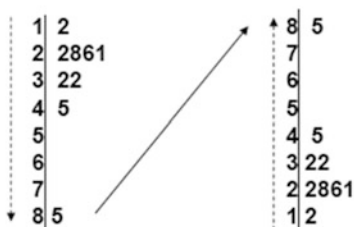


No, these two do not show the stems of 50's 60's and 70's that were described by (1) and (2), i.e., "absence" of such ages.

Fig. 1.8 Common mistakes in stem-and-leaf plot

Fig. 1.9 Two stem-and-leaf plots describing the same data sets by ascending and descending orders

Are these two describing the same data?



Yes, the plot can be turned up-side-down.

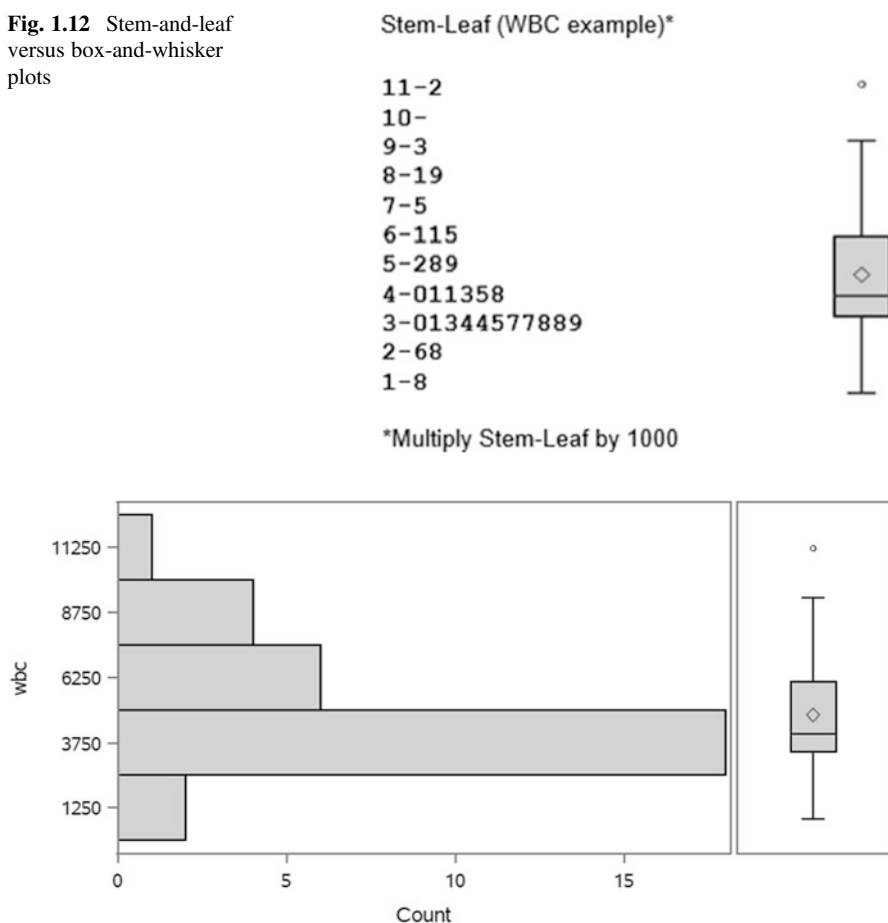
1.3 Descriptive Statistics

In addition to the visual description such as stem-and-leaf plot, further description of the distribution by means of a few statistical metrics is useful. Such metrics are called descriptive statistics, which indicate where the most frequently observed values are concentrated (i.e., representative location) and how much the occurring chance distribution is scattered around that concentrated location (i.e., dispersion).

1.3.1 Statistic

A **statistic** is a function of data, wherein a function usually appears as a mathematical expression that takes the observed data and reduces (processes) to a single

Fig. 1.12 Stem-and-leaf versus box-and-whisker plots



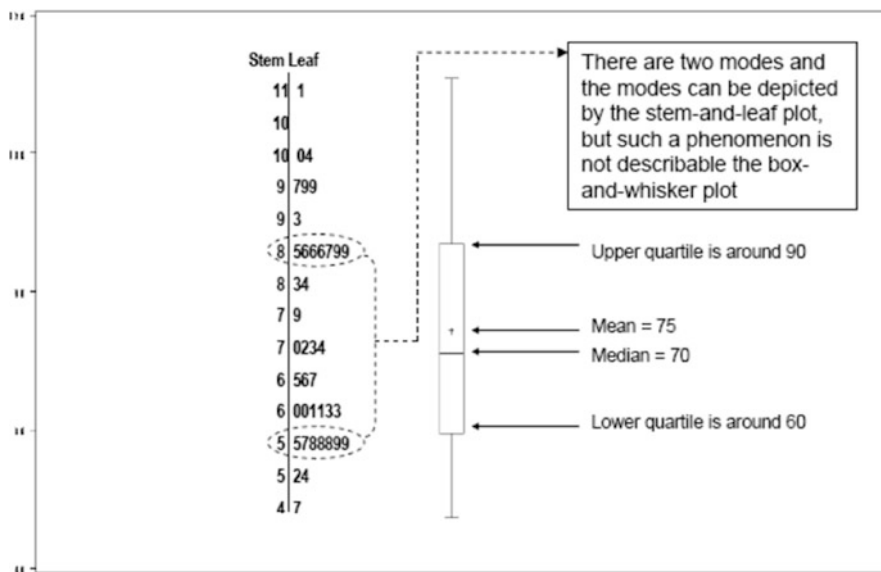


Fig. 1.14 Stem-and-leaf and box-and-whisker plots of a skewed data set

1.3.3 Dispersion Descriptive Statistics for Quantitative Outcomes

The range (i.e., maximum value – minimum value) and interquartile range (i.e., 75th–25th percentile) are very simple to generate by which the dispersion of a data set is gauged. Other commonly used dispersion descriptive statistics are variance, standard deviation, and coefficient of variation, and these statistics describe the dispersion of data (particularly when the data are symmetrically scattered around the mean), and the variance and standard deviation are important statistics that play a pivotal role in the formal statistical inferences, which will be discussed in Chap. 2.

1.3.4 Variance

The variance of a distribution, denoted by σ^2 , can be conceptualized as an averaged squared deviation (explained in detail below) of the data values from their mean. The more dispersed the data are, the more the variance. Standard textbooks present *definitional* and *computational* mathematical expressions. Until the modern computer was not widely used, finding a shortcut for manual calculations and choosing the right tool for a quick and easy calculation had been a major issue of statistical education and practice. Today's data analysis utilizes computer software, and manual calculation skill set is not highly important. Nonetheless, understanding

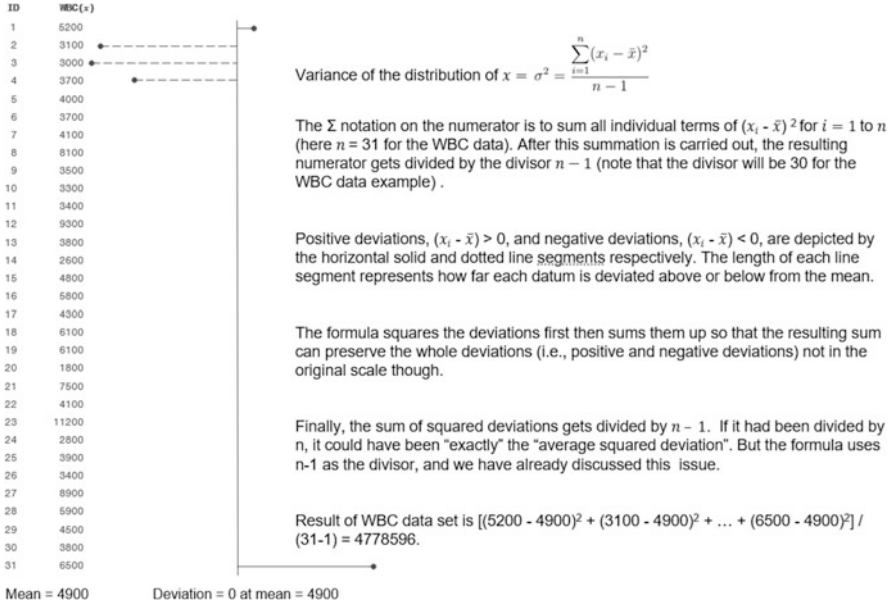
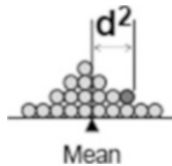


Fig. 1.15 Definitional formula of variance

the genesis of definitional expression, at least, is important. The following is the demonstration of the definitional expression of the variance:

$$\sigma^2 = \frac{\sum_{i=1}^n (x_i - \bar{x})^2}{n - 1}$$

where x_i 's, for $i = 1, 2, \dots, n$ (i.e., the sample size) are the individual data values and $\bar{x}$ is their mean. The $\sum_{i=1}^n$ notation on the numerator is to sum over all individual terms, $(x_i - \bar{x})^2$, for $i = 1$ to n (e.g., $n = 31$ for the WBC data). The term $(x_i - \bar{x})^2$ for i is the squared deviation of an i th individual data value from its mean and is depicted by d^2 in the following visual demonstration.



After this summation is carried out, the resulting numerator is divided by the divisor $n - 1$ (note that the divisor will be 30 for the WBC data example).

As depicted in Fig. 1.15, deviations (i.e., $x_i - \bar{x}$) are presented by horizontal solid, for $(x_i - \bar{x}) > 0$, or dashed, for $(x_i - \bar{x}) < 0$, line segments. The length of each line segment represents how far each datum is displaced above or below the mean. How do we cumulate and preserve the deviations of the entire group? If straight summation is considered, the positive and negative individual deviations may cancel out each other, and the resulting sum may not retain the information. Thus, straight summation is not a great idea. The individual deviations are squared first and then summed up so that the resulting sum can retain the information (i.e., positive and negative deviations) although the retained quantity is not in the original scale. Then, the sum of squared deviations is divided by $n - 1$. If it had been divided by n , it could have been the average squared deviation. Instead, the used divisor is $n - 1$. Normally, an average is obtained by dividing the sum of all values by the sample size n . However, when computing the variance using sample data, we divide it by $n - 1$, not by n . The idea behind is the following. If the numerator (i.e., sum of squared deviations from the mean) is divided by the sample size, n , then such a calculation will slightly downsize the true standard deviation. The reason is that when the deviation of each individual data point from the mean was obtained, the mean is usually not externally given to us but is generated within the given data set (i.e., referencing to an internally obtained mean value), and the observed deviations could become slightly smaller than what it should be. So, to make an appropriate adjustment for the final averaging step, we divide it by $n - 1$. You may be curious why it needs to be 1 less than the sample size, not 2 less, or some other values. Unlike other choices (e.g., $n - 2$, $n - 3$, ..., etc.), $n - 1$ can handle any sample size that is 2 or greater (note that the smallest sample size that will have a variance is 2). There is formal proof that the divisor of $n - 1$ is the best for any sample size but it is not necessary to cover it in full detail within this introductory level context.

The calculated variance of the WBC data set is $[(5200-4900)^2 + (3100-4900)^2 + \dots + (6500-4900)^2]/(31-1) = 4,778,596$. Note that variance's unit is not the same as the raw data unit (because of squaring the summed deviations).

1.3.5 Standard Deviation

The standard deviation of a distribution, denoted by σ , is the square root of variance (i.e., $\sqrt{\text{variance}}$), and the scale of the standard deviation is the same as that of the raw data. The more the data are dispersed, the greater the standard deviation is. If the dispersed data form a particular shape (e.g., bell curve), then one standard deviation unit symmetrically around the mean (i.e., above and below the mean) will cover about middle two thirds of the data range (see standard normal distribution in Sect. 1.4.3).

1.3.6 Property of Standard Deviation After Data Transformations

The observed data often require transformations for the purpose of analysis. One example is to shift the whole data set to a new reference point by simply adding a positive constant to or subtracting it from the raw data values. Such a simple transformation does not alter the distances between the individual data values, thus the standard deviation remains unchanged (Fig. 1.16).

Another example is to change the scale of the data with or without changing the reference point. In general, if data vector $\mathbf{x} = (x_1, x_2, \dots, x_n)$ whose mean $= \mu$ and standard deviation $= \sigma_x$ is transformed to $y = a \cdot x + b$, where a is the scaling constant and b is the reference point, then the mean of y remains the same as $y = a \cdot (\text{mean of } x) + b = a \cdot \mu_x + b$ and the standard deviation y is $\sigma_y = a \cdot \sigma_x$. Note that adding a constant does not alter the original standard deviation, and only the scaling factor does.

The following example (see Fig. 1.17) is to demonstrate how the means and standard deviations are changed after transformation. The first column lists a set of body temperature of eight individuals recorded in $^{\circ}\text{C}$, the second column lists their deviations from the normal body temperature 36.5°C (i.e., $d = C - 36.5$), and the third column lists their values in $^{\circ}\text{F}$ (i.e., $F = 1.8C + 32$). The mean of the deviations from the normal temperature is 0.33 (i.e., 0.33 degrees higher than the normal temperature on average), which can be reproduced by the simple calculation of the difference between the two mean values 36.83 and 36.5 without having to recalculate the transformed individual data. The standard deviation remains the

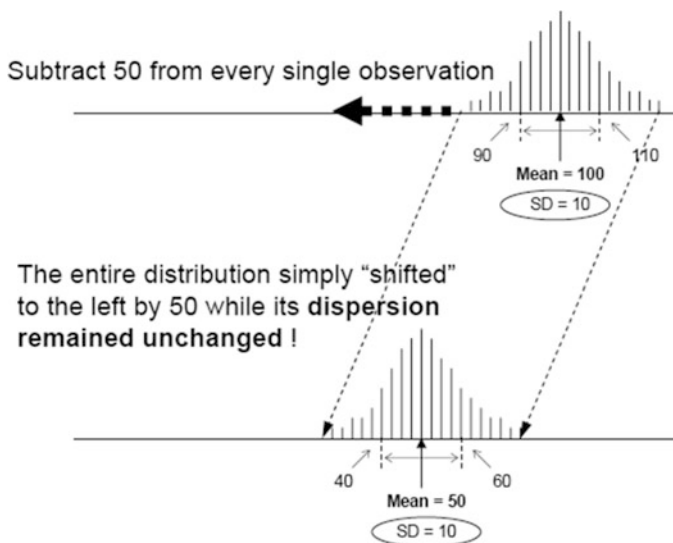


Fig. 1.16 Shifted data without changing dispersion

	Body Temperature °C (Raw Data)	Body Temperature Deviation from 36.5 °C Reference Point (Transformation: d = C - 36.5)	Body Temperature °F (Transformation: F = 1.8C + 32)
	36.40	-0.10	97.52
	36.50	0.00	97.70
	36.50	0.00	97.70
	36.50	0.00	97.70
	36.60	0.10	97.88
	37.20	0.70	98.96
	38.10	1.60	100.58
Stem and Leaf	38.(0~4) 1 37.(5~9) 2 37.(0~4) 36.(5~9) 5556 36.(0~4) 4	1.(5~9) 6 1.(0~4) 0.(5~9) 7 0.(0~4) 0001 -0.(0~4) 1	100. 6 99. 0* 98. 97. 0005 * 98.96 was rounded to 99.0
Mean	36.83	0.33 (subtract 36.5 from the original mean)	98.29
Std. Dev.	0.62	0.62 (recalculation is unnecessary)	1.12 (0.62 was multiplied by .8)

Fig. 1.17 Scale invariant and scale variant transformations

same because this transformation is a just shifting of the distribution to the reference point 32. The mean of the transformed values to °F scale is 98.29, which can be obtained by the simple calculation of 1.8 times the mean of 36.83; then add 32 without having to recalculate using the transformed individual observations. This transformation involves not only the distribution shifting but also the rescaling where the rescaling was to multiply the original observations by 1.8 prior to shifting the entire distribution to the reference point of 32. The standard deviation of the data transformed to °F scale is 1.12, which can be directly obtained by multiplying 1.8 to the standard deviation of the raw data in °C scale, i.e., $1.12 = 0.62 \times 1.8$.

1.3.7 Other Descriptive Statistics for Dispersion

Figure 1.18 illustrates the asymmetrical distribution of the WBC that was illustrated in Figs. 1.11, 1.12 and 1.13. The mean, median, and mode are not very close to each other. What would be the best description of the dispersion? The calculated standard deviation is 2186, which can be interpreted that a little less than one-thirds of the data

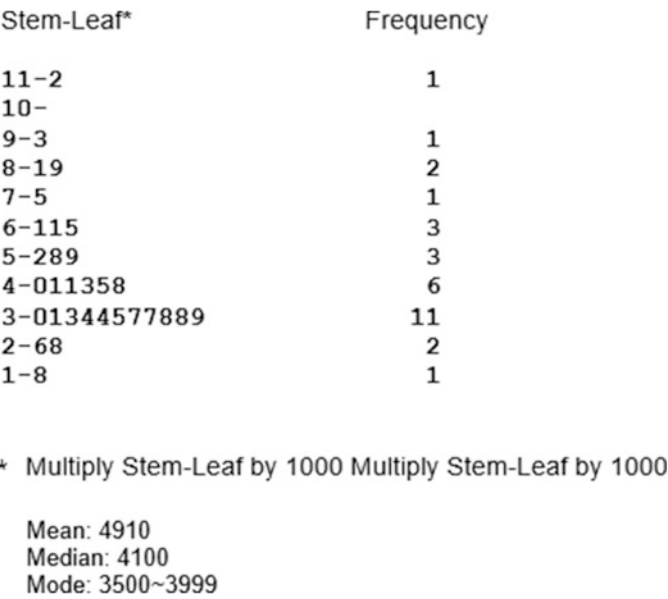


Fig. 1.18 An asymmetrical distribution depicted by stem-and-leaf plot

are between 2724 and 7096 (i.e., within $\text{mean} \pm 1$ standard deviation) if the contour of the distribution had appeared to a bell-like shape. Because the distribution was not symmetrical, the interquartile range may describe the dispersion better than the standard deviation. The 25th and 75th percentiles (i.e., first and third quartiles) are 3400 and 6100, respectively, and this tells literally that half of the group is within this range, and the width of the range is 2700 (i.e., $\text{interquartile range} = 6100 - 3400 = 2700$).

1.3.8 *Dispersions Among Multiple Data Sets*

Figure 1.19 presents two data sets of the same measurement variable in two separate groups of individuals. The two group-specific means are the same, but the dispersion of the first group is twice as the second group's. The difference in the dispersions is not only visible but is also observed in the standard deviations of 10 and 5.

The comparison of the dispersions may become less straightforward in certain situations. What if the two distributions are from either the same characteristics (e.g., body temperatures) from two distinct groups or different characteristics measured in the same unit but of the same individuals (e.g., fat mass and lean mass in the body measured in grams, or systolic blood pressure (SBP) and diastolic blood pressure (DBP) measured in mmHg). In Fig. 1.20, can we say the systolic blood pressure (SBP) values are more dispersed than DBP solely by reading the two standard

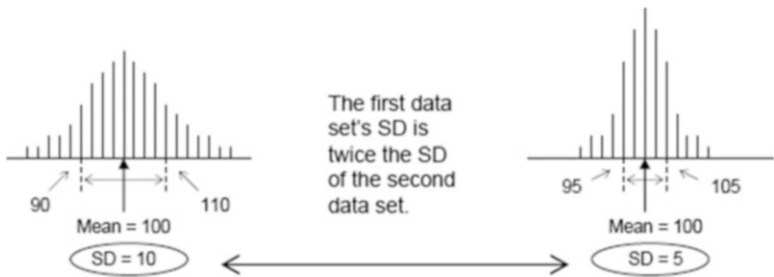


Fig. 1.19 Two data sets with unequal dispersions and equal means

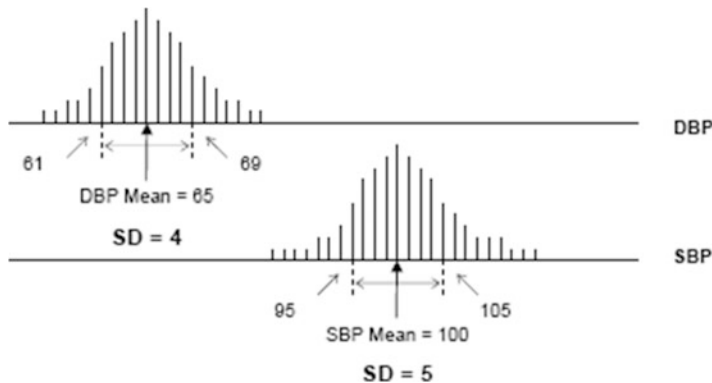


Fig. 1.20 Two data sets with unequal dispersions and unequal means

Table 1.2 Application of CV to compare the dispersions of two different characteristics, measured in the same unit, of the same individuals

	N	Mean	Standard deviation	CV
Body fat mass (g)	160	19783.28	8095.68	40.9%
Body lean mass (g)	160	57798.63	8163.56	14.1%

deviations? Although the standard deviation of SBP distribution is greater than that of DBP, the mean SBP is also greater, and the interpretation of the standard deviations needs to take into account the magnitudes of the two means. Coefficient of variation (CV) is a descriptive statistic that is applicable for such a circumstance by converting the standard deviation to a universally comparable descriptive statistic.

CV is defined as a standard deviation to mean ratio expressed in percent scale (i.e., $CV = 100 \times \text{standard deviation}/\text{mean}$). This statistic is useful for comparing the dispersions of two or more distributions of the same variable in two or more different data sets of the means are not identical, or those of two or more different variables measured in the same unit in the same data set. Table 1.2 demonstrates the situation of comparing the dispersions of two different characteristics measured from the same

Table 1.3 Application of CV to compare the dispersions of the same characteristics, measured in the same unit, of two distinct groups

		N	Mean	Standard deviation	CV
Body fat mass (g)	Group 1	80	21118.04	8025.78	38.0%
	Group 2	80	18448.53	7993.01	43.3%

individuals in the same unit. The standard deviation of the fat mass in grams is smaller than that of the lean mass in grams of the same 150 individuals, but the CV of the fat mass is greater describing that the fat mass distribution is more dispersed (CV 43.0% compared to 14.4%).

Table 1.3 compares the dispersions of the same characteristic measured from the same individuals. The standard deviations appeared greater within group 1, but the CV was greater within group 2 describing that the dispersion of fat mass was greater within group 2.

1.3.9 Caution to CV Interpretation

CV is a useful descriptive statistic to compare dispersions of two or more data sets when the means are different across the data sets. However, the CV should be applied carefully. When the dispersions of two distributions are compared, we need to ensure that the comparison is appropriate. A comparison of the dispersions of the same or compatible kinds is appropriate (e.g., CVs of body weights obtained from two separate groups or CVs of SBP and DBP obtained from the same group of persons). However, a comparison of two dispersions of which one is a result of a certain transformation of the original data is that not appropriate. For example, in the case of the body temperature example in 1.3.6, the CV of the original $^{\circ}\text{C}$ is $100 \times (0.62/36.82) = 1.68\%$ and the CV of the transformed data via $^{\circ}\text{C} - 36.5$ is $100 \times (0.62/0.33) = 187.88\%$. Did the dispersion increase this large after the whole distribution simple shift? No, the dispersion did not differ, and the standard deviations remained the same. However, the CV of scale data distribution is different from the original $^{\circ}\text{C}$ scale.

1.4 Statistics for Describing Relationships Between Two Outcomes

1.4.1 Linear Correlation Between Two Continuous Outcomes

Previous sections discussed how to summarize the data observed from a single variable (univariate). This section discusses how to describe a relationship between a set of pairs of continuous outcomes (e.g., a collection of heights measured from

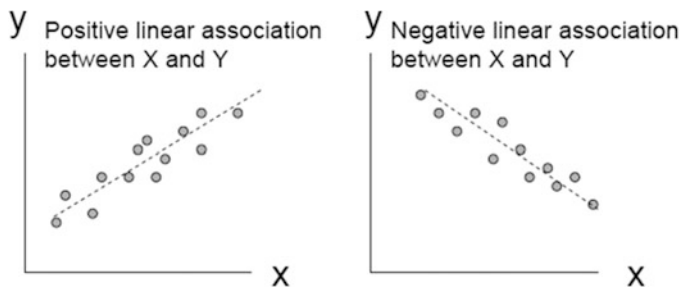


Fig. 1.21 Linear relationships between two continuous outcomes

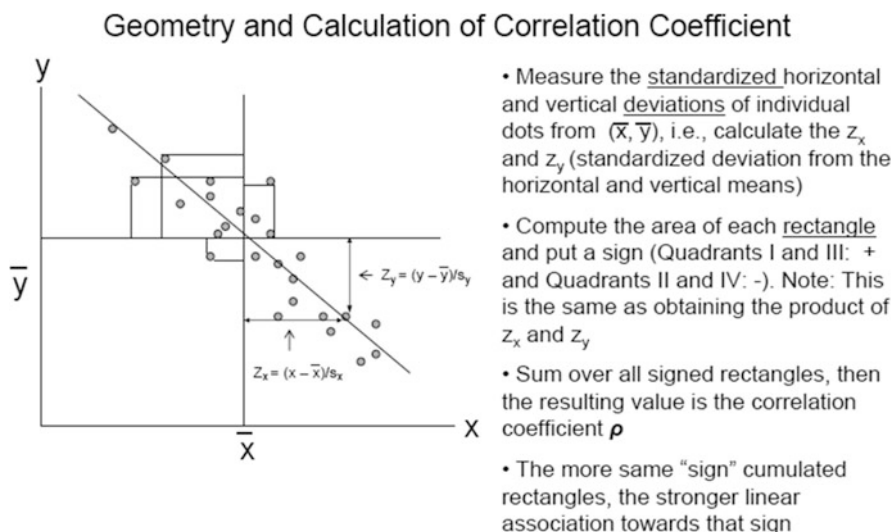
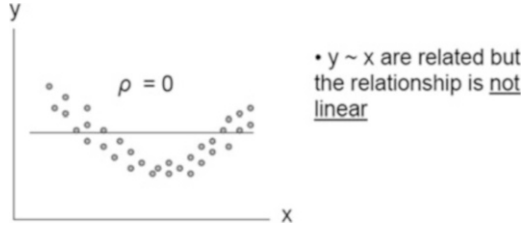


Fig. 1.22 Geometry of correlation coefficient

biological mother and her daughter pairs). The easiest way to describe such a pattern is to create a scatter plot of the paired data (Fig. 1.21). Correlation coefficient, ρ , is a descriptive statistic that summarizes the direction and strength of a linear association. The correlation coefficient exists between -1 and 1 (geometry of the correlation coefficient is demonstrated by Fig. 1.22). Negative ρ values indicate a reverse linear association between the paired variables, and positive ρ values indicate the same directional linear association. For example, ρ between x and y , $\rho_{xy} = -0.9$, indicates a strong negative linear association between x and y , and $\rho_{xy} = 0.2$ indicates a weak positive linear association. Note that the correlation coefficient measures only the linear association. Figure 1.23 illustrates a situation that the correlation coefficient is 0, but there is a clear relationship between the paired variables. The definitional expression is:

Fig. 1.23 Nonlinear relationship between two continuous outcomes



$$\rho = \frac{\Sigma(x_i - \bar{x})(y_i - \bar{y})}{(n-1)} \cdot \frac{1}{s_x s_y},$$

using the sample pairs of (x_i, y_i) , for $i = 1, 2, \dots, n$. The denominator $(n-1)s_x s_y$ can be rewritten as $\sqrt{\Sigma(x_i - \bar{x})^2 \Sigma(y_i - \bar{y})^2}$, and finally:

$$\frac{\Sigma(x_i - \bar{x})(y_i - \bar{y})}{(n-1)s_x s_y} = \frac{\Sigma(x_i - \bar{x})(y_i - \bar{y})}{\sqrt{\Sigma(x_i - \bar{x})^2 \Sigma(y_i - \bar{y})^2}},$$

where $\bar{x}$ and $\bar{y}$ are the means of x and y , respectively. This calculation can be cumbersome if done manually, but computer software is widely available, and Excel can also be used (see Chap. 7 for details).

1.4.2 Contingency Table to Describe an Association Between Two Categorical Outcomes

Qualitative categorical outcomes cannot be summarized by the mean and standard deviation of the observed categories even if the categories were numerically coded (i.e., mean value of such a codified data is meaningless). It is also true that an association of a pair of the numerically categorized outcomes cannot be assessed by the correlation coefficient because the calculation of the correlation coefficient involves the mean value and deviations from the means (see Fig. 1.22). A scatter plot is not applicable for a visual description between a pair of categorical outcomes. To describe the pattern of a set of pairs obtained from two categorical outcomes, the contingency table is used (Fig. 1.24, where each cell number is the observed frequency of the study subjects). The number appeared in each cell (i.e., cell frequency) provides you the information about the association between two categorical variables. Figure 1.24 illustrates the perfect association, moderate association, and complete absence of the association between a disease status and a deleterious risk factor. Figure 1.25 illustrates what data pattern is to be recognized for a summary interpretation. There are 20% (i.e., 10 out of 50) of mothers who are ≤ 20 years old and delivered low-weight babies, whereas only 10% (i.e., 15 out of

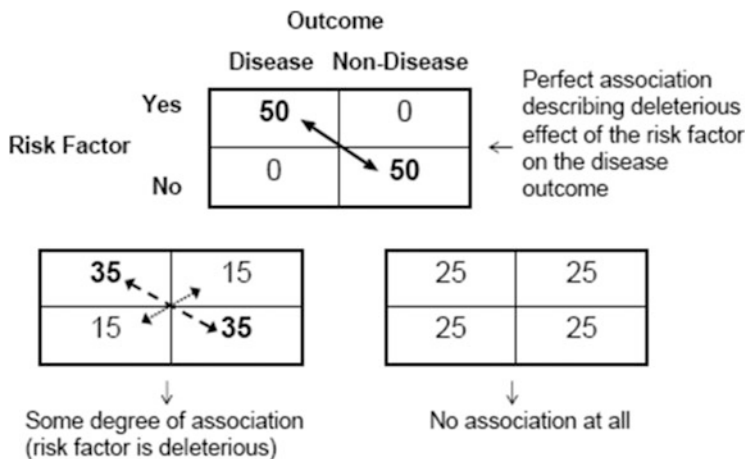


Fig. 1.24 Patterns of association between two binary outcomes

Fig. 1.25 Data summary by a contingency table

		Birth Weight		
		≤2500 g	>2500 g	
Mother's	≤20	10 (20%)	40 (80%)	Row Total 50 (100%)
Age	>20	15 (10%)	135 (90%)	150 (100%)
Column Total		25 (12.5%)	175 (87.5%)	200

150) of the >20 years old mothers did so. It is also noted that the 20% is greater (and the 10% is smaller) than the marginal proportion of the ≤2500 grams (i.e., 12.5%). This observed pattern is interpreted as a twofold difference in proportion of ≥2500 grams between the two mother groups.

1.4.3 Odds Ratio

Odds ratio (*OR*) is a descriptive statistic that measures the direction and strength of an association between two binary outcomes. It is defined as a ratio of two odds. The odds is the ratio of a probability of observing an event of interest, π , to the probability of not observing that event, $1 - \pi$, i.e., $odds = \pi / (1 - \pi)$. In practical applications, the odds can be calculated simply by taking the ratio between the number of events of interest and the number of events not of interest (e.g., number of successes divided by the number of failures). Thus, the odds ratio associated with a presented risk factor versus the absence of the risk factor for the outcome of interest is defined as:

$$OR = \frac{\pi_1/(1 - \pi_1)}{\pi_2/(1 - \pi_2)}.$$

The odds ratio ranges from 0 to infinity, and a value $0 < OR < 1$ is a protective effect (in a setting of association between a risk factor and a disease outcome) of the factor (i.e., the outcome is less likely to happen within the risk group), $OR = 1$ being neutral, and $OR > 1$ is a deleterious effect of the risk factor (i.e., the outcome is more likely to happen within the risk group). According to the definition, the odds ratio associated with the mother's age ≤ 20 years versus >20 years for the offspring's birth weight ≤ 2500 grams is $[0.2/(1-0.2)]/[0.1/(1-0.1)] = 2.25$. The same result is obtained simply by the cross-product ratio, i.e., $[(10/40)]/[(15/135)] = (10 \times 135) / (40 \times 15) = 2.25$. The interpretation of this is that the odds to deliver the offspring with ≤ 2500 grams of birth weight among the mothers aged ≤ 20 years is 2.25 times of that of the mothers >20 years. It is a common mistake to make the following erroneous interpretation that the risk of having low-birth weight delivery is 2.25 times greater. Here, the odds ratio was interpreted as if it were a risk ratio. Note that a risk is the probability of observing an event of interest, whereas an odds is the ratio between the two probabilities whose numerator is the probability of observing an event of interest and the denominator is the probability of not observing it.

1.5 Two Essential Probability Distribution

The remaining part of this chapter concentrates on two probability models that are essential and instrumental to the statistical inference, which are the Gaussian model and binomial model. Let us go over some definitions first. A **distribution** is a complete description of a set of data, which species the domain of data occurrences and the corresponding relative frequency over the domain of occurrence. Note that the object being distributed is the relative frequency. A **probability model** (e.g., Gaussian model, binomial model, etc.) is the underlying mathematical rule (i.e., mechanism) about event occurrence (i.e., observed data).

The probability model is describable by means of a random variable and its distribution function. A **random variable** is a variable that maps an event onto its probability of occurrence. Uppercase letters are used to denote random variables. Let us make a brief review about the random variable through a small experiment. The experiment is to toss a biased coin twice in a row, wherein at each toss, the chance for "heads" falling is 0.7. Let X denote the number of heads (note that an uppercase letter is used to denote a random variable). Then X can be 0, 1, or 2. By denoting $P(X = x)$, the probability of observing x heads: $P(X = 0) = 0.3 \times 0.3 = 0.09$ (i.e., TT), $P(X = 1) = 0.3 \times 0.7 + 0.7 \times 0.3$ (i.e., HT or TH) = 0.42, and $P(X = 2) = 0.7 \times 0.7 = 0.49$ (i.e., HH). Note that the variable X connects each event (i.e., total number of heads = 0, 1, 2) and its corresponding probability (0.09, 0.42, 0.49). This variable is called a random variable. For a discrete random variable

Fig. 1.26 Probability mass function of a discrete random variable

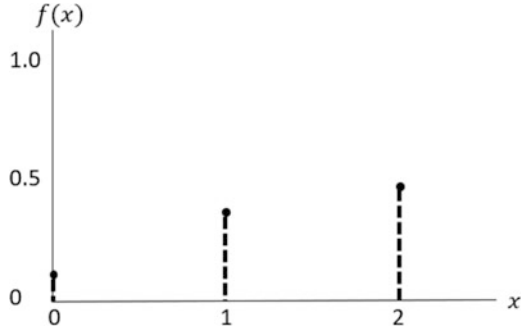
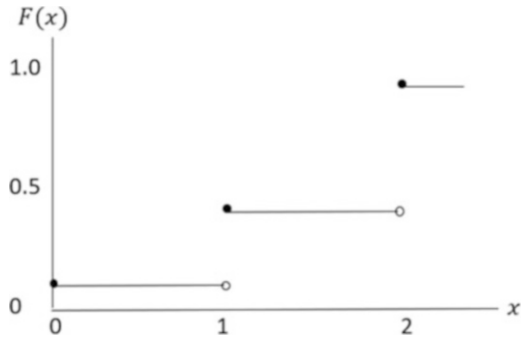


Fig. 1.27 Cumulative distribution function of a discrete random variable



x (i.e., the variable whose values can only be integers numbers such as this example) with its corresponding probability of occurrence, the set of ordered pairs $(x_i, f(x_i))$ for all i is called a **probability mass function** (Fig. 1.26).

A **cumulative distribution function** (in short, a distribution function) of a random variable, denoted by $F(x)$, is a function that gives, for each of a value of the random variable (lowercase letters are used to denote the specific values of the random variable), the probability that a newly observed value of the random variable falls at or below that value. In the case of the above biased coin flipping example, $F(x \leq 0) = 0.09$, $F(x \leq 1) = 0.09 + 0.42 = 0.51$, and $F(x \leq 2) = 0.09 + 0.42 + 0.49 = 1$ (Fig. 1.27). For a continuous random variable X , a **probability density function** of X (in short, density function), $f(x)$, is a positive function whose total area under the graph of the function is 1, and the area between two numbers (e.g., $a \leq x < b$) is the probability, denoted by $\Pr(a \leq X < b)$, that the values of the random variable are observable in this interval. In the case of the above discrete random variable, the density function will appear as a three-point function (probability mass function, Fig. 1.26) because there are only three possible values of the random variable, i.e., $x = 1, 2$, or 3 .

The **mean** and **variance** of a random variable are $\mu = \int_{-\infty}^{\infty} xf(x)dx$ and $\sigma^2 = \int_{-\infty}^{\infty} (x - \mu)^2 f(x)dx$. A **covariance** between two random variables, $X, Y \sim f_{xy}(x, y)$, is $\sigma_{xy} = \int_{-\infty}^{\infty} (x - \mu_x)(y - \mu_y)f_{xy}(x, y)dxdy$, where $f_{xy}(x, y)$ is the density function of the joint distribution of X and Y .

The **variance of a sum of two correlated** (with its correlation coefficient of ρ) **random variables**, X and Y , can be decomposed as variance of X + variance of Y + $2 \cdot$ covariance (X, Y), and its algebraic expression is $\sigma_{x+y}^2 = \sigma_x^2 + \sigma_y^2 + 2\sigma_{xy} = \sigma_x^2 + \sigma_y^2 + 2\rho\sigma_x\sigma_y$.

1.5.1 Gaussian Distribution

The Gaussian distribution describes the continuous data generation mechanism, and it has important mathematical properties on which the applications of event probability computations and inference (see Chap. 2) rely. The name Gaussian refers to the mathematician Carl Friedrich Gauss who developed its mathematical model, which is also called normal distribution because the model describes the probability distributions of typical normal behaviors of continuous outcomes (bell curve) well. This distribution has characteristics that the mean, median, and mode are identical, and the data are largely aggregated around the central location and gradually spread symmetrically. A Gaussian distribution is completely characterized by the mean and standard deviation, and its notation is $N(\mu, \sigma^2)$ where μ and σ denote the values of mean and standard deviation (σ^2 denotes the variance), respectively.

1.5.2 Probability Density Function of Gaussian Distribution

Probability density is the concentrated quantity of a particular value of the possible data range of a continuous outcome, and this quantity is proportional to the probability of occurrence within a neighborhood of that value. The mathematical expression is:

$$f(x) = \frac{1}{\sqrt{2\pi}\sigma} e^{-\frac{(x-\mu)^2}{2\sigma^2}}, \text{ for } -\infty < x < \infty.$$

The height of the symmetric bell curve is the size of density (not the actual probability) concentrated over the values of the continuous outcome random variable x . The value of the random variable x where the density peaks (central location) and the dispersion (i.e., how the loaded density gets spread out) are completely determined by the mean and standard deviation of the distribution, respectively. The area under the density curve which is calculated by integrating the curve from one standard deviation below the mean to one standard deviation above the mean, i.e., $\int_{\mu-\sigma}^{\mu+\sigma} \frac{1}{\sqrt{2\pi}\sigma} e^{-\frac{(x-\mu)^2}{2\sigma^2}} dx$, is approximately 68.3%, meaning that a little bit over middle two thirds of the group is aggregated symmetrically within one standard deviation around the mean of any Gaussian distribution. The area under the entire density

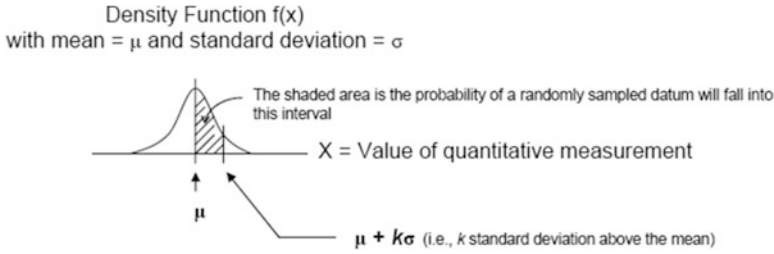
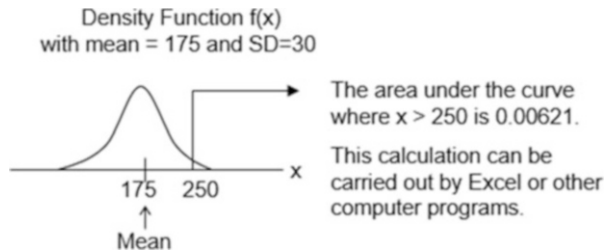


Fig. 1.28 Area under a Gaussian density curve within a closed interval

Fig. 1.29 Evaluation of an upper tail probability (a Gaussian density curve example)



curve is 1, i.e., $\int_{-\infty}^{\infty} \frac{1}{\sqrt{2\pi}\sigma} e^{-\frac{(x-\mu)^2}{2\sigma^2}} dx = 1$. Figure 1.28 describes the probability that a Gaussian random variable $x \sim N(\mu, \sigma^2)$ will be observed between its mean and k times the standard deviation above the mean, i.e., $\mu \leq x \leq \mu + k\sigma$, which is evaluated as the area under the density curve with this interval, i.e., $\int_{\mu}^{\mu+k\sigma} \frac{1}{\sqrt{2\pi}\sigma} e^{-\frac{(x-\mu)^2}{2\sigma^2}} dx$.

1.5.3 Application of Gaussian Distribution

The Gaussian (normal) distribution model is a very useful tool to approximately calculate the probability of observing a certain numerical range of events. The example shown in Fig. 1.29 is to find out the proportion of a large group of pediatric subjects whose serum cholesterol level is above 250 mg/mL if the group's cholesterol distribution follows a normal distribution with a mean of 175 and a standard deviation of 30. Because the standard deviation is 30, the value of 250 is 2.5 times the standard deviation above the mean (i.e., $250 = 175 + 2.5 \times 30$). The area under the curve that covers the cholesterol range > 250 is 0.625%, which indicates the subjects with cholesterol level > 250 are within top the 0.625% portion. The calculation requires the integration of the Gaussian density function equation. However, we can obtain the result using Excel or standard probability tables of Gaussian distribution. The next section discusses how to calculate the probability using the tables by transforming any Gaussian distribution to the standard normal distribution.

1.5.4 Standard Normal Distribution

The standard normal distribution is the Gaussian distribution whose mean is 0 and standard deviation is 1, i.e., $N(0, 1)$. A random variable x following a Gaussian distribution can be standardized by the following transformation. In the following equation, x is the random variable that represents a value of the original Gaussian distribution, its mean μ , and standard deviation σ , and z represents the new random variable resulting from the following transformation:

$$z = \frac{x - \mu}{\sigma}.$$

This transformation shifts the entire data set uniformly by subtracting μ from all individual values and rescales the already shifted data values by dividing them by the standard deviation; thus, the transformed data will have mean 0 and standard deviation 1.

The standard normal distribution has several useful characteristics on which the statistical inference rely (this will be discussed in Chap. 2). First, as shown above, the density is symmetrically distributed over the data range resembling a bell-like shape. Moreover, one standard deviation below and above the mean, i.e., the interval from -1 to 1 on z , covers approximately 68.3% of the distribution symmetrically. The interval of z from -2 to 2 (i.e., within two standard deviation units symmetrically around the mean) covers approximately 95.5% of the distribution. The normal range, -1.96 to 1.96 on z which covers 95% of distribution around the mean, is frequently sought (Fig. 1.30).

Figures 1.31 and 1.32 illustrate the utilization of Table 10.1 to compute the probability evaluated within a certain interval without using a computer program. For example, let's calculate the probability that an observed value (x) will be 250 or greater if its occurrence mechanism follows a Gaussian probability model with a mean (μ) of 175 and a standard deviation (σ) of 30. The probability is evaluated by a $z = \frac{x - \mu}{\sigma}$ transformation, i.e., $Pr(x \geq 250) = Pr(\geq \frac{250 - 175}{30}) = Pr(z \geq 2.5) = 0.0062$.

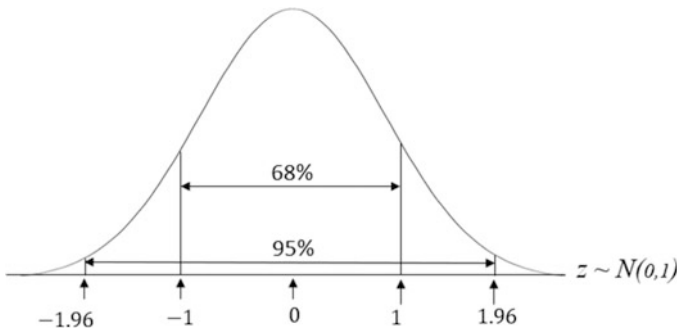


Fig. 1.30 Proportion of standard normal distribution covered by 1 (and 1.96) unit of standard deviation above and below the mean

Fig. 1.31 List of selected normal random variates and cumulative probabilities up to those values

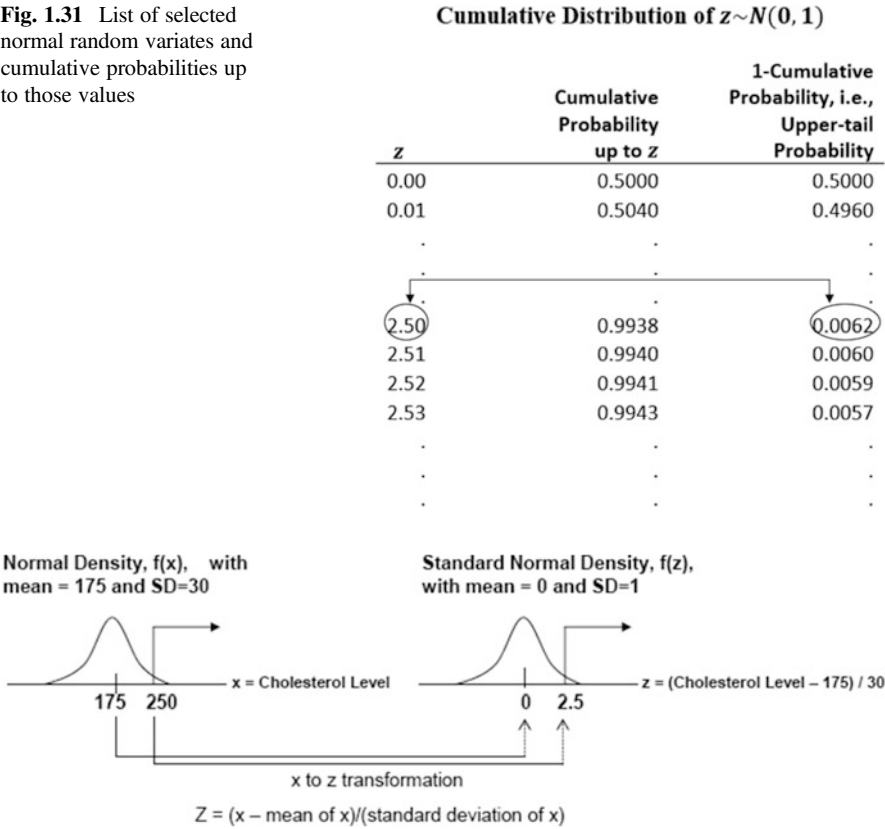


Fig. 1.32 Standardization of an observed value $x = 250$ from $N(\mu = 175, \sigma^2 = 30^2)$ to $z = 2.5$ of standardized normal distribution, i.e., $N(0, 1)$

1.5.5 Binomial Distribution

The probability values that are distributed to the finite counts of dichotomous event outcomes (e.g., success or failure) are typically modeled by binomial distribution. For demonstration purpose, let us consider the following situation. Suppose that it is known that a new investigative therapy can reduce the volume of a certain type of tumor significantly, and the average success rate is 60%. What will be the probability of observing four or more successful outcomes (i.e., significant tumor volume reduction) from a small experiment treating five animals with such a tumor if the 60% average success rate is true? First, calculate the probabilities of all possible outcomes under this assumption, i.e., no success, 1 success, 2, 3, 4, or all 5 successes given the true average success rate is 60%. Note that a particular subject’s single

result should not alter the next subject's result, i.e., the resulting outcomes are independent on experimental animals. In this circumstance, a probability distributed to the single dichotomous success event outcome (shrunk tumor as the success or no response as the failure) of each animal is characterized by Bernoulli distribution with its parameter π , which is the probability of success in a single animal treatment (i.e., the two probabilities are π , the success rate, and $1 - \pi$, the failure rate). The single trial, in this case, each trial is a treatment given to each animal, is called Bernoulli trial. The resulting probability distribution of the total number of successes out of those five independent treatment series (i.e., five independent Bernoulli trials) is then described by binomial distribution, which is characterized by two parameters in which the first parameter is the total number of Bernoulli trials, n , and the second parameter is the Bernoulli distribution's parameter of the success rate, π . In this example, the total number of independent trials, n , is 5 and the parameter of the success rate, π , on each single trial Bernoulli distribution is 0.6. Table 1.4 lists all possible results and their probabilities (0 = failure with its single occurring chance of 0.4, 1 = success with its single occurring chance of 0.6). As shown in the last column of the table, these calculated probabilities are 0.0102 for 0 successes (i.e., all failures and the probability of such an event is $0.4 \times 0.4 \times 0.4 \times 0.4 \times 0.4 = 0.0102$), 0.0768 for 1 success, 0.2304 for 2 successes, 0.3456 for 3 successes, 0.2592 for 4 successes, and 0.0778 for all 5 successes. The general notation of a binomial distribution is $X \sim Bi(n, \pi)$. In this example, by letting X denote the number of successes, the notation is $X \sim Bi(5, 0.6)$. Let us also note that the Bernoulli distribution is a special case of binomial distribution, and its general notation is $Bi(1, \pi)$. Figure 1.33 displays $Bi(5, 0.6)$. Thus, the probability of observing four or more successes out of the treatments given to five independent animals is $0.2592 + 0.0778 = 0.3370$. The probability value of $X = x$ of $Bi(n, \pi)$ is probability of ($X = x$ successes out of n independent Bernoulli trials) $= K\pi^x(1 - \pi)^{n-x}$, where $K = \binom{n}{x}$ is an integer value multiplier that is the count of all possible assortments of the number of success events x ($x = 0, 1, \dots, n$). Readers who are familiar with combinatorics can easily figure out $\binom{n}{x} = n! / [x!(n-x)!]$. In Table 1.4, ($n = 5$), $K = 1$ for $x = 0$, $K = 5$ for $x = 1$, $K = 10$ for $x = 2$, $\dots$, and $K = 1$ for $x = 5$. It is straightforward that the expression of $Bi(1, \pi)$ is the probability of ($X = x$ out of 1 Bernoulli trials) $= \pi^x(1 - \pi)^{1-x}$, where $x =$ either 1 (for success) or 0 (failure).

While the binomial distribution fits the probability of success counts arising from a fixed number of independent trials, when the event of interest is not rare (i.e., π is not very small) and the size of the trial, n , becomes large, then the probability calculation for a range of a number of success events can be conveniently approximated by using the Gaussian distribution even if the number of successes is not continuous. Figure 1.34 demonstrates the rationale for such an application. In general, for $n \times \pi \geq 5$ (i.e., the expected number of successes is at least 5), if n becomes large for a given π , or π becomes large for a given n , then the distributed probability pattern of binomial distributions becomes closer to $N(\mu = n \times \pi, \sigma^2 = n \times \pi \times (1 - \pi))$.

Table 1.4 Binomial distribution of with $n = 5$ and $\pi = 0.6$

Number of successes $X = x$ out of $n = 5$ independent trials, $K: \binom{n=5}{x=0, 1, 2, 3, 4, 5}$	Number of assortments	Events of Bernoulli trials					Probability
		1st	2nd	3rd	4th	5th	
$\binom{5}{0}$	1 assortment	0	0	0	0	0	$0.4 \times 0.4 \times 0.4 \times 0.4 \times 0.4 = 0.6^0 \times 0.4^5$ (subtotal = 0.0102)
$\binom{5}{1}$	5 assortments	1	0	0	0	0	$0.6 \times 0.4 \times 0.4 \times 0.4 \times 0.4 = 0.6 \times 0.4^4$
		0	1	0	0	0	$0.4 \times 0.6 \times 0.4 \times 0.4 \times 0.4 = 0.6 \times 0.4^4$
		0	0	1	0	0	$0.4 \times 0.4 \times 0.6 \times 0.4 \times 0.4 = 0.6 \times 0.4^4$
		0	0	0	1	0	$0.4 \times 0.4 \times 0.4 \times 0.6 \times 0.4 = 0.6 \times 0.4^4$
		0	0	0	0	1	$0.4 \times 0.4 \times 0.4 \times 0.4 \times 0.6 = 0.6 \times 0.4^4$ (subtotal = 0.0768)
$\binom{5}{2}$	10 assortments	1	1	0	0	0	$0.6 \times 0.6 \times 0.4 \times 0.4 \times 0.4 = 0.6^2 \times 0.4^3$
		1	0	1	0	0	$0.6 \times 0.4 \times 0.6 \times 0.4 \times 0.4 = 0.6^2 \times 0.4^3$
		1	0	0	1	0	$0.6 \times 0.4 \times 0.4 \times 0.6 \times 0.4 = 0.6^2 \times 0.4^3$
		1	0	0	0	1	$0.6 \times 0.4 \times 0.4 \times 0.4 \times 0.6 = 0.6^2 \times 0.4^3$
		0	1	1	0	0	$0.4 \times 0.6 \times 0.6 \times 0.4 \times 0.4 = 0.6^2 \times 0.4^3$
		0	1	0	1	0	$0.4 \times 0.6 \times 0.4 \times 0.6 \times 0.4 = 0.6^2 \times 0.4^3$
		0	1	0	0	1	$0.4 \times 0.6 \times 0.4 \times 0.4 \times 0.6 = 0.6^2 \times 0.4^3$
		0	0	1	1	0	$0.4 \times 0.4 \times 0.6 \times 0.6 \times 0.4 = 0.6^2 \times 0.4^3$
		0	0	1	0	1	$0.4 \times 0.4 \times 0.6 \times 0.4 \times 0.6 = 0.6^2 \times 0.4^3$
		0	0	0	1	1	$0.4 \times 0.4 \times 0.4 \times 0.6 \times 0.6 = 0.6^2 \times 0.4^3$ (subtotal = 0.2304)
$\binom{5}{3}$	10 assortments	0	0	1	1	1	$0.4 \times 0.4 \times 0.6 \times 0.6 \times 0.6 = 0.6^3 \times 0.4^2$
		0	1	0	1	1	$0.4 \times 0.6 \times 0.4 \times 0.6 \times 0.6 = 0.6^3 \times 0.4^2$
		0	1	1	0	1	$0.4 \times 0.6 \times 0.6 \times 0.4 \times 0.6 = 0.6^3 \times 0.4^2$
		0	1	1	1	0	$0.4 \times 0.6 \times 0.6 \times 0.6 \times 0.4 = 0.6^3 \times 0.4^2$
		1	0	0	1	1	$0.6 \times 0.4 \times 0.4 \times 0.6 \times 0.6 = 0.6^3 \times 0.4^2$

(continued)

Table 1.4 (continued)

Number of successes $X = x$ out of $n = 5$ independent trials, $K: \binom{n=5}{x=0, 1, 2, 3, 4, 5}$	Number of assortments	Events of Bernoulli trials					Probability
		1st	2nd	3rd	4th	5th	
$\binom{5}{4}$	5 assortments	1	0	1	0	1	$0.6 \times 0.4 \times 0.6 \times 0.4 \times 0.6 = 0.6^3 \times 0.4^2$
		1	0	1	1	0	$0.6 \times 0.4 \times 0.6 \times 0.6 \times 0.4 = 0.6^3 \times 0.4^2$
		1	1	0	0	1	$0.6 \times 0.6 \times 0.4 \times 0.4 \times 0.6 = 0.6^3 \times 0.4^2$
		1	1	0	1	0	$0.6 \times 0.6 \times 0.4 \times 0.6 \times 0.4 = 0.6^3 \times 0.4^2$
		1	1	1	0	0	$0.6 \times 0.6 \times 0.6 \times 0.4 \times 0.4 = 0.6^3 \times 0.4^2$ (subtotal = 0.3456)
		1	1	1	1	0	$0.6 \times 0.6 \times 0.6 \times 0.6 \times 0.4 = 0.6^4 \times 0.4^1$
$\binom{5}{3}$	5 assortments	1	1	1	0	1	$0.6 \times 0.6 \times 0.6 \times 0.4 \times 0.6 = 0.6^4 \times 0.4^1$
		1	1	0	1	1	$0.6 \times 0.6 \times 0.4 \times 0.6 \times 0.6 = 0.6^4 \times 0.4^1$
		1	1	0	1	1	$0.6 \times 0.6 \times 0.4 \times 0.6 \times 0.6 = 0.6^4 \times 0.4^1$
		1	0	1	1	1	$0.6 \times 0.4 \times 0.6 \times 0.6 \times 0.6 = 0.6^4 \times 0.4^1$
		0	1	1	1	1	$0.4 \times 0.6 \times 0.6 \times 0.6 \times 0.6 = 0.6^4 \times 0.4^1$ (subtotal = 0.2592)
		1	1	1	1	1	$0.6 \times 0.6 \times 0.6 \times 0.6 \times 0.6 = 0.6^5 \times 0.4^0$ (subtotal = 0.0776)
	1 assortment	1	1	1	1	1	

Fig. 1.33 Distribution of Bi
($n = 5, \pi = 0.6$)

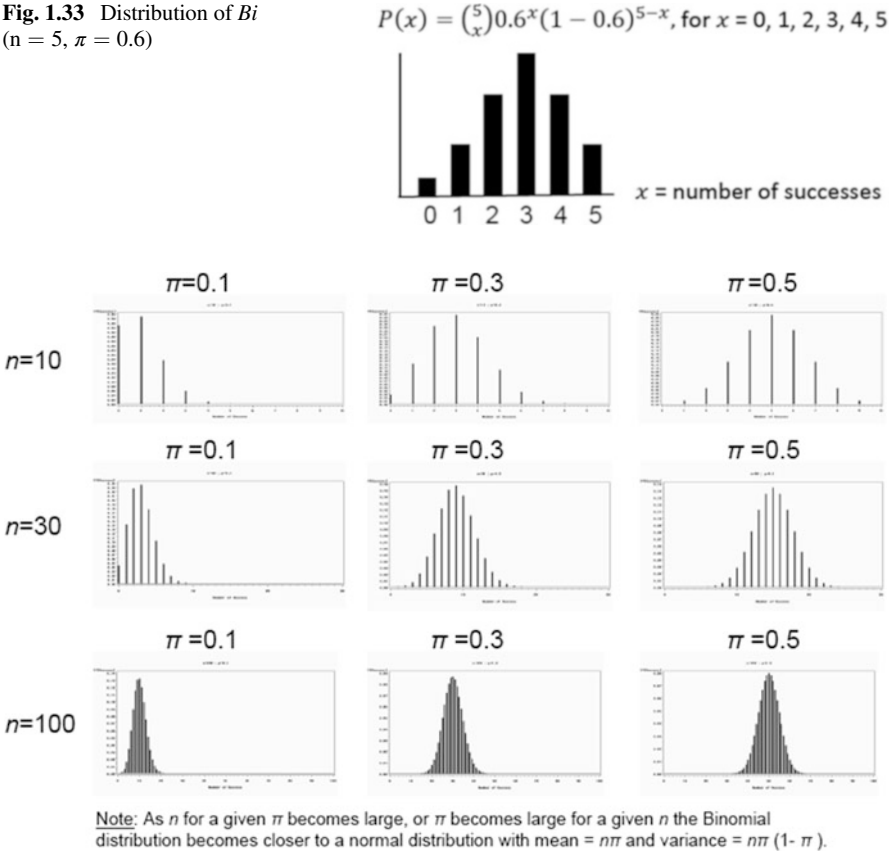


Fig. 1.34 Large sample behavior of binomial distributions with various trial sizes and success rates

Suppose that we now increased the number of animal experiment to 100, and we want to compute the probability of observing 50 to 75 successes arising from 100 independent trials. Because $n \times \pi = 100 \times 0.6 = 60$, and $n \times \pi \times (1 - \pi) = 100 \times 0.6 \times 0.4 = 24$, this task can be resorted to the normal approximation for which the used distribution is $N(\mu = 60, \sigma^2 = 24)$. Then as depicted in Fig. 1.35, the first step is to transform the interval from 50 to 75 on $N(\mu = 60, \sigma^2 = 24)$ to a new interval on $N(0, 1)$. Letting x denote the number of successes, according to $z = \frac{(x - \mu)}{\sigma}$ transformation, $x = 50$ is transformed to $z = (50 - \mu)/\sigma = (50 - 60)/\sqrt{24} = -2.05$, and $x = 75$ to $z = (75 - \mu)/\sigma = (75 - 60)/\sqrt{24} = 2.05$. So, the probability to observe between 50 and 75 successes is the area under the density curve of $N(0, 1)$ covering from -2.05 and 2.05 on z , which is 0.98.

On the other hand, when the event of interest is rare and the size of the trial becomes very large, then the computation can be approximated by Poisson model in which the number of trials is no longer an important constant (i.e., parameter) that

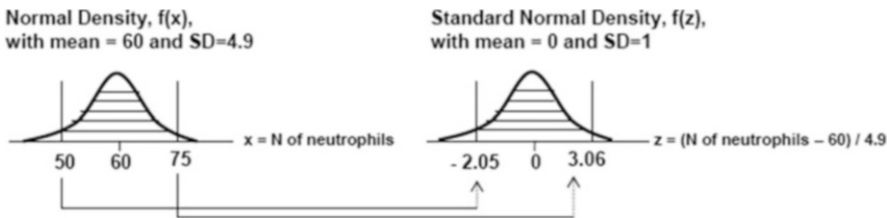


Fig. 1.35 Normal approximation to calculate the probability of observing 50–75 successes arising from 100 independent trials wherein the single Bernoulli trial’s success rate is 0.6

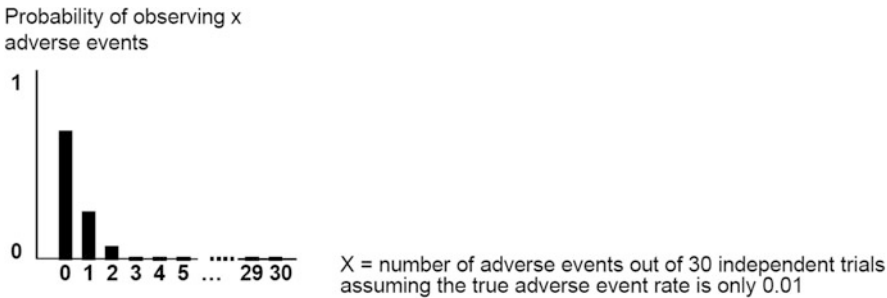


Fig. 1.36 Distribution (probability mass function) of Bi ($n = 30, \pi = 0.01$)

characterizes the Poisson distribution. The notation is $Pois(\lambda)$, where λ denotes the number of average successes of the rare event out of a large number of independent trials. An exemplary outcome that is well characterized by the Poisson model is the number of automobile accidents on a particular day in a large metropolitan city. The rare events can be the ones whose binomial characteristic constants are $n \times \pi < 5$ (i.e., expected number of successes is less than 5). The next example is a binomial distribution for which the probability calculation can be approximated by a Poisson distribution. Figure 1.36 displays the probabilities of observing 0, 1, 2, ..., 30 adverse events among 30 independent clinical trials of a new drug if the true adverse event rate = 0.01 (i.e., 1%). The typical pattern of Poisson distribution is that the probability value decreases exponentially after certain number of successes, and as the expected number of successes, $n \times \pi$, becomes smaller, the value decreases faster. If we let a computer calculate the probability of observing three or more adverse events out of 30 trials, then the result will be 0.0033. If we approximate this distribution to $Poisson(\lambda = 30 \times 0.01 = 0.3)$ and let a computer calculate such an event, the result will be 0.0035, which is not much different from the binomial model-based calculation.

Study Questions

1. What are the similarity and dissimilarity between the interval scale and ratio scale?
2. What is the definition of a distribution? What are being distributed in the distribution?
3. In a box-and-whisker plot, what proportion of the population is contained in the “box” interval? Is such a plot useful to describe a bimodal (i.e., two modes) distribution?
4. Please explain the definition of standard deviation.
5. What proportion of the data values are within one standard deviation above and below the mean if the data are normally distributed?
6. Can a correlation coefficient measure the strength of linear and nonlinear relationship between two continuous observations?
7. What are the definitions of odds and odds ratio?
8. What are the two parameters that completely characterize a Gaussian distribution?
9. What are the two parameters that completely characterize a binomial distribution?
10. Under what condition can a Gaussian model approximate the proportion of a population that lies within a certain range of number of events describable by a binomial model?

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Chapter 2

Statistical Inference Concentrating on a Single Mean



Statistical inference is to infer the population characteristics of interest through the observed sample data. If the whole collection of the population data were observed, then there would be no room for uncertainty about the true knowledge of the population, the statistical inference would be unnecessary, and the data analysis would be totally descriptive. For a real-world investigation, a smaller size of the sample data set than that of the whole population is gathered for inference. Since the sample data set does not populate the entire population and is not identical to the population, the inference using the sample data set becomes necessary. This chapter discusses the relationship between the population and sample by addressing (1) the uncertainty and errors in the sample, (2) underpinnings that are necessary for a sound understanding of the applied methods of statistical inference, (3) areas and paradigms of drawing inference, and (4) good study design to minimize/avoid inherent errors in the sample.

2.1 Population and Sample

2.1.1 *Sampling and Non-sampling Errors*

Let us first discuss, before addressing the statistical inference, several important phenomena and statistical concepts arising from using sample data. Suppose that our objective is to discover the average (i.e., mean) body weight of a large population ($N > \text{one million}$) of men and women. It is impractical to measure every individual in the population. Nonetheless, one can probably investigate by using a sufficiently large yet manageable size, n , of well-represented random sample.

Let us assume that a statistician helped you determine the study sample size, $n = 1000$ individuals, whose sample mean is to serve as a good estimate of the population's mean body weight. The following are two major sources of uncertainty involved in the sample. The first is the sampling error. These 1000 subjects were

randomly drawn with an equal chance to be selected. If another separate random sample set of 1000 subjects is taken, then these new 1000 randomly drawn subjects will not be exactly the same individuals as the initial 1000 subjects. The mean values of these two sets will differ from each other, thus both means will differ from the population mean. A discrepancy between the sample means and the true population mean explained solely by this supposition is understood as a sampling error. This sampling error will eventually disappear as the sample size becomes very large. One extreme example is that if the sample consisted of the entire population, then there is no such error.

The next is a non-sampling error. No matter how well the randomly drawn sample subjects represent the whole subjects in the population, it is still possible that the sample mean and true population mean can be discrepant due to other reasons. For instance, if an electronic scale that was used to measure weight systematically underestimated (e.g., always reads 0.1 lb. less than the truth) the true value (by accident) because of a mechanical problem, then the discrepancy between the observed sample mean and the true population mean due to such machine error is understood as the non-sampling error. This kind of error will never disappear even if the sample size becomes very large. While sampling error can be reduced by increasing the sample size, non-sampling error prevention would need a prudent sampling design (Fig. 2.1).

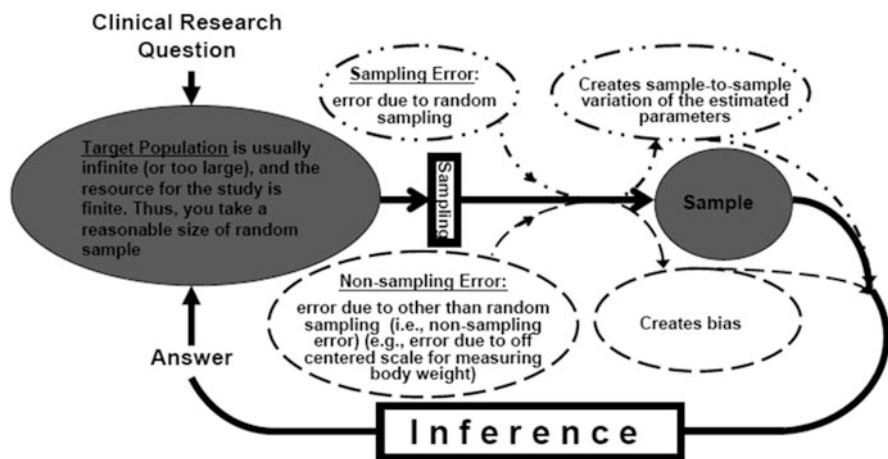


Fig. 2.1 Overview of inference using sample data

2.1.2 *Sample Distribution and Sampling Distribution*

The concepts of sample distribution and sampling distribution are the important bases of statistical inference. The following example demonstrates the sample and sampling distributions.

The following example demonstrates the sample- and sampling distributions. We are interested in the average neutrophil counts per 100 white blood cells of healthy adults in a large population. Assume that the healthy adults' neutrophil counts per 100 white blood cells are normally distributed with a mean of 60 and standard deviation of 5 in this population. Let us also assume that there is no non-sampling error in this sampling.

A random sample of 30 individuals was taken from a large healthy adult population, and the sample mean was calculated. Would this sample mean be exactly 60? Moreover, if 20 research groups, including yours, had taken their random samples with the same sample size (i.e., $n = 30$) independently from the same population, then those sample means of yours and your colleagues will differ. Figure 2.2 A.1 illustrates how the histograms of these 20 individual sample sets would appear, and Fig. 2.2 A.2 illustrates how the 20 respective sample means would vary. A result from the same kind of experiment except for choosing a larger sample size ($n = 300$) is also demonstrated (Fig. 2.2 B.1 and B.2).

In Fig. 2.2 A.1, 20 sample distributions are described by histograms demonstrating the observed distribution of each random sample set with a sample size of 30 (so-called sample distribution). Each of the other 20 histograms in Fig. 2.2 B.1 summarizes each random sample's observed distribution with an increased sample size of 300. Note that the sample distributions with 10-fold larger increased sample size are more reflective of the population's distribution.

Figure 2.2 A.2 depicts the sampling distribution of the 20 sample means in Fig. 2.2 A.1, and Fig. 2.2 B.2 depicts those of 20 sample means in Fig. 2.2 B.1. Note that the observed sampling distribution of the sample means drawn from the 20 repeated samples of size 300 (Fig. 2.2 B.2) is much less dispersed than that from the 20 repeated samples of size 30 (Fig. 2.2 A.2). The standard deviations of these two sampling distributions provide very good sense of how the sample means vary over the repeated random samplings.

2.1.3 *Standard Error*

The standard deviation of a sampling distribution measures the variability of the sample means that vary over the independently repeated random samplings. This standard deviation is called the standard error of the sample mean, $s_{\bar{x}}$. Note that in real-life research, it is unrealistic to draw multiple samples in order to describe the sampling distribution as demonstrated in Section 2.1.2. The above example was to facilitate conceptualization of the sample- and sampling distributions. The

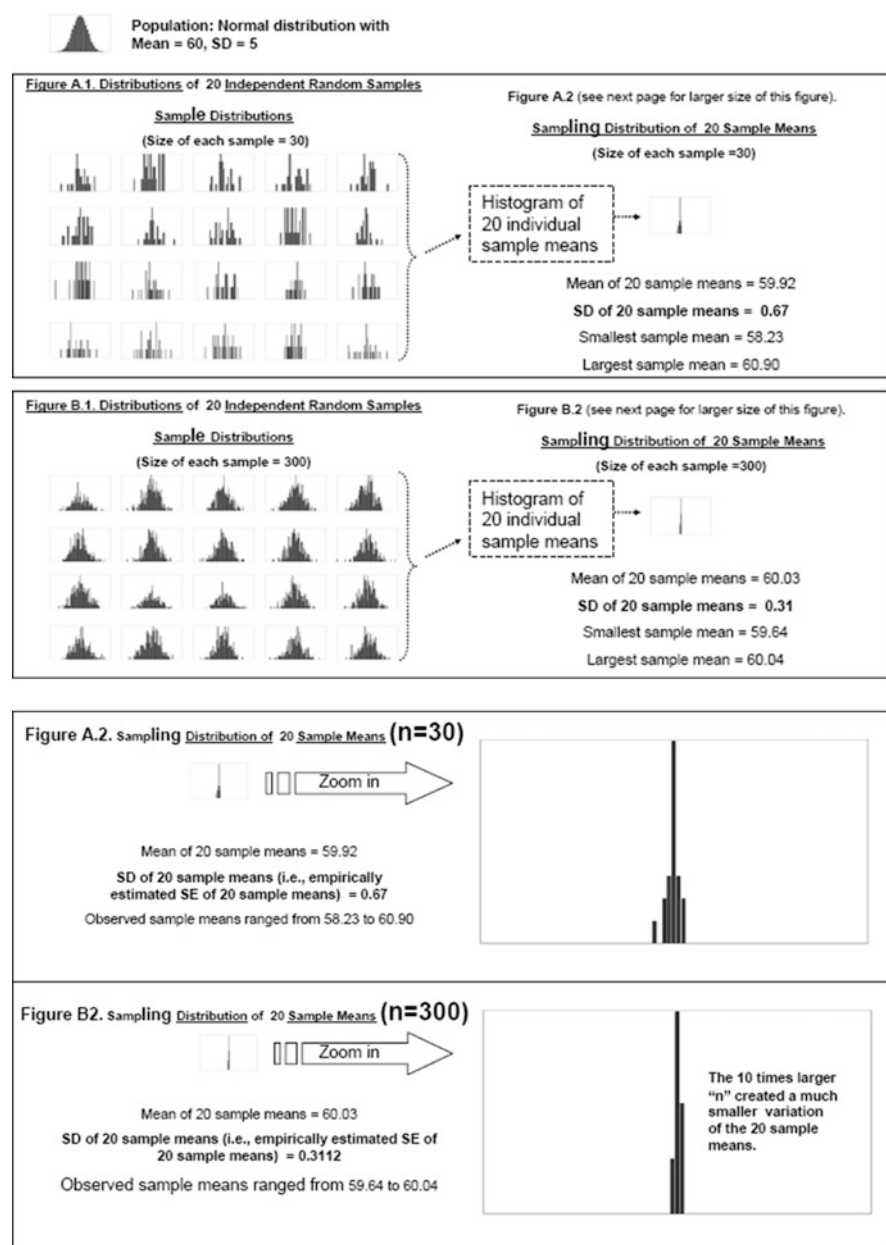


Fig. 2.2 Sample distributions of random sampled data sets from a normal distribution and the sampling distribution of their sample means

Table 2.1 Sample- and sampling distributions

	Sample distribution	Sampling distribution
Distribution of	Only one sample data set	Sample means being obtained over multiple sets of random samples
As the sample size increases the shape of distribution	Becomes closer to the population distribution	Becomes symmetrical and narrower
Name (and notation) of the dispersion statistic is	Sample standard deviation (s)	Standard error (SE) of the mean, $s_{\bar{x}}$
Relationship between s and $s_{\bar{x}}$	$s = \sqrt{n}s_{\bar{x}}$	$s_{\bar{x}} = s/\sqrt{n}$

investigators draw only one sample set, and there is uncertainty about how far/close the obtained sample mean is from/to the true population mean (i.e., such a presentation depicted by Fig. 2.2 cannot be made). There is a mathematical equation (but it is not derived in this monograph) that allows the investigators to be able to utilize to estimate the standard error of mean based solely on the observed sample data set, which is to simply divide the sample standard deviation, s , by the square root of its sample size, n , i.e., $s/\sqrt{n}$. In the above illustration, a standard error of the sample mean can be estimated by directly applying $s/\sqrt{30}$ to any of the 20 sample distributions with sample size of 30, which would be very close to 0.67 (or 0.3112 for $n = 300$) with a small sample-to-sample variation. Such an estimated standard error is very close to the standard deviation of the 20 observed sample means from the experiment (such an experiment is called a Monte Carlo simulation). Table 2.1 summarizes the distinction between the sample- and sampling distributions as well as the relationship between the sample standard deviation and standard error.

2.1.4 Sampling Methods and Sampling Variability of the Sample Means

What makes a sampling method unique is the sampling probability, i.e., the probability that an individual sampling unit is assigned to be sampled. For example, the sampling methods can be classified into the equal probability sampling and the unequal probability sampling. Intuitively, random sampling with replacement (i.e., a sampling unit is sampled with equal probability) is equal probability sampling, and the random sampling without replacement is unequal probability sampling. There can be many forms of unequal probability sampling methods. Each sampling method will offer a unique latitude of accuracy and precision of the estimator. This section discusses some useful relationships between the sampling probability and accuracy and precision of the sample mean estimate, i.e., how the bias and sampling variance of the sample mean can be reduced by unequal probability sampling techniques.

Consider a small experiment in which you place four balls each marked with two-digit numbers 12, 23, 34, and 45 in an urn container. You draw a ball, read the

Table 2.2 Sampling two balls out of four in a jar with replacement and the calculated means of the observed ordered pairs

$(y_{(1)}, y_{(2)})$	$\bar{y}$	$(y_{(1)}, y_{(2)})$	$\bar{y}$
(12, 12)	12	(34, 12)	23
(12, 23)	17.5	(34, 23)	28.5
(12, 34)	23	(34, 34)	34
(12, 45)	28.5	(34, 45)	39.5
(23, 12)	17.5	(45, 12)	28.5
(23, 23)	23	(45, 23)	34
(23, 34)	28.5	(45, 34)	39.5
(23, 45)	34	(45, 45)	45

number, return it to the container (i.e., random sampling with replacement), then draw a ball again to read the drawn number (i.e., sampling sequentially two balls with replacement, i.e., a random sampling with replacement with its sample size of 2), and then finally calculate the mean of the two observed numbers. Table 2.2 is the complete list of all possible 16 samples and their means of the ordered observed pairs $(y_{(1)}, y_{(2)})$.

The mean of those 16 sample means is $(12.00 + 17.00 + \dots + 45.00)/16 = 28.5$, and it is the same as the population mean of the four numbers in the container, i.e., $(12 + 23 + 34 + 45)/4 = 28.5$. It is indeed expected that the mean of all possible 16 sample means observed through the above experiment must be the same as the true population mean. The arithmetic mean of those sample means over the whole sampling distribution is called the **expectation** (or expected value) of the sample mean. Likewise, the variance of the sample means can be calculated as the arithmetic mean over the squared deviations of every sample mean from the expectation, i.e., $[(12-28.5)^2 + (17.5-28.5)^2 \dots + (45-28.5)^2]/25 = 75.625$.

Now, let us consider another experiment, but at this time let us increase the chance for 45 being sampled (this will be possible if we duplicate this ball so that 12, 23, 34, 45, and an extra 45 will be placed in the container). Then, let us conduct the experiment of sampling two balls with replacement again. With respect to the unique face value of the drawn ball, this experiment is an unequal chance sampling, in that the chances for the four unique values to be sampled are unequal, i.e., each draw will have a 20% chance to pick 12, 23, or 34, but 40% chance to pick 45. Table 2.3 is the complete list of all possible 25 sample means of the ordered pairs $(y_{(1)}, y_{(2)})$.

This presentation can be modified as is shown below (see Table 2.4), in that the events are combined commutatively (e.g., 12, 23 represents two events (12, 23) and (23, 12)).

The arithmetic mean of all these 25 sample means is $(17.5 \times 2 + 23 \times 2 + 28.5 \times 4 + 28.5 \times 2 + 34 \times 4 + 39.5 \times 4 + 12 \times 1 + 23 \times 1 + 34 \times 1 + 45 \times 4)/25 = 31.8$. It is obvious that this result differs from the result of the first experiment because 45 was drawn twice more frequently than other numbers. The variance (i.e., arithmetic mean of the squared deviations of the sample means from the expected value) of these means is $[(17.5-31.8)^2 \times 2 + (23-31.8)^2 \times 2 + (28.5-31.8)^2 \times 4 + (28.5-31.8)^2 \times 2 + (34-31.8)^2 \times 4 + (39.5-31.8)^2 \times 4 + (12-31.8)^2$

Table 2.3 Sampling two balls, with unequal chances, out of four in a jar with replacement and the calculated means of the observed ordered pairs

$(y_{(1)}, y_{(2)})$	$\bar{y}$	$(y_{(1)}, y_{(2)})$	$\bar{y}$
(12, 12)	12	(34, 45)	39.5
(12, 23)	17.5	(34, 45)	39.5
(12, 34)	23	(45, 12)	28.5
(12, 45)	28.5	(45, 23)	34
(12, 45)	28.5	(45, 34)	39.5
(23, 12)	17.5	(45, 45)	45
(23, 23)	23	(45, 45)	45
(23, 24)	28.5	(45, 12)	28.5
(23, 45)	34	(45, 23)	34
(23, 45)	34	(45, 34)	39.5
(34, 12)	23	(45, 45)	45
(34, 23)	28.5	(45, 45)	45
(34, 34)	34		

Table 2.4 Commutatively combined ordered pair events listed in Table 2.3

y_1, y_2	$\bar{y}$	Frequency
12, 23	17.5	2
12, 34	23	2
12, 45	28.5	4
23, 34	28.5	2
23, 45	34	4
34, 45	39.5	4
12, 12	12	1
23, 23	23	1
34, 34	34	1
45, 45	45	4

times; $1 + (23-31.8)^2 \times 1 + (34-31.8)^2 \times 1 + (45-31.8)^2 \times 4/25 = 82.28$. Can we invent a method of the sample mean estimation so that it would offer the sample mean as if the chance for 45 to be sampled were the same as that of the other four balls (i.e., 20%)? Because the chance for each unique number to be sampled is known (i.e., 20% for 12, 23, and 34, and 40% for 45), can the sample mean calculation of the two numbers at each draw be modified by assigning less weight to 45 than the weight assigned to 12, 23, or 34? One option is to assign such a weight to every drawn number that is inversely proportional to the chance for each unit to be sampled (so-called inverse probability weight, IPW). The new mean, $\bar{y}_{ipw}$ is the inverse probability weighted mean of the two numbers while ignoring the drawing order. This new weighted mean is $(y_1/p_1 + y_2/p_2)/2$ instead of $(y_1 + y_2)/2$, where p_1 and p_2 are the sample probability for each draw. This calculation is intuitively straightforward but produces the weighted sum, not the mean. Thus, to obtain the weighted mean, it is necessary for the weighted sum to be multiplied by 0.25, which is the fair chance per draw from the original four balls (i.e., the chance to draw 12, 23, 34, or 45), and $\bar{y}_{ipw} = 0.25 \times (y_1/p_1 + y_2/p_2)/2$ (See Table 2.5).

Table 2.5 Application of inverse probability weight to calculate the mean of each observed event

y_1, y_2	$\bar{y}_{ipw}$	Frequency
12, 23	21.875	2
12, 34	28.75	2
12, 45	21.5625	4
23, 34	35.625	2
23, 45	28.4375	4
34, 45	35.3125	4
12, 12	15	1
23, 23	28.75	1
34, 34	42.5	1
45, 45	28.125	4

Note that the expected value of the $\bar{y}_{ipw}$ is now 28.5, i.e., $\bar{y}_{ipw} = (21.875 \times 2 + 28.75 \times 2 + \dots + 28.125 \times 4)/25$, and this is the same as the expectation of the sample means demonstrated in Table 2.2. What about the variance of the IPW-based 25 sample means? It is $[(21.875 - 28.5)^2 \times 2 + (28.75 - 28.5)^2 \times 2 + \dots + (28.125 - 28.5)^2 \times 4]/25 = 37.86$. In this example, the IPW estimate of the sample mean was unbiased (i.e., the expected mean was the same as that of the first experiment). Moreover, the sampling variance decreased as well. However, it is not always true that IPW can decrease the sampling variance.

2.2 Statistical Inference

In the remaining sections of this chapter, we will discuss necessary data reduction procedures for the inference. We will discuss two pillars of inference of which one is hypothesis testing, and the other is estimation. Then, we will also discuss briefly about two paradigms of statistical inference, frequentist inference and Bayesian inference.

2.2.1 Data Reduction and Related Nomenclatures

The procedure of the statistical inference can be viewed as an itinerant avenue that connects the sample to population. With a given sample data set, for both the hypothesis testing and estimation, the very first step to walk through that avenue is to reduce the sample data set into several descriptive summary statistics (i.e., extract the summary statistics out of the data set). Such an intervening step of operation is called data reduction. A descriptive data analysis being applied in the sample data for the purpose of making a statistical inference is a good example of data reduction.

A **parameter** is a measured characteristic of a population (e.g., mean age, mean blood pressure, proportion of women, etc.). A **statistic** is a measured characteristic

as a function (i.e., processed form) of sample data (e.g., sample mean age, sample mean of blood pressure, sample proportion of women, etc.). **Estimation** is the procedure to know the value of the population parameter of interest using the sample data. An **estimator** is a mathematical function of sample data that is used to estimate a parameter (e.g., $\bar{x} = [x_1 + x_2 + \dots + x_n]/n$, where $x_1 + x_2 + \dots + x_n$ is the sum of all observed values of variable x and n is the sample size). Note that an estimator is also a statistic. An **estimate** is a certain observed value of the estimator (e.g., mean year age estimate = 23, etc., i.e., a resulting value from data reduction).

2.2.2 Central Limit Theorem

The central limit theorem (CLT) is one of the important theoretical bases for inference. It describes the typical phenomenon of sample means (probably the most common central tendency statistic) arising from random sampling.

The demonstrated sampling experiments below will help the readers comprehend the meaning and usefulness of the CLT. The next two experiments are slightly different from the ones illustrated in Sect. 2.1.2 in that the population distribution of these sampling experiments is a continuous non-Gaussian distribution. The population distribution from which the samples are drawn is a bimodal (i.e., two modes) distribution, which often characterizes a distribution as a mixture of two subgroup distributions resulting in two subgroups clustered around two different central locations (Fig. 2.3).

Experiment 1: Draw 30 distinct random sample sets from the given population set with sample size = 25 and make 30 separate histograms for these individual sample distributions. Then make a histogram of the 30 sample means that are obtained from individual sample distributions.

Experiment 2: Repeat the same kind of experiment by increasing the sample size to 100 and create the histograms the same way as the previous experiment.

In Fig. 2.4, as expected, each sample distribution appeared similarly to the population distribution, and those from the sample size of 100 resembled the original population distribution more closely. Notably, the sampling distribution of the sample means drawn from the sample size of 100 appeared unimodal (i.e., one mode) and symmetrical (i.e., bell-like) although the population distribution was

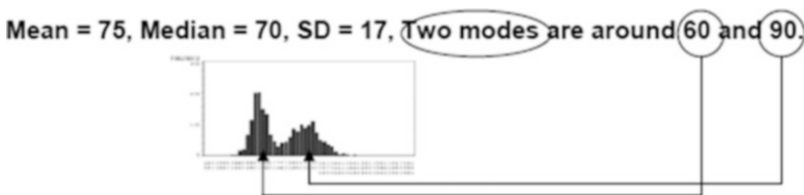
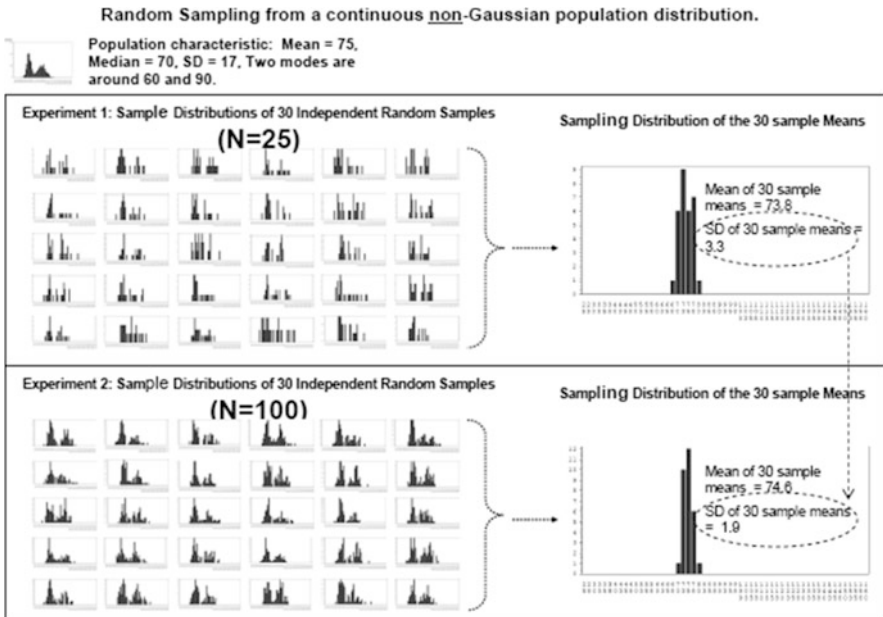


Fig. 2.3 Histogram of a bimodal distribution



1) With the larger sample size of $N=100$ the sample distributions became more reflective of the population distribution that is bimodal; 2) The two sampling distributions of the sample means are "bell" shaped.

Fig. 2.4 Sample distributions of randomly sampled data sets from a non-normal distribution and the sampling distributions of their sample means

bimodal. The dispersion of the sampling distribution of $n = 100$ decreased (standard error decreased from 3.3 to 1.9), which is the same phenomenon that was already observed in the previous experiment in Sect. 2.1.2 (i.e., decreased sampling error for the increased sample size).

Central Limit Theorem (CLT)

For a random variable that follows any continuous distribution having its finite** mean = μ and standard deviation = σ , if random samples are repeatedly taken over many (m) times independently where each sample size $n_k = n$ (for $k = 1, 2, \dots, m$), then the sampling distribution of the m sample means $\bar{x}_k$ (for $k = 1, 2, \dots, m$) will approach to a normal distribution with its population mean = μ , population mean, and standard deviation = $\sigma/\sqrt{n}$ as each sample size n increases infinitely.

** *Cauchy distribution does not have its finite mean and variance. The sampling distribution does not converge to a normal distribution as the sample size becomes large, and it is still Cauchy distribution.*

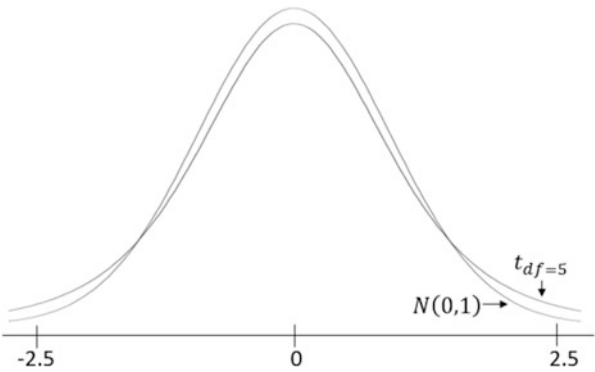
2.2.3 The *t*-Distribution

When the sample size, n , is not large, the sampling distribution of sample means $\bar{x}_k$ arising from the random sampling from a normally distributed population may not be well approximated by the normal distribution (more spread out than the normal distribution, and the CLT may not be fully applicable). Moreover, in many real-world research studies, their population standard deviations can be unknown. When the sample size is not large enough and the population standard deviation is also unknown, the random variable defined below:

$$t = \frac{(\bar{x} - \mu)}{s/\sqrt{n}} = \frac{\sqrt{n}(\bar{x} - \mu)}{s},$$

where s is the sample standard deviation, which will follow a Gossett’s student t -distribution (simply called t -distribution) with a parameter that is so-called degrees of freedom, df , which is not defined in the Gaussian distribution, wherein $df = n - 1$. The probability density curves of t -distributions have heavier tails than that of $N(0, 1)$ when df is small (see Fig. 2.5), and those become very close to it as the df becomes large (see Fig. 2.6).

Fig. 2.5 Probability density curve of $t_{df=5}$ and $N(0, 1)$



97.5th Percentiles of t -distributions with various degrees of freedom

Degree of Freedom	97.5 th Percentile
5	2.571
10	2.228
20	2.086
30	2.042
120	1.980
240	1.970
∞	1.96

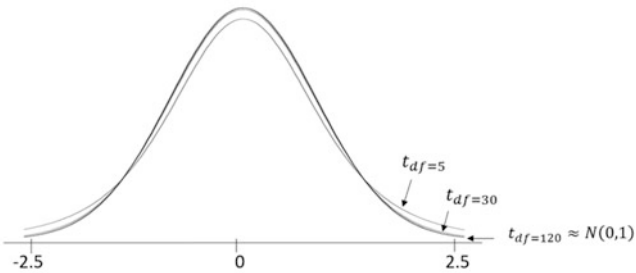


Fig. 2.6 t -distributions with small and large df_s versus $N(0, 1)$

In-depth understanding of the degrees of freedom requires ample knowledge of mathematical statistics and linear algebra, and the following is a simple explanation for the applied users. The degrees of freedom (df), whenever appears in this book, is understood as a parameter that characterizes a unique probability distribution (e.g., t , F , or χ^2 will be discussed later). In practice, finding out df is necessary for inference. The upcoming chapters and sections will concentrate the discussion of the degrees of freedom only on the minimally necessary knowledge about it while leaving out from this book the details of the calculation and letting the statistical analysis software package programs handle it. Besides the degrees of freedom, there is an additional parameter to characterize a t -distribution, but it has not been dealt with yet and will be introduced in Chap. 8 because the involvement of the non-centrality parameter is unnecessary until the power and sample size topic appears in Chap. 8. Until then, all t -distributions being dealt with are assumed to have their non-centrality parameter of 0 (central t -distribution).

2.2.4 Hypothesis Testing

As mentioned earlier in this chapter, the two pillars of statistical inference are **hypothesis testing** and **estimation**.

2.2.4.1 Frame of Hypothesis Testing

The main objective of a scientific investigation is to convince that a new discovery is different (improved) from what has been discovered in the past. Scientific investigations usually involve formal procedures consisting of articulating a research hypothesis about the anticipated new finding, designing a study, conducting the study (i.e., experiment or observation) and gathering data, and performing data analysis (i.e., making statistical inference) to reveal the data evidence that is beyond the reasonable doubt. The primary reason for performing data analysis is to conduct statistical inference. We will discuss the hypothesis testing first and then discuss the estimation (concentrating especially on the interval estimation).

Hypothesis testing requires stated hypotheses (a statement of the hypothesized facts) and a test rule in that the researcher states mutually contradictory hypotheses pair of which the first is to be ruled out if data evidence is not strong and the second is to be favorably admitted if data evidence is strong. For example, “a new intervention A is as efficacious as the currently available intervention B” can be such a form of the first hypothesis of the pair and “a new intervention A is more efficacious than the currently available intervention B” can be the second of the pair. Traditionally, the first hypothesis is denoted by H_0 and the second by H_1 . This is like a challenger H_1 is challenging H_0 with the observed data evidence.

Having stated H_0 and H_1 , the next step is to test if the data evidence favors H_0 or H_1 based on a rigorous and objective statistical rule. The test result can either rule out H_0 so that the study can finally pronounce that H_1 wins H_0 , or vice versa. The logical framework of claiming H_1 given the data evidence is not to let the data evidence prove the proposition H_1 directly, but it is rather to rule out H_0 if the observed data showed significant counter-evidence against H_0 . On the other hand, if the counter-evidence against H_0 was not significantly strong, then this logical framework lets the researcher return to H_0 (i.e., the previous scientific finding remains the same, so go back to the drawing board). Using such a procedure, a researcher can set H_1 as a new milestone if the data evidence was significant to reject H_0 (i.e., advancement over previous finding). Thus, H_0 is called the null hypothesis (i.e., back to null) and H_1 is called the alternative hypothesis.

In this procedure, the test rule-based final decision is to reject the null hypothesis, H_0 , or to fail to reject it. Here, “fail to reject” H_0 is not synonymous with “accept” H_1 in that the observed data cannot be absolutely certain and perfect because the observed sample data always involve uncertainty (i.e., sampling and non-sampling errors), and the sample data can never be able to prove either hypothesis. One may argue why we do not carry out an inference involving only H_1 as a single hypothesis and let the study data directly prove it? Justifications can be made from purely mathematical to very pragmatic manners. One pragmatic justification can be that the H_0 versus H_1 approach always directs the researchers to the next study plan because the approach offers two options either returning to H_0 or proceeding with H_1 . The analogy is the logical framework of the courtroom trial. A defendant remains innocent (H_0) if there are insufficient factual evidence, otherwise, they become guilty (H_1) if there were sufficient factual evidence.

2.2.4.2 Step-by-Step Overview of Hypothesis Testing

This is a formulae-less overview of the hypothesis testing. For the sake of convenience, the flow can be broken down into five steps. Note that such a breakdown is quite arbitrary and made for convenience.

Step 1: Stating null (H_0) and alternative (H_1) hypotheses. This step usually takes place at the beginning of the study (i.e., protocol development stage). The study investigator translates the research hypotheses into the statistical hypotheses and writes them in the statistical analysis plan section of the protocol.

Step 2: Establishing the test rule (decision rule to determine the significance of the observed data evidence to reject the null hypothesis). This step also usually takes place at the protocol development stage. The study investigator articulates the decision rule (i.e., method of a test) in the statistical analysis plan section of the protocol.

Step 3: Collecting data (i.e., conducting clinical study according to the written protocol) and data reduction (i.e., performing data analysis to obtain sample statistics).

- Step 4:** Data analysis (i.e., apply the test rule established in Step 2 to the collected data).
- Step 5:** Making interpretation and report writing.

2.2.4.3 Stating Null and Alternative Hypotheses

The following four pairs of null and alternative hypotheses are typical setups of the hypotheses for one mean inference (Table 2.6).

2.2.4.4 How to Phrase the Statistical Hypotheses

A hypothesis is a statement about a fact in the population (i.e., not a statement about the sample), and the standard format of a written statistical hypothesis is that its tense is present and the language “statistically significant” should not be in the sentence. Table 2.7 exhibits some examples of improper phrases.

2.2.4.5 Significance of the Test

Having stated the null and alternative hypotheses, the researcher collects data that would or would not disprove the null hypothesis. While there are many important

Table 2.6 Null and alternative hypotheses

Null hypothesis	Alternative hypothesis	Simple or composite ^a		Directionality of composite hypothesis
		Null	Alternative	
$H_0: \mu = \mu_0$ Mean is equal to μ_0	$H_1: \mu = \mu_1$ Mean is equal to μ_1	Simple	Simple	N/A
$H_0: \mu = \mu_0$ Mean is equal to μ_0	$H_1: \mu \neq \mu_0$ Mean is not equal to μ_0	Simple	Composite	Nondirectional (two-sided)
$H_0: \mu = \mu_0$ Mean is equal to μ_0	$H_1: \mu > \mu_0$ Mean is greater than μ_0	Simple	Composite	Directional (one-sided)
$H_0: \mu = \mu_0$ Mean is equal to μ_0	$H_1: \mu < \mu_0$ Mean is smaller than μ_0	Simple	Composite	Directional (one-sided)

Note: In some tests, the null hypothesis can be directional and composite. For example, $H_0: \mu \leq \mu_0$ vs. $H_1: \mu > \mu_0$ is such a case. For certain tests, $H_0: \mu = \mu_0$ vs. $H_1: \mu > \mu_0$ is used interchangeably without loss of generality. The readers may consult with an intermediate or advanced theory literature for more discussion

^aSimple hypothesis involves a single value of parameter; and composite hypothesis involves more than one value (e.g., an interval) of the parameter

Table 2.7 Examples of improper phrases for statistical hypotheses

Improper phrases	Reason
H ₀ : The sample mean is not different from 150 H ₁ : The sample mean is different from 150	Hypotheses are statements about the population, not about the sample
H ₀ : The mean is not statistically significantly different from 150 H ₁ : The mean is statistically significantly different from 150	The sentence must not include the wording “statistically significantly”
H ₀ : The mean will not be different from 150 H ₁ : The mean will be different from 150	Although research hypotheses are written in the future tense, the statistical hypotheses should be written in present tense because they are statements about the fact of the population of interest

constituents of the test, introducing all at once may create confusion. This section discusses the testing procedure as untechnically possible.

A **level of significance** (or significance level in short), α , is chosen by the researcher before the data analysis (i.e., this is not a result value from the observed data), and it determines the resilience of the test rule to reject the null hypothesis. This level is the maximally allowed error probability size that the test would reject the null hypothesis erroneously even though it should not be rejected. Such a decision error is called a Type 1 error. The level of significance, α , is also called test size. A common choice is $\alpha = 5\%$.

A **test statistic** is a sample statistic (see Sects. 1.3.1 and 2.2.1 for its definition) to gauge the strength of evidence opposing H_0 . Its value is calculated from the observed sample data and it varies from sample to sample (i.e., study to study). The phenomenon of this sample-to-sample variation of the test statistic is measured by the standard error of the test statistic (i.e., standard deviation of the sampling distribution of the test statistic). The strength of this evidence opposing H_0 is assessed by the relative extremity (i.e., how unlikely that the calculated test statistic value is observed) of the test statistic according to its sampling distribution (see Sect. 2.1.2 for its definition). The typical (with few exceptions) formulation of the test statistic is constructed by *Observed Estimate* ~ *Null Value* ~ *Standard Error (SE)* trio shown below:

$$\text{Test Statistic} = \frac{\text{Observed Estimate} - \text{Null Value}}{SE(\text{Observed Estimate} - \text{Null Value})} = \frac{\text{Signal}}{\text{Noise}}.$$

The numerator, *Observed Estimate* – *Null Value*, is the deviation of the sample estimate from the hypothesized parameter value specified in the null hypothesis. In this expression, for instance, for a test in which H_0 : mean $\mu = \mu_0$ versus H_1 : mean $\mu \neq \mu_0$, the *Observed Estimate* is the sample mean $\bar{x}$ and the *Null Value* is the parameter value μ_0 specified in the null hypothesis. The $SE(\cdot)$ is the standard error of the numerator, which is the sampling error of the observed difference of (*Observed*

Estimate – Null Value). The test statistic is the ratio of the former to the latter. The denominator of this expression can be further simplified in that since the *Null Value* is a hypothesized value (i.e., constant with no variation), the sampling variability of the *Observed Estimate – Null Value* will be the same as that of only the *Observed Estimate*. Thus, $SE(\text{Observed Estimate} - \text{Null Value}) = SE(\text{Observed Estimate})$, and:

$$\text{Test Statistic} = \frac{\text{Observed Estimate} - \text{Null Value}}{SE(\text{Observed Estimate})}$$

A test statistic with a value of 0 indicates the observed mean estimate is equal to the null value, 1 indicates the observed mean estimate's deviation from the null value is as large as its average random sampling error size (such a value does not indicate a significant deviation from H_0), -1 indicates the same degree of deviation from the null but to the opposite direction of the value 1 (such a value does not indicate a significant departure from H_0 either), and 3 indicates the deviation is three-fold greater than the average sampling error size (such a large value may indicate a significant deviation from H_0) (Fig. 2.7).

Note that this formulation unifies most of the common test statistics that are used for many kinds of comparison situations. This formulation can easily be extended to comparing a difference in two means to a specified null value of 0 (see Sect. 3.1).

Usually, the **name of a test** comes from the name of the sampling distribution of the test statistic. For example, *t*-test is the test in which the test statistic follows the *t*-distribution with a unique degree of freedom (it will be explained later that the

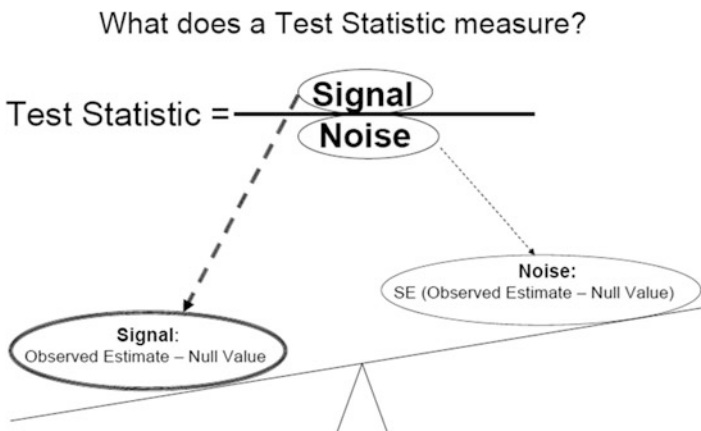


Fig. 2.7 Illustration of what a test statistic measures

degrees of freedom of the t -distribution are uniquely determined by the sample size, see Sect. 2.2.4.5).

A **p -value** is the probability of observing the test statistic values that are as or more extreme than the currently calculated test statistic value if the null hypothesis H_0 was true. This probability is calculated by evaluating the tail area under the density curve of the sampling distribution of the test statistic (see Sects. 1.5.2, 1.5.3, and 1.5.4 for area under a density curve). If the p -value is less than the significance level, α , then it is interpreted that the observed data evidence is significant to suggest that data have not been gathered from the population that is specified by the null hypothesis H_0 , and the probability that such a deviation could have been solely due to a chance alone is less than the adopted significance level, α .

Technically, the evaluation of the tail part area under the density curve will be a daunting numerical integration without utilizing a computer. However, resorting the significance assessment to the idea of **critical region** can replace such a numerical integration. A critical region of a test statistic is a collection of all possible test statistic values (i.e., an interval or multiple intervals of the test statistic values on the sampling distribution) of which the total probability for encountering all such values is less than the significance level when the null hypothesis H_0 is true. The critical region of a test is primarily determined by the adopted significance level (i.e., the critical region becomes narrower as the adopted significance level becomes more stringent). H_0 is then rejected at the adopted significance level if the observed test statistic value falls into this region. Note that the critical region is also called rejection region. Checking if the test statistic resides inside or outside of the rejection region can be done using a statistical table of the sampling distribution of the test statistic. The statistical table is a collection of the intervals on the possible range of the test statistic and their corresponding probability of occurrence (use of the tables will be introduced in Sect. 2.2.4.6). Note that if the test statistic value is equal to the critical value, then the p -value is equal to the adopted significance level; if it fell into (outside) the critical region, then p -value is less (greater) than the significance level.

2.2.4.6 One-Sample t -Test

This section introduces one-sample t -test for the inference about a single mean of a population. The following example is used for introducing the procedure, particularly why this test is called t -test, and how the aforementioned constituents covered in Sect. 2.2.4.5 are put into operation in the previously mentioned five steps.

Example 2.1

This is an example of a laboratory-based animal study to test if a synthetic hormone that is delivered via dermal gel cream applied directly to the thigh muscle of mouse. The investigator hypothesized that the hormone level circulating in mouse blood measured at 1 hour after proper application is about 15% of the total volume contained in the prescribed gel dose. Furthermore, the circulating hormonal volumes are known to follow a normal distribution. The investigator plans that if the current experiment shows the mean circulating volume is at least 15%, then the gel's

hormone concentration will not be increased, otherwise a new experiment with an increased concentration rate will be conducted.

Step 1 – Stating null (H_0) and alternative (H_1) hypotheses:

The first step for the hypothesis test inference is to state the null and alternative hypotheses. The investigational objective can be translated as below. Null hypothesis: the mean is 15 (i.e., $H_0: \mu = 15$); alternative hypothesis: the mean is less than 15 (i.e., $H_1: \mu < 15$).

Step 2 – Establishing the test rule (decision rule to determine the significance of the observed data evidence to reject the null hypothesis):

A 5% significance level, i.e., $\alpha = 0.05$, is adopted. The laboratory scientists would reject the null hypothesis if $p < 0.05$, or equivalently if the observed test statistic falls into the critical region of the test static's sampling distribution (i.e., the test statistic falls outside the interval determined by the 5% alpha level. The test statistic will gauge the ratio of (*Observed mean* – *Null mean*)/(*Standard error of the numerator*).

Step 3 – Collecting data and data reduction:

The investigator randomly selected 10 mice with the same body weight and then applied exactly the same dose of the gel to each experimental animal. The following data are the circulating volumes measured in percentage value of the delivered gel volume:

14.45, 14.40, 14.25, 14.27, 14.57, 14.99, 12.97, 15.29, 15.07, 14.67.

Step 4 – Data analysis (i.e., apply the test rule established in Step 2 to the collected data):

The objective of data analysis of the hypothesis test is to calculate the test statistic (i.e., data reduction) and make a decision to reject or not to reject H_0 based on the rule. The rule had been outlined briefly in step 2. With the observed data, the rule can be completed now. The first task is to formulate the test statistic and calculate its value. The second task is to evaluate the significance of it either by directly calculating the p -value or by determining if it falls into/outside (of) the critical region of the test statistic. The third task is to make a decision. The test statistic is constructed as (the observed mean – null 1 value)/ SE (of the numerator). The observed mean of the 10 observations is 14.493, and its null value is 15. Note that the null value is the parameter value that is expected if the null hypothesis is true. The numerator part, which is to measure the signal of observed data departure from the null value, is $14.493 - 15 = -0.507$ (approximately half minute faster than 15 minutes). The denominator, which is the random sampling noise (i.e., the standard error of the signal), should be sought. Note that the null value is a fixed constant, and it does not add additional sampling variation to the signal *Observed Estimate* – *Null Value*. Therefore, the standard error of the whole numerator remains the same as that of only the observed sample mean. The standard error of the observed sample mean can be calculated by dividing the sample standard deviation

by the square root of the sample size (see Sect. 2.1.3). The sample standard deviation is 0.640, thus the standard error is $0.640/\sqrt{10} = 0.203$. Finally, the test statistic value is obtained, i.e., $-0.507/0.203 = -2.503$. Let us take a close look at the test statistic. The mathematical expression of this test statistic is congruent to the quantity $(\bar{x} - \mu)/(s/\sqrt{n})$, which follows a t -distribution introduced in Sect. 2.2.3, where, the numerator, $\bar{x} - \mu$, is to gauge how much the observed sample mean is deviated from the hypothesized null value, and the denominator, $s/\sqrt{n}$, is to gauge the sampling variability of the numerator. The test statistic will follow a t -distribution with $df = n - 1$, if the raw data were drawn from a normally distributed population. Since circulating hormonal volumes are known to follow a normal distribution, this test statistic calculated from these 10 observations will follow the t -distribution with $df = 10 - 1 = 9$. The naming convention of a test is to give the name the sampling distribution of the test statistic. Having known the sampling distribution of this test statistic is t -distribution, we call this a t -test. It is worth mentioning here at least briefly that test statistics can be derived from many situations that are not exactly the same as the above inference (i.e., testing if a single mean is equal to a specific value). For example, a test can be devised to compare two means. As we will mention such a situation and other variants in the later chapters (e.g., independent samples t -test, paired sample t -test, etc.), there are test statistics derived from various situations that will also follow t -distribution. To uniquely identify the t -test being applied to this single mean inference illustrative example, we specifically identify the t -test applied to a single mean inference as one-sample t -test.

Lastly, we can calculate the p -value determined by the test statistic value $t = -2.503$ based on the t -distribution with its $df = 9$ (see Fig. 2.8). This calculation can be carried out by using Excel's TDIST function, which delivers a calculated value of an area under the t -distribution's density curve's upper tail or both tails specified by the user. The user specification TDIST($|t|$, df , 1 or 2) includes the absolute value of the t -statistic value, df , and the choice of 1- or 2-tail in this order. For this exercise, TDIST($| -2.503 |$, 9, 1) is necessary. Note that the first input value of $| -2.503 |$ should be either 2.503 (i.e., without notation) or the absolute value of (-2.503) because the computer program utilizes the symmetrical feature of t -distributions. The actual code is TDIST(2.503, 9, 1). Figure 2.9 illustrates how the raw data are organized, the sample mean and its standard error as well as the t -statistic are calculated, and how

Fig. 2.8 Determination of p -value of a test statistic in a one-sample directional t -test

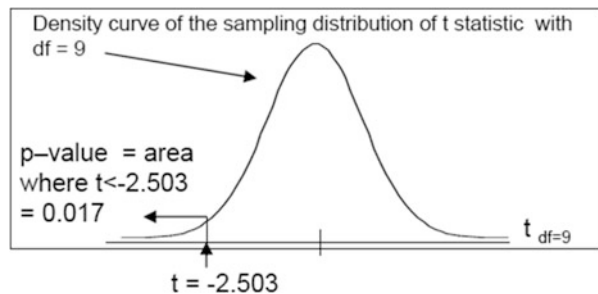


Fig. 2.9 Numerical illustration of calculating test statistic in a one-sample *t*-test

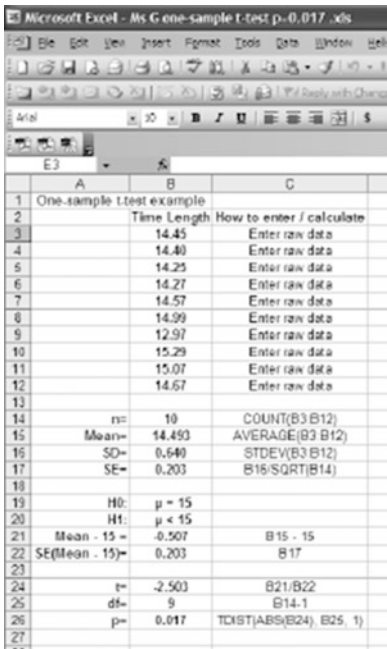


Table 10.2 (1-*p*)th Percentiles of *t*-Distributions

df	p=0.00025	p=0.0005	p=0.0125	p=0.025	p=0.0375	p=0.05
1*	1273.2393	636.6192	25.4517	12.7062	8.4490	6.3138
2*	44.7046	31.5991	6.2053	4.3027	3.4428	2.9200
3	16.3263	12.9240	4.1765	3.1824	2.6808	2.3534
4	10.3063	8.6103	3.4954	2.7764	2.3921	2.1318
5	7.9757	6.8688	3.1634	2.5706	2.2423	2.0150
6	6.7883	5.9588	2.9687	2.4469	2.1510	1.9432
7	6.0818	5.4079	2.8412	2.3646	2.0897	1.8946
8	5.6174	5.0413	2.7515	2.3060	2.0458	1.8595
9	5.2907	4.7809	2.6850	2.2622	2.0127	1.8331
10	5.0490	4.5869	2.6338	2.2281	1.9870	1.8125
11	4.8633	4.4370	2.5931	2.2010	1.9663	1.7959
12	4.7165	4.3178	2.5600	2.1788	1.9494	1.7823
13	4.5975	4.2208	2.5326	2.1604	1.9354	1.7709
14	4.4992	4.1405	2.5096	2.1448	1.9235	1.7613
15	4.4166	4.0728	2.4899	2.1314	1.9132	1.7531
⋮	⋮	⋮	⋮	⋮	⋮	⋮
⋮	⋮	⋮	⋮	⋮	⋮	⋮

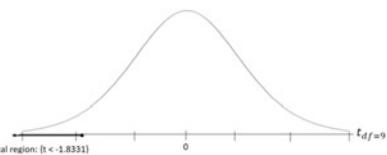


Fig. 2.10 Determination of the critical region of a one-sample directional *t*-test

the *p*-value is obtained. The calculated *p*-value =0.017 is much less than the significance level 0.05. Therefore, H_0 is rejected.

We can also take an approach to check if the $t = -2.503$ falls into or outside of the critical region. Figure 2.10 illustrates how the critical region of this one-sided one-sample *t*-test is at a 5% significance level using a *t*-distribution table (Table 10.2) and visualizes it on the density curve of the sampling distribution of the test statistic t ($df = 9$). This critical region is where the test statistic is less than the fifth percentile of the t ($df = 9$) distribution. The table in Fig. 2.10 shows the 95th percentile rather than the fifth percentile because the density curve of a *t*-distribution is symmetrical. The 95th percentile is 1.8331, thus the fifth percentile is -1.8331 . The critical region

$\{t \leq -1.8331\}$ is also graphically depicted in the figure. The observed $t = -2.503$ is less than -1.8331 and falls into this critical region. Thus, H_0 is rejected.

Step 5 – Making interpretation and report writing:

From the above steps, this test can be summarized as “These data showed that the volume of the circulating hormone is less than 15% of the prescribed total volume at a 5% significance level.”

2.2.4.7 Comments on Statistically Significant Test Results

In a clinical study, statistically significant evidence with a tiny size of the signal may not necessarily comprise a clinical significance. Such a result can be observed in a study with a very large sample size (i.e., unnecessarily larger than the adequate sample size). On the other hand, clinically significant evidence can be statistically insignificant due to an inadequately small sample size or very large data variability. Such a result could have been significant if a larger sample size or better error controls (e.g., better study design, etc.) had been considered. See Chap. 6 for more discussions.

Reporting format in the results section of clinical research journals is also important. Table 2.8 shows a few examples of recommended and not recommended formats.

2.2.4.8 Types of Errors in Hypothesis Tests

Hypothesis tests cannot be completely free of decision errors. The first type of such error is an error that H_0 was rejected although it is true, and the second type is the one that H_0 was not rejected although it was not true. The first kind is called Type 1 error, and the second kind is called Type 2 error.

The adopted significance level of a test, α , predetermined by the investigator is the maximum allowed probability size of Type 1 error. The maximum allowed probability of Type 2 error size is called β and $1 - \beta$ is called the power of the test. Figure 2.11 illustrates the probabilities of Type 1 and Type 2 errors of a test in which the null and alternative hypotheses are both simple hypotheses (i.e., a single-valued hypothesis and the test is directional). As depicted, Type 1 error is the area under the density curve of the sampling distribution of the test statistic within the rejection region under the assumption that H_0 is true. Type 2 error is the area under the density curve of the sampling distribution of the test statistic within the non-rejection region under the assumption that H_0 is not true (i.e., H_1 is true). Chapter 6 will address the relationship between the sizes of Type 1 and Type 2 errors and the sample size, and will discuss how to determine adequate study sample size that keeps Type 1 and Type 2 error sizes small (Table 2.9).

Note that the following fire alarm system metaphor helps the understanding of Type 1 and Type 2 errors, level of significance, and power (Table 2.10).

Table 2.8 Examples of recommended and not recommended formats of summary sentences to appear in the results sections of clinical research journal articles

Recommended summary sentences	
“These data showed that the mean is significantly different from 15 min ($p = 0.031$)”	Show the actual p -value
“These data showed that the mean is significantly different from 15 min ($p < 0.05$)”	Simply report that the p -value was smaller than the significance level
“These data showed that the mean is not significantly different from 15 min at a 5% significance level”	A nonsignificant test result at the level of significance = 0.05
“These data showed that the mean is not significantly different from 15 minutes (NS at 5% significance level)”	A nonsignificant test result at the level of significance = 0.05
Not recommended summary sentences	
“The null hypothesis is rejected because $p < 0.05$. These data showed that the mean is significantly different from 15 minutes ($p = 0.031$)”	Do not say in the report “. . . the null hypothesis was rejected because $p < 0.05$. . .” storytelling of technical details of the procedure is unnecessary
“These data showed that the mean is not significantly different from 15 min ($p > 0.05$)”	Do not report the meaningless p -values in the concluding sentences when the p -value that is greater than your significance level. However, in tables summarizing multiple results together, such format is allowed
“These data showed that the mean is not significantly different from 15 min ($p = 0.278$)”	Do not report the meaningless p -values in the concluding sentences when the p -value that is greater than your significance level. However, in tables summarizing multiple results together, such format is allowed
“The null hypothesis is rejected because $p < 0.05$. These data showed that the mean is significantly different from 15 minutes ($p = 0.031$)”	Do not say in the report “. . . the null hypothesis was rejected because $p < 0.05$. . .” storytelling of technical details of the procedure is unnecessary

2.2.5 Accuracy and Precision

The concepts of accuracy and precision are illustrated in Fig. 2.12. Archery Player 1 performed more accurately and less precisely than Player 2. Player 3 performed more accurately and precisely than the other two players. This illustration can be considered as three researchers (Researchers 1, 2, and 3 correspond, to Players 1, 2, and 3 respectively) having repeated 10 random samplings with a choice of fixed sample size (i.e., n_1 by Researcher 1, n_2 by Researcher 2, and n_3 by Researcher 3) to obtain each point estimate (represented by each “x” mark) using each sample data set and the same computational method. Intuitively, Researcher 1 might have chosen a smaller sample size than that of Researcher 2 but might have chosen a better sampling method that prevented the bias that was presented in the estimates of Researcher 2. How Researchers 1 and 2 could improve their results to expect a result like that of Researcher 3? Researcher 1 would increase the sample size without

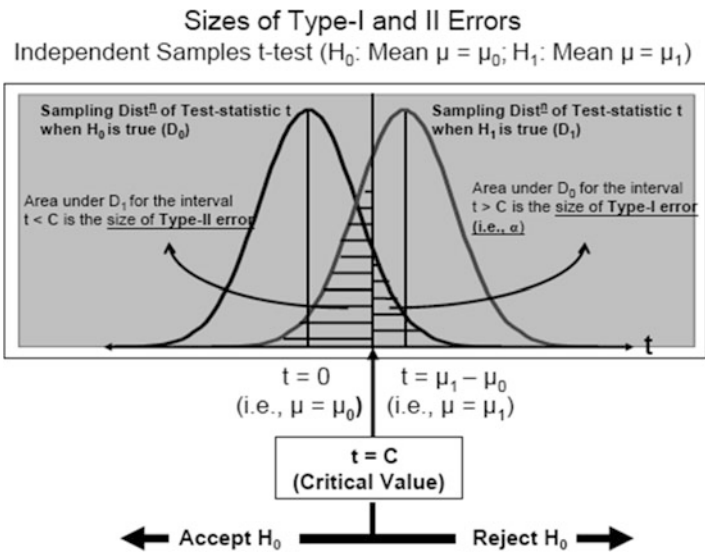


Fig. 2.11 Determination of sizes of Type 1 and Type 2 errors in a one-sample *t*-test

Table 2.9 Errors in decisions and their probabilities in a hypothesis test

	H_0 is true	H_1 is true
Accept H_0	Correct decision Probability of this correct decision is called operating characteristic	Type 2 error Probability to commit Type 2 error = β
Reject H_0	Type 1 error Maximum allowed probability to commit Type 1 error in a test = α level (e.g., 5%)	Correct decision Probability of this = $1 - \beta$ = Power

Table 2.10 Alarm system metaphor of testing hypotheses

Hypothesis test	Alarm system metaphor
Type 1 error	Alarm turns on even if there is no fire breakout
Type 2 error	Alarm does not turn on even if there is a fire breakout
Level of significance, α	The level of false alarm system sensitivity.
Power, $1 - \beta$	Performance level of the alarm system (i.e., turns on whenever it should be) after the sensitivity level is set

having to reconsider the sampling technique, and Researcher 2 would examine possible sources of systematic non-sampling error and eliminate such causes in the future sampling procedure without having to increase the sample size. Connection of study design and sample size to the accuracy and precision is discussed in Sect. 2.2.8.

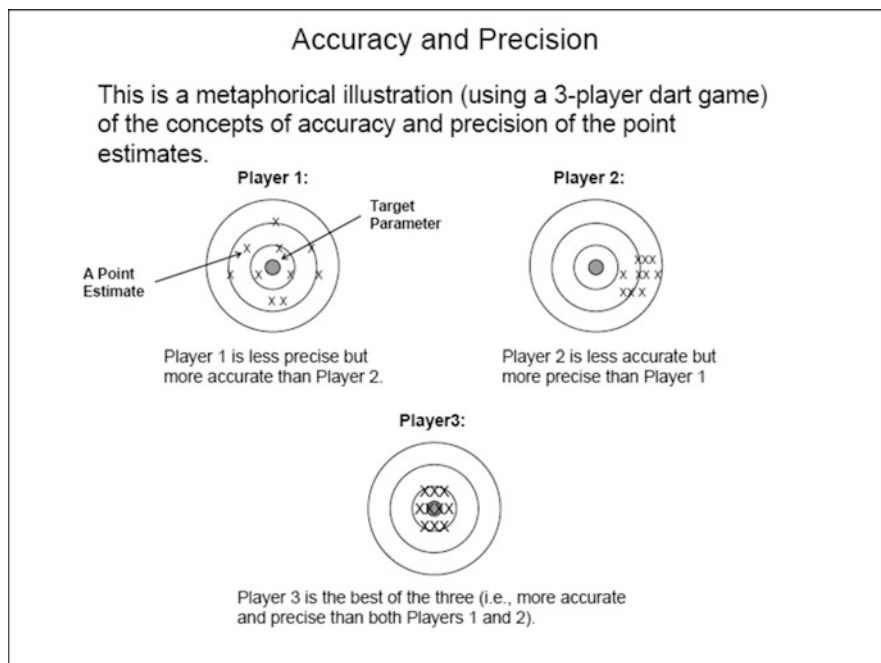


Fig. 2.12 Illustration of accuracy and precision

2.2.6 Interval Estimation and Confidence Interval

2.2.6.1 Overview

In Sect. 2.2.4, hypothesis testing was applied to a single mean inference. Having rejected H_0 and pronounced that the population mean is significantly different from the null value at the adopted significance level (α), it may be of further interest to find out a set of possible range of the point estimates that would not exclude the unknown population mean so that the interval can be considered as a collection of the possible mean values that are different from the unknown population mean with a certain level of confidence. On the other hand, if a test had not been able to reject H_0 , could a range be found to include the unknown parameter value? The interval estimation is an avenue to make such an inference, which is linked to the precision.

A popular approach is to make an interval estimation by constructing a confidence interval. This approach relies on the theory of sampling distribution of the estimator (e.g., sample mean). For instance, the sample mean is a point estimate of the unknown population mean of interest (see Sect. 2.2.1 for the definition of an estimate), and the sample standard error measures the sampling variability of the estimated sample mean. With these, an interval around the point estimate (i.e., the sample mean in the case of mean inference) can be constructed based on the sampling distribution of the sample mean. Section 2.2.6.2 demonstrates a rendering

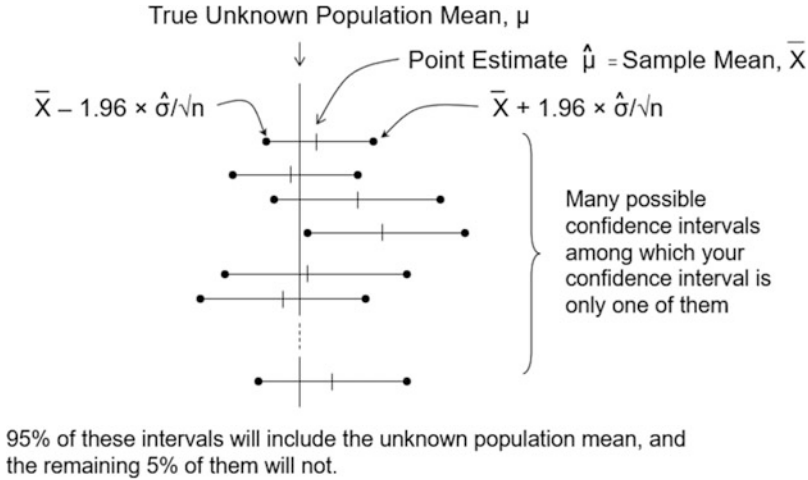


Fig. 2.13 Illustration of the concept of normal distribution-based 95% confidence interval of a mean

idea of the 95% confidence interval of the mean, while the derivation is illustrated when the sampling distribution is Gaussian with known-unknown standard deviation of the distribution of the population characteristic of interest. The derived lower and upper limits of the interval are indeed the 2.5th and 97.5th percentiles, respectively, of the sampling distribution of the sample mean, and the standard deviation of this sampling distribution is the standard error of the sample mean. Such a derived interval is one case of many intervals obtainable from many possible repeated random sampling experiments with the same sample size (see Sect. 2.1.2). Of those large number of experiments, about 95% of the time, the individual intervals may contain the unknown population mean (see Fig. 2.13). Such an interval is called a 95% confidence interval of the mean.

2.2.6.2 Confidence Interval for a Mean

If a random sample of size n , $(x_1, x_2, \dots, x_n)$, is taken from a population in which the probability distribution of the population characteristic of interest is a Gaussian distribution with its mean μ and standard deviation σ , then the sampling distribution of the sample mean $\bar{x}$ will follow a normal distribution with mean μ and standard deviation $(\sigma/\sqrt{n})$ when n is large (i.e., central limit theorem). Or equivalently, the sampling distribution of $(\bar{x} - \mu)/(\sigma/\sqrt{n})$ will follow the standard normal distribution (i.e., the Gaussian distribution with mean 0 and standard deviation 1).

One can find out an interval that covers the middle 95% of the observable sample means based on the standard normal distribution:

Probability $(2.5\text{th percentile} \leq (\bar{x} - \mu)/(\sigma/\sqrt{n}) \leq 97.5\text{th percentile}) = \{-1.96 \leq (\bar{x} - \mu)/(\sigma/\sqrt{n}) \leq 1.96\} = 0.95$. Then solving $-1.96 \leq (\bar{x} - \mu)/(\sigma/\sqrt{n}) \leq 1.96$ for

μ will offer the expression to obtain the 95% confidence interval of the population mean: $\bar{x} - 1.96 \times \sigma / \sqrt{n} \leq \mu \leq \bar{x} + 1.96 \times \sigma / \sqrt{n}$. If σ is unknown (it is usually unknown in clinical studies), then utilize the sample standard deviation, $\hat{\sigma} = s$, i.e., $\bar{x} - 1.96 \times s / \sqrt{n} \leq \mu \leq \bar{x} + 1.96 \times s / \sqrt{n}$.

Example 2.2

A small pilot study of healthy women's systolic blood pressure was conducted. The sample mean and standard deviation estimated from a random sample of 10 women are 115 and 10.31, respectively. What is the 95% confidence interval of the mean systolic blood pressure of this healthy women population?

$\bar{x} = 115$, and the population standard deviation is unknown; thus the sample standard deviation will be used for calculating the standard error, i.e., $s / \sqrt{n} = 10.31 / \sqrt{10} = 3.26$.

$$\begin{aligned} 95\%CI &= (115 - 1.96 \times 3.26, 115 + 1.96 \times 3.26) \\ &= (115 - 6.39, 115 + 6.39) \\ &= (108.61, 121.39) \end{aligned}$$

Result summary: These data showed that the estimated mean systolic blood pressure was 115 mmHg (95% confidence interval–CI: 108.61–121.39). Note that the format of this sentence is mostly recommended in applied research articles.

With a sample size of 10, it is skeptical to assume the sampling distribution of the sample mean with unknown population standard deviation follows a Gaussian distribution with mean of 115 mmHg and its standard error of 3.26 mmHg. Alternatively, a t -distribution can be applied, in that the 2.5th and 95th percentiles of the t -distribution with $df = n - 1 = 10 - 1 = 9$ are -2.685 and 2.685 , respectively (see Sect. 2.2.3 for df). In this way, the resulting 95% confidence interval using the t -distribution with $df = 9$ will become slightly wider (because $|2.865| > |1.96|$).

Let us discuss how to report the result of interval estimation and why a certain phrasing is recommended. Once a confidence interval (CI) has been constructed, the true parameter (e.g., the mean) is either within or outside this interval (i.e., the parameter is not a moving target but a fixed constant, and the constructed interval is a varying interval depending on the sampled data). The probabilistic argument makes sense only prior to the construction of the CI. If many equal-sized random samplings are conducted independently from a normally distributed population, then the sample means of these random sample sets will form a normal distribution (i.e., the sampling distribution). If we obtain, from each sample data set, an interval whose lower limit is sample mean $- 1.96 \times SE$ and the upper limit is sample mean $+ 1.96 \times SE$ (i.e., a 95% CI based on the normal approximation), then each interval can either include or exclude the true population mean. What is being probabilistic is that about 95% of these anticipated intervals will include the true mean. And this argument will not make sense once a researcher's sample data already produced a confidence interval. What has already happened is either this interval did or did not include the unknown population value. So, it is not stated that "We are 95% confident ..." but

stated to simply focus on the point estimate accompanied with the numerical interval: “These data showed that the estimated mean SBP was 115 (95% CI: 108.61–121.39)”.

2.2.6.3 Confidence Interval for a Proportion

It is often of interest to make inferences about a single population proportion. If we are interested in knowing the proportion of the men and women in a large metropolitan population who were vaccinated with the influenza vaccine in the past 6 months, we may possibly take a certain size of random sample of men and women and obtain the information about the vaccination by a survey. The question can be translated into “What is the probability that a man or woman in this population received an influenza vaccination in the past 6 months?” As explored earlier (see Sect. 1.5.5), this question can be well articulated by the Bernoulli distribution in that a single person’s survey answer (i.e., coded with 1 if “Yes,” or 0 if “No”) being the event outcome and p being the population vaccination rate. We can assume that the answers of the sample of the men and women are independent. The actual inference happens as below. With the sample of n men and women, we first count the number of “Yes” answers, which is simply the sum of all 1 s and 0 s (i.e., $1 + 0 + 1 + 1 + 0 + 1 + 0 + 0 + \dots$), then figure out what the sampling distribution of that sum value (let x denote this value). It is obvious that this sampling distribution is $Bi(n, \pi)$, and the inference is all about π given n and x . The inference about π is to find its point estimate, perform a one-sample test if π is equal to a certain proportion (e.g., $H_0: \pi = 0.3$ versus $H_1: \pi \neq 0.3$), and/or find its 95% confidence interval. Straightforwardly, the point estimate of π can be obtained by x/n . The one-sample test and the interval estimation can be resorted to the normal approximation (see Sect. 1.2.2) if n is large, or directly to the sampling distribution, i.e., binomial distribution. For example, if the survey sample size was 1000 and there were 270 persons who answered “Yes,” then $\hat{\pi} = 270/1000 = 0.27$; the test statistic for a one-sample normal approximation test (i.e., z -test) with $H_0: \pi = 0.3$ versus $H_1: \pi < 0.3$ is $(0.27 - 0.3)/\sqrt{0.27 \times (1 - 0.27)/1000} = -0.03/0.014 = -2.137$ whose two-sided test’s p -value is 0.0163 (NORMSDIST(−2.137) by Excel), and the lower and upper 95% confidence limits are $0.27 - 1.96 \times 0.014 = 0.242$ and $0.275 + 1.96 \times 0.014 = 0.298$, respectively (i.e., 95% CI of the vaccination rate: 24.2% - 29.8%).

What if the sample size is small and a normal approximation cannot be resorted to the inference? If n_1 events were observed out of n Bernoulli trials, i.e., $\hat{\pi} = n_1/n$, then Clopper–Pearson exact $100 \times (1 - \alpha)\%$ lower and upper confidence limits, π_L and π_U , respectively, can be found by equating the probability that there would have been n_1 or more successes with the lower bound of π to $\alpha/2$, and also equating the probability that there would have been n_1 or fewer successes with the upper bound of π to $\alpha/2$ (i.e., let the tail probabilities of both sides be α):

$$\sum_{x=n_1}^n \binom{n}{x} \pi_L^x (1 - \pi_L)^{n-x} = \alpha/2, \text{ and}$$

$$\sum_{x=0}^{n_1} \binom{n}{x} \pi_U^x (1 - \pi_U)^{n-x} = \alpha/2.$$

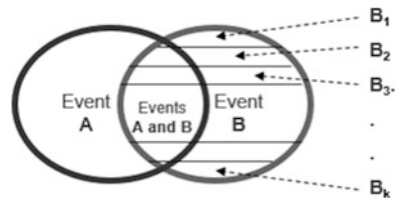
Take an example with three observed successes out of $n = 10$ (i.e., $\hat{\pi} = 0.3$). The exact 95% confidence limits for a proportion π are found by solving $\sum_{x=3}^{10} \binom{10}{x} \pi_L^x (1 - \pi_L)^{10-x} = 0.025$ and $\sum_{x=0}^3 \binom{10}{x} \pi_U^x (1 - \pi_U)^{10-x} = 0.025$. The first estimating equation is to find the lower limit by which the tail probability becomes 0.025 and the second equation is to find the upper limit by which the tail probability becomes 0.025 as well. The solved limits (calculated by a computer) $\hat{\pi}_L$ and $\hat{\pi}_U$ are 0.0667 and 0.6525, respectively. Note that this exact confidence interval ($0.0667 \leq \pi \leq 0.6525$) is not symmetrical with respect to $\pi = 0.3$, whereas the normal approximation-based interval, ($0.0160 \leq \pi \leq 0.5840$), is symmetrical.

2.2.7 Bayesian Inference

The aforementioned inference in this chapter is called frequentist inference. Its paradigm is to let the parameter be a fixed constant and perform either the hypothesis testing (i.e., reject or do not reject the fixed null hypothesis by the chosen rule) or perform the point and interval estimations without the prior probabilistic description about the parameter. Bayesian inference is a method of statistical inference in which the paradigm is different from the frequentist in that this inference lets the parameter of interest be a random variable (i.e., moving target) and lets the observed data determine the probability that a hypothesis is true. The hypothesis testing in this setting is informal, and the typical format of Bayesian inference is estimation. The term “Bayesian” comes from Bayes, the statistician who popularized the rule (i.e., Bayes’ rule) of conditional probability as described below.

As illustrated in Fig. 2.14, event A can also be conceived as the collection of an overlapping B and a nonoverlapping B. The following intuitive algebra is useful for calculating the conditional probability of occurrence event B conditional on A (i.e., event A is already occurred):

Fig. 2.14 Two overlapping event sets A and B



$$P(B|A) = P(A \text{ and } B) / P(A) - \{1\}$$

$$P(A|B) = P(A \text{ and } B) / P(B) - \{2\}$$

$$\text{From (2), } P(A|B) \times P(B) = P(A \text{ and } B) - \{3\}$$

Plug (3) into (1), then

$$P(B|A) = [P(B) \times P(A|B)] / P(A) - \{4\}$$

Furthermore, if event B is partitioned into k disjoint sub-events, then $P(A)$ can be written as $P(B_1) \times P(A|B_1) + P(B_2) \times P(A|B_2) + \dots + P(B_k) \times P(A|B_k)$. So, {4} can be written as $P(B_i|A) = [P(B_i) \times P(A|B_i)] / [P(B_1) \times P(A|B_1)] + [P(B_2) \times P(A|B_2) + \dots + P(B_k) \times P(A|B_k)]$, for i ($i = 1, 2, \dots, k$). Typical application to the Bayesian inference is to consider $B_1, B_2, \dots, B_k$ as the hypothesized parameter values and event A as the observed data. With such an application, we can calculate $P(B_i|A)$, i.e., the posterior probability distribution of B_i given observed data, with the prior distribution (i.e., $P(B_1), P(B_2), \dots, P(B_k)$) and the conditional probabilities to observe data set A given $B_1, B_2, \dots, B_k$ (i.e., $P(A|B_1), P(A|B_2), \dots, P(A|B_k)$).

Technically, the Bayesian inference picks a prior probability distribution (before observing the data) over hypothesized parameter values that vary with chance (i.e., the parameter is viewed as a random variable) depending on the observed data. We then determine the so-called likelihoods of those hypothesized parameter values using the information contained in the observed data. Finally, we determine the likelihoods that are expected over the prior distribution, which is called posterior probabilities of the hypotheses. The ultimate decision is to pick the hypothesized parameter value that gained the greatest posterior probability and identify the narrowest interval that covers 95% of the posterior distribution (i.e., Bayesian 95% confidence interval).

The following example demonstrates how the Bayesian estimation is different from the frequentist estimation.

Example 2.3

Let's consider a cross-sectional study that estimates the prevalence rate of osteoporosis in an elderly (age 65+ years) women population. The sample size was 500 and the observed number of subjects with osteoporosis diagnosis was 159.

The frequentist inference approach to this problem is to find out the point estimate and its 95% confidence interval based on the normal approximation of the sampling distribution of the estimated proportion. The point estimate is 0.32 (i.e., 159/500). The upper and lower limits of its 95% CI are $(159/500) \pm 1.96 \times \sqrt{[(159/500)(1 - 159/500)/500]}$, thus the 95% CI: 0.28–0.36.

The Bayesian approach to construct a 95% confidence interval is different as described in a few steps below. Step 1. The investigator views the population prevalence rate is a random variable (i.e., the prevalence rate can vary with chance) that varies within the [0.01, 1.00) interval. This is articulated as the “prior” (i.e., prior to data collection) distribution of the population prevalence rate is uniform

Table 2.11 Distribution of posterior probabilities

Hypothesized range of π	Picked value for likelihood evaluation, π^*	Prior Probability	Probability of observing 159 cases out of 500 random sampled subjects if the population prevalence = π^* is true	Posterior Probability which is proportional to [Prior probability] x [Probability of observing 159 cases out of 500 random sampled subjects if the population prevalence = π^* is true]
$0.01 \leq \pi < 0.02$	0.01	$1/99 = 0.010$	$\binom{500}{159} (0.01)^{159} (1-0.01)^{(500-159)} = 0.0000$	$(1/99) \binom{500}{159} (0.01)^{159} (1-0.01)^{(500-159)} = 0.000$
$0.02 \leq \pi < 0.03$	0.02	$1/99 = 0.010$	$\binom{500}{159} (0.02)^{159} (1-0.02)^{(500-159)} = 0.0000$	$(1/99) \binom{500}{159} (0.02)^{159} (1-0.02)^{(500-159)} = 0.000$
.	.	.	.	.
$0.29 \leq \pi < 0.30$	0.29	$1/99 = 0.010$	$\binom{500}{159} (0.29)^{159} (1-0.29)^{(500-159)} = 0.0150$	$(1/99) \binom{500}{159} (0.29)^{159} (1-0.29)^{(500-159)} = 0.00015$
$0.30 \leq \pi < 0.31$	0.30	$1/99 = 0.010$	$\binom{500}{159} (0.30)^{159} (1-0.30)^{(500-159)} = 0.0261$	$(1/99) \binom{500}{159} (0.30)^{159} (1-0.30)^{(500-159)} = 0.00026$
$0.31 \leq \pi < 0.32$	0.31	$1/99 = 0.010$	$\binom{500}{159} (0.31)^{159} (1-0.31)^{(500-159)} = 0.0355$	$(1/99) \binom{500}{159} (0.31)^{159} (1-0.31)^{(500-159)} = 0.00036$
$0.32 \leq \pi < 0.33$	0.32	$1/99 = 0.010$	$\binom{500}{159} (0.32)^{159} (1-0.32)^{(500-159)} = 0.0381$	$(1/99) \binom{500}{159} (0.32)^{159} (1-0.32)^{(500-159)} = 0.00038$
$0.33 \leq \pi < 0.34$	0.33	$1/99 = 0.010$	$\binom{500}{159} (0.33)^{159} (1-0.33)^{(500-159)} = 0.0325$	$(1/99) \binom{500}{159} (0.33)^{159} (1-0.33)^{(500-159)} = 0.00033$
.	.	.	.	.
$0.99 \leq \pi < 1.00$	0.99	$1/99 = 0.010$	$\binom{500}{159} (0.99)^{159} (1-0.99)^{(500-159)} = 0.0000$	$(1/99) \binom{500}{159} (0.99)^{159} (1-0.99)^{(500-159)} = 0.00000$

distribution (i.e., the prior probability for each prevalence rate is equally likely within this interval). Step 2. Invoking Bayes' rule, i.e., the [posterior probability of population prevalence = π] is calculated by [prior probability of population prevalence = π] $\times$ [probability of observing 159 cases out of 500 random sampled subjects if π is the true value of population prevalence]. According to the theory, the posterior probability is proportional to (not always equal to because the actual likelihood evaluation may take place only within a selective subset of all possible parameter values) the product of these two quantities. Note that there are infinitely many values within $[0.01, 1.00)$, and the researcher may get to evaluate only 99 equally spaced discrete values, e.g., 0.01, 0.02, ..., 0.99, and such a calculated probability at each discrete value π is called likelihood. Step 3. The ultimate goal is to find the prevalence rate that gives the greatest posterior probability and to construct the narrowest interval that covers 95% of the posterior distribution.

The estimated π of 0.32 is the Bayesian estimate that had the greatest posterior probability. To construct a Bayesian confidence interval, let us note that the sum of column 5 in Table 2.11 is not 1 but 0.002 because the likelihood value was obtained only within 99 values of all possible values, and that according to the theory, the posterior distribution is proportional to the calculated value of the last column. The posterior distribution can be standardized to the last column so that the sum of those 99 posterior probability values can be a probability distribution (i.e., to make the sum = 1). Because the sum of column 5 is 0.002, we can divide every posterior probability value by 0.002.

The interval that is close to 95% around the Bayesian point estimate 0.32 (i.e., find where the 95% of the posterior probabilities are distributed around the point estimate) can now be found out. The values of [individual posterior probability]/

Table 2.12 Rescaled posterior probabilities of the prevalence near the 95% Bayesian confidence interval

Hypothesized Prevalence value	Rescaled (i.e., divided by 0.002) Posterior Probability
0.28	0.033
0.29	0.075
0.30	0.131
0.31	0.178
Interval I ← 0.32 → Interval II	0.191
0.33	0.163
0.34	0.111
0.35	0.061
0.36	0.027
Interval I covers	→ 93.6% of the posterior distribution
Interval II covers	→ 94.2% of the posterior distribution

[sum of the all individual posterior probabilities] = [individual posterior probability]/ 0.002 for hypothesized prevalence value for the following values cover about 95% of the posterior distribution are listed in Table 2.12.

The meaning of the frequentist confidence interval might not have been very clear (see Sect. 2.2.6.2), and appreciation of the Bayesian confidence interval can be helpful. The Bayesian 95% CI is the interval where 95% of the actual posterior probability is distributed, thus this CI is interpreted as “based on the observed data we are 95% confident that the unknown true and varying population parameter exists in this interval,” whereas the requentist confidence interval is not such an interval as we discussed in the earlier part of this chapter.

2.2.8 Study Design and Its Impact to Accuracy and Precision

Good study designs will minimize sampling errors (i.e., increase precision) and non-sampling errors (i.e., increase accuracy and decrease bias) (Fig. 2.15).

2.2.8.1 Sampling and Non-sampling Error Control

Sampling error is the random error involved in the sample statistics (i.e., estimates of the population parameters of interest) arising from sampling (i.e., sample-to-sample random fluctuation), and it becomes very small when the sample size becomes very large. An extreme example is that if the observed sample is the whole population,

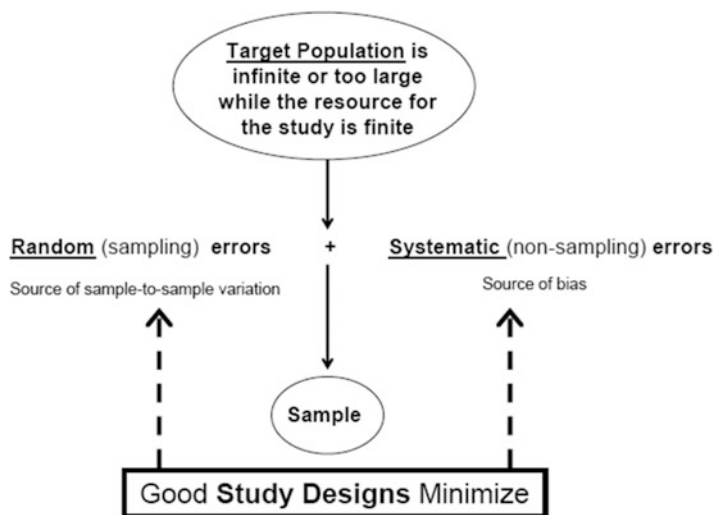


Fig. 2.15 Population, sample, sampling, and non-sampling errors

then there is no sampling error (see Sect. 2.1.1). How can a study design control (i.e., reduce) the sampling error? An effort to increase the sample size, as much as it can be, will reduce the sampling error. Determination of the sample size is a very important part of the study design, and its ultimate goal is to allow the minimally tolerable sampling error size of the sample estimate to the extent that the investigator can find significant data evidence to answer the study question. However, it is unrealistic to make the sample size very close to that of the population size. Chapter 8 will discuss the statistical sample size determination based on the statistical power of the hypothesis test.

Non-sampling error is the systematic error that causes the bias involved in the sample statistics due to a poor study design. Unlike the sampling error, it is not sample size dependent. How can the non-sampling error be controlled? A proper study design is necessary to identify the source of the bias causing non-sampling errors and prevent it before data collection. The next sections will briefly introduce the types of studies and popular approaches to study design.

2.2.8.2 Study Types and Related Study Designs

In clinical research settings, studies can be categorized into either observational or experimental study. In the observational study, the outcome-causing factors are not controlled by the study investigator. For example, in a study of gender difference in health-seeking behavior, the researchers cannot assign sex to the study subject because sex is not a condition that can be created by the researcher. On the other hand, in the experimental studies, the study investigator controls the outcome-

causing factor. For example, the dose levels of a dose response study of a certain medication are determined by the researcher.

2.2.8.3 Observational Study Designs

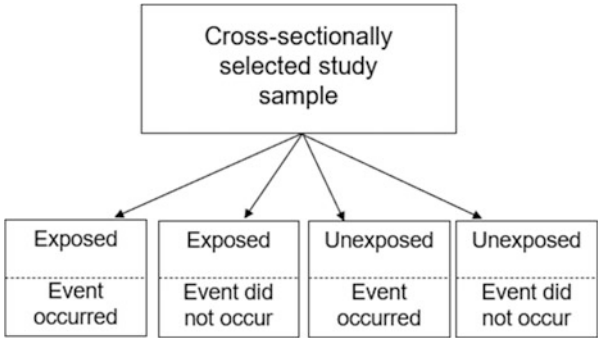
Case series design is applied in a small-scale medical investigation to describe patient information seen over a relatively short period of time. This is a purely descriptive study design, and it does not involve between-group comparisons and there are no control subjects. Because of such a descriptive nature, this design does not construct hypotheses. The result of a study by this design cannot be generalized.

Cross-sectional design is used to examine a selected sample of subjects from a defined population at one point in time (Fig. 2.16). The examples are disease-diagnostic test performance study, opinion survey in an election year, etc. Such a design cannot serve as a good design to investigate a causal determinant of the outcome because it takes time for a resulting outcome is manifested by its causal factor.

Cohort design is to monitor the cohorts of exposed (to a causal factor) and unexposed subjects prospectively over time and to examine the effect of the exposure on the long-term outcome by comparing between the exposed and unexposed sub-cohorts (Fig. 2.17). This design is useful for studies of relatively common outcomes. However, it requires extremely large cohort sizes to study rare outcomes and generally requires a long study period. The potential bias can be prevented relatively easily by this design because the inclusion/exclusion criteria for the cohorts are definable at the outset.

Case-control design is to overcome the limitation of the cohort design, particularly the long study period and large cohort size requirement for rare outcomes. Unlike the cohort design, the case-control design looks back-sampled cases and controls retrospectively to verify the risk factor exposure status in the past by which sufficient size of rare events can be ascertained through large registry, etc. For example, if a cohort design is considered to study whether men with BRCA 1/2 mutation have an increased lifetime risk of breast cancer would require large cohorts

Fig. 2.16 Cross-sectional study design



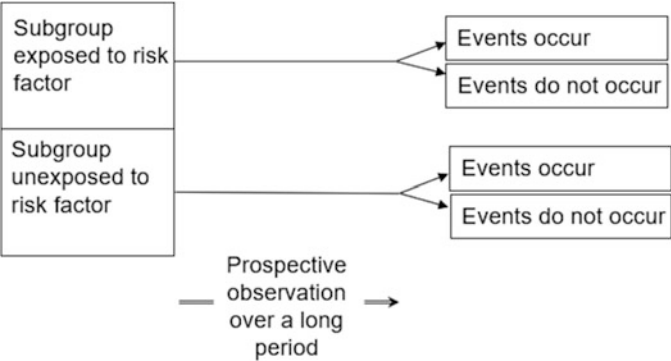


Fig. 2.17 Cohort study design

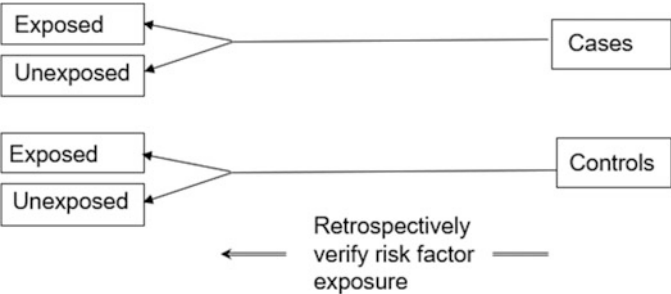


Fig. 2.18 Case-control study design

of cancer-free men with and without the mutation and must wait until the study observes enough number of cancer incidence cases in both groups for reliable comparison (Fig. 2.18). However, the case-control design can reduce the study burden by collecting enough number of male breast cancer patients from an already accessible large cancer registry that has been established over many decades and the same number of healthy control men then collect their DNA and perform the comparative analysis. Although the case-control design is less burdensome than the cohort design, the chance of bringing in non-sampling errors is much greater than the cohort design. For example, if the exposure is not measurable by an objective biological material such as DNA mutation, then the collection of the past exposure history may induce recall bias (see Table 2.13). Long-term radiation exposure, smoking history, etc., can be examples of such exposures.

2.2.8.4 Experimental Study Designs

Experimental designs can rule out potential sources of non-sampling errors. In a clinical research setting, the randomized study design is such a kind. The concurrent

Table 2.13 Comparison of advantages (+) and disadvantages (–) between cohort and case–control designs

	Cohort design		Case–control design
–	Long study period	+	Short study period
–	Very costly	+	Relatively inexpensive
–	Suitable for relatively common disease	+	Suitable for rare disease
+	Less selection bias in control group	–	Selection bias in control group
+	No recall bias	–	Recall bias in both case and control groups
–	Direct monitoring of study volunteers is needed	+	Medical chart review (only paper documents or computer records) is possible
–	Attrition problem	+	No attrition problems
+	Incidence rates (i.e., probability of an outcome within a certain time, e.g., annual cancer incidence rate) can be determined	–	Cannot determine incidence rate
+	Relative risk is accurate	–	Relative risk is approximate

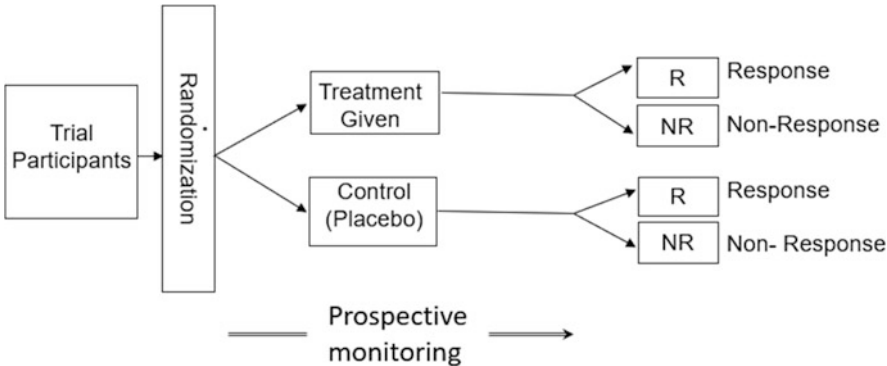


Fig. 2.19 Concurrent (parallel) controlled randomized design

(parallel) randomized design (Fig. 2.19) is commonly applied to clinical trials to evaluate the efficacy of new therapy by comparing the average results of the subjects who received the treatment and those of untreated subjects. Well-carried out randomization will keep the distributions of demographic and clinical characteristics of the treated and untreated subjects balanced so that the estimated efficacy would not be confounded with uneven demographic and clinical characteristics, i.e., a randomized study avoids potential bias causing non-sampling errors.

However, the randomized clinical design can be infeasible in some situations such as when the placebo control group cannot be the comparison group (e.g., urgent medical problems that worsen the subject if remain untreated). The single group self-controlled cross-over design is to examine whether the difference in the within-participant changes over Period 1 and Period 2 are equal (Fig. 2.20).

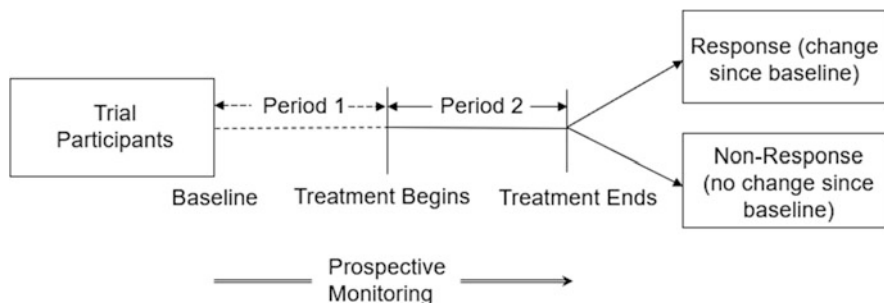


Fig. 2.20 Self-controlled (cross-over) design

Study Questions

1. What are the mean and standard deviation of the sampling distribution (i.e., standard error) created from the sample means of three numbers randomly drawn from 1, 2, 3, 4, and 5? Note that there will be 10 sample means.
2. What are the two pillars of the frequentist inference?
3. In central limit theorem, what becomes a Gaussian distribution when the sample size becomes large?
4. Under what circumstance a t -distribution is resorted (instead of the standard Gaussian distribution) to the inference of a mean?
5. Explain the idea of *Observed* ~ *Expected* ~ *Standard Error* trio formulation of the test statistic t of a one-sample t -test for a single-mean inference.
6. In this chapter, what is the parameter being inferred in a one-sample t -test?
7. What is the numerator of the test statistic in a one-sample t -test?
8. What is the denominator of the test statistic in a one-sample t -test?
9. Why is the t -test named as such?
10. Letting alone the statistical significance, how should these three values be interpreted (hint: signal to noise): $t = 1.5$ ($df = 10$); $t = 2.5$ ($df = 10$); $t = -3.5$ ($df = 10$)?
11. Is the 95% confidence interval around the mean the same as the 2.5th to 97.5th percentile range of a sample distribution?
12. Please criticize the following sentences: “These data showed that the outcome was significantly different from 50 ($p < 0.05$).” Is the word “outcome” appropriate?
 “These data showed that the mean age was significantly different from 50 years.” Does this sentence present a p -value?
 “These data showed that the sample mean age was significantly different from 50 years ($p < 0.05$).” Was the inference made about the sample mean?

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Chapter 3

t-Tests for Two-Mean Comparison



In Chap. 2, the one-sample *t*-test was introduced to test whether a single mean of a population is equal to a certain value. This chapter will introduce the extension of the *t*-test to examine whether the difference between two population means is equal to a certain value. Two situations will be discussed: the first is when the two means are from independent (i.e., unrelated) populations, and the second is when the two means are from related populations.

3.1 Independent Samples *t*-Test for Comparing Two Independent Means

Example 3.1: Does Medication X Prevent the Low Birthweight Delivery?

A prospective two-arm randomized controlled clinical trial (Arm 1: Medication X and Arm 2: Placebo control) was conducted. Because the study design was a randomized clinical trial, the two study groups are independent. The data raw data and result summary of a descriptive analysis are shown below. Birth weights (lb) of 15 newborns from mothers treated with medication X (denoted by Tx) and 15 from mothers with placebo are listed. Let us assume the data are normally distributed and the dispersions of the two distributions are not much different.

Tx (n = 15): 6.9 7.6 7.3 7.6 6.8 7.2 8.0 5.5 5.8 7.3 8.2 6.9 6.8 5.7 8.6
Placebo (n = 15): 6.4 6.7 5.4 8.2 5.3 6.6 5.8 5.7 6.2 7.1 7.0 6.9 5.6 4.2 6.8
(Fig. 3.1)

How can we tackle this problem? The approach to use an extension of the *t*-test will be introduced first, and then the applied result will be illustrated.

Below is the framework for testing the equality of two population means using two independently drawn random samples from normally distributed populations.

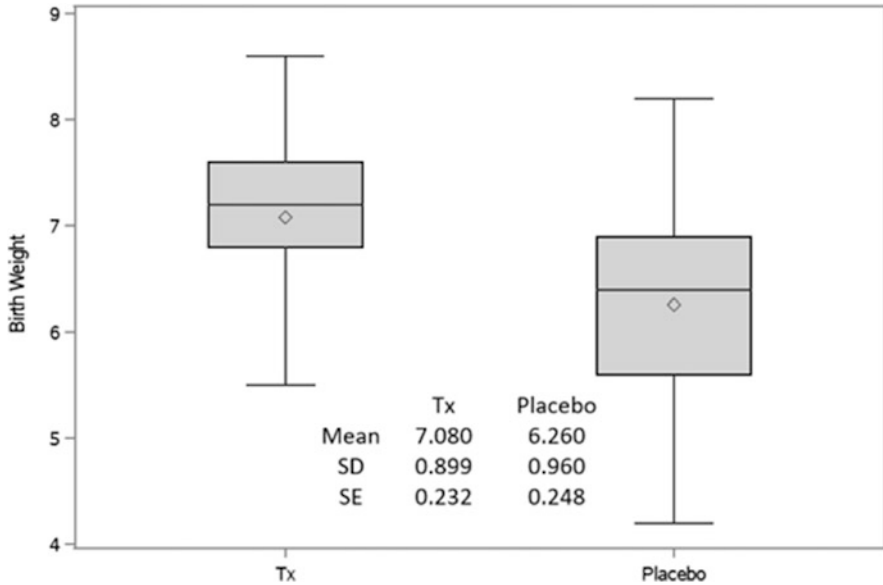


Fig. 3.1 Listed and visualized data from a two-arm randomized clinical trial

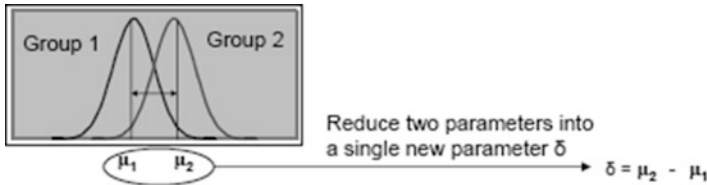


Fig. 3.2 Illustration of two means of normally distributed paired continuous outcomes and difference of the two means

The null hypothesis is that the two population means are equal (i.e., $H_0: \mu_1 = \mu_2$). Note that “two population means are equal” can be translated into “difference of the two population means = 0 (i.e., $H_0: \mu_2 - \mu_1 = 0$)” by which the dimension of the argument is reduced into one (i.e., by letting δ denote the single translated parameter, see Fig. 3.2, where $\delta = \mu_2 - \mu_1$) so that the corresponding sample statistic to work with is the observed difference between the two sample means (i.e., by letting d denote $\bar{x}_2 - \bar{x}_1$). Finally, the null and alternative hypotheses are expressed as $H_0: \delta = 0$ and $H_1: \delta \neq 0$ for a nondirectional test (or $H_1: \delta > 0$ for a directional test examining $\mu_2 > \mu_1$) (Table 3.1).

The idea of *Observed Estimate* ~ *Null Value* ~ *standard error (SE)* trio (see Sect. 2.2.4.5) is applied to derive the test statistic:

$$t = \frac{\text{Observed Estimate} - \text{Null Value}}{\text{SE}(\text{Observed Estimate} - \text{Null Value})} = \frac{(d - \delta)}{\text{SE}(d)} = \frac{(d - 0)}{\text{SE}(d)} = \frac{d}{\text{SE}(d)}.$$

Table 3.1 Null and alternative hypotheses for comparing two means

Null hypothesis	Alternative hypothesis	Simple or composite ^a		Directionality of composite hypothesis
		Null	Alternative	
$H_0: \mu_2 - \mu_1 = 0$ (i.e., $\mu_1 = \mu_2$) Means are equal	$H_1: \mu_2 - \mu_1 = \delta$ Mean difference is equal to δ	Simple	Simple	N/A
$H_0: \mu_2 - \mu_1 = 0$ (i.e., $\mu_1 = \mu_2$) Means are equal	$H_1: \mu_2 - \mu_1 \neq 0$ Means are not equal	Simple	Composite	Nondirectional (two-sided)
$H_0: \mu_2 - \mu_1 = 0$ (i.e., $\mu_1 = \mu_2$) Means are equal	$H_1: \mu_2 - \mu_1 > 0$ Mean 2 is greater than mean 1	Simple	Composite	Directional
$H_0: \mu_2 - \mu_1 = 0$ (i.e., $\mu_1 = \mu_2$) Means are equal	$H_1: \mu_2 - \mu_1 < 0$ Mean 2 is smaller than mean 1	Simple	Composite	Directional

^aSimple hypothesis involves a single value of the parameter, and composite hypothesis involves more than one value (e.g., an interval) of the parameter

The numerator is the deviation of the observed sample mean difference (i.e., $d = \bar{x}_2 - \bar{x}_1$) from the null value of population mean difference (i.e., $\delta = \mu_2 - \mu_1 = 0$). Note that δ is set to 0 for the equality test. The denominator, $SE(d)$, is the estimated standard error of the numerator. The simplified form of this test statistic is $t = d/SE(d)$. Note that a large value of this ratio (positive, negative, or either direction depending on the alternative hypothesis) indicates that the observed deviation of d from the null value of $\delta = 0$ may not be due to the chance alone. According to the sampling theory, under the null hypothesis, if the two population variances are equal, this test statistic will follow the *t*-distribution with $df = (n_1 - 1) + (n_2 - 1)$, where n_1 and n_2 are the sample sizes of the two groups respectively. We can then find out how extreme (i.e., unlikely to observe) the observed test statistic t is, if it results from the sample data gathered from the two population distributions under null hypothesis.

The derivation of the standard error, $SE(d)$, may seem complex for beginners. As this book intends not to let the readers plug and play formulae, the conceptual steps of its derivations are shown below. The definition of the standard error is the

standard deviation of the sampling distribution of a sample statistic (see Sect. 2.1.3 for its definition). The sample statistic of this case is d , which is the difference between the two observed sample means, i.e., $\bar{x}_2 - \bar{x}_1$. So, the standard deviation of the sampling distribution, $SE(d)$, of $\bar{x}_2 - \bar{x}_1$ is to be derived. Because the standard deviation is the square root of the variance, the main derivation is to derive its variance. Note that the sample variances of $\bar{x}_1$ and $\bar{x}_2$ are s_1^2/n_1 and s_2^2/n_2 , respectively, where s_1^2 and s_2^2 are notations for the variance of the sample distributions (see Sect. 2.1.2 definition of the sample distribution). The sample variance of the difference $\bar{x}_2 - \bar{x}_1$ would increase (see Example 3.2). Note that $\bar{x}_1$ and $\bar{x}_2$ vary from sample to sample with its variance s_1^2/n_1 and s_2^2/n_2 , so the difference $\bar{x}_2 - \bar{x}_1$ would vary greater, and the resulting variance of difference is the addition of the individual variances s_1^2/n_1 and s_2^2/n_2 . Finally, the denominator of the test statistic is derived, which is the square root of the weighted sum of the two sample variances, i.e., $\sqrt{s_1^2/n_1 + s_2^2/n_2}$.

Having derived the test statistic, the inference of Example 3.1 continues as given below.

The test statistic $t = (6.26 - 7.08) / \sqrt{0.2322 + 0.2482} = 0.82/0.34 = 2.41$ will follow the *t*-distribution with $df = 28$. The critical value of this nondirectional test for a 5% significance level is 1.701, and the critical region is $\{t > 1.701\}$. The observed test statistic from the data, $t = 2.41$, falls into the critical region (Fig. 3.3).

The *p*-value can also be calculated using the Excel function TDIST(2.41, 28, 1), which gives the result 0.0113.

Summary: These data showed that there was a significant effect of medication X to prevent low birthweight delivery ($p = 0.0113$).

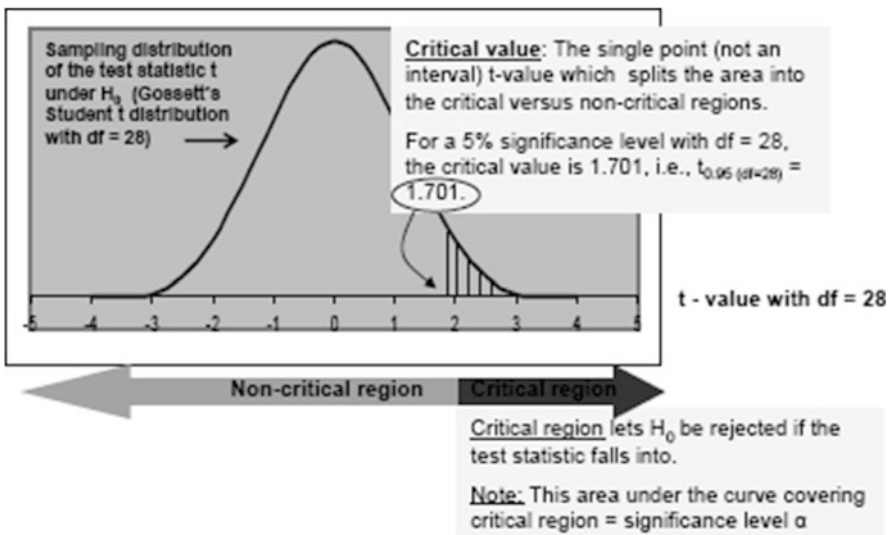


Fig. 3.3 Illustration of the critical region of the directional *t*-test (applied to Example 3.1)

3.1.1 *Independent Samples t-Test When Variances Are Unequal*

When the two population variances are not equal, the sampling distribution of the aforementioned test statistic will not perfectly follow the *t*-distribution with $df = n_1 + n_2$. This book does not describe computational details but rather discusses how we can diagnose such a phenomenon, and how we can make inferences and interpret the results provided that the intermediate computational results for the diagnosis and the calculated SE are made by computer programs or expert statisticians. If heteroskedasticity is detected, then we resort to a theoretical *t*-distribution with a slightly modified *df*. In this case, a modified *df* becomes little bit smaller than $(n_1 + n_2 - 2)$, which is obtained under the homoskedasticity (i.e., equal variance) assumption.

How can we diagnose whether the variances are equal before carrying out the independent samples *t*-test? There is a test that evaluates it. The null and alternative hypotheses of this test are $H_0: \sigma_1^2 = \sigma_2^2$ and $H_1: \sigma_1^2 \neq \sigma_2^2$ (note that the nondirectional alternative hypothesis is its obvious choice). Compute the two sample variances and take the ratio of the two. This ratio will follow an *F*-distribution under the null hypothesis. The *F*-distribution has two *dfs* where $df_1 = n_1 - 1$, and $df_2 = n_2 - 1$. Then we resort to the *F*-distribution to determine the *p*-value (details for computing a *p*-value from a *F*-distribution will be shown in Chap. 4). A *p*-value < 0.05 indicates unequal variances, and you would need to take into account this condition during the *t*-test.

If the data show that the variances are unequal, then a modified version of the independent samples *t*-test is recommended. The main work is to make adjustment to the *df* as well as to estimate the *SE* of the mean difference, which could become biased if not well taken care of. This monograph does not introduce the mathematical/computational details, but the following is the idea of *df* adjustment. The idea is that as the *df* becomes smaller, the tail of the *t*-distribution becomes thicker, meaning that if the *df* is adjusted downward for heteroskedasticity, then the observed *t*-statistic will produce a larger *p*-value than that would have resulted from the *t*-distribution with the unadjusted *df* because the modified distribution has a heavier tail; therefore, the test becomes conservative.

3.1.2 *Denominator Formulae of the Test Statistic for Independent Samples t-Test*

Other books offer you plug-and-play formulae for calculating the denominator of the test statistic, which prompt you to plug in the sample sizes and the sample variances of the two groups being compared. The readers do not need to master the computational details of the standard error derivation. However, those are shown in Table 3.2.

Table 3.2 Derivation of the denominator of the test statistic for independent samples *t*-test

Sample size	Population variances	Derived standard error (i.e., denominator of the test statistic <i>t</i>)
Unequal	Unequal	$\sqrt{s_1^2/n_1 + s_2^2/n_2}$
Equal	Unequal	Substitute <i>n</i> for <i>n</i> ₁ and <i>n</i> ₂ , so $\sqrt{(s_1^2/n + s_2^2/n)} = \sqrt{(s_1^2 + s_2^2)/n}$
Unequal	Equal	Substitute <i>s</i> _{<i>p</i>} ² * * for <i>s</i> ₁ ² and <i>s</i> ₂ ² , so $\sqrt{s_p^2/n_1 + s_p^2/n_2} = \sqrt{s_p^2(1/n_1 + 1/n_2)}$
Equal	Equal	Substitute <i>n</i> for <i>n</i> ₁ and <i>n</i> ₂ , and <i>s</i> _{<i>p</i>} ² * * for <i>s</i> ₁ ² and <i>s</i> ₂ ² , so $\sqrt{(s_1^2/n + s_2^2/n)} = \sqrt{2s_p^2/n}$

** $s_p^2 = [(n_1 - 1)s_1^2 + (n_2 - 1)s_2^2]/(n_1 - 1 + n_2 - 1)$. This weighted average is called a pooled sample variance

3.1.3 Connection to the Confidence Interval

Having concluded a testing hypothesis to merely claim if the two independent means are equal, the researchers may further be interested in estimating the size of the mean difference and its confidence interval. The following numerical illustration is to construct a 95% confidence interval for the mean difference using Example 3.1.

The observed mean difference (i.e., the point estimate of the mean difference between the groups with treatment and placebo) = 7.08–6.26 = 0.82, and the standard error of that observed mean difference = 0.339. Finding the 2.5th and 97.5th percentiles of *t* with *df* = 28 using Excel, i.e., TINV (0.05, 28) = 2.048 (or using a table for the percentiles of *t*-distribution). The lower limit of the 95% confidence interval (CI) based on *t*-distribution with *df* = 28 is 0.82–2.048 × 0.339 = 0.126, and its upper limit is 0.82 + 2.048 × 0.339 = 1.514, thus 95% confidence interval is 0.126–1.514. Note that although the test was directional, it is a tradition to construct the 95% nondirectionally disregarding the directionality of testing the hypotheses. Summary: These data showed that the mean of the treated group was greater by 0.82 (95% CI: 0.126–1.514).

3.2 Paired Sample *t*-Test for Comparing Paired Means

Pairing helps a study design control the subject-to-subject outcome variation in that the responses to a study medication being studied may vary among different subjects and testing whether the average of the within-subject longitudinal changes in response would eliminate the source between subject variation. The following example illustrates the paired sample *t*-test that is applied to such a clinical investigation (Fig. 3.4).

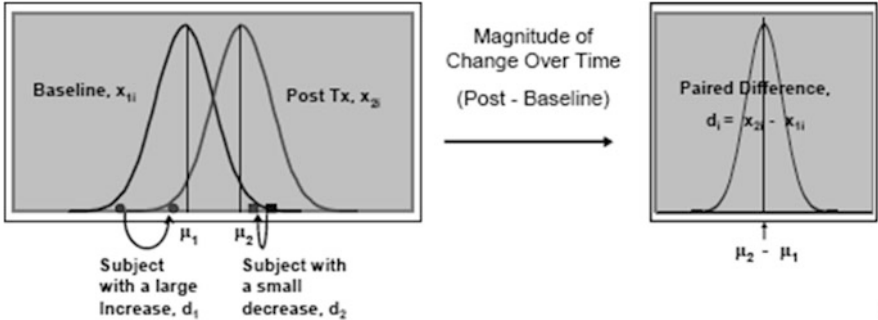


Fig. 3.4 Illustration of distributions of paired normally distributed continuous outcomes and their paired differences

Example 3.2

Does the use of oral contraceptives (OC) affect systolic blood pressure (SBP)? A self-control (pre-measurement and post-measurement pairs) design is applied, and the following test for $H_0: \mu_2 - \mu_1 = 0$ and $H_1: \mu_2 - \mu_1 \neq 0$ is considered.

The following data are the systolic blood pressure measurements in mmHg from 10 study volunteers before using OC (i.e., baseline) and those taken from the same 10 women after the use of OC for a certain period (i.e., follow-up). Let us assume that the distribution of the SBP values in the population is a Gaussian distribution.

Subject	Baseline SBP	Follow-up SBP
1	115	128
2	112	115
3	107	106
4	119	12
5	115	122
6	138	145
7	126	132
8	105	109
9	104	102
10	115	117

The following preliminary data analysis (i.e., data reduction) was performed and summarized in Table 3.3. Do these summary statistics describe the phenomenon clearly?

While the summary statistics in Table 3.3 and the graphical display of the data depicted by Fig. 3.5 described that the mean SBP at follow-up was slightly elevated and the variability was also slightly increased, the paired nature of the design was not reflected. However, Fig. 3.6 illustrates a typical graphical description of the paired data in that most subjects showed an increase in the follow-up SBP, and the second panel described that the blood pressure range of the subjects was larger than the average within-subject change.

Table 3.3 Sample statistics of the SBP at baseline and follow-up

	Baseline SBP	Follow-up SBP
n	10	10
Mean	115.00	120.40
Median	115.00	119.50
Standard deviation	10.31	13.22
Standard error of the mean	3.26	4.18



Fig. 3.5 Illustration of inappropriate box-and-whisker plot for paired continuous outcomes

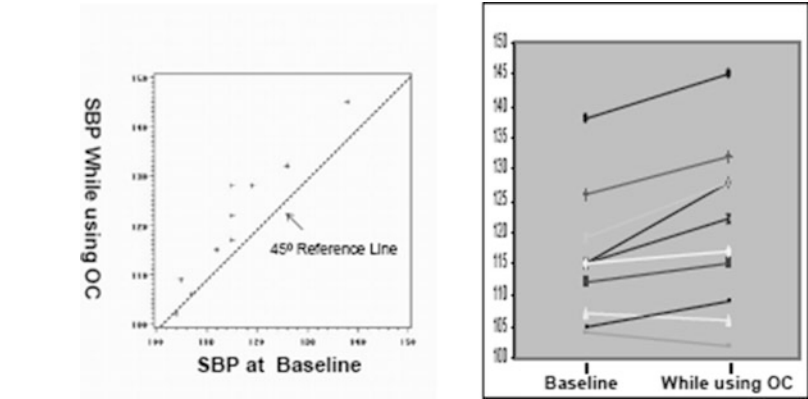


Fig. 3.6 Illustration of properly displayed paired continuous outcomes

A significance level of 5% is adopted. The test is derived based on the *Observed* ~ *Null Value* ~ *SE* trio. To complete this task, let us revisit the articulated hypotheses. By letting δ denote $\mu_2 - \mu_1$, the null and alternative hypotheses are rewritten as $H_0: \delta = 0$ and $H_1: \delta \neq 0$. The observed estimate of δ , $d = 4.80$, is the sample mean of 10 changes in SBP between the baseline and follow-up, and the null value of δ is 0. The $SE(d - \delta) = SE(d)$ is directly estimated from the 10 within-

Subject	Baseline SBP	Follow up SBP	Difference (<i>d_i</i>) of the repeated measures
1	115	128	13
2	112	115	3
3	107	106	-1
4	119	128	9
5	115	122	7
6	138	145	7
7	126	132	6
8	105	109	4
9	104	102	-2
10	115	117	2
Mean	115.60	120.40	4.80
SD	10.31	13.22	4.57
SE	3.26	4.18	1.44

Fig. 3.7 Illustration of calculated paired differences that can be analyzed by a one-sample *t*-test



Fig. 3.8 Box-and-whisker plot applied to describe the distribution of differences calculated from pairs of continuous outcomes

subject longitudinal changes. The resulting test statistic is the same as that of the one-sample *t*-test as the within-subject change is treated as the unit of analysis, which is depicted by Figs. 3.7 and 3.8. The test statistic $t = (d - \delta) / SE(d - \delta) = d / SE(d)$ follows the *t*-distribution with $df = n \text{ of pairs} - 1 = 10 - 1 = 9$.

The resulting $t = (4.8 - 0) / 1.44 = 3.324$ and *p*-value is 0.0089, which indicates that the mean change of 4.8 was significant at 5% significance level.

The paired sample *t*-test, as the baseline and follow up data (i.e., two outcome variables) are reduced into the pairwise within-subject differences (i.e., transformed one outcome variable), is the same as the one-sample *t*-test, and *df* is the number of unique individuals - 1.

A common misapplication of the *t*-test to the paired data is to apply the independent samples *t*-test. As illustrated below, such an erroneous application would mislead the study investigation. If the equal population variance independent samples *t*-test (see Table 3.2 in Sect. 3.1.2) had been applied to the above example, then

the test statistic’s denominator, $SE(\bar{x}_2 - \bar{x}_1)$, where $\bar{x}_1$ and $\bar{x}_2$ are the baseline and follow-up mean SBPs, respectively, turns out to be $\sqrt{s_1^2/n + s_2^2/n} = \sqrt{2s_p^2/n}$, where $s_p^2 = [(n_1 - 1)s_1^2 + (n_2 - 1)s_2^2]/(n_1 - 1 + n_2 - 1)$, $n_1 = n_2 = 10$, $s_1^2 = 10.31^2$, and $s_2^2 = 13.22^2$. The calculated value of this standard error is 5.3028. Then the test statistic $t = (4.8 - 0)/5.3028 = 0.91$ with $df = 18$. The p -value is 0.3773, which is contradictory to the result from the paired sample t -test.

3.3 Use of Excel for *t*-Tests

The one-sample t -test, independent samples t -test, and paired sample t -test can easily be carried out using Excel (Figs. 3.9, 3.10, and 3.11).

Use of Excel to perform a **diagnostic test for heteroskedasticity** before choosing the proper independent samples t -tests between the equal- versus unequal variance versions

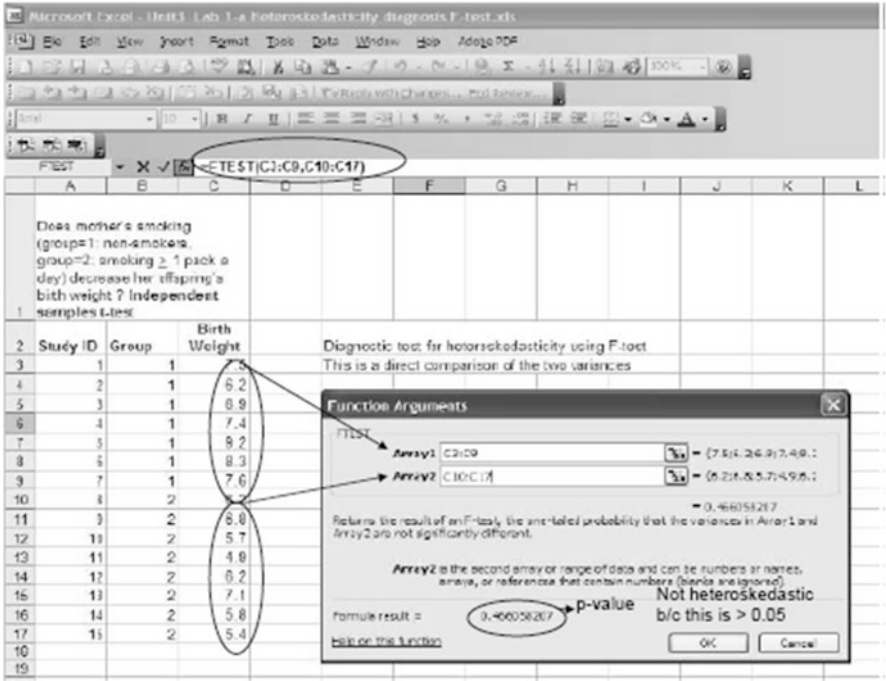


Fig. 3.9 Use of Excel function FTEST for testing equality of two variances

Use of Excel to perform **Independent Samples t-test**

Independent *t*-test, i.e., Two Sample Unpaired *T*-test using Excel

Please see how the data are entered into the spread sheet; and see the array 1 and array 2, and the options given to TTEST function.

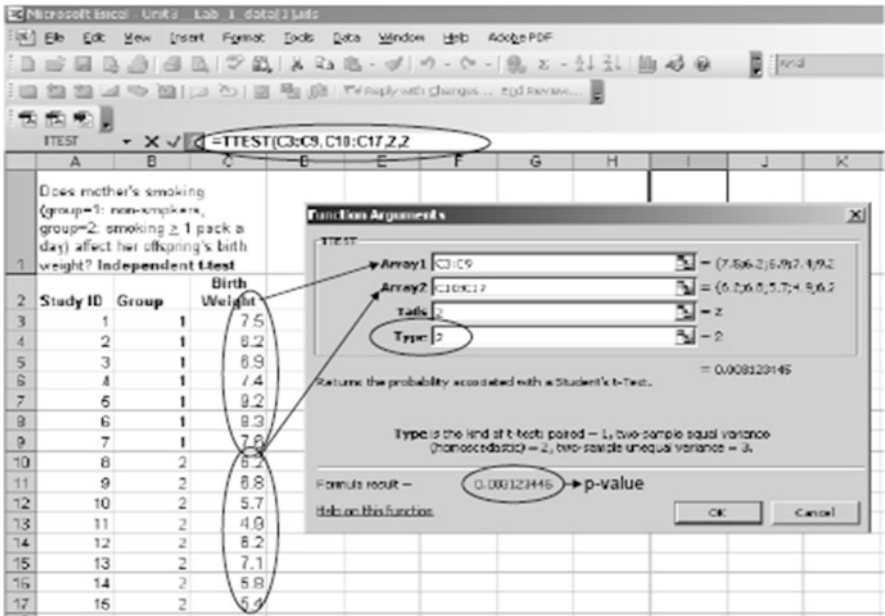


Fig. 3.10 Use of Excel function for independent samples *t*-tests

Study Questions

1. Explain the idea of *Observed ~ Expected ~ SE* trio formulation of the test statistic *t* of the independent samples *t*-test.
2. Can the paired sample *t*-test be conceived as a one-sample *t*-test? Why?
3. What is the idea behind to adjust (i.e., decrease) the degrees of freedom for the independent samples *t*-test when the variances of the two populations are not equal?
4. Please criticize the following awkward sentences:
 - “These data showed that the two groups were significantly different ($p < 0.05$).”
 - “These data showed that the two group-specific means were significantly different.”
 - “The calculated *p*-value was less than 0.05, thus we rejected the null hypothesis.”

Use of Excel to perform **Paired Samples t-test**

Paired t-test

Please see how the data are entered into the spread sheet, and see the array 1 and array 2, and the options given to TTEST function.

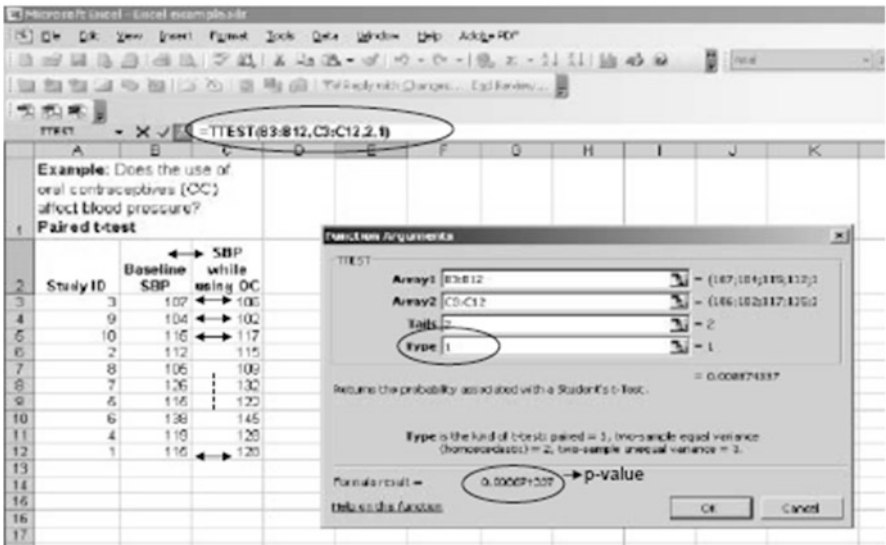


Fig. 3.11 Use of Excel function for paired sample *t*-test

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Chapter 4

Inference Using Analysis of Variance (ANOVA) for Comparing Multiple Means



This chapter discusses single-factor analysis of variance (ANOVA), which is mainly applied to compare three or more independent means. The term “single factor” refers that the means are compared across levels of a “single” classification variable (i.e., classification of means by a single categorical variable). The classification variable is called the independent variable or factor (thus, the method is also called single-factor ANOVA), and the outcome variable whose means are compared is called the dependent variable. This method requires certain assumptions: (1) the dependent variable values are the observations sampled from a normal distribution and (2) the population variances are equal (homoskedasticity) across the levels of the independent variable (Fig. 4.1).

Having mentioned that ANOVA is mainly applied to compare three or more means, is it obvious why the variance is analyzed to compare the means? The ANOVA is a two-step procedure. The first step is to measure two partitioned amounts of outcome data variations due to two sources of variations of which the first is the variation of the outcome variable explained by the groups being compared, and the second is the unexplained residual (error) variation. The next step is to utilize these two amounts of data variations to carry out the hypothesis testing for comparing the means. These amounts of data variations are measured by means of sums of squares (see Sect. 1.3.4).

4.1 Sums of Squares and Variances

In Fig. 4.2, three groups of sample data that are clearly separated (for illustrative purposes) by the underlying group effects are illustrated, and the symmetrically scattered outcomes around their group means reveal the random sampling error. ANOVA is to examine whether the group effect depicted by the distances among the three group means can separate the group-wise data cluster in the presence of the random sampling error. The group effect (i.e., signal) and the random sampling error

Fig. 4.1 Illustration of homoskedastic density curves of three continuous outcomes with unequal means

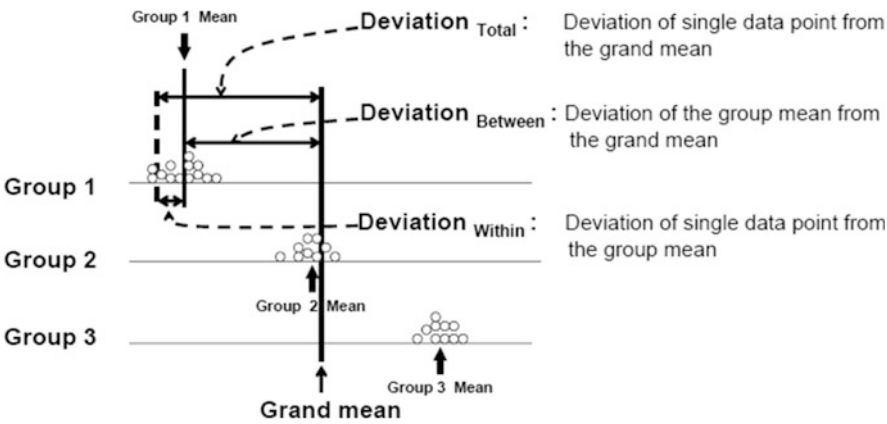
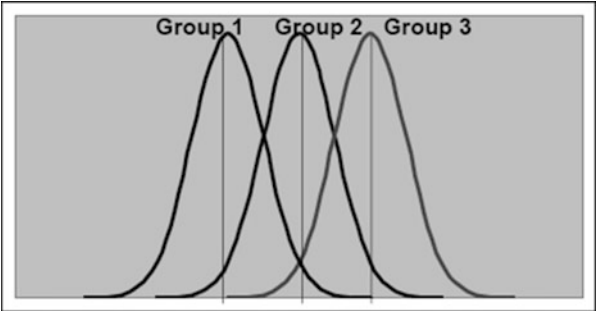


Fig. 4.2 Illustrated deviations arising from comparing three independent means

(i.e., noise) are measured by two kinds of sums of squares first and then transformed into kinds of average sum of squares (so-called mean squares).

Figure 4.3 demonstrates the concepts of derivations and these sums of squares (SS) that are the underpinnings of ANOVA. For instance, the first observed outcome value 5 (Sample Number 1 in Group 1) is conceived as a value deviated by -15 from the single grand mean of 20; this outcome is also conceived as a value deviated by -5 from its group mean 10 wherein this group mean is deviated by -10 from the grand mean 20. It is obvious that the three corresponding squared deviations are $(-15)^2 = 225$, $(-5)^2 = 25$, and $(-10)^2 = 100$. Repeating the same calculation over every observation within each group and cumulating the resulting individual squared deviations into the group sum totals, and then summing the group sum totals over the three groups finally produces three kinds of sums of squares. These three sums of squares are called total-sum, between-sum, and within-sum of squares, and the resulting values are 950, 800, and 150, respectively. Intuitively, the total sum of squares of 950 is partitioned into between-group sum of squares of 800 and within-group sum of squares of 150 (Fig. 4.4).

Dividing a sum of squares by a divisor (i.e., degrees of freedom, see Sect. 2.2.3) is a kind of variance. In ANOVA, two such variances are compared, and these are the

Sample Number	Group	Observed Outcome Value	Deviation of Each Single Data Point from Grand Mean	Deviation of Each Group Mean from Grand Mean	Deviation of Each Single Data Point from Group Mean	Squared Deviation of Each Single Data Point from Grand Mean	Squared Deviation of Each Group Mean from Grand Mean	Squared Deviation of Each Single Data Point from Group Mean
1	1	5	-15	-10	-5	225	100	25
2	1	10	-10	-10	0	100	100	0
3	1	10	-10	-10	0	100	100	0
4	1	15	-5	-10	5	25	100	25
Group 1 Mean = 10						Sub total = 450	Sub total = 400	Sub total = 50
5	2	15	-5	0	-5	25	0	25
6	2	20	0	0	0	0	0	0
7	2	20	0	0	0	0	0	0
8	2	25	5	0	5	25	0	25
Group 2 Mean = 20						Sub total = 50	Sub total = 0	Sub total = 50
9	3	25	5	10	-5	25	100	25
10	3	30	10	10	0	100	100	0
11	3	30	10	10	0	100	100	0
12	3	35	15	10	5	225	100	25
Group 3 Mean = 30						Sub total = 450	Sub total = 400	Sub total = 50
Grand Mean = 20						SS Total = 950	SS Between = 800	SS Within = 150

Fig. 4.3 Formulae-less numerical illustration of sum of squares in single-factor ANOVA

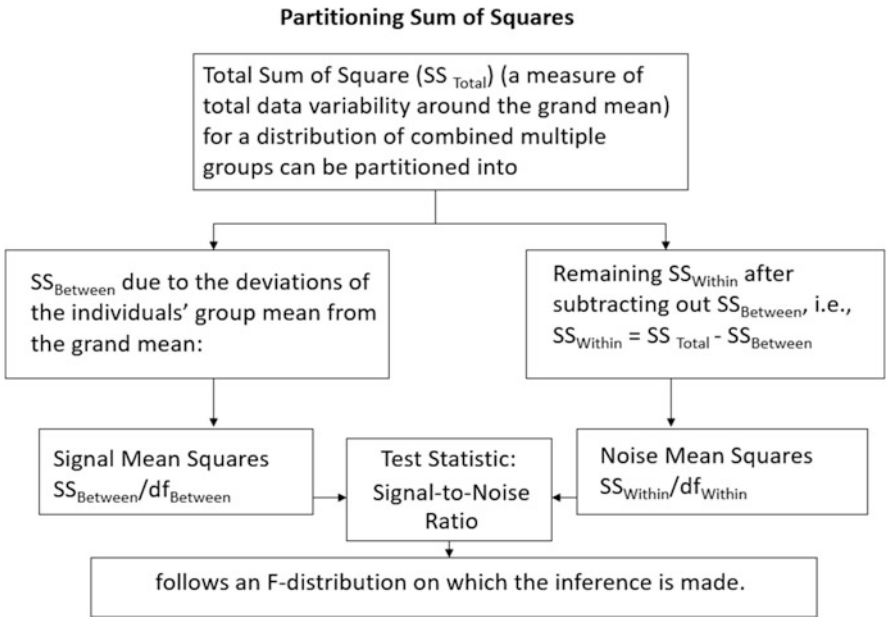


Fig. 4.4 Partitioning sum of squares for single-factor ANOVA

variance due to the group difference and the variance due to the sampling error (i.e., unexplained residual variance by the systematic group difference). To distinguish the meaning of the ordinary variance, these variances obtained in AVOVA are termed “mean squares” (MS). The two major mean squares in the single-factor ANOVA are the between-group mean square (MS_{between}) and the within-group mean square (MS_{within}). MS_{between} is the sum of squared deviations from the grand mean for all data values divided by its divisor for which the divisor is one less than the number of unique deviations of the group means from the referenced overall grand mean. Note that as demonstrated in Fig. 4.3, there are only three unique deviations of the group specific means from the grand mean, and the MS_{between} becomes $(400 + 0 + 400)/(3-1) = 800/2 = 400$. MS_{within} is $(50 + 50 + 50)/[(\text{number of unique deviations of the individual data points from Group 1 mean} - 1) + (\text{number of unique deviations of the individual data points from Group 2 mean} - 1) + (\text{number of unique deviations of the individual data points from Group 3 mean} - 1)] = 150/[(4-1) + (4-1) + (4-1)] = 150/9 = 16.67$.

4.2 F-Test

With the calculated between-group and within-group mean squares, the hypothesis test to compare means can begin. The null and alternative hypotheses are $H_0: \mu_1 = \mu_2 = \mu_3, \dots, \mu_k$ (all means are equal) and H_1 : At least one mean is different (not all means are equal). The test statistic is derived under the assumption that the data are drawn from normally distributed populations and that the population variances are equal across the groups. Unlike the t -tests, the test statistic of this test is not derived from the “trio” (see Sects. 2.2.4.5 and 3.1). Instead, the ratio of the two mean squares (see Sect. 4.1) is used as the test statistic. By letting F denote this ratio, $F = MS_{\text{between}}/MS_{\text{within}}$ is the signal (of the between-group difference)-to-noise (random sampling error)-ratio test statistic. Under the null hypothesis, this test statistic follows an F -distribution, which is characterized by two degrees of freedom, df_{between} and df_{within} . Each df is the divisor that is used for calculating the mean squares, and each divisor is one less than the number of unique squared deviations from the referenced mean that are summed into the SS calculations.

$F = 400/16.67 = 23.99$ is a specific value of the test statistics from the observed data that will follow the F -distribution with $df_{\text{between}} = 2$ and $df_{\text{within}} = 9$. The method to identify the rejection region using the F -distribution table and the graphical illustration are illustrated in Fig. 4.5. Evidently, such a large value 23.99 fell into the rejection region (i.e., $F > 4.26$); thus, these data showed the evidence to reject H_0 at a 5% significance level.

Example 4.1

A cross-sectional study design is applied to examine if a mother’s smoking affected the offspring’s birth weight. The null and alternative hypotheses for the inference are

Upper 95 th Percentiles of F-Distributions										
df ₁	1	2	3	4	5	6	7	8	9	10
df ₂										
2	18.51	19.00	19.16	19.25	19.30	19.33	19.35	19.37	19.38	19.40
3	10.13	9.55	9.28	9.12	9.01	8.94	8.89	8.85	8.81	8.79
4	7.71	6.94	6.59	6.39	6.26	6.16	6.09	6.04	6.00	5.96
5	6.61	5.79	5.41	5.19	5.05	4.95	4.88	4.82	4.77	4.74
6	5.99	5.14	4.76	4.53	4.39	4.28	4.21	4.15	4.10	4.06
7	5.59	4.74	4.35	4.12	3.97	3.87	3.79	3.73	3.68	3.64
8	5.32	4.46	4.07	3.84	3.69	3.58	3.50	3.44	3.39	3.35
9	5.12	4.26	3.86	3.63	3.48	3.37	3.29	3.23	3.18	3.14
10	4.96	4.10	3.71	3.48	3.33	3.22	3.14	3.07	3.02	2.98
11	4.84	3.98	3.59	3.36	3.20	3.09	3.01	2.95	2.90	2.85
12	4.75	3.89	3.49	3.26	3.11	3.00	2.91	2.85	2.80	2.75

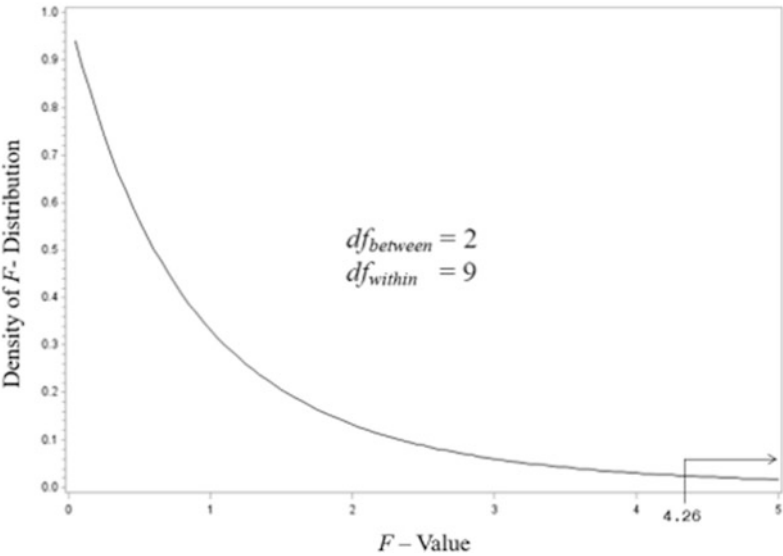


Fig. 4.5 Determination of the rejection region of the *F*-distribution for an ANOVA *F*-test for comparing three means

$H_0: \mu_1 = \mu_2 = \mu_3 = \mu_4$ (all four means are equal) and H_1 : At least one mean is different (not all means are equal).

The following data are birth weights (lb) of 27 newborns classified by their maternal smoking status (i.e., one-way classification). The data normality is assumed.

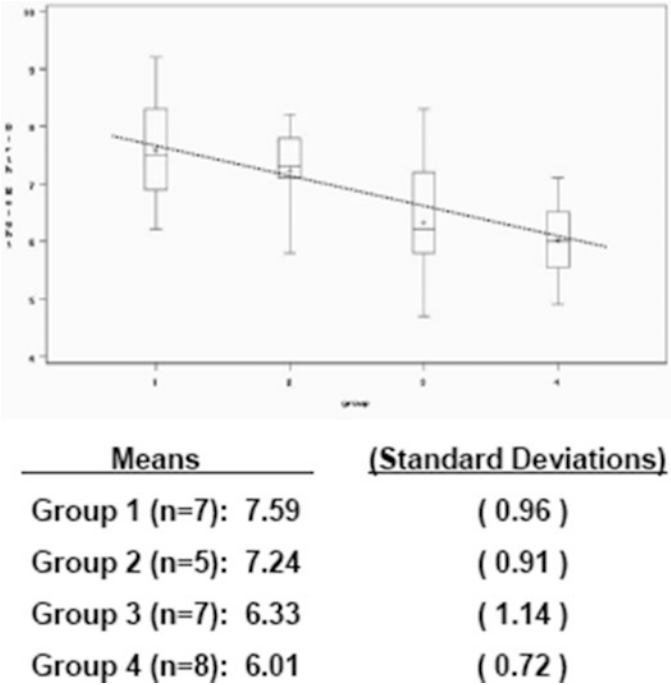


Fig. 4.6 Box-and-whisker plot of four distributions whose mean are compared by single-factor ANOVA F -test

Table 4.1 ANOVA table

Source of variation	SS	Df	MS	F	p-value
Between group	11.74	3	3.91	4.20	0.017
Within group	21.28	23	0.93		
Total	33.02	26			

Group 1 – Mother is a nonsmoker (n = 7): 7.5 6.2 6.9 7.4 9.2 8.3 7.6
Group 2 – Mother is an ex-smoker (n = 5): 5.8 7.3 8.2 7.1 7.8
Group 3 – Mother smokes <1 pack/day (n = 7): 5.9 6.2 5.8 4.7 8.3 7.2 6.2
Group 4 – Mother smokes ≥1 pack/day (n = 8): 6.2 6.8 5.7 4.9 6.2 7.1 5.8 5.4
(Fig. 4.6)

The means and standard deviations are obtained as the descriptive summary statistics. Besides these descriptive statistics, a summary table (so-called ANOVA table) is traditionally to present the sum of square and mean square for each source of variation as well as the test statistic F and its p -value (see Table 4.1).

Computation of the sums of squares and mean squares can be done either by a computer package program or manually. As this chapter does not offer directly usable computational formulae, this illustration is made in order to solely walk you

through the essential computations of the variance partitioning and the derivation of the test statistic.

The first quantity to calculate is how much the group means deviate from the grand mean, where the grand mean, $\bar{x}_G$, is the weighted average of the four means, where the weights are the group sample sizes, i.e., $\bar{x}_G = (7 \times 7.59 + 5 \times 7.24 + 7 \times 6.33 + 8 \times 6.01)/27 = 6.73$.

$SS_{between}$ is the summation of seven of the squared unique Group 1 sample mean's deviation from the grand mean, i.e., $7 \times (\bar{x}_1 - \bar{x}_G)^2$, summation of sum of five of such squared value from Group 2, i.e., $5 \times (\bar{x}_2 - \bar{x}_G)^2$, $7 \times (\bar{x}_3 - \bar{x}_G)^2$ from Group 3, and $8 \times (\bar{x}_4 - \bar{x}_G)^2$ from Group 4, which is $7 \times (7.59 - 6.73)^2 + 5 \times (7.24 - 6.73)^2 + 7 \times (6.33 - 6.73)^2 + 8 \times (6.01 - 6.73)^2 = 11.74$.

SS_{within} is the summation of four within-group sum of squares whose each within-group sum of squares is obtainable from the already calculated within-group standard deviation. Because the variance within k th subgroup s_k^2 is $SS_{k\ within}/(n_k - 1)$, where $k = 1, 2, 3$, and 4 indicating the group, the sum of squared deviation within the k th subgroup $SS_{k\ within} = s_k^2 \times (n_k - 1)$. Therefore, SS_{within} from all four groups $= (7-1) \times 0.962 + (5-1) \times 0.912 + (7-1) \times 1.142 + (8-1) \times 0.722 = 21.28$.

Finally, $F = MS_{between}/MS_{within} = [11.74/(4-1)]/[21.28/(7-1) + (5-1) + (7-1) + (8-1)] = (11.74/3)/(21.28/23) = 3.91/0.93 = 4.20$. Finally, the p -value was directly evaluated by using Excel (not by determining the critical region and seeing if the observed F value fell into the critical region), i.e., $p = \text{FDIST}(4.20, df_{between} = 3, df_{within} = 23) = 0.017$. Figure 4.7 is the graphical demonstration for this calculation wherein the density function is an F -distribution characterized by the two required degrees of freedom and 3 and 23 appeared differently from the one with degrees of freedom 2 and 9 (Fig. 4.5). Note that the shape of the density curve of an F -distribution is characterized by the degrees of freedom (i.e., not all F -distributions look similar).

Although the p -value was directly calculated in this example, the significance can also be determined by checking if the test statistic fell into the critical region. The critical region of this test is $F > 3.03$, which is found from the critical value table of F -distribution at 5% significance level with the between- and within-group degrees of freedom of 3 and 23.

The suggestive format of the summary sentence is either "These data showed that at least one age group's mean birth weight is significantly different from the means of the other age groups ($p = 0.017$)" or "These data showed that at least one age group's mean birth weight is significantly different from the means of the other age groups at 5% significance level."

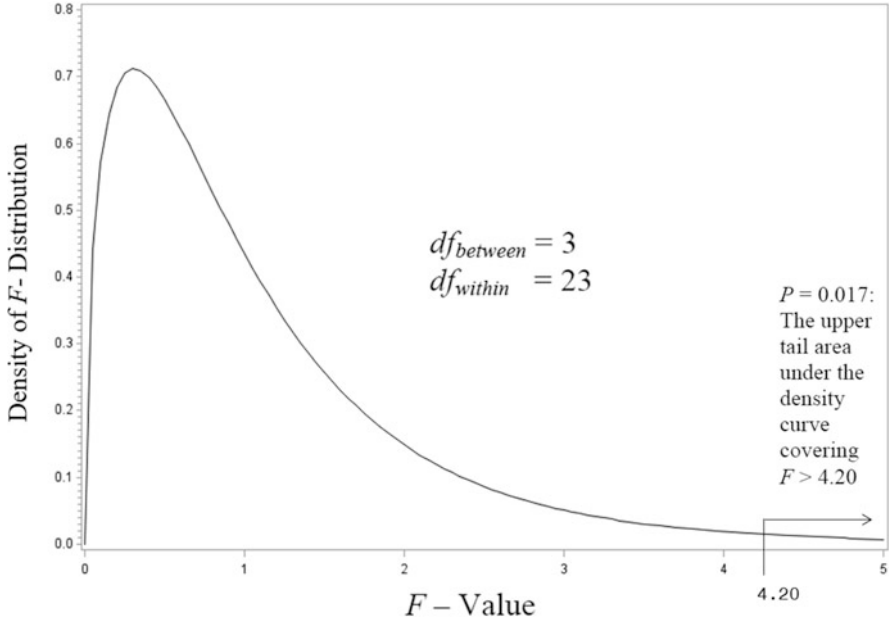


Fig. 4.7 *p*-value calculation of the *F*-statistic obtained from an ANOVA *F*-test comparing three means using the data set illustrated in Fig. 4.6

4.3 Multiple Comparisons and Increased Chance of Type 1 Error

Suppose $H_0: \mu_1 = \mu_2 = \mu_3 = \mu_4$ in favor of H_1 : At least one mean differs from the others was rejected by a one-way ANOVA *F*-test. It remains ambiguous which specific means differed until all possible pair-wise tests (six independent samples *t*-tests can be performed in this case) comparing two groups at a time. In doing so, the increased number of tests increases the potential Type 1 error. To protect from such an increased chance of committing Type 1 error, a stringent criterion (i.e., modified significance level) for these tests needs to be adopted. One of such options is to lower the significance level of each test by dividing it by the number of comparisons (Bonferroni's correction, i.e., adjusted $\alpha = 0.05/6 = 0.0083$, or to inflate the calculated *p*-value by multiplying it by the number of comparisons, i.e., inflated *p*-value = $6 \times$ observed *p*-value). However, this tends to be too conservative as the number of tests increases. Many “*not-too-conservative but powerful*” tests have been invented. The least significance difference (LSD), highly significant difference (HSD), student-Newman-Keuls (SNK), and Duncan's multiple range test procedures are popularized procedures applying pair-wise multiple tests for comparing means, and Dunnett's procedure is a procedure to compare ordered groups with a baseline group using modified critical values of the test statistic.

4.4 Beyond Single-Factor ANOVA

4.4.1 Multi-factor ANOVA

As the number of categorical independent variables increases and the outcomes are classified by these independent variables, the ANOVA F -tests will involve the partitioning of the total variance into the between-group variances due to the effects of these individual independent variables, the between-group variances due to the interactions of two or more independent variables, and the remaining variance that is not explained by those sources of variations that are already taken into account. The testing procedure for comparing subgroup means (e.g., the difference of the means among the levels of the first independent variable, that among the levels of the second independent variable, and that among the levels arising from the interaction of the first and second independent variables) is F -test whose test statistic's numerator is the mean square due to the between-group effect of interest, and the denominator is the unexplained error mean square. With a firm understanding of the single-factor ANOVA, it can be a trivial calculation problem to extend the method to the multifactor ANOVA. However, it would be worth addressing the definition and interpretation of the two topics discussed in Sects. 4.4.2 and 4.4.3.

4.4.2 Interaction

The following example focuses on illustrating the definition of interaction as well as the marginal means, main effects, and simple means arising in the two-factor ANOVA. The example is self-explanatory and does not necessitate verbal definitions.

Example 4.2 A study of abdominal fat reduction (measured in % reduction) after 8-week programs of exercise alone or diet and exercise conducted involving two age groups of age < 50 years and age $\geq$ 50 years. The sample size of each subgroup was 2000. Let us assume that every observed difference was statistically significant (Table 4.2).

Table 4.2 Means of % fat reduction by age group and program

	Program 1: Exercise alone	Program 2: Exercise and diet
Age < 50	n = 2000 Mean reduction = 5%	n = 2000 Mean reduction = 8%
Age $\geq$ 50	n = 2000 Mean reduction = 3%	n = 2000, Mean reduction = 4%

Q1. Please calculate (a) marginal means, (b) main effects, and (c) simple effects.

- (a) Marginal means: marginal mean of Program 1 = $(5 + 3)/2 = 4\%$; marginal mean of Program 2 = $(8 + 4)/2 = 6\%$ regardless of the age; marginal mean of age < 50 is $(5 + 8)/2 = 6.5\%$, and marginal mean of age ≥ 50 is $(3 + 4)/2 = 3.5\%$ regardless of the program type.
- (b) Main effects (effects toward the healthier outcome): program main effect (marginal mean of Program 2 – marginal mean of Program 1) = $(8 + 4)/2 - (5 + 3)/2 = 6 - 4 = 2\%$, meaning that Program 1 showed a 2%-point greater fat reduction in all ages; age main effect (mean of age < 50 – mean of age ≥ 50) = $(5 + 8)/2 - (3 + 4)/2 = 6.5 - 3.5 = 3\%$, meaning the younger age group showed a 3%-point greater reduction on average regardless of the program.
- (c) Simple effects: The four specific means by themselves simply exhibit the age-specific and program-specific subgroup mean outcomes and do not directly describe the “effects (i.e., mean differences).” Simple effect is the size of mean difference within each group. Program 2 versus 1 simple effect within age < 50 is 3% (i.e., $8 - 5 = 3\%$); Program 2 versus 1 simple effect within age ≥ 50 is 1% (i.e., $4 - 3 = 1\%$); also, age < 50 versus ≥ 50 simple effect within Program 1 is 2%, and that of Program 2 is 4%.

Q2. Was there an interaction? If so, please describe the observed phenomenon.

There was an interaction in that the younger age group’s Program 2 versus Program 1 effect in mean reduction was greater than that of the older age group. The younger age group showed three times better performance (i.e., $8 - 5 = 3\%$ reduction in the younger age group by adding diet to the exercise, versus $4 - 3 = 1\%$ reduction in the older age group by adding diet to the exercise). Yes, there was an interaction (i.e., age interacted with the programs). Note that the observed interaction is a phenomenon, and its measured quantity (i.e., the interaction effect) is the difference in simple effects between the two age groups (i.e., $(8 - 5 = 3\% \text{ in age } < 50) - (4 - 3 = 1\% \text{ in Age } \geq 50) = 2\%$). If both age groups had shown the same mean fat reduction differences, we would have pronounced that there was no interaction.

4.4.3 Repeated Measures ANOVA

The importance of choosing the appropriate method to take into account the source of variation arising from the study design and experiment cannot be overemphasized. The paired sample t -test is applied to compare two related means (e.g., pre-mean to post-mean difference) because the independent samples t -test does not take into account the related with individual paired measures. Many approaches as a means to the companion to the paired sample t -test are available for comparing more than two related means. Depending on the patterns of the within-subject

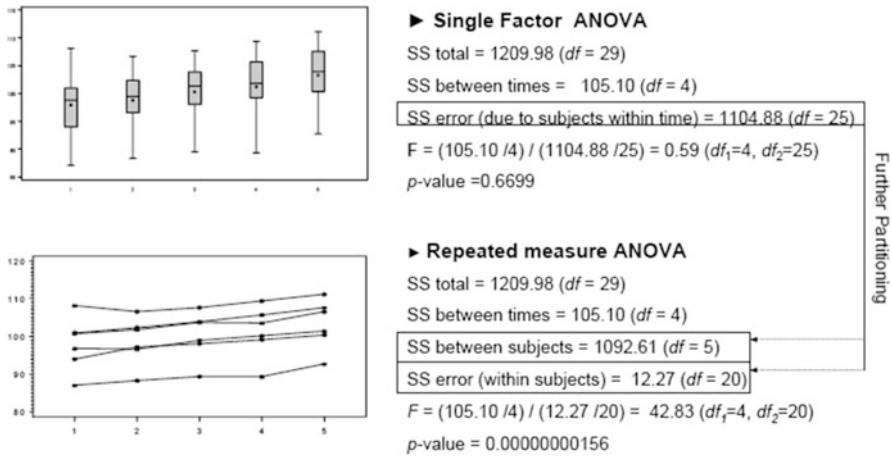


Fig. 4.8 Illustration of improved result by repeated measures ANOVA compared to single-factor ANOVA

correlation among more than two repeated measures, there are various technical options to deal with the situations. The repeated measures ANOVA is one such option, and its idea is illustrated in Fig. 4.8. This was done to study whether or not the means of the repeatedly measured outcomes changed over time, i.e., $H_0: \mu_1 = \mu_2 = \mu_3 = \mu_4 = \mu_5$ versus H_1 : Not all five means are equal (i.e., at least one mean at a certain time is different from the rest), by measuring the longitudinal (i.e., those five serial outcomes are related within a subject) outcomes from six ($n = 6$) individual subjects. The first part of the illustration is a result of applying the single-factor ANOVA F -test, which was not appropriate because it did not take into account the repeated measurements. Let us discuss which explainable source of variation could have been taken into account if the analysis had been done by an appropriate method. The box-and-whisker plot showed a gradually increasing pattern of the means over time, but the single-factor ANOVA F -test's p -value was 0.6699 (i.e., not significant at 5% alpha level). Keeping in mind that these data are repeated measures, the individual subject-specific unique patterns (i.e., trajectory over time) and their difference between the subjects could be a new explainable source of outcome variation. As depicted by the second plot (so-called spaghetti plot), the visualized six individual trajectories describe the data variation differently than what was possible solely by the box-and-whisker plot. What can be pointed out by this "spaghetti plot" is that there was a common trend of increasing pattern among the subjects and a non-ignorable portion of the overall outcome spread-out (i.e., total variance) was clearly due to the initial baseline variation among the subjects (i.e., the trajectories did not cross over very much). For this reason, it is reasonable to add the between-subject variation as a newly explainable source of data variation. As shown in the two ANOVA tables, $SS_{\text{between subjects}}$ in the repeated measures ANOVA summary was partitioned from the previous SS_{error} (between and within subjects had been combined) of the single-factor ANOVA (i.e., $1104.88 = 1092.61 + 12.27$).

Finally, the smaller portion that has not still been explained by time and between-subject variability sources remained as SS_{error} (only within subjects), thus the p -value of the repeated measures ANOVA F -test for the time effect resulted much smaller than that of the single-factor ANOVA F -test. Note that this approach is indeed the same as to apply a two-factor ANOVA in which the between-subject effect is considered the second factor. This illustration skipped the detailed calculation involved in the analysis because it is unnecessarily laborious without using computers. The objective of this illustration was to focus mainly on the rationale for dealing with the repeated measures and its impact on the improved the result and interpretation.

4.4.4 Use of Excel for ANOVA

Excel offers the ANOVA procedure for single, two-factor, and repeated measures ANOVA. The features allow only the data sets whose group sample sizes are equal. However, such data sets can still be handled by Excel via regression analysis with dummy variables (see Chap. 5).

Study Questions

1. Using an Excel sheet, please carry out the computation of sums of squares for the one-way ANOVA and F -test ($H_0: \mu_1 = \mu_2 = \mu_3$) of these data (similar demonstration is available in Fig. 4.3):

Sample data from group 1: 1, 2, 3, 4, 5

Sample data from group 2: 4, 5, 6

Sample data from group 3: 6, 7, 8

2. Given these descriptive statistics, please figure out what are the $SS_{between}$ and SS_{within} for the one-way ANOVA.

Group 1: $n = 9$, sample mean = 10, sample variance = 36

Group 2: $n = 10$, sample mean = 15, sample variance = 36

Group 3: $n = 11$, sample mean = 20, sample variance = 36

3. If the independent samples t -test is applied to compare two means, and the population variances are equal, can the one-way ANOVA be applied to this inference? If so, will the p -value from the applied one-way ANOVA be the same as that resulted from the independent samples t -test?

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Chapter 5

Inference of Correlation and Regression



In Chap. 1, Pearson's correlation coefficient, as a means to describe a linear association between two continuous measures, was introduced. In this chapter, the inference of the correlation coefficient using sample data will be discussed first, and then the discussion will extend to a related method and its inference to examine a linear association of the continuous and binary outcomes with one or more variables using sample data.

5.1 Inference of Pearson's Correlation Coefficient

A linear association measured by Pearson's correlation coefficient between two continuous measures obtained from a sample, r , requires an inference. The two areas of inferences are hypothesis testing and interval estimation (i.e., construction of the confidence interval, CI). Testing hypothesis is to state the null and alternative hypotheses, compute the test statistic and determine if it is significant. Let us discuss the hypothesis testing first. The null hypothesis states that there is no linear association between two continuous outcomes (i.e., $H_0: \rho = 0$), and the alternative hypothesis either a nondirectional alternative hypothesis (i.e., $H_1: \rho \neq 0$) or a directional alternative hypothesis (i.e., $H_1: \rho > 0$, or $H_1: \rho < 0$), depending on the researcher's objective. For such an inference, we need a test statistic. A typical test statistic involves an arithmetic transformation of the sample correlation coefficient r because the sampling distribution of r is not approximately normal even when the sample size becomes large (i.e., the central limit theorem (CLT) is not applicable for the sample correlation coefficient). Nonetheless, it is noted that the sampling distribution of the following transformation (Fisher's z transformation) $z = 0.5[\ln(1 + r) - \ln(1 - r)]$ follows $N(0, 1/\sqrt{n-3})$ under the null hypothesis as the number of observed data pairs, n , becomes sufficiently large. The idea of *Observed Estimate* $\sim$ *Null Value* $\sim$ *Standard Error (SE)* trio (see Sect. 2.2.4.5) is then applied to

derive the test statistic. Instead of directly plugging in the observed sample correlation r , Fisher's z transformation is substituted for the observed estimate, i.e.:

$$\begin{aligned}\lambda &= (z - 0) / \left(1 / \sqrt{n-3} \right) = \{ 0.5 [\ln(1+r) - \ln(1-r)] - 0 \} / \left(1 / \sqrt{n-3} \right) \\ &= 0.5 [\ln(1+r) - \ln(1-r)] \sqrt{n-3}.\end{aligned}$$

The significance of this statistic z is then determined by the standard normal distribution, $N(0, 1)$. For small sample size, the testing can be performed using a t -distribution (i.e., one-sample t -test) for which the test statistic is $t_{df=n-2} = r \sqrt{(n-2)/(1-r^2)}$.

The interval estimation can be made by using this z statistic. The lower and upper 95% confidence limits of the population correlation coefficient can be obtained in two steps, of which the first step is to find the lower and upper 95% confidence limits (i.e., 2.5th and 97.5th percentiles of the sampling distribution, i.e., $\lambda = -1.96$ and 1.96 , respectively) of z , and the second step is to solve the following two equations for ρ_l and ρ_u , where the subscripts l and u indicate lower limit and upper limit, respectively:

$$\begin{aligned}-1.96 &= 0.5 [\ln(1+\rho_l) - \ln(1-\rho_l)] \sqrt{n-3}, \text{ and} \\ 1.96 &= 0.5 [\ln(1+\rho_u) - \ln(1-\rho_u)] \sqrt{n-3}.\end{aligned}$$

The solutions for the upper and lower limits are expressed as:

$$\begin{aligned}\rho_l &= \frac{e^{2[\ln(1+r) - \ln(1-r) - 1.96\sqrt{1/(n-3)}]} - 1}{e^{2[\ln(1+r) - \ln(1-r) - 1.96\sqrt{1/(n-3)}]} + 1} \\ \rho_u &= \frac{e^{2[\ln(1+r) - \ln(1-r) + 1.96\sqrt{1/(n-3)}]} - 1}{e^{2[\ln(1+r) - \ln(1-r) + 1.96\sqrt{1/(n-3)}]} + 1}\end{aligned}$$

The inference can be extended to compare two correlation coefficients using Fisher's z -test. Let us consider r_1 and r_2 to be the two independently estimated correlation coefficients from sample sizes of n_1 and n_2 , respectively. The null and alternative hypotheses are $H_0 : \rho_1 - \rho_2 = 0$ and $H_{01} : \rho_1 - \rho_2 \neq 0$. The Fisher's z transformations will be $z_1 = \ln [(1+r_1)/(1-r_1)] \sim N(0, 1/\sqrt{n_1-3})$, and $z_2 = \ln [(1+r_2)/(1-r_2)] \sim N(0, 1/\sqrt{n_2-3})$. According to the *Observed Estimate* $\sim$ *Null Value* $\sim$ *SE* trio, the test statistic will be $\lambda = [(z_1 - z_2) - 0] / \sqrt{SE_1^2 + SE_2^2} = (z_1 - z_2) / \sqrt{1/(n_1-3) + 1/(n_2-3)}$.

Example 5.1

Make an inference if the strength of the linear relationship between two variables in Population 1 is equal to that in Population 2, i.e., are $H_0: \rho_1 - \rho_2 = 0$ and $H_{01}: \rho_1 - \rho_2 \neq 0$. Please construct the 95% confidence interval for each correlation as well as the difference of the two using the normal approximation. The observed sample statistics are $n_1 = 100$, $r_1 = 0.32$ and $n_2 = 100$, $r_2 = 0.35$.

$z_1 = \ln(1 + 0.32)/(1 - 0.32) = 0.66$, $z_2 = \ln(1 + 0.35)/(1 - 0.35) = 0.73$,
 $SE_1 = SE_2 = \sqrt{97} = 9.85$, $\lambda = (0.66 - 0.73)/\sqrt{1/9.85 + 1/9.85} = -0.15$,
 two-sided p -value = Area under the standard normal density curve where is value is < -0.15 or $> 0.15 = 2 * \text{NORM.S.DIST}(-0.15, \text{TRUE}) = 0.88$. These data showed that the strength of the linear relationship between two variables in Population 1 is not different from that in Population 2 ($p = 0.88$).

5.2 Linear Regression Model with One Independent Variable: Simple Regression Model

Linear regression model with one independent variable describes how a value of the dependent variable is represented conjointly by a given value of the independent variable and a random error. The following expression is a typical linear regression model with one independent variable:

$$y_i = (\beta_0 + \beta_1 x_i) + \epsilon_i,$$

where ϵ_i , for individual i , is a random error term that follows a normal distribution with mean = 0 and variance = σ_ϵ^2 , i.e., $N(0, \sigma_\epsilon^2)$. The observed outcome y_i (e.g., annual income) of the i th individual consists of two parts of which the first part in parentheses is a nonrandom deterministic function and the other part is a random error term. The function $\beta_0 + \beta_1 x_i$ is a value determined deterministically by a given x_i (e.g., years of education). The added ϵ_i is a random deviation from the value of the deterministic function given x_i . In Fig. 5.3, $\beta_0 + \beta_1 x_i$ is the regression mean of the population whose value of the independent variable is x . This model is called a *simple linear* regression model. It is called *simple* because there is only one independent variable and called *linear* because the deterministic function is a linear combination of regression parameters (i.e., β_0 and β_1 with the observed value of the independent variable as its coefficient of linear combination). As discussed later in this chapter, a *multiple* (as opposed to simple) linear regression model is a linear model that includes more than one independent variable, e.g., $y_i = \beta_0 + \beta_1 x_{1i} + \beta_2 x_{2i} + \beta_3 x_{3i} + \dots + \beta_k x_{ki} + \epsilon_i$.

The simple regression model can have its variants, and the following is such an example: $y_i = \beta_0 + \beta_1 x_i^2 + \epsilon_i$. First, how many independent variables are there? Only one, so the *simple* part makes sense. Having x^2 in the model as the independent variable does not mean this is a nonlinear model. Let us note that the word *linear*

means that the deterministic function, $\beta_0 + \beta_1 x_i^2$, is linearly determined by a given value of the independent variable (i.e., the rate of linear change is β_1) for a unit change of x^2 , and the amount $\beta_1 x_i^2$ is determined first and then it is added to β_0 (i.e., a linear combination of β_0 and β_1 wherein the combination coefficients are 1 and x_i^2 , respectively). To make it clearer, one can rename x^2 to a new name z , i.e., $y_i = \beta_0 + \beta_1 z_i + \epsilon_i$.

5.3 Simple Linear Regression Analysis

Simple regression analysis is to utilize data to find the best estimate of the deterministic regression function that determines the mean value of the outcome variable given a certain value of the independent variable. The widely applied computational procedure is the least squares method.

In Fig. 5.1, the drawn line is the line of estimated regression function, which is usually derived by the so-called least squares method; the residual e_i is the deviation of the observed y_i from the predicted value $\hat{y}_i$ by the estimated regression line. Note that the residual e_i is not the same random error term ϵ_i that was in Sect. 5.3 in that the term ϵ_i is the difference between the observed and the true (but unknown) population regression line. The least squares method is to estimate the intercept and slope of the regression line that minimize the sum of squared residuals, e_i^2 . The resulting estimated line is indeed the whole collection of the predicted means of the outcome variable y given the values of the independent variable x when the normality assumption of the ϵ_i error term's distribution is true (see Fig. 5.3). Computer programs (even Excel software has the feature) are widely available for estimating each regression equation parameter (i.e., intercept β_0 , and slope β_1) and the standard

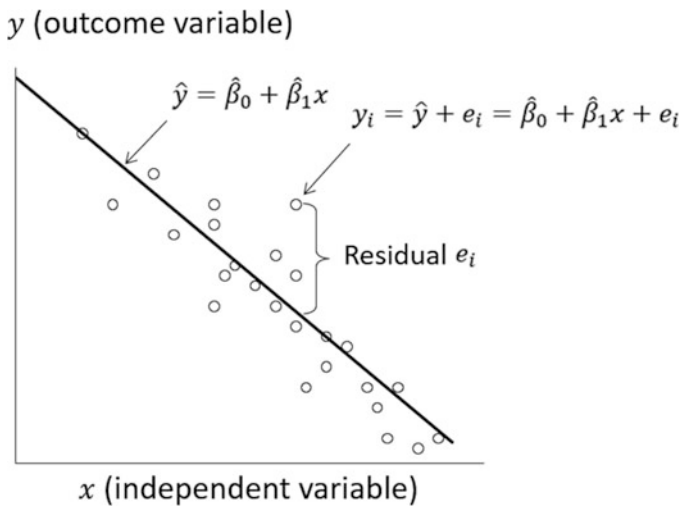


Fig. 5.1 Estimated least squares linear regression function and residuals

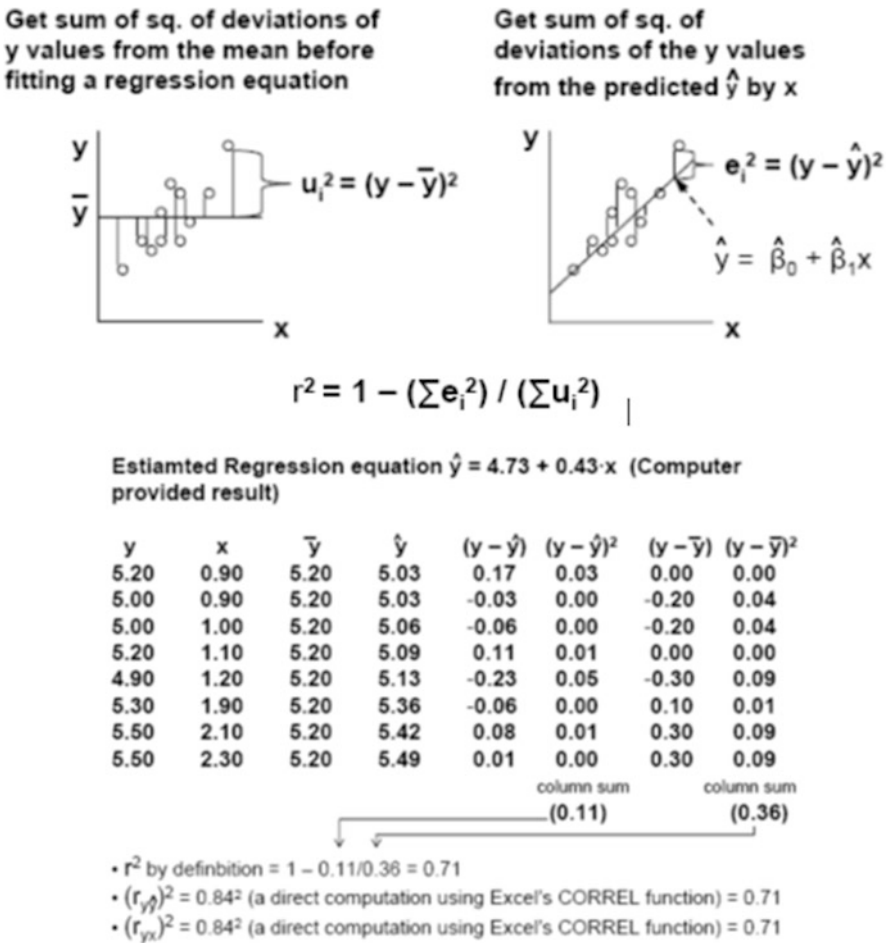


Fig. 5.2 Numerical illustration of r^2

error of each estimated regression parameter and for providing the test statistic to test if each population regression coefficient is different from zero, as well as the 95% confidence interval of each regression parameter.

A goodness of fit for the estimated simple linear regression equation is measured by r^2 . This metric is the same as the squared value of the sample linear correlation coefficient calculated from the observed y and x pairs. It is also the same as the proportion of the explained variation of the dependent variable by the estimated regression equation. The possible range is from 0 (0% is explained) to 1 (100% is explained). The r^2 is 1- (sum of squares of the residuals/sum of squares deviations of the observed outcome values from the overall mean of the outcome values). Fig. 5.2 illustrates the concept of r^2 and demonstrates the computational details. The first plot depicts the r^2 in the absence of a fitted regression equation for which the horizontal line represents the mean of y irrespective of the values of the independent variable.

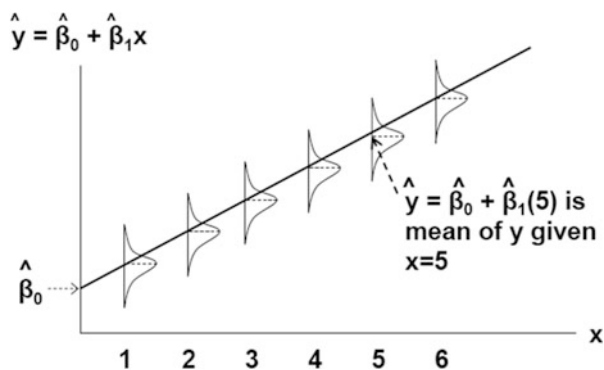
The second plot depicts the r^2 of the fitted regression equation. It is also noted that r^2 is the squared value of the correlation coefficient between y and $\hat{y}$, i.e., and it is also the same as the squared value of the correlation coefficient between y and x . This can be shown algebraically and numerically.

Let us use an example of a simple linear regression equation estimated from an analysis, $y = 64.30 + 1.39 \cdot x$, where y denotes systolic blood pressure (SBP) and x denotes age. The interpretation is that mean SBP increases linearly by 1.39 as a person's age increases by 1 year. For instance, a mean SBP of 30-year-old persons is predicted as $64.30 + 1.39 \cdot 30 = 106$. This value 106 is the common systematic rule to everyone whose age = 30. Note that this regression equation should be applied for a meaningful interval of the predictor variable x (e.g., age = 200 or age = -10 is nonsense). $y = 64.30$ when $x = 0$ is indeed the y -intercept, and this may not be a value of interest (i.e., for age = 0).

It is important to know that what is being predicted by this linear regression equation is the mean value of the dependent variable given a certain value of the independent variable (conditional mean). In the above blood pressure prediction example, the predicted SBP value = 106 for a given age = 30 is indeed the estimated mean SBP of all subjects with age = 30. In Fig. 5.3, the estimated regression line represents the collection of predicted means (i.e., conditional means) of the dependent variable y given certain values of independent variable x .

All values on the predicted regression line are the means over the range of the given independent variable values, and a single point on that line is the estimated mean value given a certain value of the independent variable. Many computer software programs can find the best (*unbiased minimum variance estimates*) estimates of the regression coefficients of the specified model. Such programs also provide the estimated standard errors (SEs) of the estimated regression coefficients for drawing inference. The hypothesis tests and interval estimations for the regression coefficients can be either directly available or easily completed by utilizing the computer-generated estimates, $\hat{\beta}_s$ and $SE(\hat{\beta}_s)$. The hypothesis test for the slope, β_1 , is usually performed by a z - or t -test depending on the sample size. The practical choice of z -test is when the sample size is large enough (e.g., 30 or greater),

Fig. 5.3 Illustration of regression mean



otherwise a t -test is usually applied. The null hypothesis usually states that the regression slope is 0, i.e., $H_0: \beta_1 = 0$ (i.e., independent variable is not predictive of the outcome). The alternative hypothesis can either be nondirectional or directional depending on the research question, i.e., $H_1: \beta_1 \neq 0$ for a nondirectional test and $H_1: \beta_1 > 0$ for a directional test to claim a positive slope, etc. For both the z - and t -tests, the test statistic is derived by the aforementioned trio, i.e., $(\hat{\beta}_1 - 0)/SE(\hat{\beta}_1)$ (see Sect. 2.2.4.5). The degrees of freedom for a t -test is $n - 2$.

The interval estimation for each regression coefficient, i.e., the slope can be constructed using z - or t -distribution depending on the sample size. For example, the 95% confidence interval for the regression slope β_1 with a sample size of 20 is derived as $[\hat{\beta}_1 - 2.101 \times SE(\hat{\beta}_1), \hat{\beta}_1 + 2.101 \times SE(\hat{\beta}_1)]$, where 2.101 is the t value whose the tail area below -2.101 is 0.025 (i.e., 2.5th percentile) and the area above 2.101 is 0.025 (i.e., 97.5th percentile) with $df = 18$. The 95% confidence interval using z -distribution when the sample size is large enough is derived as $[\hat{\beta}_1 - 1.96 \times SE(\hat{\beta}_1), \hat{\beta}_1 + 1.96 \times SE(\hat{\beta}_1)]$.

The confidence interval (band) for the entire regression mean response line (i.e., whole collection of individual regression means given the individual values of independent variable) can also be constructed. The algebraic expression becomes more complex than that of the slope because the interval estimation for the regression line involves issues of the underlying correlation between the intercept and slope estimates that are not independent on each other. The technical details may be beyond the level of the knowledge of most of the readers. Letting alone the details, Fig. 5.4 demonstrates how the confidence band appears in that the bandwidth around the mean of the independent variable is the narrowest and becomes wider as the values of the independent variable departs from its mean. Figure panel A demonstrates a special situation that only the intercept is estimated and not the slope (assumed to be known and fixed during the estimation). Intuitively, the confidence band is parallel to the estimated regression line because the slope is always fixed to

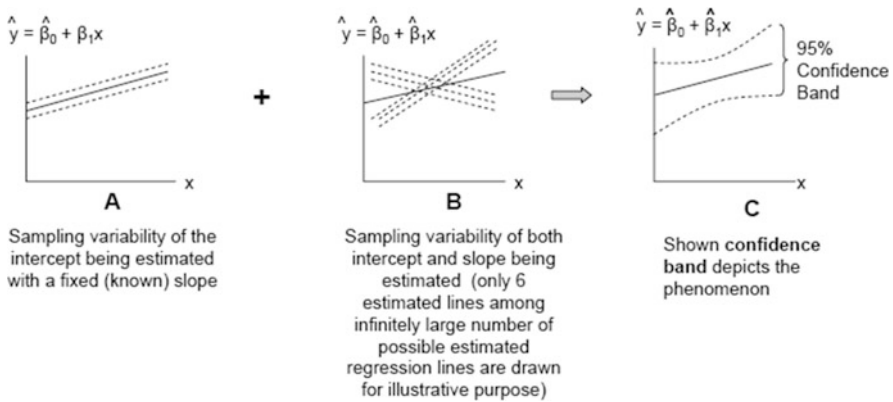


Fig. 5.4 Illustration for aiding to understand confidence interval of the estimated linear regression equation

one value. Panel B demonstrates the variability of estimated regression line whose intercept and slope are estimated (the shown lines are only several of infinitely large number of regression lines that are estimated and fluctuating due to the sampling variability of the raw data). Then, panel C illustrates the actual band of a regression line. The actual calculation of this is usually done by computer software.

Another interval estimation problem is to construct a confidence interval for predicted individual outcomes. Intuitively, the point estimate of an individual outcome value given a certain value of independent variable is the regression mean evaluated at that value of the independent variable. However, the confidence band of the predicted individual outcome values turn out to be a little bit wider than that of the regression line (i.e., the regression mean response line) because for a point on the regression line, many individual values are surrounded randomly above and below that single mean value on a certain point of the regression line. Such a band is called a prediction band (e.g., 95% prediction band), and its computational details take into account the additional random variability of these surrounded individual observations. The actual calculation of this is usually done by computer software.

Before we proceed to the topic in the next section, a very important issue needs to be discussed. In many applications, the data are observed at multiple time points within one subject, and the observations are correlated within a subject. Health science studies may include a long time series data of only a single subject (e.g., a long time series of weekly incidence of an infectious disease in a certain place over many years) where that place can be viewed as a single study subject, and the dependent variable is the number of new cases and the independent variable is the number of weeks since week 0 or multiple subjects with relatively short time series data (monthly height growth pattern of a group of infants over the first 6 months of life, i.e., the dependent variable is height and the independent variable is the month after birth). The method of least squares estimation assumes all data are uncorrelated (i.e., there is no autocorrelation). If this assumption is violated, then the standard error of the regression coefficient estimate will become inaccurate. Some techniques to deal with such issues will be discussed in the last section of this chapter.

5.4 Linear Regression Models with Multiple Independent Variables

The outcome (dependent) variable of a regression model may need to be explained by more than one explanatory (independent) variable. For example, gray-haired people may show higher blood pressure than the rest, but the association between age and blood pressure is probably confounded with gray hair and age association, and such a phenomenon needs to be taken into account. If multiple independent variables are additionally entered into the model, the model will decrease the residual variation of the dependent variable that had not been explained solely by the primary

independent variable of interest. Such a model can be expressed as a linear combination (i.e., a value of the dependent variable given a set of values of all independent variables in the model is expressed as a weighted sum of the independent variables where the regression coefficients β 's being the weights), $y_i = \beta_0 + \beta_1 x_{1i} + \beta_2 x_{2i} + \beta_3 x_{3i} + \dots + \beta_k x_{ki} + \epsilon_i$, where the assumption about ϵ_i is the same as what is specified in Sect. 5.2. The predicted value of the estimated regression equation for the i th individual, i.e., $\hat{y}_i = \hat{\beta}_0 + \hat{\beta}_1 x_{1i} + \hat{\beta}_2 x_{2i} + \hat{\beta}_3 x_{3i} + \dots + \hat{\beta}_k x_{ki}$, is the mean value of dependent variable y given the observed values of $x_{1i}, x_{2i}, x_{3i}, \dots$, and x_{ki} . The model fitting usually requires computer software. Below is a brief overview of how to perform such an analysis for model fitting (i.e., estimation of regression coefficients) and related inference.

The goodness of fit for a linear regression with multiple independent variables is measured by R^2 , which is interpreted as the proportion of the explained variation of the dependent variable by the estimated regression equation. The least squares estimation seeks the regression coefficients that maximizes R^2 . This R^2 is the squared value of the correlation coefficient between y and $\hat{y}$. To distinguish it from the case of simple linear regression's case (i.e., r^2), the notation is capitalized, i.e., R^2 .

Multicollinearity is a phenomenon due to a set of correlated independent variables in a multiple regression setting. It affects the estimated regression equation adversely. For an estimated multiple regression, $\hat{y}_i = \hat{\beta}_0 + \hat{\beta}_1 x_{1i} + \hat{\beta}_2 x_{2i} + \hat{\beta}_3 x_{3i} + \dots + \hat{\beta}_k x_{ki}$, if two independent variables (for instance, x_1 and x_2) are highly correlated, then the uncertainty about $\hat{\beta}_1$ and $\hat{\beta}_2$ increases and the standard errors of these two estimated regression coefficients are inflated. In such a case, R^2 value can be high (i.e., the independent variables, as a whole set, predict the mean outcomes well), but the test results for some individual coefficients may not be significant (due to the inflated standard error of the regression coefficient estimate) and the interpretation of such regression coefficients in conjunction with other regression coefficient (s) becomes dubious (Fig. 5.5).

Exclusion of the independent variables that are highly correlated (i.e., redundant to certain variables) will prevent such an adverse consequence. A formal diagnosis can be made by using the tolerance statistic, which is the proportion of unexplained variance of the independent variable being diagnosed by all other remaining independent variables (i.e., $1 - R^2$ of the estimated regression of the independent variable being diagnosed on all other variables). The inverse of tolerance is called

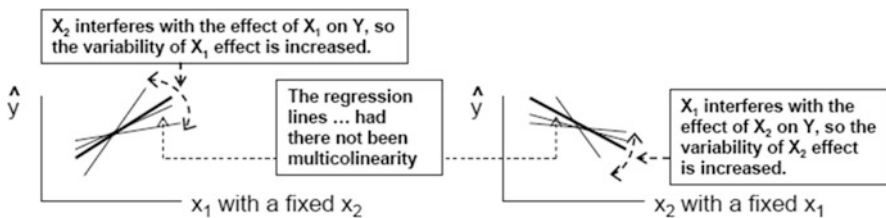


Fig. 5.5 Illustration of multicollinearity in a multiple regression with two independent variables

the variance inflation factor (VIF). A common criterion is to exclude the independent variable if tolerance is less than 0.1 (or VIF is greater than 10).

5.5 Logistic Regression Model with One Independent Variable: Simple Logistic Regression Model

Modeling a binary outcome variable by a regression is different from that of the continuous outcome that was introduced in the previous sections. Let us discuss the following example.

Figure 5.6 illustrates a set of raw data of a set of binary outcomes y (e.g., certain disease; illness if $y = 1$ and $y = 0$ if illness free) versus a continuous measure of x (e.g., $x =$ age in years) in the same way that was adopted to demonstrate the single independent variable linear regression model. The linear regression line stretches out above 1 and below 0, which is unrealistic. So, the idea of forcing the feasible range lies between 0 and 1; the logistic function is adopted, and Fig. 5.7 illustrates this idea. It is noted that the observations take values of either 0 or 1, but the regression curve does not exceed either 0 or 1, and it is also noted that that the vertical axis is the probability of observing $y = 1$ given a value of x . This is called logistic regression model because the shape of the response curve resembles the cumulative distribution function of logistic distribution (simply called logistic function). The mathematical expression of this function (where e is the base of natural logarithm) is:

Fig. 5.6 Illustration of inappropriate linear function to predict the event probability of binary outcome given independent variable X

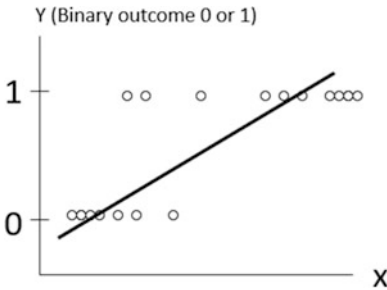
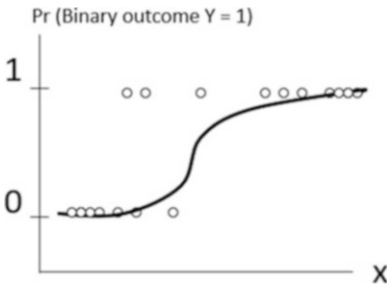


Fig. 5.7 Illustration of logistic function to predict the event probability of binary outcome with independent variable X



$$\text{Probability } (y = 1 \text{ given } x) = \frac{e^{\beta_0 + \beta_1 x}}{1 + e^{\beta_0 + \beta_1 x}},$$

$$\text{and probability } (y = 0 \text{ given } x) = 1 - \frac{e^{\beta_0 + \beta_1 x}}{1 + e^{\beta_0 + \beta_1 x}}.$$

Unlike the linear regression model, the logistic regression model does not need the random error term because the transformed outcome variable of this logistic regression model specifies the probability of the event ($y = 1$), and this completely characterizes the probability distribution of the original outcomes of $y = 1$ and $y = 0$ (i.e., no other random error terms are necessary). Consider an example in which the outcome variable y is binary (1 = had an event, 0 = did not have an event) and independent variable x is a risk scale (1 = low risk, 2 = moderate risk, and 3 = high risk); the following probabilities of interests can be expressed via logistic equations:

$$\text{Probability } (y = 1 \text{ for moderate risk}) = \frac{e^{\beta_0 + \beta_1 \cdot 2}}{1 + e^{\beta_0 + \beta_1 \cdot 2}},$$

$$\text{Probability } (y = 0 \text{ for moderate risk}) = 1 - \frac{e^{\beta_0 + \beta_1 \cdot 2}}{1 + e^{\beta_0 + \beta_1 \cdot 2}},$$

$$\text{Probability } (y = 1 \text{ for high risk}) = \frac{e^{\beta_0 + \beta_1 \cdot 3}}{1 + e^{\beta_0 + \beta_1 \cdot 3}}, \text{ and}$$

$$\text{Probability } (y = 0 \text{ for high risk}) = 1 - \frac{e^{\beta_0 + \beta_1 \cdot 3}}{1 + e^{\beta_0 + \beta_1 \cdot 3}}.$$

In addition to these predicted probabilities, the odds ratios (OR, see Sect. 1.4.3) are also of interest. If we were interested in the odds ratio of the event with high risk versus moderate risk, then this odds ratio can be derived by the following simple algebra:

$$\text{OR} = \frac{\text{Probability } (y = 1 \text{ for high risk}) / \text{Probability } (y = 0 \text{ for high risk})}{\text{Probability } (y = 1 \text{ for moderate risk}) / \text{Probability } (y = 0 \text{ for moderate risk})}$$

$$= \frac{\left(\frac{e^{\beta_0 + \beta_1 \cdot 3}}{1 + e^{\beta_0 + \beta_1 \cdot 3}} \right) / \left(1 - \frac{e^{\beta_0 + \beta_1 \cdot 3}}{1 + e^{\beta_0 + \beta_1 \cdot 3}} \right)}{\left(\frac{e^{\beta_0 + \beta_1 \cdot 2}}{1 + e^{\beta_0 + \beta_1 \cdot 2}} \right) / \left(1 - \frac{e^{\beta_0 + \beta_1 \cdot 2}}{1 + e^{\beta_0 + \beta_1 \cdot 2}} \right)} = e^{\beta_1 \cdot 3} - e^{\beta_1 \cdot 2} = e^{\beta_1}.$$

Likewise, the OR of moderate versus low risk is $e^{\beta_1 \cdot 2} - e^{\beta_1 \cdot 1} = e^{\beta_1}$, and the OR of high versus low risk is $e^{\beta_1 \cdot 3} - e^{\beta_1 \cdot 1} = e^{2\beta_1}$.

While the OR is the measure of association of our ultimate interest, its inference is made on the regression coefficient, β_1 , because the OR is merely the transformed value of the regression coefficient (i.e., $\text{OR} = e^{\beta_1}$). The standard method for estimating the regression coefficients (i.e., fitting the logistic regression function) is the maximum likelihood (ML) method. The method is to find β_0 and β_1 values that maximize the following likelihood function. The likelihood function, denoted by L , in a simple description, is a function that gauges the joint probability of the

occurrence of all observed outcomes in the collected sample as a function of the parameters, β_0 and β_1 . In the example, it is simply the product of all probabilities of $y = 1$ given x for all observations with the outcome value 1 and all probabilities of $y = 0$ given x for all observations with the outcome value 0. The following is the spelled-out expression of the illustrative observation set listed below:

Observation no.	Outcome y (0 or 1)	Predictor x (0 or 1)
1	1	1
2	1	0
3	1	1
.	.	.
.	.	.
.	.	.
.	.	.
$n - 1$	0	1
n	0	0

$$L \approx \left(\frac{e^{\beta_0 + \beta_1(x=1)}}{1 + e^{\beta_0 + \beta_1(x=1)}} \right) \times \left(\frac{e^{\beta_0 + \beta_1(x=0)}}{1 + e^{\beta_0 + \beta_1(x=0)}} \right) \times \dots \times \left(1 - \frac{e^{\beta_0 + \beta_1(x=1)}}{1 + e^{\beta_0 + \beta_1(x=1)}} \right) \times \left(1 - \frac{e^{\beta_0 + \beta_1(x=0)}}{1 + e^{\beta_0 + \beta_1(x=0)}} \right)$$

$\uparrow$
 Observation 1
 $y=1, x=1$

$\uparrow$
 Observation 2
 $y=1, x=0$

$\dots$

$\uparrow$
 Observation $n-1$
 $y=0, x=1$

$\uparrow$
 Observation n
 $y=0, x=0$

Each term in the above product is the logistic model-based probability of either $y = 1$ or 0 given x . The maximum likelihood estimation procedure is to find the solutions for β_0 and β_1 that maximize this likelihood function. For convenience, the actual computation uses its natural logarithm, $\ln(L)$, instead of L to lessen the computation burden (i.e., the multiplicative operation becomes addition). Letting alone the further details of computational aspect that is not explicitly spelled out here, it is practically useful to remind the property of the regression coefficients that are obtained from the ML method (ML estimators). The property is that the sampling distribution of such an estimator follows Gaussian (i.e., normal) distribution as long as the sample size is sufficiently large. Relying on this property, similar to the simple linear regression case (see Sect. 5.2), a one-sample z -test (Wald's z -test) or t -test, if sample size is not large, is a common method for a regression coefficient β_1 to be tested for $H_0: \beta_1 = 0$ versus $H_1: \beta_1 \neq 0$. For interval estimation, the lower and upper limits of 95% confidence interval for the regression coefficient (see Sect. 5.2) are obtained first; then these limits are transformed to OR limits, i.e., the limits are $e^{\text{Lower limit of the regression coefficient}}$ and $e^{\text{Upper limit of the regression coefficient}}$.

Because the maximum likelihood method does not resort to the least squares method, the goodness of fit is not assessed by r^2 (for one independent variable) or R^2 (for multiple independent variables). The goodness of fit for an estimated logistic regression equation can be assessed by several options. The most popular option is Hosmer–Lameshow statistic, which measures the disagreement between observed versus expected events of interest in partitioned deciles (or three to nine if fewer than 10 observed patterns of the independent variable(s) existed) of the predicted probabilities, and this measured agreement is transformed to a chi-square statistic with $g-2$ degrees of freedom where g is the number of ordered partitions of the predicted probabilities (see Sect. 6.1).

5.6 Consolidation of Regression Models

5.6.1 General and Generalized Linear Models

Linear regression models that have more than one independent variable are called general linear models. If the regression models with more than one independent variable are not linear but are transformed into a linear form, then such transformed models are called generalized linear models. The meaning of “linear” is that the predicted mean value given the independent variables is expressed as a linear combination of the regression coefficients (i.e., simple addition of more than one term whose individual term is the product of a regression coefficient and the corresponding independent variable) (see Sect. 5.2). For example, $a + bx$ is a linear combination of the two terms a and bx , and $c + dx^2$ is also a linear combination if c and dx^2 . In the case of $c + dx^2$, the linearity is held between c and d . What is often confusing is that the resulting value of $c + dx^2$ turns out as a quadratic function with respect to x . However, by definition, such a regression equation is a linear model rather than a nonlinear model because the linearity between c and d is held as long as the observed x^2 value is viewed as the weight of the linear combination.

Unlike the linear models, nonlinear models are the ones that the model equation cannot be expressed by the linear sum of the products created by the regression coefficients and their corresponding independent variables. For example, the logistic regression equation is a nonlinear function called logistic function (see Sect. 5.5). Nevertheless, the nonlinear function often can be converted to a linear function via algebra (i.e., linearization), and such transformed models are called generalized linear models. In the case of logistic regression, the logistic function to predict the probability of events can be transformed into a linear function to predict the log of the odds.

For the logistic regression equation, probability ($y = 1$ given x) = $\frac{e^{\beta_0 + \beta_1 x}}{1 + e^{\beta_0 + \beta_1 x}}$,

By letting *logit* (p) denote the transformation $\ln(\text{odds}) = \ln[p/(1 - p)] = \ln(\text{probability of } y = 1 \text{ given } x) / (1 - \text{probability of } y = 1 \text{ given } x)$, the resulting

equation becomes $\text{logit}(p) = \beta_0 + \beta_1 x$, which is now a linear function to predict the *logit* (i.e., natural logarithm of the odds) while preserving the interpretation of both β_0 and β_1 the same as that were made in the original form, i.e., $\text{OR}(x = 1 \text{ versus } 0) = e^{\beta_1}$. Such a linearization makes the computation of the estimation less burdensome. The computational detail is beyond the objective of this monograph and is not described.

5.6.2 *Multivariate Analysis Versus Multivariable Model*

The terminology *multivariate analysis* is often misused by applied researchers. A multivariate analysis is the simultaneous analysis of two or more related outcome variables (i.e., dependent variables). Some popular methods are T^2 -test for simultaneous comparison of two or more related means between two groups (e.g., simultaneous comparison of mean weight and mean height between men and women), multivariate analysis of variance (MANOVA) for simultaneous comparison of two or more related means among three or more groups (e.g., comparison of mean weight and mean height among three ethnic groups), multivariate regression analysis to fit more than one correlated dependent variables by means of more than one related regression equations, factor analysis (FA), principal component analysis (PCA) to reduce a large dimension of linearly correlated variables into a small dimension, canonical correlation analysis to examine a set of correlated variables with another set of correlated variables, and linear discriminant analysis to build a linear equation by a set of linearly correlated random variables to differentiate the individuals into two or more groups, etc.

A *multivariable model* refers exclusively to a regression model of a single outcome variable with two or more independent variables (e.g., multiple linear regression models, analysis of covariance [ANCOVA] models, etc.), and this method belongs to the univariate analysis because there is only one outcome (i.e., dependent) variable. Note that multiple independent variables in a model do not mean the statistical method is multivariate.

5.7 Application of Linear Models with Multiple Independent Variables

Figures 5.8 and 5.9 demonstrate a type of general linear model to predict the mean of dependent variables using multiple independent variables. In the first case, the predicted mean given the independent variables (i.e., regression equation) is determined by two independent variables whose first variable is continuous x and the second, z , is to take either 1 or 0. Such a dichotomized independent variable to take either 0 or 1 is called a dummy variable.

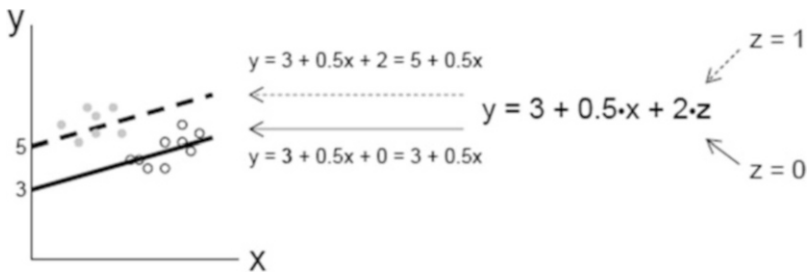


Fig. 5.8 Illustration of the dummy variable technique without modeling an effect of interaction

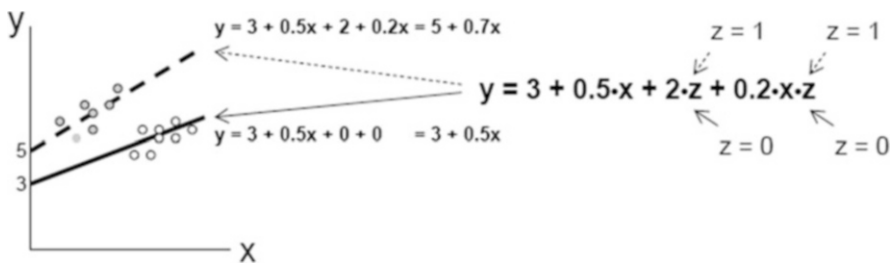


Fig. 5.9 Illustration of the dummy variable technique applied to model a main effect and an effect of interaction

In Fig. 5.8, the dummy variable was used to fit the two regression lines with the same slopes but different intercepts.

In Fig. 5.9, the dummy variable was used to fit the two regression lines with two different slopes and intercepts. The last term of the regression equation is $0.2 \cdot x \cdot z$ whose variable that takes data values is the product of x and z . Such a term is called an interaction term. The corresponding regression coefficient is the size of the difference in slopes between the two subgroups of having $z = 1$ and having $z = 0$. Note that the product term variable, $x \cdot z$, is considered a single variable (e.g., it can be renamed as any one letter variable name such as w).

5.8 Worked Examples of General and Generalized Linear Models

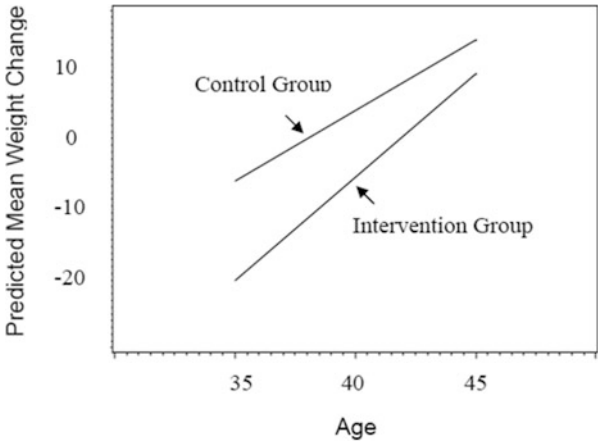
5.8.1 Worked Example of a General Linear Model

Four hundred ($n = 400$) overweighted adults with age between 35 and 45 years participated in a 1:1 randomized 1-year study of a weight loss intervention program (i.e., 200 on the invention arm and 200 on control arm). The study collected the baseline weight and the weight change (WC) after the completion of the study.

Table 5.1 Summary of general linear model analysis: weight loss intervention study

Model	Independent variables	$\hat{\beta}$	$SE(\hat{\beta})$	p -value
Model 1	Intercept	4.82	0.59	<0.0001
	Intervention (I)	−8.56	0.83	<0.0001
Model 2 $R^2 = 0.76$	Intercept	−78.55	4.77	<0.0001
	Intervention (I)	−49.32	6.92	<0.0001
	AGE (AGE)	2.09	0.12	<0.0001
	Interaction of intervention and AGE ($I \times AGE$)	0.98	0.17	<0.0001

Fig. 5.10 Illustration of interaction between intervention and age



The baseline mean ($\pm$ standard deviation) weight (in lb) among all participants was 201.6 ($\pm$ 32.9) and their mean values of the 1-year weight changes were -3.74 (± 9.34) and 4.82 (± 7.12) in the intervention and control groups, respectively. A general linear model analysis was applied to determine the intervention effects on the mean weight change without and with adjusting the individual participant’s age (Table 5.1 and Fig. 5.10). The dependent variable was the 1-year weight change (WC: post-1-year weight $-$ baseline weight), and the independent variables were intervention (I: 1 = yes, 0 = no) and age (AGE: continuous). Note that intervention (I) is the dummy variable.

In Model 1, the estimated intercept value of 4.82 lb. is the mean weight change of the control participants and the estimated parameter value of the intervention variable (I) -8.56 ($p < 0.0001$) directly offers the significant estimated difference (i.e., effect) in the mean weight changes between the two groups. With these two regression coefficients, the mean change in the intervention group can be estimated by calculating the difference of between these two coefficients, i.e., $4.82 - 8.56 = -3.74$, which is the same as the group-specific descriptive summary statistics presented above (before performing the general linear model analysis). Model 2 was constructed to predict the mean weight change not only by the given intervention status but also by the age. The main effect of age as well as its interaction with the intervention (i.e., whether the age effects were different between the intervention and control subjects) was added to this model. Note that the

estimated parameter value of the intervention (I) does not directly offer the difference in the mean weight changes between the intervention and control groups because the additional variables are included now, and those effects must be taken into account simultaneously. The estimated parameter value of -49.32 ($p < 0.0001$) is the group difference of the mean weight changes only for persons with age 0. The age 0 is unrealistic. So, if we choose $\text{age} = 40$ for a meaningful interpretation, then the intervention group's mean weight change is predicted as $-78.55 - 49.32 \times 1 + 2.09 \times 40 + 0.98 \times 1 \times 40 = -5.07$ and that of the control group is $-78.55 - 49.32 \times 0 + 2.09 \times 40 + 0.98 \times 0 \times 40 = 5.05$, thus the estimated effect (i.e., the mean difference) at age 40 is $-5.07 - (-5.05) = -10.12$, which is the conditional effect of the intervention for 40-year-old participants. As shown in Fig. 5.10, the conditional effect decreases as age increases.

5.8.2 *Worked Example of a Generalized Linear Model (Logistic Model) Where All Multiple Independent Variables Are Dummy Variables*

A large survey study investigated whether college students in California are less involved in binge drinking (Wechsler et al. 1997). The survey sample comprised 1864 college students from California and 17,592 from elsewhere in the United States. The logistic regression analysis was performed as below:

Dependent variable – Binge drinking (1 vs. 0)

Independent variables – California student (1 vs. 0); age < 24 (1 vs. 0); male gender (1 vs. 0); never married (1 vs. 0); White ethnicity (1 vs. 0); non-commuter (1 vs. 0); smoker (1 vs. 0)

Unlike the result summary of the general linear model (Table 5.1), the result summary of this generalized linear model analysis did not show the estimated parameter values of the intercepts (Table 5.2) because the intercept is the nuisance parameter for the odds ratio (see Sect. 5.5). The simple logistic regression model of binge drinking solely on the California residency indicator variable (1 = live in California, 0 = elsewhere) showed that there was significant decrease in binge drinking among college students in California ($\widehat{OR} = 0.52$, $p < 0.0001$). However, after simultaneously adjusting the other demographic variables and other risk factors (every variable was dichotomized as 1 = yes and 0 = no), this effect was no longer significant (adjusted $\widehat{OR} = 1.20$, not significantly different from 1 at a 5% significance level), while all the other covariates were significantly associated with binge drinking in that students under 24 years (adjusted $\widehat{OR} = 2.24$, $p < 0.0001$), male students (adjusted $\widehat{OR} = 1.56$, $p < 0.0001$), never married students (adjusted $\widehat{OR} = 3.58$, $p < 0.0001$), students with White ethnicity (adjusted $\widehat{OR} = 2.95$, $p < 0.0001$), non-commuters (adjusted $\widehat{OR} = 1.97$, $p < 0.0001$), and smokers (adjusted $\widehat{OR} = 4.38$, $p < 0.0001$) were involved more in binge drinking.

Table 5.2 Summary of generalized linear model analysis: binge drinking study of college students in California

Model	Independent variables	$\hat{\beta}$	$SE(\hat{\beta})$	$\widehat{OR} = e^{\hat{\beta}}$	p -value
Model 1	California	-0.66	0.053	0.52	<0.0001
Model 2	California	0.18	0.19	1.20	0.353
	Age < 24	0.81	0.05	2.24	<0.0001
	Male	0.44	0.03	1.56	<0.0001
	Never married	1.27	0.06	3.58	<0.0001
	White	1.08	0.05	2.95	<0.0001
	Non-commuter	0.68	0.04	1.97	<0.0001
	Smoker	1.54	0.04	4.38	<0.0001

5.9 Measure of Agreement Between Outcome Pairs: Concordance Correlation Coefficient for Continuous Outcomes and Kappa (κ) for Categorical Outcomes

In experimental studies, it is often needed to test if a newly invented readout of an experimental result produces the same result as the readout from the traditional gold standard. To fulfil such a goal, the following data analytic strategies can be the options by examining (1) if the reading data pairs align tightly around a straight line by calculating a correlation coefficient, (2) if the mean of the paired differences is zero by conducting a paired sample t -test, or (3) if a 45° line passing through the origin (i.e., the line of identity) can be fit to the paired data by a simple least squares linear regression analysis. These approaches are attractive, but there are drawbacks to all these options (see Fig. 5.11).

In the case of applying the correlation coefficient as a single measure of agreement, even a perfect value of Pearson's correlation (i.e., $r = 1$) can neglect a situation that all pairs are perfectly aligned on a straight line but that line does not pass through the origin or the slope is not 1 (see Fig. 5.11a–c). The paired sample t -test alone can even neglect $r = 0$ (see Fig. 5.11d). When the regression analysis is applied, the inference will involve testing two regression parameters simultaneously (i.e., a joint test for intercept=0 and slope = 1). None of these approaches (alone) can fulfill the goal. To assess the variation of observed pairs $(X_1, Y_1), (X_2, Y_2), \dots, (X_n, Y_n)$ around a straight line of slope 1 with its vertical intercept of 0 (i.e., the line of identity), concordance correlation coefficient (C) (Lin 1989), a single-valued statistic (i.e., simple to use), can be applied. The expression of this statistic is:

$$C = \frac{2\Sigma(X_i - \bar{X})(Y_i - \bar{Y})}{\Sigma(X_i - \bar{X})^2 + \Sigma(Y_i - \bar{Y})^2 + n(\bar{X} - \bar{Y})^2}.$$

The numerator is the total squared deviations, cumulated over the whole (Y_i, X_i) pairs (for $i = 1, 2, \dots, n$), of the rectangular deviations of the pairs from $(\bar{X}, \bar{Y})$, and

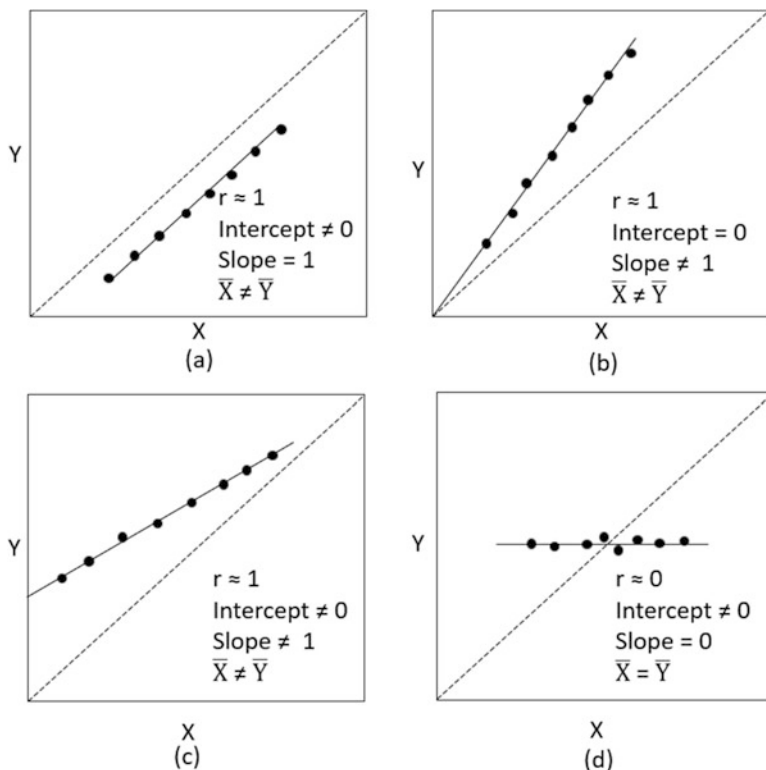
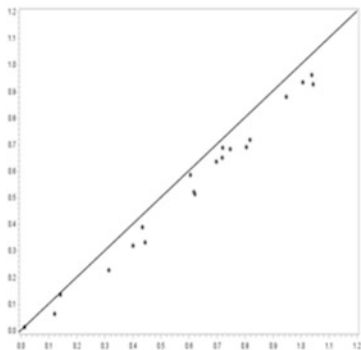


Fig. 5.11 Examples of limitations in assessing agreement using correlation coefficient and simple linear regression

the denominator is the total horizontal and vertical squared deviations plus the sum, over the whole pairs, of squared deviations due to the mean shift. The C closer to 1 indicates that the pairs are aligned closer on the line of agreement. If the two means are equal, i.e., $\bar{X} = \bar{Y}$, then the last term of the denominator, $n(\bar{X} - \bar{Y})^2$, becomes 0; moreover, if the pairs are perfectly aligned on the line of agreement, then the numerator, $2\sum(X_i - \bar{X})(Y_i - \bar{Y})$, can be expressed as $2\sum(X_i - \bar{X})^2$ or $2\sum(Y_i - \bar{Y})^2$, which is indeed the same as the remaining terms of the denominator, $\sum(X_i - \bar{X})^2 + \sum(Y_i - \bar{Y})^2$; thus $C = 1$. The inference to test the significance of C can be made in a similar fashion as how the inference of Pearson's correlation is made (see Sect. 5.1), in that the sampling distribution of the following test statistic:

$$Z = \frac{1}{2} \ln \frac{1+C}{1-C}$$

X_1	X_2
0.97	1.05
0.93	1.06
0.23	0.39
0.69	0.84
0.88	0.97
0.51	0.67
0.72	0.85
0.39	0.50
0.14	0.24
0.94	1.02
0.69	0.76
0.52	0.67
0.01	0.12
0.65	0.76
0.06	0.22
0.32	0.47
0.59	0.66
0.33	0.51
0.64	0.74
0.69	0.79



$$\begin{aligned}\bar{X}_1 &= 0.54, \bar{X}_2 = 0.67, \\ 2\Sigma(X_i - \bar{X})(Y_i - \bar{Y}) &= 0.29, \\ \Sigma(X_i - \bar{X})^2 &= 1.60, \\ \Sigma(Y_i - \bar{Y})^2 &= 1.47, \\ n(\bar{X} - \bar{Y})^2 &= 0.29, \text{ and} \\ C &= 0.907\end{aligned}$$

Fig. 5.12 Calculation of sample concordance correlation coefficient

will be an approximate normal distribution with its mean = $\{1/2\} \ln (1 + C)/(1 - C)$ with a finite variance.

The next illustration is the calculation of sample concordance correlation coefficient (Lin’s C) of 20 pairs of measurements produced by two tested laboratory devices (X_1, X_2) (Fig. 5.12).

Next, let us discuss measuring the agreement between two readings (e.g., pair of the readings from two different readers or single rater’s initial and second reading) with categorical values. For such an agreement assessment, Kappa coefficient (κ) is a

Table 5.3 Joint distribution of disease severity levels of 15 patients assessed by two raters

Counts (cell %)		Rater 1			
		Low	Medium	High	Total
Rater 2	Low	5 (33.33%)	0 (0.00%)	0 (0.00%)	5 (33.33%)
	Medium	0 (0.00%)	4 (26.67%)	1 (6.67)	5 (33.33%)
	High	0 (0.00%)	2 (13.33%)	3 (20.00%)	5 (33.33%)
	Total	5 (33.33%)	6 (40.00%)	4 (26.67%)	15 (100%)

popular statistic for a pair of binary outcomes. Cohen invented Kappa for a pair of binary outcomes, and Fleiss’ Kappa extended for more polytomous (i.e., having three or more categories) outcomes. The following discussion concentrates on the inference of Fleiss’ Kappa (κ). For a $k \times k$ contingency table, let p_o denote the overall proportion of observed agreement, p_e the overall proportion of chance-expected agreement, p_{ii} the proportion of observed agreement of a specific category, p_i , and p_j the row and column sum of p_{ii} , respectively, then the sample estimator ($\hat{k}$) is:

$$\hat{k} = \frac{p_o - p_e}{1 - p_e},$$

where $p_o = \sum_{i=1}^k p_{ii}$ and $p_e = \sum_{i=1}^k p_i.p_{.i}$. This estimates the proportion of the degree of agreement that achieved the above chance alone out of the degree of agreement that is attainable above chance. It ranges from 0 (complete absence of agreement) to 1 (perfect agreement), and its standard error to be used for the inference is:

$$SE(\hat{k}) = \frac{1}{\sqrt{n}(1 - p_e)} \sqrt{p_e + p_e^2 - \sum_{i=1}^k p_i.p_{.i}(p_i + P_{.i})}.$$

The following tabulated data (Table 5.3) illustrate the calculation of Fleiss’ Kappa to measure the agreement between two raters’ assessments of disease severity of 15 patients.

The estimated proportions of p_o and p_e are $p_o=0.3333 + 0.2667+ 0.2000 = 0.8$ and $p_e=0.3333 \times 0.3333 + 0.4000 \times 0.3333 + 0.2667 \times 0.3333 = 0.3333$, respectively. Thus $\hat{k}=0.7$, $SE(\hat{k})=0.15$, and 95% CI: 0.41–0.99.

5.10 Handling of Clustered Observations

The independence among the observations pertaining to the same outcome in a sample (i.e., the phenomenon that the value of one observation does not depend on that of other observations) is an important condition that many statistical methods (e.g., *t*-test, ANOVA, least squares regression, etc.) rely on. If this condition is not held, the sampling variance of the sample estimate of interest calculated by the method introduced in Sect.1.3.4 will be biased. This section will briefly discuss such a bias and the strategies to avoid it.

Consider there are *n* independent units of sample (e.g., persons), and each unit constitutes *m* repeated interdependent observations. The within-cluster interdependence can be a simple structure that any paired measures outcomes among *m* share a common degree of correlation, or otherwise the interdependence structure depends on the relationship between the pairs (e.g., stronger correlation between certain pairs, etc.).

The following example is a collection of five consecutive longitudinal outcomes that are measured from seven subjects in Group 1 and seven in Group 2. The total 70 (14 × 5) observations are conceived as 14 clusters wherein each cluster is a set of five observations that are interdependent within each study subject. Figure 5.13 illustrates the longitudinal observations over five time points of these individual subjects in the two groups.

We want to compare if the mean levels are equal between Group 1 and Group 2. These data show a specific pattern that the values of the subjects with a high starting value remain high (and vice versa) in both groups (i.e., strong positive within-cluster longitudinal correlation). In the presence of such a pattern, ignoring it and assuming (erroneously) that the entire observations in the sample are

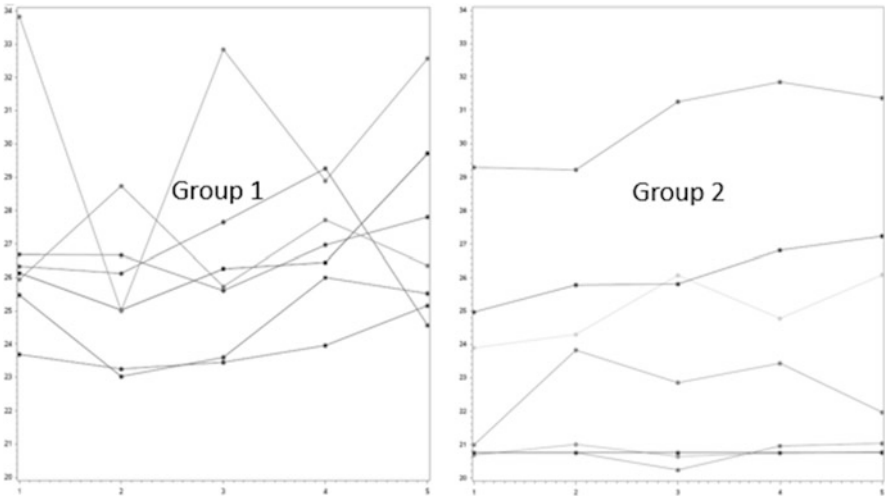


Fig. 5.13 Illustration of longitudinal observations

independent and applying the t -test seems problematic, in that such a naïve standard error estimate of each sample mean will be biased (underestimated) and the p -value will be inaccurately smaller than what it should be if the standard error of the mean in each group was obtained utilizing the formula for the independent sample t -test. One of the appropriate methods for this mean comparison will be the repeated measures ANOVA (see Sect. 4.4.3).

Although the least squares method-based repeated measure ANOVA is a method to handle the within-cluster correlation, this method can handle only certain within-cluster correlation structures such as when the equal cluster size and the correlation between any within-cluster pairs are equal (compound symmetry). The within-cluster correlation structure can be more complex than what it can be handled by the repeated measure ANOVA, such as variable cluster sizes with each cluster size of m_k rather than the size being fixed with m for all clusters and nonhomogeneous complex within-cluster correlation structure. One of the variable cluster size examples can be a collection of multiple plaque volumes measured from m_k coronary arteries of each of n patients. Another such example can be a heritable trait collection of m_k available extended family members of each of n proband subjects. The correlation structure of such an extended family data set involves not only the variable number of clusters, but it can also involve nonhomogeneous within-cluster correlation structure due to a stronger correlation between first-degree relatives than that between the second-degree relatives, etc. The maximum likelihood-based inference will offer unbiased estimates of the between-group mean difference if the within-cluster correlation structures in the likelihood function for inference is correctly specified. The use of linear mixed effects (LME) model with the maximum likelihood-based inference is a popular approach to fulfill this analysis goal. An LME model specifies each observed value of the dependent variable via a linear combination of the model parameters that are associated with either fixed or random effects. For example, the repeated measures y_{it} over five time points $t = 1, 2, \dots, 5$ for the i^{th} ($i = 1, 2, \dots, 14$) of the 14 individuals who are from either Group 1 ($x = 0$) or Group 2 ($x = 1$) can be expressed by the following *linear* model (i.e., linear combination of the model parameters μ , β_{0i} , β_1 , and ϵ_{it} with their respective coefficients of 1, 1, x , and 1):

$$y_{it} = \mu + \beta_{0i} + \beta_1 x + \epsilon_{it}.$$

In this model, μ is the overall mean for all i and t . There will be two values of $\beta_1 x$, i.e., $\beta_1 \cdot (x = 1) = 0$ for all the subjects in Group 1 and $\beta_1 \cdot (x = 1) = \beta_1$ for all the subjects in Group 2, which are the group-specific effects added to μ . There will be 14 β'_{0i} s for $i = 1, 2, \dots, 14$, which are subject-specific effects added to $\mu + \beta_1 x$. We are not primarily interested in quantifying these 14 unique individual subject-specific effects, β'_{0i} s, because the focal point is the inference about the group-level difference in the means. The parametrization of the 14 β'_{0i} s is to account each individual-level subject-specific effect as a random deviation term around each group-specific mean

$\mu + \beta_1 x$ following $N(0, \sigma_{\beta_0}^2)$; thus β_{0i} is called a random effect. The variance, $\sigma_{\beta_0}^2$, of these 14 β_{0i} s gauges the random subject-level variability around each group-specific mean. Because the model contains both fixed and random effects, such a linear model is called a linear mixed effects (LME) model. Regarding the within-subject correlations among the five time points, those can be fully specified via correlation structure of 60 (i.e., 14 subjects $\times$ 5 time points) ϵ_{it} that are either uncorrelated (i.e., independent of each other among the five time points) random disturbances following $N(0, \sigma_\epsilon^2)$, otherwise the within-subject correlations can be described as a 5×5 symmetrical (i.e., $r_{12} = r_{21}, \dots, r_{45} = r_{54}$) correlation matrix that is the same for all 14 subjects:

$$\begin{pmatrix} r_{11} & r_{12} & r_{13} & r_{14} & r_{15} \\ r_{21} & r_{22} & r_{23} & r_{24} & r_{25} \\ r_{31} & r_{32} & r_{33} & r_{34} & r_{35} \\ r_{41} & r_{42} & r_{43} & r_{44} & r_{45} \\ r_{51} & r_{52} & r_{53} & r_{54} & r_{55} \end{pmatrix}$$

The description of such within-subject correlation structure is incorporated into the likelihood function specification for the maximum likelihood-based inference. The detailed description of the specified likelihood function is not the focal point of this book, and its presentation is skipped. In summary, the mixed effects model involves fixed and random effects on the overall mean, and the within-subject correlations structure is specified in the model. The parameters of these group-level fixed- and subject-level random effects on the means as well as the within-subject correlations are inferred by the normal distribution maximum likelihood method (provided that the population distribution of the outcome is a normally distribution), which offers unbiased estimates of the model parameters if the within-cluster correlation structure is correctly specified.

The correct specification of the within-subject correlation structure may become a difficult task if it involves more than one within-cluster correlation (e.g., stronger longitudinal correlations between closer time points than farther time points; stronger correlation between first-degree relatives than second-degree relatives; etc.). In real-world data analysis, the correlation structure is usually unknown, thus the accurate estimation of the standard errors can be challenging. There are alternative approaches to obtain robust results even if the correlation structures are inaccurately specified. Generalized estimating equations (GEE) approach is one of such alternative methods. This method also allows the within-subject correlation structure to be specified but neither does introduce the subject-level random effect nor its inference resorts to the maximum likelihood method. The resulting estimates of the parameters to determine the means are unbiased, and their standard errors are robust even in some degree of misspecification of the within-subject correlation structures. Most

Table 5.4 Summary results of t -test, linear mixed effects model (LME), and genderized estimating equations (GEE) concurrently applied to a two-group comparison of the means of clustered longitudinal observations

	t -Test		Linear mixed effects model		GEE approach	
	Group 1	Group 2	Group 1	Group 2	Group 1	Group 2
No. of clusters, n	7	7	7	7	7	7
Cluster size, m	5	5	5	5	5	5
$\bar{X}$	26.63	23.80	26.63	23.80	26.63	23.80
$S_{\bar{X}}$	0.44	0.60	0.80	1.40	0.74	1.30
95% CI	25.73–27.53	22.58–25.02	25.78–28.20	22.32–26.54	25.17–28.09	21.26–26.34
p -value	0.0003		0.0852		0.0584	

statistical software programs (e.g., SAS, SPSS, STATA, R, etc.) provide LME and GEE.

Table 5.4 summarizes the result of the discussed example. The comparisons were made by using t -test, linear mixed effects model (LME), and generalized estimating equations (GEE) approach. As expected, all three methods produced the same means but different standard errors, in that t -test underestimated $S_{\bar{X}}$ and produced its p -value smaller than the other two methods (it was significant at a 5% alpha unlike the other two methods).

Study Questions

1. The estimated least squares linear regression equation (simple or multiple regression) does not predict an individual's specific outcome value given the subject's value(s) of the independent variable(s). What value does the regression equation predict?
2. What is the quantitative interpretation of a regression coefficient of a linear regression equation?
3. What value does a logistic regression equation predict given an individual's value (s) of the independent variable(s)?
4. What is the quantitative interpretation of the regression coefficient of a logistic regression equation?
5. What are the definitions of the following?

Odds ratio

General linear model

Generalized linear model

6. Explain why the multiple linear regression analysis and multiple logistic regression analysis are not multivariate analyses.
7. What is the best method of following statistics to assess the agreements between paired continuous measurements?
 - (a) Pearson's correlation coefficient
 - (b) Simple linear regression slope
 - (c) Paired sample *t*-test
 - (d) Lin's concordance correlation coefficient

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Chapter 6

Normal Distribution Assumption-Free Nonparametric Inference



Methods of inference not requiring normality of the outcome distribution are discussed.

6.1 Comparing Two Proportions Using a 2×2 Contingency Table

If a researcher needs to compare the binary outcome frequencies (e.g., disease rates) between two groups (e.g., risk group versus risk-free group), then they will count the numbers and calculate the proportions (%) of observed outcomes of interest (i.e., $100 \times [\text{number of responses}]/[\text{number of subjects in each group}]$) and compare them. The first step is to tabulate the observed data in a 2×2 contingency table whose four cells represent the observed number of subjects.

Figure 6.1 illustrates 2×2 contingency tables that provide information about the association between two categorical variables. The number in each cell is the observed number of subjects, i.e., cell frequency.

Example 6.1

Does it seem that there is an association between the mother's age and the offspring's low birth weight? This example illustrates how to interpret a 2×2 contingency table. In the tabular summary in Fig. 6.2, the observed data of 200 neonates are classified into one of the four cells in the 2×2 contingency table.

The noticeable pattern is that 10 out of 50 (20%) mothers who were ≤ 20 years old delivered low-weight babies, whereas only 15 out of 150 (10%) of the >20 -year-old mothers did so. Note that the shaded cells are the ones whose row percentage value is greater than the marginal total column percentage. These data revealed a two-fold increase in the risk (20% versus 10%) by this descriptive analysis. The next section will discuss its formal statistical inference.

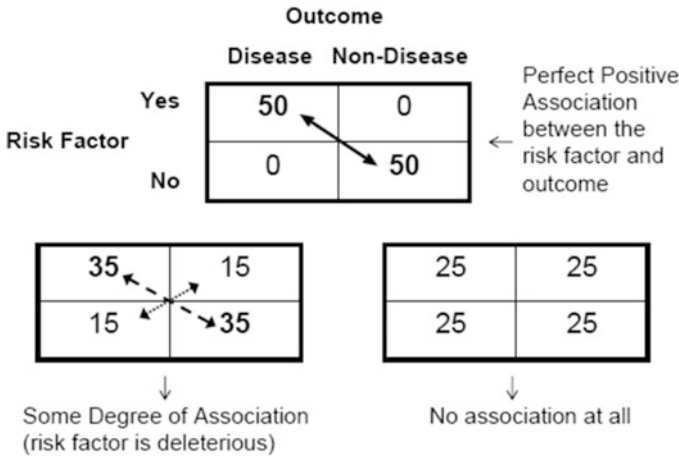


Fig. 6.1 Patterns of association between two binary outcomes

Counts (Row %)		Birth Weight		
		≤ 2500 grams	>2500 grams	Total
Mother's Age	≤ 20 years	10 (20%)	40 (80%)	50 (100%)
	>20 years	15 (10%)	135 (90%)	150 (100%)
	Total	25 (12.5%)	175 (87.5%)	200

Fig. 6.2 Illustration of exploratory data analysis by a 2×2 contingency table

6.1.1 Chi-Square Test for Comparing Two Independent Proportions

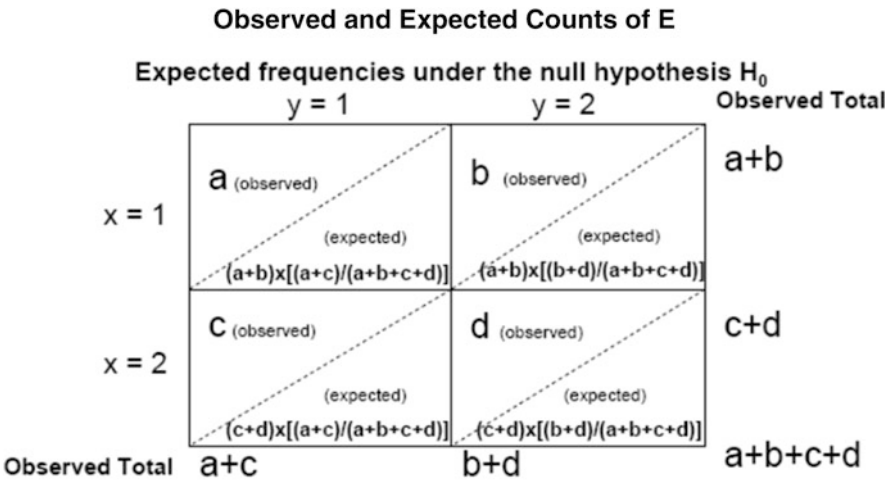
The statistical inference to make a comparison of the two independent proportions can be performed by the chi-square test. The null and alternative hypotheses, in general, are H_0 : There is no association and H_1 : There is an association, respectively, or more formally:

$H_0: \pi_1 = \pi_2$ (There is no association between the two categorical variables).

$H_1: \pi_1 \neq \pi_2$ (There is an association between the two categorical variables),

where π_1 and π_2 denote the two population proportions. Note that the chi-square test is always nondirectional (this will be demonstrated in the later part of this section). The idea is to measure the discrepancy between the observed frequencies and the frequencies that are expected under the null hypothesis (i.e., there is no association) and transform the discrepancy measure to the test statistic.

The interpretation of a 2×2 contingency table according to the row (or column) percentage values is discussed (Fig. 6.2). The following discussion focuses on



Observed and Expected Counts of Example 6.1 Data

Observed Counts [Expected Counts]		Birth Weight		
		≤ 2500 grams	>2500 grams	Total
Mother's Age	≤ 20 years	10 [6.25]	40 [43.75]	50
	>20 years	15 [18.75]	135 [131.25]	150
	Total	25	175	200

Fig. 6.3 Calculation of observed and expected counts under the null hypothesis in a 2 × 2 contingency table

deriving the test statistic. As presented in Fig. 6.3, the expected frequencies of the four cells under H_0 is obtained as below. The expected frequency for the cell with $y = 1$ and $x = 1$ is obtained by multiplying the proportion of $y = 1$, i.e., $(a + c)/(a + b + c + d)$ by the observed total frequency with $x = 1$, i.e., $(a + b)$. Thus, the expected frequency of that cell under H_0 is $(a + b) \times [(a + c)/(a + b + c + d)]$. Likewise, the expected frequencies of all four cells under H_0 are expressed as the lower entries of the cells. In the second table, the cells are shaded if their observed counts are greater than expected.

The next step is to derive a test statistic that gauges the degree of discrepancy between the observed and expected frequencies under the null hypothesis, by which we find out how unlikely this value could have been under the null hypothesis. The test statistic is obtained by summing over:

$$\frac{(\text{Observed cell count, } O_{ij} - \text{Expected cell count, } E_{ij})^2}{\text{Expected cell count, } E_{ij}}$$

$$= \sum_{i,j} \frac{(O_{ij} - E_{ij})^2}{E_{ij}}, \text{ for all } i, j.$$

The sampling distribution (see Sect. 2.1.2) of this test statistic is approximately $\chi^2_{(1)}$ (chi-square distribution with $df = 1$). This statistic is always nonnegative, and it does not reflect the “directionality” of the deviation of the observed values from the expected values (i.e., which observed that the cell frequency is “more or less” than the corresponding expected frequency). Thus, chi-square test cannot handle directional hypotheses (i.e., always nondirectional).

In the case of Example 6.1, $\chi^2_{(1)} = (10 - 6.25)^2/6.25 + (40 - 43.75)^2/43.75 + (15 - 18.75)^2/18.75 + (135 - 131.25)^2/131.25 = 3.42$, and p -value = 0.064 is calculated based on $\chi^2_{(1)}$ (with Excel, the tail probability where $\chi^2_{(1)} \geq 3.42$ can be calculated by $1 - \text{CHISQ.DIST}(\chi^2, df, \text{cumulative}) = 1 - \text{CHISQ.DIST}(3.42, 1, \text{TRUE}) = 0.064$).

Please note, in the above calculation, that the expected cell counts are non-integers (those were not avoidable), whereas the observed counts are obviously integers. This phenomenon lets the sampling distribution of this test statistic only approximate (close to but not the same as) $\chi^2_{(1)}$. To reduce such an approximation error, it is suggested to subtract 0.5 from the difference between each observed frequency count and its expected frequency count in a 2×2 contingency table (proposed by F. Yates). Such a correction prevents the researchers from overstating (i.e., reducing the chance to commit Type 1 error). In the case of Example 6.1, the continuity corrected $\chi^2_{(1)} = [(10 - 6.25) - 0.5]^2/6.25 + [(40 - 43.75) - 0.5]^2/43.75 + [(15 - 18.75) - 0.5]^2/18.75 + [(135 - 131.25) - 0.5]^2/131.25 = 2.58$ ($df = 1$), and its p -value = 0.109 (with Excel, the tail probability where $\chi^2_{(1)} \geq 2.58$ can be calculated by $1 - \text{CHISQ.DIST}(2.58, 1, \text{TRUE}) = 0.109$).

The application of continuity correction to Example 6.1 data analysis offered a conservative result. Nonetheless, both the continuity corrected and uncorrected chi-square statistics revealed that the association was not significant at a 5% significance level. Thus, the conclusion can be summarized as “These data showed that there was no association between the mother’s age and offspring’s low birth weight at a 5% significance level.”

In Chap. 1, the odds ratio (OR) was introduced. The OR is applicable in the 2×2 contingency table analysis. Note that if $\pi_1 = \pi_2$, then $\text{OR} = 1$ because $[\pi_1/(1 - \pi_1)]/[\pi_2/(1 - \pi_2)] = 1$, if $\pi_1 \neq \pi_2$ then $\text{OR} \neq 1$. Thus, the following four pairs of null and alternative hypotheses are exchangeable.

H_0 : Two proportions are equal
 H_1 : Two proportions are unequal

$H_0: \pi_1 = \pi_2$
 $H_1: \pi_1 \neq \pi_2$

H_0 : There is no association (between the two categorical variables y and x)
 H_1 : There is an association

$H_0: OR = 1$
 $H_1: OR \neq 1$

In the case of Example 6.1, the estimated odds ratio is $[0.2/(1 - 0.2)]/[0.1/(1 - 0.1)] = 2.25$ describing that the odds for an underage mother having a low-birth-weight baby is two and quarter times greater than that of an of-age mother. However, this was not statistically significant at a 5% significance level.

When a directional test in a 2×2 contingency table analysis setting (e.g., $H_0: \pi_1 = \pi_2$ and $H_1: \pi_1 < \pi_2$) is necessary, the test can be carried out by means of a z-test. This test can be conceived as to apply independent samples z-test to compare two means where data are binary (i.e., 0 or 1). Unless π_1 and π_2 are not too close to 0 or 1 and the sample size is not small, this method is valid for a similar reason that the probability distribution of a binominal distribution can be approximated by the normal distribution (see Sects. 1.5.5 and 2.2.6.3).

6.1.2 Fisher’s Exact Test

If there are any cells in a contingency table whose expected frequency counts under H_0 are less than 5 (Fig. 6.4), then the sampling distribution of *Sum over all cells of $[(Observed - Expected)^2/(Expected)]$* does not follow $\chi^2_{(1)}$ distribution, and the chi-square test is no longer a valid test.

A special method to deal with such a situation is to directly calculate the exact probability of observing as or more extreme (i.e., deviating from the null hypothesis) outcomes than the observed outcome, and then reject H_0 if the calculated exact probability is smaller than the adopted significance level (e.g., 0.05). The calculation

Fig. 6.4 Observed and expected counts in a 2×2 contingency table

Observed counts [Expected counts]	Deleterious Event	No Event	Total
Exposed to Risk	5 [2.52]	4 [6.48]	9
Unexposed	2 [4.48]	14 [11.52]	16
Total	7	18	25

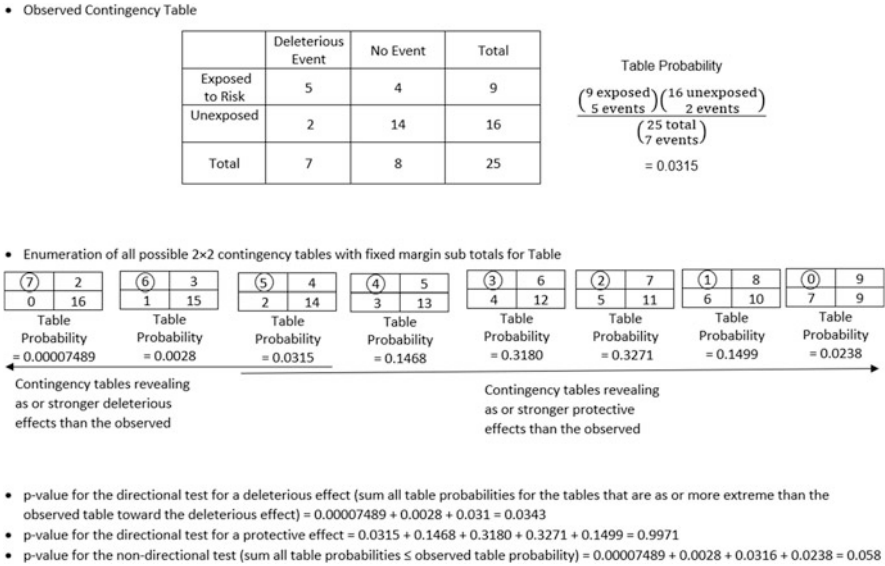


Fig. 6.5 Illustration of calculating *p*-values of Fisher’s exact test

of the exact probability (this is indeed the *p*-value of this exact test method) is a daunting task if carried out manually. Figure 6.5 illustrates the exact *p*-value calculation for the data set presented in Fig. 6.4, in that among the depicted eight contingency tables, the first table summarizes the observed data, and the other seven are complete enumeration of all possible more extreme outcomes that could have occurred. This enumeration is made in both directions that are departing from the null hypothesis (i.e., deleterious and protective effects). The individual probability values that were attached to the corresponding tables are the probability of observing the specific outcomes conditional on both margins (i.e., row and column totals being fixed by the observed data). This probability is called a table probability. For example, the calculation for the observed data can be spelled out as finding the chance to observe simultaneously, 5 events out of 9 subjects with the risk factor and 2 events out of 16 subjects without the risk factor given 7 total events out of 25 study subjects, i.e., $\binom{9}{5} \times \binom{16}{2} / \binom{25}{7} = 126 \times 120 / 480700 = 0.0316$, where $\binom{m}{n}$ for $m \geq n$ is the notation for the number of combinations of observing *n* events out of *m*, i.e., $m! / [n! (m - n)!]$. The remaining seven remaining table probabilities can be calculated in the same manner.

With the calculated table probabilities of all possible enumerated tables, the *p*-values of the tests are calculated as follows: For the directional test with its alternative hypothesis *H*₁: Effect is deleterious, it is intuitive (as illustrated in Fig. 6.5) to cumulate three table probabilities that are as or more extreme to the direction of this alternative hypothesis, i.e., 0.00007489 + 0.0028 + 0.0315 = 0.0343 indicating that there was a statistically significant deleterious effect of the risk factor at a 5% significance level. The *p*-value for the other directional test, i.e., *H*₁: Effect is

protective, was 0.9971 (i.e., $0.0315 + 0.1468 + 0.3180 + 0.3271 + 0.1499 + 0.0238$), indicating that the risk factor was not protective. For the nondirectional test, i.e., H_1 : Effect is either deleterious or protective, it is also intuitive that finding the p -value is to cumulate all table probabilities that were less than or equal to that of the observed table probability in both directions, i.e., $0.00007489 + 0.0028 + 0.0315 + 0.0238 = 0.0581$.

6.1.3 Comparing Two Proportions in Paired Samples

As we apply the paired sample t -test to compare two paired means (see Sect. 3.2), when the binary outcomes result from paired samples, then the application of the method introduced in Sect. 6.1.1 would no longer be valid, and McNemar's test is a valid method.

Consider a study that investigates the agreement between two antibody status testing laboratory techniques, Test 1 and Test 2, which will result in either positive or negative. The summary table in Fig. 6.6 is the result of 100 pairs. Let n_{00} , n_{10} , n_{01} , and n_{11} denote the number of pairs with both tests negative, only Test 1 positive, only Test 2 positive, and both tests positive results, respectively. The ratio n_{01}/n_{10} (if $n_{01} \geq 1$ and $n_{10} \geq 1$) will depart from 1 as evidence to the discrepancy between the two tests. These data showed that the ratio was 3 (i.e., $n_{01}/n_{10} = 30/10 = 3$) meaning that the discrepant outcome of Test 1 negative and Test 2 positive results occurred three times more frequently than those of Test 1 positive and Test 2 negative. Note that the number of concordant pairs is not involved in this calculation.

Let us discuss hypothesis testing. The null hypothesis that the two tests perform the same can be written as H_0 : population proportion $n_{01}/(n_{00} + n_{01} + n_{10} + n_{11}) =$ population proportion $n_{10}/(n_{00} + n_{01} + n_{10} + n_{11})$. Under the null hypothesis, if $n_{01} + n_{10} > 0$, the statistic $(n_{01} - n_{10})^2/(n_{01} + n_{10})$ follows the chi-square distribution with $df = 1$, i.e., $\chi^2_{(1)}$. This test is called McNemar's chi-square test. Figure 6.6

	Test-2 Negative Pairs	Test-2 Positive Pairs	Total
Test-1 Negative Pairs	$n_{00} = 30$	$n_{01} = 30$	60
Test-1 Positive pairs	$n_{10} = 10$	$n_{11} = 30$	40
Total	40	60	100

$H_0: n_{01}/(n_{00} + n_{01} + n_{10} + n_{11}) = n_{10}/(n_{00} + n_{01} + n_{10} + n_{11})$;
 McNemar's χ^2 statistic = $(30 - 10)^2/(30 + 30) = 6.67$, thus p -value
 based on $\chi^2_{(1)}$ is 0.0098 (with Excel 1-CHISQ.DIST(6.67,1,TRUE)).

Fig. 6.6 McNemar's chi-square statistic for comparing two proportions in paired samples

illustrates an example with data showing a statistically significant discrepancy between the performances of the two related test methods ($p = 0.0098$).

6.2 Normal Distribution Assumption-Free Rank-Based Methods for Comparing Distributions of Continuous Outcomes

It is true that by virtue of the central limit theorem (see Sect. 2.2.2), the sample means will follow normal distributions when the sample size is large even if the sample data are drawn from a non-normally distributed (with a finite mean and a variance) population. Thus, an inference about the mean using a large sample can resort to the z -test if the population standard deviation is known. Otherwise, if the population variance is unknown, then a practical choice is to substitute with the sample standard deviation and apply the t -test. Unfortunately, such an application of t -test is valid only if the sample data were drawn from a normally distributed population. If the sample data are not drawn from a normally distributed population, then the sampling distribution of the t -statistic calculated from such data will not follow a t -distribution, and the t -test is invalid. Figure 6.7 describes a distribution of a population whose

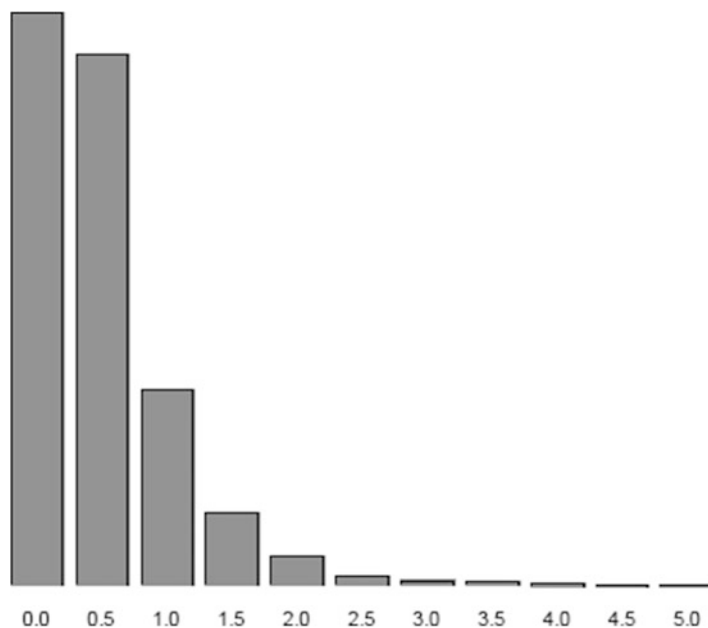


Fig. 6.7 Distribution of an exponentially distributed population with mean = 0.5 and standard deviation = 0.5

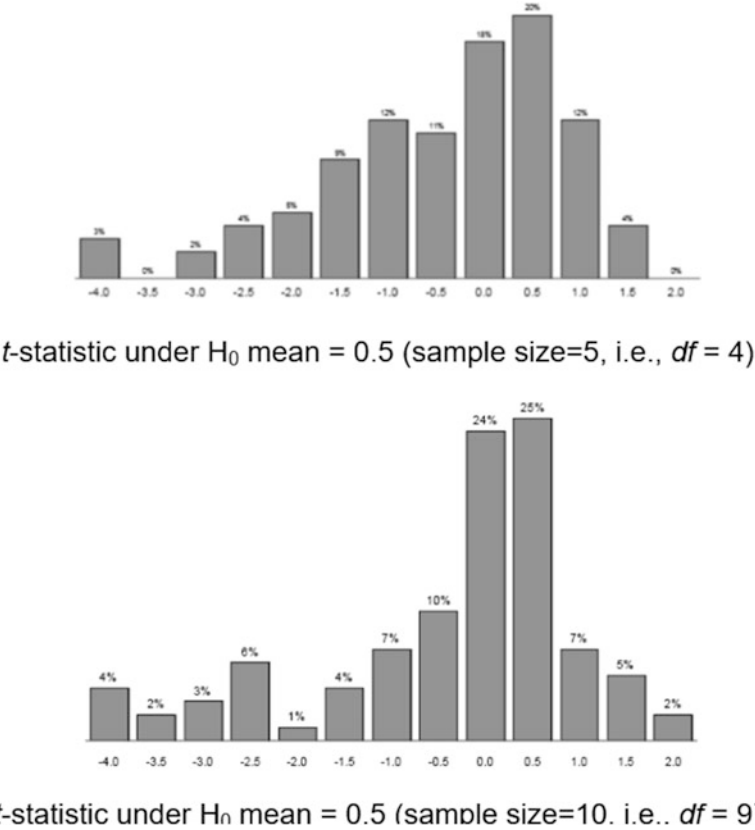


Fig. 6.8 Sampling distributions of t -statistics with $n = 5$ and $n = 10$ based on 100 random sampled data sets drawn from exponentially distributed population with mean = 0.5 and standard deviation = 0.5

distribution is not a Gaussian distribution (this is indeed an exponential distribution with its mean = 0.5 and standard deviation = 0.5).

Figure 6.8 demonstrates two sampling distributions of the t -statistic calculated under H_0 ; mean = 0.5, which are not close enough to t -distributions. The upper histogram is the result out of 100 distinct random samples, with each sample size of 5, drawn from the above population described in Fig. 6.7, and the other is that with sample size of 10. Both sampling distributions of the t -statistics calculated from small sample sizes turned out to be severely skewed to the left (i.e., not symmetrical around 0, which is the expected mean of t -statistic under H_0); thus, the one-sample t -test (H_0 : Mean = 0.5) is not applicable. For data sampled from such a distribution, one may consider making a natural logarithmic transformation that will turn the sample distribution symmetrical (as if the data were sampled from a normal distribution) so that the one-sample t -test can be applied to test H_0 : $\ln(\text{Mean}) = \ln(0.5)$. This example was one of many possible challenging problems of the continuous

outcome samples taken from non-normal distributions, and in most cases, finding out a proper transformation is even more challenging.

The rest of this chapter will introduce alternative methods that do not require the normality assumption about the population distribution.

6.2.1 Permutation Test

The following example demonstrates the permutation test, one of the normal distribution assumption free methods, to measure how unlikely that the samples of Group A and Group B were drawn from the same population.

Group A sample: 96, 102, 108, 126 (mean = 108).
Group B sample: 120, 128, 138, 156 (mean = 135.5).

Figure 6.9 enumerates all possible ordered permutations that could have occurred if four observations had been drawn independently first as Group A from a population and then another four drawn independently as Group B from the same population. For each permutation, the sum of the ranks of the observed values that were observed in Group B is calculated. The sampling distribution of these rank sums is depicted by the histogram in the figure. Among the listed 70 (i.e., $\frac{8!}{4!4!}$) possible permutations, the only two permutations whose Group B's ranks are 4, 6, 7, 8, and 5, 6, 7, 8 would have made the two groups as or more discrepant toward the direction that Group B is larger in their values; and the corresponding probability is 0.029 (i.e.,

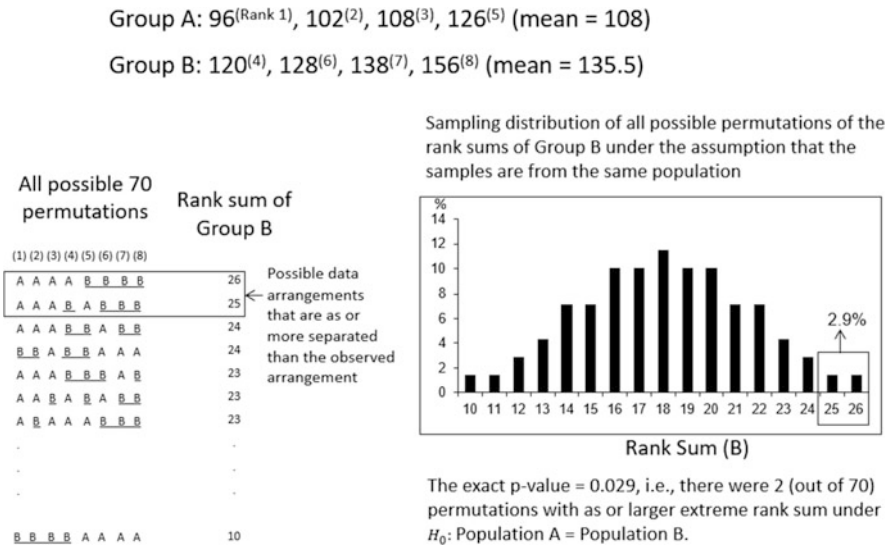


Fig. 6.9 Permutation test illustration

2/70). This probability is indeed the p -value of the directional permutation test in which the null H_0 : Populations A and B are identical.

6.2.2 Wilcoxon’s Rank Sum Test

If continuous outcome data contain extreme values and the sample mean is not meaningful and the normality assumption is not held, then comparing two means using the independent samples t -test will not be an appropriate method. What can be an alternative solution other than the permutation test? The application of the t -test after normalization by an appropriate transformation of the data can be an option. For instance, if the sample distribution is skewed to the right with a longer upper tail, then a logarithmic transformation can (but not always) normalize the distribution so that the independent samples t -test can be applied. However, it cannot be a universal solution. Instead, we rank the pooled (two samples being compared are combined) data, and then apply the independent samples t -test. The ranking de-skews and de-extremizes the distribution so that the independent samples t -test can be applied.

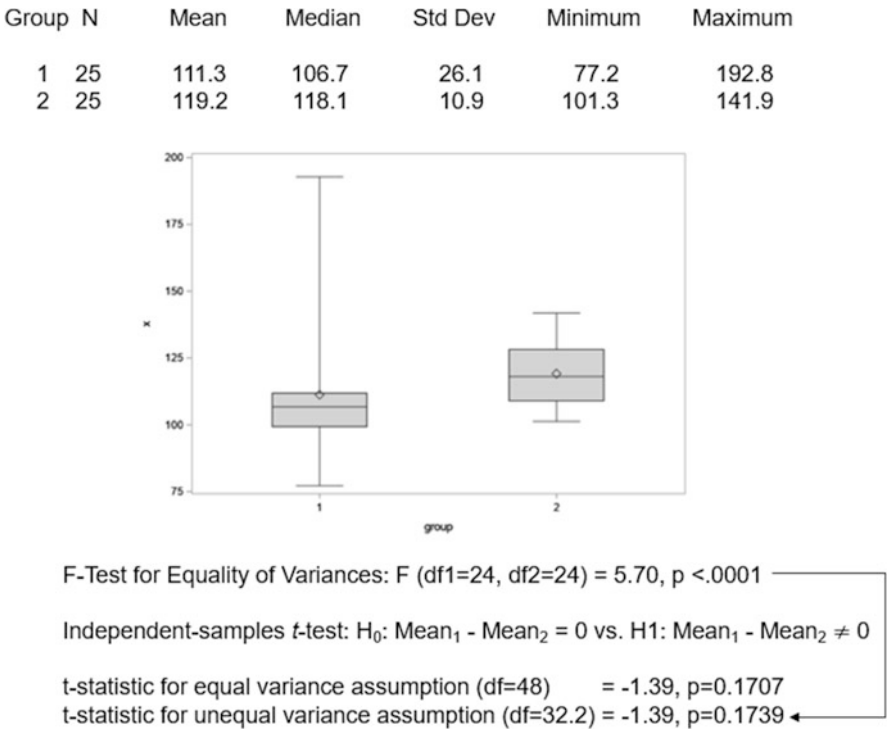


Fig. 6.10 The independent samples t -test did not show a difference in the two population means when applied to the raw (i.e., unranked) data

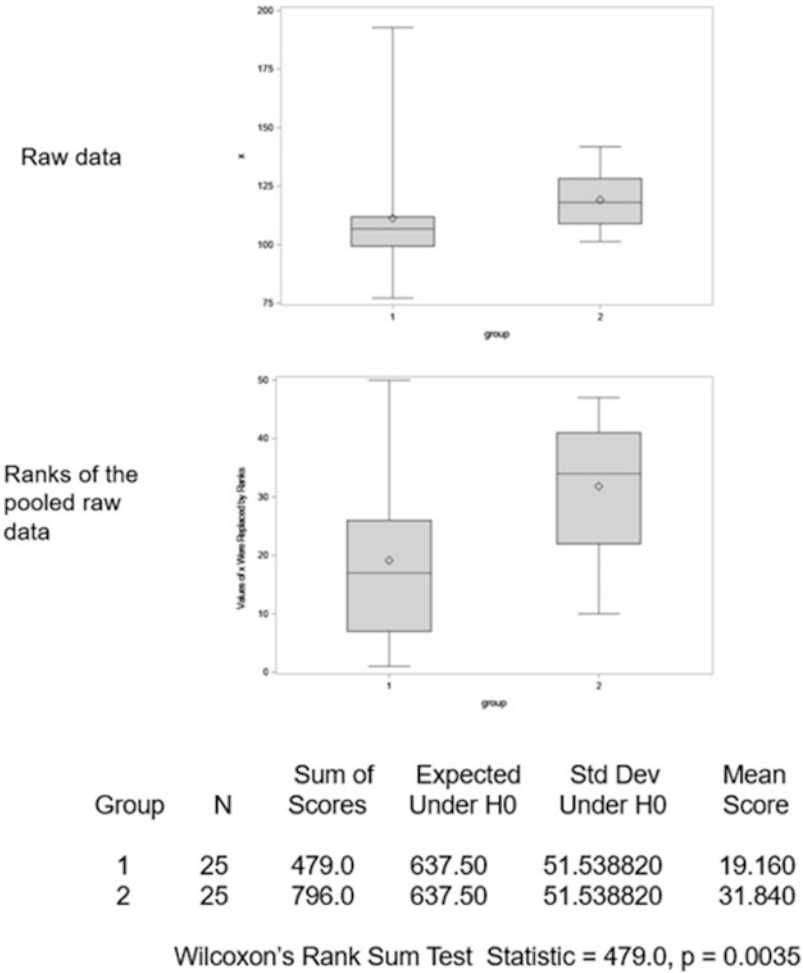


Fig. 6.11 The Wilcoxon’s rank sum test (i.e., rank-based independent samples *t*-test) applied to the same data set (see Fig. 6.10) showed a difference ($p = 0.0035$) in the two population distributions, whereas the independent samples *t*-test that was applied to the raw (i.e., unranked) data did not

Such a rank-based independent samples *t*-test is known as the Wilcoxon’s rank sum test. Wilcoxon discovered this, and Mann and Whitney added more mathematical work; it is also called Mann–Whitney U-test.

Figure 6.10 illustrates a data set for which the independent samples *t*-test was applied as an inappropriate method. The independent samples *t*-test showed that there was no significant difference in means at a 5% significance level. Note that the observed mean difference ($\text{Mean}_1 - \text{Mean}_2$) was -7.9 , but the difference in the medians was -11.4 , which may still suggest that the two sample sets could have been drawn from two different populations.

Figures 6.11 illustrates that the rank-based analysis (i.e., Wilcoxon's rank sum test) would be more appropriate. The Wilcoxon's rank sum test showed that there was a significant difference in the two distributions at a 5% significance level, whereas the independent samples t -test could not detect such a difference.

6.2.3 *Kruskal–Wallis Test*

If data are not normally distributed, the single-factor analysis of variance (ANOVA) F -test for comparing more than two means will not serve well. In such a case, we can rank the pooled data, and then perform the single-factor ANOVA F -test. The rank-based single-factor ANOVA is known as Kruskal–Wallis test.

6.2.4 *Wilcoxon's Signed Rank Test*

If the distribution of the paired difference ($d_i = B_i - A_i$) is heavily skewed or contains extreme values, the mean of the paired difference would not be meaningful, and the paired sample t -test may be an inappropriate method. In such cases, we generate the signed ranks of the pair-wise differences ($d_i = B_i - A_i$, for all i), i.e., $(\text{sign of } d_i) \times (\text{rank of } |d_i|)$, and then compute the sum of the ranks with positive sign and that of the ranks with negative sign, and then check the discrepancy between the positively and negatively signed rank sums. A larger positive (or negative) sum suggests a nonzero average difference (i.e., existence of a difference). The result of this method will be the same as that of the one-sample t -test (see Sect. 2.2.4.5) that is applied to the generated signed ranks of the paired differences (H_0 : Mean of the signed ranks = 0). Such a signed-rank based one-sample t -test is known as Wilcoxon's signed rank test.

6.3 Linear Correlation Based on Ranks

We can also handle the correlation analysis between two continuous measures, say x and y based on ranks if the data had extreme values. We assign the ranks to x and y individual observations first, and then feed these ranks into the mathematical expression or computer program for Pearson's product moment correlation (see Sect. 5.1), also known as Spearman's rank correlation.

6.4 About Nonparametric Methods

Parametric forms of inference are to estimate and test directly the parameters, and the “parameters” in this context imply the underlying probability models that generate the outcomes (i.e., beyond the constants of a particular distribution, e.g., mean and standard deviation for a Gaussian Distribution) such as Gaussian distribution, binomial distribution, etc. (see Chap. 2). For continuous outcomes data, all methods of inference that were introduced in the previous chapters (e.g., *t*-test, ANOVA, Pearson’s correlation, least squares regression) require the assumption that the raw data are gathered from a Gaussian distribution (i.e., data are drawn from a normally distributed population). The term “nonparametric” means that the inference does not require the normality assumption. The rationale behind the nonparametric methods for continuous outcomes is to make extreme values to be non-extreme so that the transformed rank data appear as a “bell-like” shape, and these data can be handled by the normal distribution assumption requiring parametric methods.

What are we testing by nonparametric tests? Did this chapter discuss how the null and alternative hypotheses should be articulated? The nonparametric inference does not involve an articulated hypothesis involving the parameters such as means, in that the inference does not resort to a parametric distribution (e.g., the normal distribution that is completely characterized by two “parameters”: (1) mean, the location parameter and (2) standard deviation, the dispersion parameter). Now, what would be the articulated null hypothesis of nonparametric counterpart of the independent samples *t*-test (i.e., Mann–Whitney’s U-test or Wilcoxon’s rank sum test)? The null hypothesis is “ H_0 : The two population distributions are identical,” which means the two samples were drawn from the same population. The alternative hypothesis is “ H_1 : The two population distributions are not identical.” These hypotheses do not involve explicit statements about the parameters, e.g., means. Moreover, the testing is always performed non-directionally. Some textbooks may introduce you to a null hypothesis like “the medians are the same” in that it is thought that the Mann–Whitney’s U-test or Wilcoxon’s rank sum test compares the medians. This is mathematically true for some situations, but there are many other situations that this is not actually true depending on the particularity of the data distribution. Advanced knowledge of mathematical statistics is necessary for full apprehension of this issue, which is beyond the level and expectation of this book.

How about the chi-square test in the 2×2 contingency table setting? Does this method require data drawn from a normal distribution? No, it does not. But the probability model that generates the outcome data is a binomial model that involves parameters (i.e., p , and n); moreover, the test statistic is chi-square with $df = 1$, which is a parametric distribution. Then, is chi-square a parametric or nonparametric method? chi-square test is classified as a nonparametric method even if the data distribution is parameterized by the binomial model and the test statistic is chi-square with $df = 1$. Generally speaking, nonparametric methods are the ones that do not

require the normality of the mechanism by which the raw data are realized. A more appealing name can be “raw data normality-free methods”.

Study Questions

1. What circumstances of the parametric inference become problematic?
2. Chi-square test for an association between two categorical variables is a nonparametric inference even if this is not a rank-based test. Why is this considered a nonparametric inference?
3. The chi-square test result applied to a 2×2 contingency table analysis will be consistent with that of the univariable (i.e., single independent variable) logistic regression analysis by which the test for a nonzero odds ratio is performed. Why? What would be the dependent and independent variables of the logistic regression?

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Chapter 7

Methods for Censored Survival Time Data



This chapter deals with the survival time data for which the inference cannot be made by any of the parametric and nonparametric methods that we learned in the previous chapters.

Typical survival time data type is time to event (e.g., time to death, time to treatment failure, time to disease relapse, time to recovery after surgery, etc.). The word “survival time” originated from the time-to-death event. The survival time distribution commonly appears as nonsymmetrical, and another unique feature of these data is censoring, which is difficult to deal with using the methods covered in the previous chapters.

Some common statistical questions are as follows: What is the probability that a subject in a group would survive longer than t years? What is the median survival time of the group (i.e., what is the time point when half of the subjects would remain alive)? Are the survival time distributions significantly different between the two groups (i.e., did Group A survive longer than Group B on average, etc.)?

This chapter walks you through two examples that illustrate the typical survival time data with censoring and shows how to handle the data analysis problem.

7.1 Censored Observations

The following 10 values are the observed survival times since diagnosis (in years) of a group of subjects with a particular cancer.

2, 4, 4+, 5, 5, 6, 6, 7+, 8+, 10

The symbol “+” indicates that the value is a censored observation meaning that the subject remained alive until that recorded time, but we do not know whether they deceased thereafter. The truth is that they remained alive longer than the censored observation time, but there is no way to ascertain the time of death. In such a situation, what is the best way to estimate the median survival time? The next

sections will guide the readers on how the censored survival time data are analyzed and interpreted by the methods that take into account the censoring.

7.2 Probability of Surviving Longer Than Certain Duration

Statistical descriptions and interpretations of the distribution of censored survival time are carried out by estimating the probability of survival exceeding a certain time T , which is denoted by $\hat{S}(T)$. The estimate of the probability that an individual survives longer than 1 year, $\hat{S}(1)$, is obviously $\hat{S}(1) = 1$ (i.e., 100% because all 10 individuals survived longer than 1 year). The censoring did not affect this calculation. It is straightforward that $\hat{S}(2) = 9/10$ because one subject died at time = 2.

Complicated calculations occur when the time of interest is 4 years or later. The idea is to calculate the probability of surviving longer than 4 years for individuals remaining at risk of death (i.e., have survived until then) at $T = 4$. This is conceived as the product of the two probabilities of which the first piece is the already calculated probability of survival up to right before the time of evaluation, and the second piece is the probability of survival longer than the evaluation time, i.e., $\hat{S}(4) = (\text{the proportion of the subjects remaining alive longer than 2 years among all subjects}) \times (\text{the proportion of the subjects remaining alive longer than 4 years among those who lived up to 4 years}) = \hat{S}(2) \times (8/9) = (9/10) \times (8/9)$. In this calculation, $\hat{S}(2)$ was used as the first piece (the already calculated probability of survival up to right before the time of evaluation) because there were no deaths between 2 and 4 years. Likewise, $\hat{S}(5)$ is estimated as the product of the three pieces, i.e., $\hat{S}(5) = \hat{S}(0) \times \hat{S}(2) \times \hat{S}(4) \times (\text{number of subjects remaining alive longer than 5 years among those who survived after the most recent observed time of death}) = 1 \times 0.9 \times 0.8 \times (5/7) = 0.5714$. Such a method was proposed by Kaplan and Meier (Kaplan–Meier method) and is also called the product limit method for estimating survival function. What is then the Kaplan–Meier estimate of $\hat{S}(5.5)$? Since the Kaplan–Meier survival distribution function is a step function in which the change of the value happens only at the observed event, $\hat{S}(5.5)$ is still 0.5714, and it remains the same until $\hat{S}(6)$ gets updated. Applying the same idea, $\hat{S}(6) = 0.5714 \times (3/5) = 0.3429$. These calculations are summarized below:

Time, t_j	Cumulative number of deaths	Number alive	Kaplan–Meier estimate $\hat{S}(t_j) = (n_1 - d_1)/n_1 \times (n_2 - d_2)/n_2 \times \dots (n_j - d_j)/n_j$
0	0	10	$\hat{S}(0) = 10/10 = 1.0000$
2	1	9	$\hat{S}(2) = 9/10 = 0.9000$
4	2	8	$\hat{S}(4) = \hat{S}(2) \times (8/9) = 0.8000$
4+	2	7	No events occur, and update is not made

(continued)

Time, t_j	Cumulative number of deaths	Number alive	Kaplan–Meier estimate $\hat{S}(t_j) = (n_1 - d_1)/n_1 \times (n_2 - d_2)/n_2 \times \dots (n_j - d_j)/n_j$
5 5	4	5	$\hat{S}(5) = \hat{S}(4) \times (5/7) = 0.5714$
6 6	6	3	$\hat{S}(6) = \hat{S}(5) \times (3/5) = 0.3429$
7+	6	2	No events occur, and update is not made
8+	6	1	No events occur, and update is not made
10	7	0	$\hat{S}(6) \times (0/1) = 0$

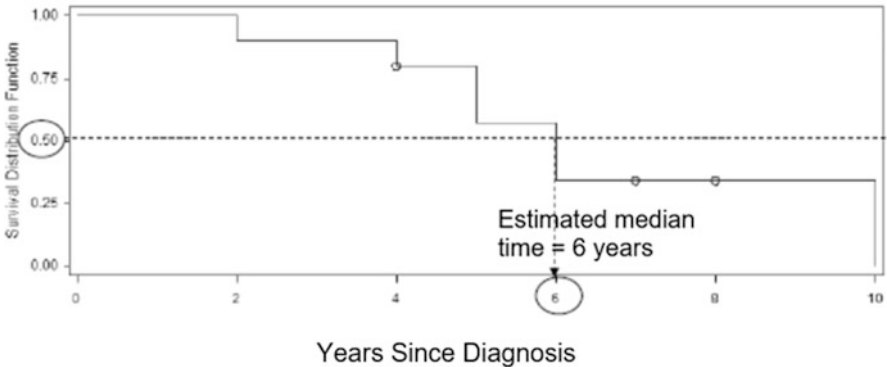


Fig. 7.1 Illustration of Kaplan Meier survival curve and estimated median survival time

For $\hat{S}(t_j) = (n_1 - d_1)/n_1 \times (n_2 - d_2)/n_2 \times \dots (n_j - d_j)/n_j$, n_j is the number of survivors before time t_j when there is no censoring. When there is censoring, it is the number of survivors less the number of censored cases. d_j is the number of deaths at t_j . Fig. 7.1 describes the estimated survival distribution, which is called the Kaplan–Meier survival curve. The graph is stepwise in that the probability of survival remains unchanged until the next death event is observed. The circles indicate the censored events.

7.3 Statistical Comparison of Two Survival Distributions with Censoring

The statistical inference to compare a difference between two mean survival times may not be performed by the independent samples t -test because of censoring even in the case that the population survival time is normally distributed. The next example describes a very commonly used nonparametric method.

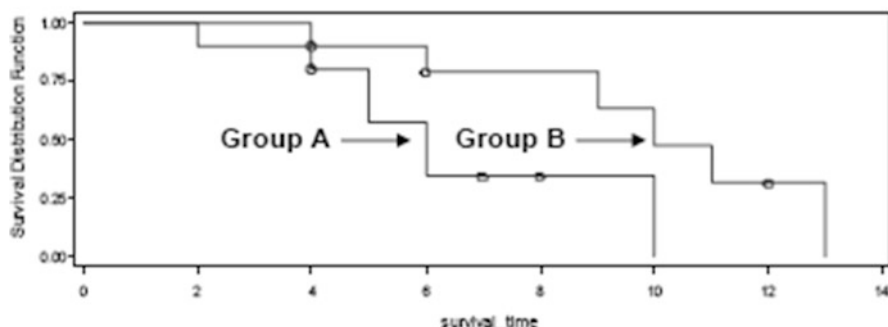


Fig. 7.2 Kaplan–Meier survival curves of two study subgroups

Example 7.1

Which group survived longer (survival times are recorded in years)?

Group A: 2, 4, 4+, 5, 5, 6, 6, 7+, 8+, 10

Group B: 4, 4+, 6, 6+, 6+, 9, 10, 11, 12+, 13 (Fig. 7.2)

Observed data summary:

Group	Total subjects	Deceased subjects	Censored subjects (%)	Median survival time
A	10	7	3 (30%)	6 years
B	10	6	4 (40%)	10 years

The inference begins with stating the null and alternative hypotheses,

H_0 : Two survival time distributions are not different versus H_1 : Two survival time distributions are different (i.e., nondirectional two-sided alternative). Note that the hypotheses did not make any statement about the means but simply stated about the distribution (i.e., the inference does not involve the location parameters of the distributions). Two statisticians, Cox and Mantel, devised a nonparametric test. The idea is to create a 2×2 contingency table at every observed event time of the combined sample set that offers evidence that the event occurrence rates between the two groups are different (e.g., for the above example, eight 2×2 tables at time = 2, 4, 5, 6, 9, 10, 11, and 13 will be created based on the numbers of each group's deceased subjects and numbers of subjects remaining at risk right before each observed event time). The test statistic cumulates the individual 2×2 table's chi-square values calculated from those 2×2 tables under H_0 . The sampling distribution of this test statistic approaches the chi-square distribution with $df = 1$ as the sample size becomes sufficiently large. This test was called log-rank test later by two statisticians, Peto and Peto. The name “*log-rank test*” reveals the technical aspect of how the test statistic is devised (Peto and Peto 1972).

The result of the test obtained from a computer software was that the log-rank chi-square ($df = 1$) was 3.9016 with its p -value of 0.0482, which can be summarized as “These data showed that the two survival time distributions were significantly different ($p=0.0482$).”

Study Question

1. Explain the product limit calculation for estimating a survival time distribution. What are the two probabilities that are being multiplied in the computation each time of the calculation?

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Chapter 8

Sample Size and Power



The idea and examples for the determination of an adequate study sample size for making inferences about one and two means and proportions are discussed.

8.1 Sample Size for Single Mean Interval Estimation

In the case of a single mean interval estimation, the width of a $100 \times (1 - \alpha) \%$ confidence interval (CI), e.g., for $\alpha = 0.05$, $100 \times (1 - 0.05) \% = 95\%$, is a function of several elements among which the pivotal one is the sample size.

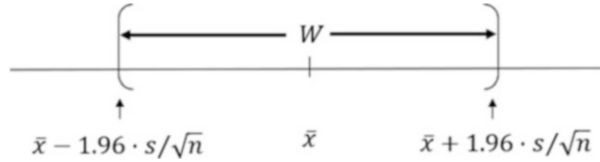
As discussed in Sect. 2.2.6.2, the following is the idea of determining the sample size for a 95% confidence interval using Gaussian approximation (i.e., resort to the central limit theorem, CLT, for approximating the sampling distribution of the sample mean to a Gaussian distribution, see Sect. 2.2.2). The first step is to set up an equation by letting one half of the interval length (an error margin of $w/2$, Fig. 8.1) be equal to the distance between 0 (i.e., the mean) and the 97.5th percentile of the standard normal distribution $\times$ the standard error of the sample mean, i.e., $w/2 = 1.96 \times s/\sqrt{n}$, where s denotes the sample standard deviation. Solving this equation for n , the sample size estimating equation will be $n = (3.92 \times s/w)^2$. As depicted by Fig. 8.1, the interval becomes narrower as the sample size increases (i.e., inversely proportional to $\sqrt{n}$).

The following example aids in understanding how the required sample size is determined for constructing a 95% confidence interval for one mean inference.

Example 8.1

The standard dosage level of medication “A” will lower the heart rate over 48 hours. A new study for a higher dose is being proposed, and the investigator wants to determine how many study subjects are needed for this study to find a 95% CI of the mean heart rate that is no wider than 5 beats per minute (BPM) if the anticipated sample standard deviation $s = 10$ BPM. Determine the sample size.

Fig. 8.1 Sample size determination for interval estimation of a single mean



Given $w = 5$ and $s = 10$;

Solve $w/2 = 1.96 \times 1/\sqrt{n}$, for n ;

$$n = (3.92 \cdot s/w)^2 = (3.92 \cdot 10/5)^2 = 61.5;$$

Sample size = 62

If the study subject recruitment is not difficult, then 62 patients can be a generously determined sample size. It is also possible to curtail the sample size to the integral part of the solution (i.e., do not need to round up to save the resource), i.e., $n = 61$, which would not have a large impact.

How many more subjects do we need if we want to shrink the length of the 95% CI down to no more than 4 bpm?

$$n = (3.92 \cdot s/w)^2 = (3.92 \cdot 10/4)^2 = 96.04$$

Sample size = 96, thus 34 more patients are required.

Note that to decrease the error margin by 20%, 34 more patients are required, which is an increase of 55%, not the same increase of 20%.

8.2 Sample Size for Hypothesis Tests

8.2.1 Sample Size for Comparing Two Means Using Independent Samples z- and t-Tests

As discussed in Sect. 2.2.4, the sample size required for a hypothesis testing to compare two independent means depends on:

1. Level of significance (α).
2. Power ($1 - \beta$), where β is the probability of committing Type 2 error.
3. Alternative hypothesis (one- or two-sided).
4. Size of detectable difference in means (δ).
5. Standard deviation of the outcome distribution under the null hypothesis (σ).

The idea of determination for the required sample size for comparing two means using independent samples z-test is illustrated in Fig. 8.2, which depicts the relationship between the sample size and the five determinants. The two density curves describe the sampling distributions of test statistic z under H_0 (left curve) and H_1 (right curve). Note that the right-hand side curve is a snapshot of the horizontally sliding curve that stopped (for illustration) where the alternative hypothesis' mean

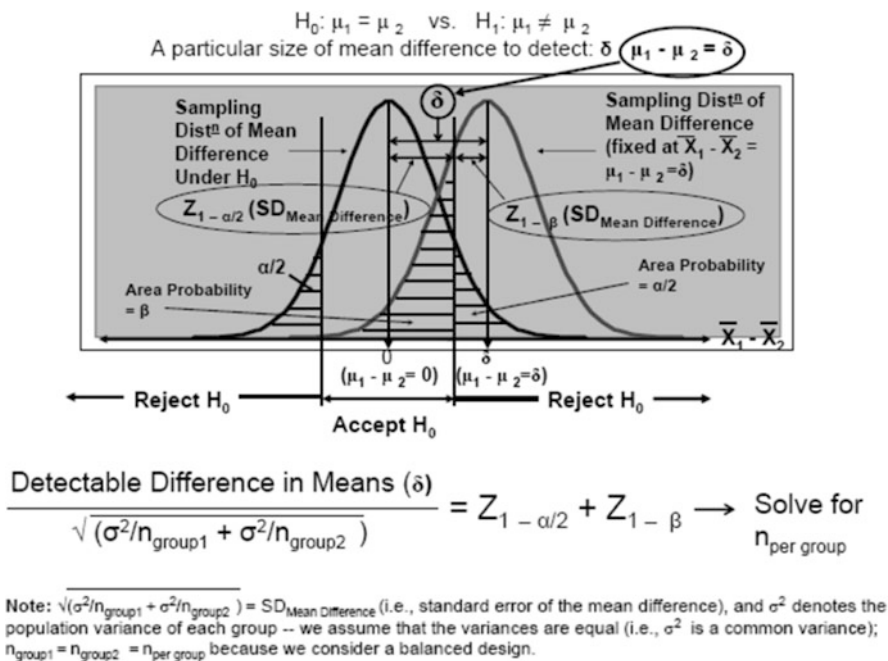


Fig. 8.2 Sample size for a nondirectional independent samples z -test for comparing two means

difference is a certain nonzero value δ . Within the rejection region, the null and alternative distributions overlapped. The shaded area under the null distribution is the size of Type 1 error, and the area under the alternative distribution is the power.

The left-hand side of the equation is the standardized (to the standard deviation unit of the sampling distribution of the mean difference) effect size. The right-hand side of the equation spells out the left-hand side into a sum of the following two parts: (1) the distance between the null mean and the critical value of the test on the z -scale and (2) the distance between the critical value and the alternative mean on the z -scale. Given the common standard deviation of the two population distributions (σ), the detectable mean difference (δ), and the sizes of Type 1 and 2 errors (α and β), this equation is solved for the sample size per group:

$$n_1 = n_2 = \frac{2\sigma^2(z_{1-\alpha/2} + z_{1-\beta})^2}{\delta^2}.$$

Note that in the numerator of this sample size estimating equation, $z_{1-\beta}$ appeared (instead of $z_{1-\beta/2}$) because with δ fixed, the left tail area of the alternative distribution covered by the other side of the rejection region is nearly zero as depicted in Fig. 8.2.

As illustrated in Fig. 8.3, the required sample size increases geometrically as the targeted power increases and the targeted effect size decreases.

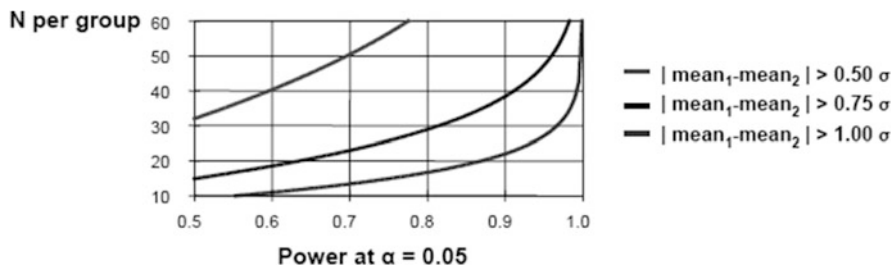


Fig. 8.3 Sample size given the power and effect size for an independent samples z -test for comparing two means

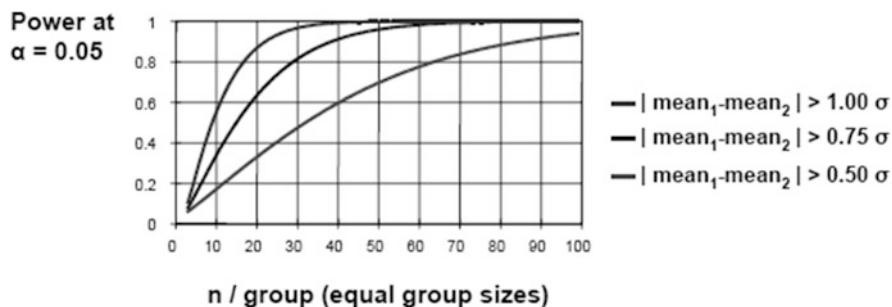


Fig. 8.4 Power given the sample size and effect size for an independent samples z -test for comparing two means

As illustrated in Fig. 8.4, the power is proportional to the square root of the given sample size and inversely proportional to the square root of the effect size.

Example 8.2

A sample size determination is needed for a nondirectional z -test, with its power = 80% at an adopted significance level, α , of 5%, to compare the mean systolic blood pressures between two independent groups, Group A and Group B. A pilot data suggested that the means are approximately 132.86 and 127.44 in Group A and Group B, respectively (i.e., difference size = $132.86 - 127.44 = 5.42$) and the known common population $\sigma = 16.8$. What should be an adequate sample size to detect a mean difference of 5.42?

For a nondirectional test with its adopted significance level, $\alpha = 0.05$, a target power to attain, $1 - \beta = 0.8$, and the common population standard deviation, $\sigma = 16.8$, the sample size per group for a balanced design (i.e., equal group size) is determined by the equation:

$$n_1 = n_2 = \frac{2\sigma^2(z_{1-\alpha/2} + z_{1-\beta})^2}{\delta^2},$$

where $z_{1-\beta} = z_{0.8} = 0.84$ (i.e., z -value whose area under the standard Gaussian density curve covers from negative infinity to 0.8, and this can be found by using Excel function $\text{NORMSINV}(0.8) = 0.84$). Similarly, $z_{1-\alpha/2} = z_{1-0.025} = z_{0.975} = 1.96$ (using Excel function $\text{NORMSINV}(0.975) = 1.96$), and the difference in means to detect is 5.42 (about one thirds of σ). The solution of the equation is 151, i.e., n per group $= 2\{16.8 \times (0.84 + 1.96)/5.42\}^2 = 151$.

Let us now consider the independent samples t -test to compare two means when the population standard deviation is not known under the assumption that the data are drawn from normally distributed populations. The principle of sample size determination is almost the same as that of the z -test except that the sampling distribution of the test statistic is t -distribution, which is more difficult to deal with. The difficult part is that the sampling distribution of the test statistic, t , under the alternative hypothesis must be characterized by not only the difference in the two means and the standard error of the mean difference but also by two more parameters. These two parameters cannot be known until the sample size is known, so, unlike the z -test case, the sample size determination cannot be achieved by solving a single closed form equation whose sample size being sought is not the one and only unknown variable. The first of these two parameters is the degrees of freedom, df . As we already learned in previous chapters, the degree of freedom is dependent on the sample size being sought. The other parameter is the non-centrality parameter, which corresponds to the standardized effect size of the test, i.e., $\frac{\delta}{\sqrt{\sigma_1^2/n_1 + \sigma_2^2/n_2}}$.

In the t -test setting, the population standard deviations, σ_1 and σ_2 , are unknown but can be replaced with the sample standard deviations s_1 and s_2 for approximation. In any event, without this parameter being known, the sampling distribution of the test statistic t under the alternative hypothesis cannot be completely characterized. Obviously, the sampling distribution under the **null** hypothesis of equal means explicitly determines its **non-centrality parameter** to be **zero**, thus the sampling distribution of t under the null hypothesis is also called a **central t -distribution**. As illustrated in Fig. 8.5, unlike the z -test, the sampling distribution of the t -statistic under the alternative hypothesis has a larger variation (degrees of freedom of 18, i.e., sample size = 20, non-centrality parameter = 3.5, reject $H_0: \mu_1 = \mu_2$ if $t < -2.1$ or $t > 2.1$ in favor of the nondirectional alternative hypothesis $H_1: \mu_1 \neq \mu_2$) than that under the null hypothesis. Of course, the degree of increase in the variation for a fixed non-centrality parameter is also dependent on the degrees of freedom (i.e., the sample size being sought). Such an interconnection makes the sample size determination more complicated than that of the z -test.

For this reason, the sample size can be determined iteratively in that a rough estimate of the sample size is fixed temporarily to calculate the resulting power based on the noncentral t -distribution characterized by the alternative hypothesis and continue to update the sample size until the resulting power is close enough to the targeted power. Computer programs as well as the calculated power tables for t -tests given a wide range of predetermined sample sizes, non-centrality parameters (or difference in means), sample standard deviations of the two groups, and level of significance, α , are widely available.

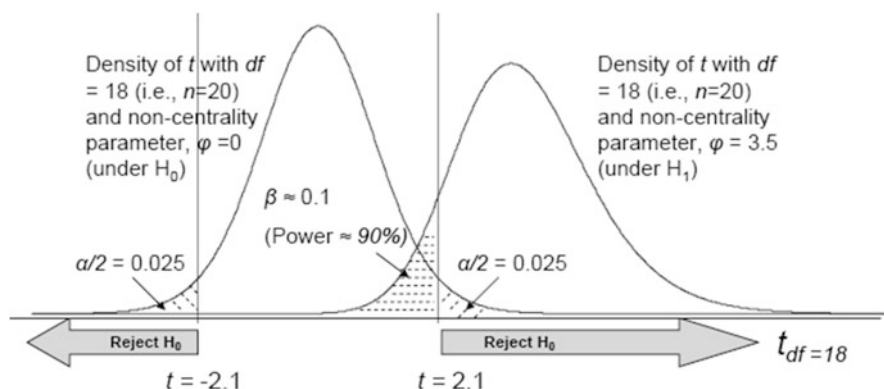


Fig. 8.5 Power of nondirectional independent samples t -test for comparing two means

Comment on non-centrality parameter and beyond: The non-centrality parameter was not introduced to the t -distribution until we encountered the situation of calculating the power of a test using a specified sampling distribution of the test statistic, t , under an alternative hypothesis. This parameter does not influence the sampling distribution under the null hypothesis, and so is the practical aspect of carrying out a t -test, i.e., calculation of the p -value and or determining the critical region as long as the directionality (one- or two-sided) of the alternative hypothesis is only specified. The non-centrality parameter was necessary in calculating the power given the significance level, α , and the known sample size. The power calculations for the chi-square- and F -test also require the non-centrality parameters of these sampling distributions under the alternative hypotheses. The involvement of the non-centrality parameter in chi-square- and F -distributions is due to the fact that both distributions are derived in relation with t -distribution. Interested readers may learn the mathematical genesis of chi-square- and F -distributions and the power calculation problem for estimating a sample size given the significance level, α , the alternative hypothesis and the known sample size from further readings such as Winer (1971).

8.2.2 Sample Size for Comparing Two Proportions

The chi-square test is commonly used to compare two independent proportions for the analysis of a 2×2 contingency table. However, because of the complexity to deal with the non-centrality discussed in Sect. 8.2.1 and also because the sampling distribution of the square root of the chi-square statistics under the null hypothesis for the 2×2 contingency table analysis is the same as that of the z -distribution (see Sect. 6.1.1), the sample size can be determined by a method very similar to that of the independent samples z -test. The following equation determines the sample size per group in a balanced design (i.e., equal group size), which is derived based on that

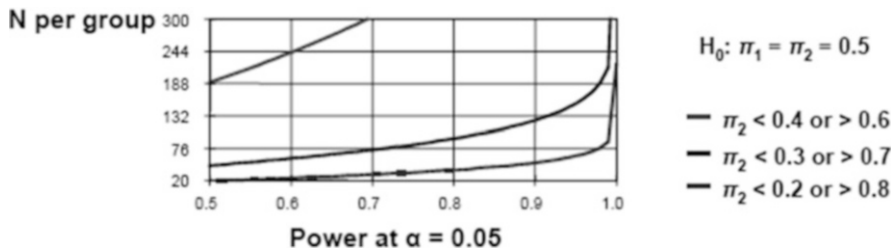


Fig. 8.6 Sample size given the power and effect size for an independent samples z -test for comparing two proportions

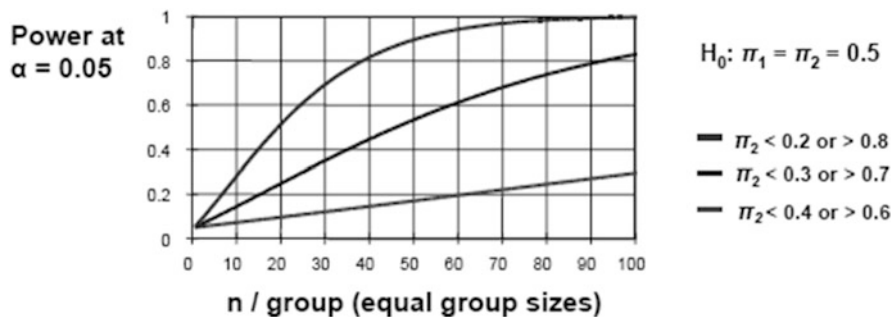


Fig. 8.7 Power given the sample size and effect size for an independent samples z -test for comparing two proportions

z -distribution (i.e., normal approximation of binomial distribution) (see Sect. 1.5.5) wherein π_1 and π_2 indicate the two independent proportions under the alternative hypothesis, α is the adopted significance level, and β is the size of the power to attain:

$$n_1 = n_2 = \frac{(\sqrt{2\pi_1(1-\pi_1)}z_{1-\alpha/2} + \sqrt{\pi_1(1-\pi_1) + \pi_2(1-\pi_2)}z_{1-\beta})^2}{\delta^2}$$

Figures 8.6 and 8.7 present the relationship among sample size, power, and effect size at an adopted 5% significance.

Study Questions

1. Explain why the graphs of Figs. 8.3 and 8.4 do not appear as straight lines.
2. An investigator determined a sample size for a new study to detect a certain effect size (difference in two means) with 80% power at a 5% significance level using an independent samples t -test. If this investigator chooses a smaller effect size, then the sample size will have to increase. If a newly chosen effect size became the half size of the previously determined sample size without altering the power and significance level of the test, then should the new sample size be doubled or quadrupled?

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Chapter 9

Review Exercise Problems



9.1 Review Exercise 1

1. A measure of body temperature ($^{\circ}\text{F}$) is _____. Which answer is correct?

- (1) an interval datum
- (2) a ratio datum
- (3) a discrete datum
- (4) a continuous datum

- (a) None of these
- (b) Only (1)
- (c) Only (2)
- (d) Only (3)
- (e) Only (4)
- (f) (1) & (2)
- (g) (1) & (3)
- (h) (1) & (4)
- (i) (2) & (3)
- (j) (2) & (4)
- (k) (3) & (4)
- (l) (1), (2), & (3)
- (m) (1), (2), & (4)
- (n) (1), (3), & (4)
- (o) (2), (3), & (4)
- (p) (1), (2), (3), & (4)

2. In a normal distribution _____. Which answer is correct?

- (1) the mean, median, and mode are the same.
- (2) the interval from half standard deviation below the mean (i.e., mean - $0.5 \times \text{SD}$) to half standard deviation above the mean (i.e., mean + $0.5 \times \text{SD}$) covers approximately 38.3% of the distribution.
- (3) the interval from 1.96 standard deviation units below the mean (i.e., mean - $1.96 \times \text{SD}$) to 1.96 standard deviation units above (i.e., mean + $1.96 \times \text{SD}$) covers 97.5% of the distribution.

- (a) None of these
- (b) Only (1)
- (c) Only (2)
- (d) Only (3)
- (e) (1) & (2)
- (f) (2) & (3)
- (g) (1) & (3)
- (h) (1), (2), & (3)

3. In the t -distribution with $df = 20$ and non-centrality parameter = 0, _____. Which answer is correct?

- (1) the mean, median, and mode are the same.
- (2) the interval from -1.725 to 1.725 covers 95% of t -values.
- (3) the interval from -2.528 to 2.528 covers 95% of t -values.
- (4) the interval from -2.086 to 0.000 covers 47.5% of t -values.

- (a) None of these
- (b) Only (1)
- (c) Only (2)
- (d) Only (3)
- (e) Only (4)
- (f) (1) & (2)
- (g) (1) & (3)
- (h) (1) & (4)
- (i) (2) & (3)
- (j) (2) & (4)
- (k) (3) & (4)
- (l) (1), (2), & (3)
- (m) (1), (2), & (4)
- (n) (1), (3), & (4)
- (o) (2), (3), & (4)
- (p) (1), (2), (3), & (4)

4. In the F -distribution (under the null hypothesis for an ANOVA F -test) with $df_{\text{numerator}} = 3$ and $df_{\text{denominator}} = 5$, _____. Which answer is correct?

- (1) the mean, median, and mode are the same.
- (2) the probability of observing a value of $F > 5.41$ is 0.025.
- (3) the smallest possible value is 0.
- (4) the square root of F ($df_{\text{numerator}} = 3$, $df_{\text{denominator}} = 5$) = t ($df = 5$).

- (a) None of these
- (b) Only (1)
- (c) Only (2)
- (d) Only (3)
- (e) Only (4)
- (f) (1) & (2)
- (g) (1) & (3)
- (h) (1) & (4)
- (i) (2) & (3)
- (j) (2) & (4)
- (k) (3) & (4)
- (l) (1), (2), & (3)
- (m) (1), (2), & (4)
- (n) (1), (3), & (4)
- (o) (2), (3), & (4)
- (p) (1), (2), (3), & (4)

5. The power refers to _____. Which answer is correct?

- (1) probability of rejecting the null hypothesis when it is false,
- (2) 1 minus the probability of Type 1 error
- (3) the size of the difference between the values stated in the null hypothesis and the alternative hypothesis,
- (4) 1 minus p -value.

- (a) None of these
- (b) Only (1)
- (c) Only (2)
- (d) Only (3)
- (e) Only (4)
- (f) (1) & (2)
- (g) (1) & (3)
- (h) (1) & (4)
- (i) (2) & (3)
- (j) (2) & (4)
- (k) (3) & (4)
- (l) (1), (2), & (3)
- (m) (1), (2), & (4)

- (n) (1), (3), & (4)
- (o) (2), (3), & (4)
- (p) (1), (2), (3), & (4)

6. Please draw manually the stem-and-leaf plot of the following 11 observations: 41 48 51 52 55 56 58 63 65 67 83

7. Dr. Z studied 28 patients with a certain chronic illness. From the patients' history, review the sample means and variances of the ages at the occurrence of the first episode for the severe, moderate, and mild groups that are summarized below:

Descriptive Statistics for Problems 7.1–7.16

Study group	n	Sample mean	Sample variance
Group 1 (severe)	10	30.00	25.0
Group 2 (moderate)	10	35.00	25.0
Group 3 (mild)	10	41.25	25.0

7.1–7.4 Based on the experience, Dr. Z suspected that the mean age of the first episode occurrence within the severe group would be 33 or less (i.e., younger). Please perform a t -test at a 5% significance level.

7.1 Stating null and alternative hypotheses

- (A) Write up a proper null hypothesis that should have been written prior to the data collection (please do not use mathematical notations but do verbalize).
- (B) Write up the alternative hypothesis that negates your stated null hypothesis that should have been written prior to the data collection (please do not use mathematical notations but do verbalize).

7.2 Test statistic

- (A) Calculate the numerator of your t -statistic under the null hypothesis, which is the [observed severe group's mean age at the occurrence of the first episode for the severe – the mean under the null hypothesis].
- (B) Calculate the denominator of the t -statistic, which is the standard error of the numerator, i.e., SE of (observed severe group's mean age at the occurrence of the first episode for the severe – the mean under the null hypothesis).
- (C) What is the value of your observed t -statistic?

7.3 Determination of the significance

You can either calculate the p -value directly or determine the critical region first and then determine if the observed test statistic is within the critical region of the sampling distribution under the null hypothesis. The degree of freedom of your test statistic's distribution is 9.

- (A) If you decided to calculate the p -value using Excel, which Excel function would you use? _____

(B) Please determine the critical region of this test (note: a critical region is not simply the critical cutoff value/values of the test statistic t).

(C) p -value of the observed test statistic t : _____

7.4 Write a single sentence that can go into the results section of a research paper. Please do not write a sentence like “The null hypothesis was rejected because ...”.

7.5–7.8 Please perform a t -test at a 5% significance level to examine whether the mean ages of the first episode occurrence are the same in both severe patients and moderate patients or otherwise the severe patients encounter their first episode sooner in life than the moderate patients.

7.5 Stating null and alternative hypotheses

(A) Write up a null hypothesis that should have been written prior to the data collection (please do not use mathematical notations but do verbalize).

(B) Write up the alternative hypothesis that negates the null hypothesis that should have been written prior to the data collection (please do not use mathematical notations but do verbalize).

7.6 Test statistic.

(A) Calculate the numerator of the t -statistic, i.e., observed mean difference – expected mean difference under the null hypothesis. Mean difference set up: [severe group’s mean – moderate group’s mean].

(B) Calculate the denominator of the t -statistic, which is the standard error of the numerator, i.e., SE (observed mean difference – expected mean difference under the null hypothesis).

(C) If you carried out (A) and (B) correctly, then the t -statistic is -2.236 . Please show your derivation (simply follow “trio” setup found in Sect. 3.1).

7.7 Determination of the significance

You can either calculate the p -value directly or determine the critical region first and then examine if the observed test statistic is within the critical region. Note: The degree of freedom of your test statistic’s distribution is 18.

(A) If you decided to calculate the p -value using Excel, which Excel function would you use? _____

(B) Please determine the critical the region of this test (note: a critical region is not simply the critical cutoff value/values of the test statistic t).

(C) p -value: _____

(D) In order for the difference to become significant at the chosen significance level, what should be the smallest value of the difference?

7.8 Confidence interval of the mean difference

(A) Using a t -distribution, please construct a 90% confidence interval on the difference between the two means (i.e., severe mean age – moderate mean age). Please make sure you carry out this calculation as the difference is defined as instructed (i.e., do not change the direction of the mean difference calculation).

(B) Write a single sentence that can go into the results section of a clinical journal paper.

7.9–7.14 The following ANOVA table can be constructed to compare the mean ages of the first episode onset among the three groups (i.e., H_0 : All three means are equal vs. H_1 : At least one mean is different from the rest).

Please complete the table. Note that you need to calculate the grand mean age, and please be careful with that calculation.

Source	df	Sum of squares	Mean squares	F -value
Between severity	2	7.9 ____	7.12 ____	7.14 ____
Within severity	25	7.10 ____	7.13 ____	
Total	27	7.11 ____		

7.15 Calculate p -value using Excel.

7.16 Write a single sentence that can go into the results section of a clinical journal paper.

7.17 Dr. Z plans on constructing a new study (Z-10 study) that will enroll only the severe and moderate patients. The plan is to enroll 180 patients in each group to detect a smaller difference than what was observed in the above small Z-09 study (note that Z-09 study and $n = 10$ per group showed a difference of 5 years in mean ages). The inference will be a directional independent samples z -test at a 5% significance level and with a 90% power. It is known that the population variances of the three groups are known and equal to 25. Please provide a reasonable approximation of the minimum detectable statistically significant difference between the two mean ages (your answer should include the wording “a difference in mean ages of at least ____ years”) for Z-10 study. Please show fully annotated work including the justification for some assumptions or subjective decisions that you might have made.

8. Dr. A at a major university-affiliated teaching hospital conducted a pilot efficacy study to investigate whether a new post-surgery rehabilitation intervention can improve patient quality of life during a 4-week period after a hip surgery (A-01 study). The study was a two-arm (conventional versus new intervention groups) randomized study. The study sample size of this A-01 study was determined to detect at least **m**% greater improvement in the mean score among the patients receiving the new intervention at a 10% significance level with a power of 80% by a directional t -test. Dr. A assumed equal variance and the common standard deviation value that Dr. A used for the sample size determination was **s**%. The

study enrolled 15 patients receiving the intervention and 15 patients receiving the traditional intervention (total sample size of 30). The result of A-01 study was quite promising in that the directional statistical test (i.e., independent samples *t*-test) was significant at the predetermined significance level of 10% (it is not uncommon that a pilot study chooses a lenient significance level than 5%), and the patients who received the new intervention showed that the mean percentage improvement of the physical strength subscale of SF-36 quality of life scale was greater by about **m%** after 4 weeks.

Based on A-01 study, Dr. A suspected that a new study could detect even a smaller difference, and he proposed a new study (A-02 study) that will enroll 60 patients per group (i.e., total study sample size of 120). In the sample size determination section of the A-02 study proposal, Dr. A justified that the proposed sample size is adequate to show a statistically significant mean percentage improvement in the subscale by at least **0.5 m%** by using a directional independent samples *t*-test at a 5% significance level with a power of 80%. The common standard deviation of the mean percentage improvement remains the same as that of study A-01.

Question: Was Dr. A's sample size justification reasonable? Why/why not?

9. Which of the following is true?

- (1) The distribution of a random sample data set drawn from a non-normal continuous population distribution will approach a normal distribution as the sample size becomes very large.
 - (2) The distribution of sample means obtained from a non-normal continuous population distribution will approach a normal distribution as the sample size becomes very large.
 - (3) The distribution of the sample means obtained from a normal population distribution will approach a normal distribution as the sample size becomes very large.
- (a) None of these
 - (b) Only (1)
 - (c) Only (2)
 - (d) Only (3)
 - (e) (1) & (2)
 - (f) (2) & (3)
 - (g) (1) & (3)
 - (h) (1), (2), & (3)

9.1.1 Solutions for Review Exercise 1

1. h
2. e
3. e

4. d

5. b

$$\begin{array}{r} 83 \\ | \end{array}$$

6. 6 356

$$\begin{array}{r} 512568 \\ | \end{array}$$

48

7. 7.1 (A) H_0 : Mean age > 33 (B) H_1 : Mean age ≤ 33 7.2 (A) $30 - 33 = -3$ (B) $SE = SD/\sqrt{10} = \sqrt{25}/\sqrt{10} = 1.58$ (C) $t = (30 - 33)/SE = -3/1.58 = -1.90$ ($df = 9$)7.3 (A) TDIST(x, df , 1), TDIST(-1.90, 9, 1)(B) $\{t < -1.8331\}$ (C) $p = 0.044944$ 7.4 These data show that the mean age of the first episode occurrence within the severe group is younger than 33 years ($p < 0.05$).

7.5 (A) The mean age of the first episode occurrence in the severe group is equal to the mean age of the first episode occurrence in the moderate group.

(B) The mean age of the first episode occurrence in the severe group is less than the mean age of the first episode occurrence in the moderate group.

7.6 Test statistic.

(A) $t = (30 - 35) - 0 = -5$ (B) $SE = \sqrt{s^2/n_1 + s^2/n_2} = (25/10 + 25/10) = 2.236$ (C) $t = [(30 - 35) - 0]/2.236 = -2.236$ 7.7 (A) TDIST(x, df , 1), TDIST(-2.236, 18, 1)(B) $\{t < -1.734\}$

(C) 0.019

(D) Let d denote the difference of the two means that we are looking for, then $t = (d - 0)/2.236 < -1.734$. $d = 3.9$. Therefore, at least there must be a 3.9-year age difference.7.8 (A) 90% CI = Mean difference $\pm t \times SE$, where $t = 1.734$ ($df = 18$, two-sided $\alpha = 90\%$), $SE = 2.236$. $-5 \pm (1.734 \times 2.236) = -5 \pm 3.9$

95% CI: (-8.9, -1.1)

(B) These data showed that the estimated mean age of the first episode occurrence in the severe group was 5 years younger than the mean age of the first episode occurrence in the moderate group (90% CI: 8.9 to 1.1 years younger).

7.9 562.5

7.10 625

7.11 1187.6

7.12 281.25

7.13 25

7.14 11.25

7.15 $F.DIST.RT(11.25, 2, 25) = 0.000328$.

$F.DIST(11.25, 2, 25, 0) = 0.000328$.

$1-F.DIST(11.25, 2, 25, 1) = 0.000328$

7.16 These data showed that at least one mean age of the first episode occurrence in one group is significantly different from the other mean ages of the other groups ($p = 0.000328$).

7.17 Using Table 10.6 at one-sided $\alpha = 0.05$, power 90%, with sample of 180 (177 per table, so this is approximated), a conservative approximation of $d = 0.35$. Let MD denote the minimum mean difference, $0.35 = MD/SD$, $MD = 0.35 \times 5 = 1.75$ years.

8. Do not approve

Note: Quadrupling the sample size is not enough because the committee also requested to change the significance level from 10% to 5%. If the chosen significance level for the statistical test of the proposed A-02 study data analysis had not been this new 5% (i.e., stay with Dr. Z's 10% significance level of the pilot study A-01), then the quadrupling idea would be perfect to downsize the effect size by half while retaining the same power of 80% with the variances remaining the same.

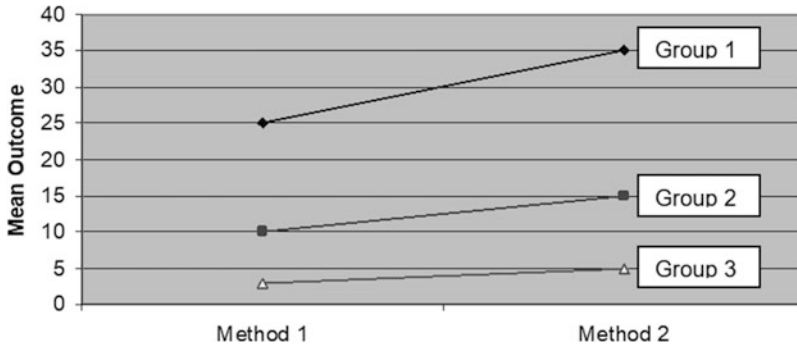
9. f

9.2 Review Exercise 2

Part A (30 points): Questions 1–15 are “True/False” questions. Please explain/criticize if you answered False

1. The coefficient of variation (CV) can measure the variability of data measured on the interval or ratio scale and is useful to compare the variations of two different characteristics of the same patient group measures in the same unit or those of the same characteristics from two or more different patient groups. (True/False).
2. A large class of statistics course's final test scores are normally distributed. The CV of the test score distribution is 10% and the average score is 80. It can be interpreted that approximately 31.7% of the students received their scores either above 88 or below 72 (i.e., outside of these two high and low scores). (True/False).

3. Mean ($\pm$ standard deviation) of pulse rate of a large group of people is known as 65 beats/min (± 10). It is also known that the pulse rate of this group is normally distributed. The proportion of the people with pulse rate over 85 beats/min is approximately less than 5% but greater than 2% (i.e., $2\% < \text{pulse rate} < 5\%$). (True/False).
4. A researcher conducted a small study ($n = 15$) and estimated a sample mean and its standard error of a continuous measurement, and then constructed a normal approximation-based 95% confidence interval around the population mean. The result was presented to a department's seminar. Another researcher made a comment that the confidence interval would be slightly wider if it had been constructed based on the t -distribution with $df = 14$. (True/False).
5. Dr. Z proposed a comparative study involving a total of 60 study subjects of whom 30 subjects will receive medication A and the other 30 will receive a placebo. He wants to detect a 10-point reduction in mean pain scale score among the subjects on medication A compared to the mean of the placebo subjects by using the directional independent samples t -test at a 5% significance level with 80% power. A reviewer agreed with the choice of the t -test. However, it is criticized that the detectable difference is too large, and a 5-point reduction is needed to be detected. Then the researcher decided to double the study sample size (i.e., a total of 120 subjects) to detect at least a 5-point reduction in mean pain score using the same test (i.e., directional independent samples t -test, the same power, the same significance level, and the same assumption about the standard deviation). Dr. Z's decision was correct. (True/False).
6. If a hypothesis test did not show a statistically significant difference in two means, but the observed difference in the two sample means was large enough, we should not exclude a possibility that the study had been underpowered. (True/False).
7. An investigator wants to perform a hypothesis test in which H_0 : Mean = 100 and H_1 : Mean > 100. It is observed that the sample mean = 100 and its standard error = 50 from a random sample set of size 10 drawn from a normally distributed population. A t -test is not applicable because the t -test is only applicable in a two-group comparison or a paired mean comparison. (True/False).
8. If a t -distribution-based 95% confidence interval on a difference in two means excluded 0 and the signs of the upper and lower confidence limits were the same and they were in the desirable direction, we can say that the nondirectional independent samples t -test for the comparison would also reject H_0 : $\text{mean}_1 = \text{mean}_2$ at a 5% significance level, the null hypothesis. (True/False).
9. The following figure reveals that there was no *Group* $\times$ *Method* interaction (please assume that every observed difference in the figures was statistically significant at a 5% significance level). (True/False).



Mean responses of the outcomes for two different methods among three study subgroups

10. An estimated multiple linear regression equation predicts the means of the dependent variable at certain values of the model's independent variables. (True or/False).
11. Odds ratio (OR) is defined as the ratio of the two event probabilities whose numerator is one group's event probability and the denominator is another group's (i.e., comparison group's) event probability. (True or/False).
12. In a 2×2 contingency table analysis setting, the alternative hypothesis of the chi-square test can be directional. (True or/False).

Please explain these:

13. Statistic
14. Sampling distribution
15. Standard error

Part B: Questions 16.1–16.3

While interested in studying the performance on a mathematics proficiency test of a very large cohort of college freshmen in a developing country, a team of investigators had no knowledge about the population distribution of the performance scores. So, they gathered a pilot random sample of 100 individual scores, created the following stem-and-leaf plot, and calculated the mean and standard deviation (observed sample mean = 57 and the observed sample standard deviation = 17).

9 57
9 0
8 55666789
8 0012234
7 889
7 004
6 556689
6 0011344
5 566666778
5 012223334444
4 555555566668889999
4 0012222244
3 566789999
3 2234
2 6

Then they came to a statistics professor and asked if the distribution of the scores in the population from which the sample set was drawn could be a normal distribution. The professor examined it and answered “Well . . . there is enough evidence that the population scores are not normally distributed because the sample data are clustered at two locations, i.e., bimodal (having two modes).”

- 16.1** – The sample median can be obtained from the information given to you: (True/False). If you answered “True,” please find out the median score. Otherwise, please explain why you cannot obtain the sample median.
- 16.2** – If a very large new random sample set is drawn (e.g., $n = 3000$), the distribution of that new sample will approach a normal (Gaussian) distribution: (True – please explain why /False – please explain why). Please explain why.
- 16.3** – If a very large number, m , of random sample sets with each individual sample size of 1000 are independently drawn, then the distribution of those m independent sample means will probably form a symmetrical bell shape distribution: (True – please explain why/False – please explain why).

Part C: Questions 17–19

The following descriptive statistics summarize the result of a small study of an effectiveness of a new intervention to improve elbow range of motion (ROM) in degrees among elderly men and women with elbow fractures previously.

	Control	Intervention A	Combined samples
Men	N = 10 Mean = 30.10 SD = 1.91	N = 10 Mean = 30.00 SD = 2.11	N = 20 Mean = 30.05 SD = 2.01
Women	N = 30 Mean = 30.90 SD = 1.91	N = 30 Mean = 36.40 SD = 2.27	N = 60 Mean = 33.65 SD = 2.10
Combined samples	N = 40 Mean = 30.70 SD = 1.91	N = 40 Mean = 34.80 SD = 2.23	N = 80 Mean = 32.75 SD = 2.08

17. Write an estimated regression equation that you would have obtained if a regression modeling (under the equal variance assumption) had been applied to women. Please let M_ROM (mean ROM) denote the dependent variable, and let I_A denote the independent variable (a dummy variable indicating Intervention A if its value = 1, and control if 0), i.e., $M_ROM = () + () \cdot I_A$.
18. The intervention was not efficacious within men (statistical analysis test was performed, and the result was not significant). Please perform a directional t -test ($df = 58$) to examine whether the intervention was efficacious within women (please use a 5% significance level and assume that the population variances are equal between the two groups) and make a statement that can appear in a clinical journal.
19. Was there an interaction? If so, please describe it; otherwise, explain why the above information did not show an interaction.

Part D: Questions 20–21

In a certain population, for a subject i ($i = 1, 2, 3, \dots$, etc.), the measurement of Y is determined by the following linear model $Y_i = \beta_0 + \beta_1 X_i + \varepsilon_i$, where ε_i is a random error with mean 0. Y and X are both continuous variables and Y follows a normal distribution. The following equation is an estimated regression equation (using a sample data $n = 122$). The number in parentheses underneath the estimated regression slope coefficient is its standard error.

$$\hat{Y} = 100 + 2.25 \cdot X$$

(0.90)

Note: It was confirmed that the intercept β_0 was significantly greater than 0 ($p < 0.00001$), and the current estimate of 100 is clinically very meaningful.

20. Perform a z -test whose $H_0: \beta_1 = 0$ versus $H_1: \beta_1 > 0$, and write a sentence that can go into a clinical journal results section. Please use a 5% significance level.
21. Please construct a z -based 90% CI around the β_1 and write a sentence that can go into a clinical journal results section.

Part E: Question 22

A small study showed that an estimated Pearson's product moment correlation coefficient between two continuous outcome variables using 12 pairs was 0.365, which was not significantly different from 0 at a 5% significance level by the directional t -test (H_0 : Population correlation coefficient $\rho = 0$ versus H_1 : Population correlation coefficient $\rho > 0$). There was another similar study result involving 22 pairs that showed $r = 0.375$ without presenting a statistical test result.

22. Please perform a t -test whose null and alternative hypotheses are H_0 : Population correlation coefficient $\rho = 0$ versus H_1 : Population correlation coefficient $\rho > 0$. Please use a 5% significance level (df for the related statistical test = 20). Please describe how you determined the significance of the result, and then write a sentence that can go into a clinical journal results section. Please do not include the explanation how you made your conclusion (i.e., do not make your answer like "... since p was less than 0.05, I rejected the null ...").

Part F: Questions 23–26

Please choose a proper nonparametric method for each analysis from the listed methods below:

23. Comparison of BMI (body mass index, i.e., $[\text{weight in pounds} \times 703]/[\text{height in inches}^2]$) distributions among three groups.
24. Test for examining if there was a change in body weight between baseline and post-6-month lifestyle modification intervention program offered to a group of obese subjects.
25. Test for an association between sex (male versus female) and smoking status (nonsmokers versus smokers).
26. Test for an association between age and heart rate.
 - (a) Wilcoxon's signed-rank test
 - (b) Mann–Whitney U-test (or Wilcoxon's rank sum test)
 - (c) Kruskal–Wallis test
 - (d) Friedman's test
 - (e) chi-square test for independence
 - (f) McNemar's test for matched pairs
 - (g) Spearman's rank correlation

9.2.1 Solutions for Review Exercise 2

1. True
2. True

$CV = SD/\text{Mean} = 10\%$
 $SD = 10\% \text{ of } 80 \text{ (and it is } 8)$
 $NORMDIST(72, 80, 8, 1) \times 2 = 0.1587 \times 2 = 0.317$
3. True

Over 85 bpm is over 2 SD above the mean (i.e., a pulse rate $> 65 + 2 \times 10$).
4. True

For a $df = 14$, the t (two-sided $\alpha = 0.05$) = 2.145 and z (two-sided $\alpha = 0.05$) = 1.96.
 z -based 95 CI: (mean $- 1.96 \times SE$, mean $+ 1.96 \times SE$)
 t -based 95% CI: (mean $- 2.145 \times SE$, mean $+ 2.145 \times SE$)
5. False

Sample size should be quadrupled in order to downsize the detectable effect size to 1/2 of the current size.
6. True

Let us not forget about the possibility of Type 2 error too.

7. False

This can be done by a one-sample *t*-test.

8. True

When the *t*-distribution-based 95% CI excluded 0, then the two-sided *t*-test would reject the null that the mean difference = 0.

9. False

Response patterns are not parallel.

10. True

11. False

OR is it not a ratio of two probabilities but is a ratio of two odds (i.e., $\text{odds1}/\text{odds2}$), where $\text{odds1} = p1/(1 - p1)$ and $\text{odds2} = p2/(1 - p2)$.

12. False

Always nondirectional because chi-square test statistics is a sum of squared discrepancies between observed and expected frequencies, and it does not have the directionality, i.e., we cannot determine which cell's observed frequency was larger or smaller than the expected frequency once this test statistic is calculated.

13. A function of sample data (e.g., sample mean age, sample mean blood pressure, sample proportion of men, etc.).

14. A distribution of a sample statistic (e.g., sampling distribution of the sample means obtained from 1000 independent random samplings with the sample size of 50).

15. A measure of sample-to-sample (random) variation of a sample statistic (e.g., estimated sample mean, estimated regression coefficient). Let us also note that SE is the standard deviation of a sampling distribution.

16.1 True

Median = average of 50th and 51st order statistics = $(53 + 54)/2 = 53.5$

16.2 False

The distribution of a sample set with a larger sample size will become close to the population distribution. Only if the population distribution is truly normal, the observed "sample" distribution of a larger size sample will be similar to a normal distribution. Otherwise, the distribution of a large sample set will resemble the population distribution, and in this case, it is not a normal distribution. Such a large size sample will reveal the bimodality (i.e., two modes) stronger.

16.3 True

Central limit theorem.

17. $M_{\text{ROM}} = 30.90 + 5.5 \cdot I_A$

18. $t = (\text{Observed mean difference} - \text{Expected mean difference}) / \text{SE}(\text{Observed mean difference} - \text{Expected mean difference})$

Observed mean difference: $36.30 - 30.90 = 5.5$

Expected mean difference: 0

Standard error, SE

$$= \sqrt{\text{SD}^2 \text{ of control women}/30 + \text{SD}^2 \text{ of women with intervention}/30}$$

$$= \sqrt{(1.91^2/30) + (2.28^2/30)} = 0.54 = 0.54$$

$$\text{So, } t = 5.5/0.54 = 10.18$$

$$p\text{-value} = \text{TDSIT}(10.18, 58, 2) = 7.97929\text{E-}15 = 0.00000000000000797929$$

These data showed that the intervention group showed a greater ROM ($p < 0.05$)

19. Yes, the intervention was not effective in men but was effective in women.

$$\begin{aligned} \text{20. } z &= (\text{Observed} - \text{Expected})/\text{SE}(\text{Observed} - \text{Expected}) = \\ &= (\text{Observed} - \text{Expected})/\text{SE}(\text{Observed}) = (2.5 - 0)/0.90 = 2.5 \\ \text{NORMSDIST}(2.5) &= 0.993790335 \end{aligned}$$

$$\text{So, } p = 2 \times (1 - 0.977249868) = 0.012419$$

These data showed that there was a significant linear increase in the mean of this outcome as the independent variable increased ($p < 0.05$).

These data showed that the estimated regression coefficient is significantly greater than 0 ($p < 0.05$).

$$\begin{aligned} \text{21. } z \text{ (2-sided } \alpha = 0.1) &= 1.645; \text{ lower and upper limits of the CI are } 2.25 - \\ &1.645 \times 0.90 = 0.77, \text{ and } 2.25 + 1.645 \times 0.90 = 3.73, \text{ respectively. } 90\% \text{ CI: } \\ &0.77 - 3.73. \end{aligned}$$

These data showed that the average increase of the outcome as one unit increase of the independent variable was 2.25 (90% CI: 0.77 – 3.73).

These data showed that the estimated regression coefficient is 2.25 (90% CI: 0.77 – 3.73).

22. H_0 : There is no linear relationship between the proximal and distal development scores.

H_1 : There is a positive linear relationship between the proximal and distal development scores.

With $df = 18$, the critical value at one-sided $= 0.05$ is 0.378. $r = 0.375$ is smaller than this critical value, so you cannot reject the null hypothesis.

These data showed that there was no statistically significant positive linear relationship between proximal and distal development scores at a 5% significance level.

23. c

24. a

25. e

26. g

Chapter 10

Statistical Tables



See Tables [10.1](#), [10.2](#), [10.3](#), [10.4](#), [10.5](#) and [10.6](#).

Table 10.1 Cumulative distribution of $z \sim N(0, 1)$

z	Cumulative probability up to z	1-Cumulative probability, i.e., upper tail probability
0.00	0.5000	0.5000
0.01	0.5040	0.4960
0.02	0.5080	0.4920
0.03	0.5120	0.4880
0.04	0.5160	0.4840
0.05	0.5199	0.4801
0.06	0.5239	0.4761
0.07	0.5279	0.4721
0.08	0.5319	0.4681
0.09	0.5359	0.4641
0.10	0.5398	0.4602
0.11	0.5438	0.4562
0.12	0.5478	0.4522
0.13	0.5517	0.4483
0.14	0.5557	0.4443
0.15	0.5596	0.4404
0.16	0.5636	0.4364
0.17	0.5675	0.4325
0.18	0.5714	0.4286
0.19	0.5753	0.4247
0.20	0.5793	0.4207
0.21	0.5832	0.4168
0.22	0.5871	0.4129
0.23	0.5910	0.4090
0.24	0.5948	0.4052
0.25	0.5987	0.4013
0.26	0.6026	0.3974
0.27	0.6064	0.3936
0.28	0.6103	0.3897
0.29	0.6141	0.3859
0.30	0.6179	0.3821
0.31	0.6217	0.3783
0.32	0.6255	0.3745
0.33	0.6293	0.3707
0.34	0.6331	0.3669
0.35	0.6368	0.3632
0.36	0.6406	0.3594
0.37	0.6443	0.3557
0.38	0.6480	0.3520
0.39	0.6517	0.3483
0.40	0.6554	0.3446
0.41	0.6591	0.3409

(continued)

Table 10.1 (continued)

<i>z</i>	Cumulative probability up to <i>z</i>	1-Cumulative probability, i.e., upper tail probability
0.42	0.6628	0.3372
0.43	0.6664	0.3336
0.44	0.6700	0.3300
0.45	0.6736	0.3264
0.46	0.6772	0.3228
0.47	0.6808	0.3192
0.48	0.6844	0.3156
0.49	0.6879	0.3121
0.50	0.6915	0.3085
0.51	0.6950	0.3050
0.52	0.6985	0.3015
0.53	0.7019	0.2981
0.54	0.7054	0.2946
0.55	0.7088	0.2912
0.56	0.7123	0.2877
0.57	0.7157	0.2843
0.58	0.7190	0.2810
0.59	0.7224	0.2776
0.60	0.7257	0.2743
0.61	0.7291	0.2709
0.62	0.7324	0.2676
0.63	0.7357	0.2643
0.64	0.7389	0.2611
0.65	0.7422	0.2578
0.66	0.7454	0.2546
0.67	0.7486	0.2514
0.68	0.7517	0.2483
0.69	0.7549	0.2451
0.70	0.7580	0.2420
0.71	0.7611	0.2389
0.72	0.7642	0.2358
0.73	0.7673	0.2327
0.74	0.7704	0.2296
0.75	0.7734	0.2266
0.76	0.7764	0.2236
0.77	0.7794	0.2206
0.78	0.7823	0.2177
0.79	0.7852	0.2148
0.80	0.7881	0.2119
0.81	0.7910	0.2090
0.82	0.7939	0.2061
0.83	0.7967	0.2033

(continued)

Table 10.1 (continued)

z	Cumulative probability up to z	1-Cumulative probability, i.e., upper tail probability
0.84	0.7995	0.2005
0.85	0.8023	0.1977
0.86	0.8051	0.1949
0.87	0.8078	0.1922
0.88	0.8106	0.1894
0.89	0.8133	0.1867
0.90	0.8159	0.1841
0.91	0.8186	0.1814
0.92	0.8212	0.1788
0.93	0.8238	0.1762
0.94	0.8264	0.1736
0.95	0.8289	0.1711
0.96	0.8315	0.1685
0.97	0.8340	0.1660
0.98	0.8365	0.1635
0.99	0.8389	0.1611
1.00	0.8413	0.1587
1.01	0.8438	0.1562
1.02	0.8461	0.1539
1.03	0.8485	0.1515
1.04	0.8508	0.1492
1.05	0.8531	0.1469
1.06	0.8554	0.1446
1.07	0.8577	0.1423
1.08	0.8599	0.1401
1.09	0.8621	0.1379
1.10	0.8643	0.1357
1.11	0.8665	0.1335
1.12	0.8686	0.1314
1.13	0.8708	0.1292
1.14	0.8729	0.1271
1.15	0.8749	0.1251
1.16	0.8770	0.1230
1.17	0.8790	0.1210
1.18	0.8810	0.1190
1.19	0.8830	0.1170
1.20	0.8849	0.1151
1.21	0.8869	0.1131
1.22	0.8888	0.1112
1.23	0.8907	0.1093
1.24	0.8925	0.1075
1.25	0.8944	0.1056

(continued)

Table 10.1 (continued)

z	Cumulative probability up to z	1-Cumulative probability, i.e., upper tail probability
1.26	0.8962	0.1038
1.27	0.8980	0.1020
1.28	0.8997	0.1003
1.29	0.9015	0.0985
1.30	0.9032	0.0968
1.31	0.9049	0.0951
1.32	0.9066	0.0934
1.33	0.9082	0.0918
1.34	0.9099	0.0901
1.35	0.9115	0.0885
1.36	0.9131	0.0869
1.37	0.9147	0.0853
1.38	0.9162	0.0838
1.39	0.9177	0.0823
1.40	0.9192	0.0808
1.41	0.9207	0.0793
1.42	0.9222	0.0778
1.43	0.9236	0.0764
1.44	0.9251	0.0749
1.45	0.9265	0.0735
1.46	0.9279	0.0721
1.47	0.9292	0.0708
1.48	0.9306	0.0694
1.49	0.9319	0.0681
1.50	0.9332	0.0668
1.51	0.9345	0.0655
1.52	0.9357	0.0643
1.53	0.9370	0.0630
1.54	0.9382	0.0618
1.55	0.9394	0.0606
1.56	0.9406	0.0594
1.57	0.9418	0.0582
1.58	0.9429	0.0571
1.59	0.9441	0.0559
1.60	0.9452	0.0548
1.61	0.9463	0.0537
1.62	0.9474	0.0526
1.63	0.9484	0.0516
1.64	0.9495	0.0505
1.65	0.9505	0.0495
1.66	0.9515	0.0485
1.67	0.9525	0.0475

(continued)

Table 10.1 (continued)

z	Cumulative probability up to z	1-Cumulative probability, i.e., upper tail probability
1.68	0.9535	0.0465
1.69	0.9545	0.0455
1.70	0.9554	0.0446
1.71	0.9564	0.0436
1.72	0.9573	0.0427
1.73	0.9582	0.0418
1.74	0.9591	0.0409
1.75	0.9599	0.0401
1.76	0.9608	0.0392
1.77	0.9616	0.0384
1.78	0.9625	0.0375
1.79	0.9633	0.0367
1.80	0.9641	0.0359
1.81	0.9649	0.0351
1.82	0.9656	0.0344
1.83	0.9664	0.0336
1.84	0.9671	0.0329
1.85	0.9678	0.0322
1.86	0.9686	0.0314
1.87	0.9693	0.0307
1.88	0.9699	0.0301
1.89	0.9706	0.0294
1.90	0.9713	0.0287
1.91	0.9719	0.0281
1.92	0.9726	0.0274
1.93	0.9732	0.0268
1.94	0.9738	0.0262
1.95	0.9744	0.0256
1.96	0.9750	0.0250
1.97	0.9756	0.0244
1.98	0.9761	0.0239
1.99	0.9767	0.0233
2.00	0.9772	0.0228
2.01	0.9778	0.0222
2.02	0.9783	0.0217
2.03	0.9788	0.0212
2.04	0.9793	0.0207
2.05	0.9798	0.0202
2.06	0.9803	0.0197
2.07	0.9808	0.0192
2.08	0.9812	0.0188
2.09	0.9817	0.0183

(continued)

Table 10.1 (continued)

z	Cumulative probability up to z	1-Cumulative probability, i.e., upper tail probability
2.10	0.9821	0.0179
2.11	0.9826	0.0174
2.12	0.9830	0.0170
2.13	0.9834	0.0166
2.14	0.9838	0.0162
2.15	0.9842	0.0158
2.16	0.9846	0.0154
2.17	0.9850	0.0150
2.18	0.9854	0.0146
2.19	0.9857	0.0143
2.20	0.9861	0.0139
2.21	0.9864	0.0136
2.22	0.9868	0.0132
2.23	0.9871	0.0129
2.24	0.9875	0.0125
2.25	0.9878	0.0122
2.26	0.9881	0.0119
2.27	0.9884	0.0116
2.28	0.9887	0.0113
2.29	0.9890	0.0110
2.30	0.9893	0.0107
2.31	0.9896	0.0104
2.32	0.9898	0.0102
2.33	0.9901	0.0099
2.34	0.9904	0.0096
2.35	0.9906	0.0094
2.36	0.9909	0.0091
2.37	0.9911	0.0089
2.38	0.9913	0.0087
2.39	0.9916	0.0084
2.40	0.9918	0.0082
2.41	0.9920	0.0080
2.42	0.9922	0.0078
2.43	0.9925	0.0075
2.44	0.9927	0.0073
2.45	0.9929	0.0071
2.46	0.9931	0.0069
2.47	0.9932	0.0068
2.48	0.9934	0.0066
2.49	0.9936	0.0064
2.50	0.9938	0.0062
2.51	0.9940	0.0060

(continued)

Table 10.1 (continued)

z	Cumulative probability up to z	1-Cumulative probability, i.e., upper tail probability
2.52	0.9941	0.0059
2.53	0.9943	0.0057
2.54	0.9945	0.0055
2.55	0.9946	0.0054
2.56	0.9948	0.0052
2.57	0.9949	0.0051
2.58	0.9951	0.0049
2.59	0.9952	0.0048
2.60	0.9953	0.0047
2.61	0.9955	0.0045
2.62	0.9956	0.0044
2.63	0.9957	0.0043
2.64	0.9959	0.0041
2.65	0.9960	0.0040
2.66	0.9961	0.0039
2.67	0.9962	0.0038
2.68	0.9963	0.0037
2.69	0.9964	0.0036
2.70	0.9965	0.0035
2.71	0.9966	0.0034
2.72	0.9967	0.0033
2.73	0.9968	0.0032
2.74	0.9969	0.0031
2.75	0.9970	0.0030
2.76	0.9971	0.0029
2.77	0.9972	0.0028
2.78	0.9973	0.0027
2.79	0.9974	0.0026
2.80	0.9974	0.0026
2.81	0.9975	0.0025
2.82	0.9976	0.0024
2.83	0.9977	0.0023
2.84	0.9977	0.0023
2.85	0.9978	0.0022
2.86	0.9979	0.0021
2.87	0.9979	0.0021
2.88	0.9980	0.0020
2.89	0.9981	0.0019
2.90	0.9981	0.0019
2.91	0.9982	0.0018
2.92	0.9982	0.0018
2.93	0.9983	0.0017

(continued)

Table 10.1 (continued)

z	Cumulative probability up to z	1-Cumulative probability, i.e., upper tail probability
2.94	0.9984	0.0016
2.95	0.9984	0.0016
2.96	0.9985	0.0015
2.97	0.9985	0.0015
2.98	0.9986	0.0014
2.99	0.9986	0.0014
3.00	0.9987	0.0013

Table 10.2 (1-p)th percentiles of *t*-distributions

<i>df</i>	p = 0.00025	p = 0.0005	p = 0.0125	p = 0.025	p = 0.0375	p = 0.05
1	1273.2393	636.6192	25.4517	12.7062	8.4490	6.3138
2	44.7046	31.5991	6.2053	4.3027	3.4428	2.9200
3	16.3263	12.9240	4.1765	3.1824	2.6808	2.3534
4	10.3063	8.6103	3.4954	2.7764	2.3921	2.1318
5	7.9757	6.8688	3.1634	2.5706	2.2423	2.0150
6	6.7883	5.9588	2.9687	2.4469	2.1510	1.9432
7	6.0818	5.4079	2.8412	2.3646	2.0897	1.8946
8	5.6174	5.0413	2.7515	2.3060	2.0458	1.8595
9	5.2907	4.7809	2.6850	2.2622	2.0127	1.8331
10	5.0490	4.5869	2.6338	2.2281	1.9870	1.8125
11	4.8633	4.4370	2.5931	2.2010	1.9663	1.7959
12	4.7165	4.3178	2.5600	2.1788	1.9494	1.7823
13	4.5975	4.2208	2.5326	2.1604	1.9354	1.7709
14	4.4992	4.1405	2.5096	2.1448	1.9235	1.7613
15	4.4166	4.0728	2.4899	2.1314	1.9132	1.7531
16	4.3463	4.0150	2.4729	2.1199	1.9044	1.7459
17	4.2858	3.9651	2.4581	2.1098	1.8966	1.7396
18	4.2332	3.9216	2.4450	2.1009	1.8898	1.7341
19	4.1869	3.8834	2.4334	2.0930	1.8837	1.7291
20	4.1460	3.8495	2.4231	2.0860	1.8783	1.7247
21	4.1096	3.8193	2.4138	2.0796	1.8734	1.7207
22	4.0769	3.7921	2.4055	2.0739	1.8690	1.7171
23	4.0474	3.7676	2.3979	2.0687	1.8649	1.7139
24	4.0207	3.7454	2.3909	2.0639	1.8613	1.7109
25	3.9964	3.7251	2.3846	2.0595	1.8579	1.7081
26	3.9742	3.7066	2.3788	2.0555	1.8548	1.7056
27	3.9538	3.6896	2.3734	2.0518	1.8519	1.7033
28	3.9351	3.6739	2.3685	2.0484	1.8493	1.7011
29	3.9177	3.6594	2.3638	2.0452	1.8468	1.6991
30	3.9016	3.6460	2.3596	2.0423	1.8445	1.6973
31	3.8867	3.6335	2.3556	2.0395	1.8424	1.6955
32	3.8728	3.6218	2.3518	2.0369	1.8404	1.6939
33	3.8598	3.6109	2.3483	2.0345	1.8385	1.6924
34	3.8476	3.6007	2.3451	2.0322	1.8368	1.6909
35	3.8362	3.5911	2.3420	2.0301	1.8351	1.6896
50	3.7231	3.4960	2.3109	2.0086	1.8184	1.6759
75	3.6391	3.4250	2.2873	1.9921	1.8056	1.6654
100	3.5983	3.3905	2.2757	1.9840	1.7992	1.6602

Table 10.3 Upper 95th and 99th percentiles of chi-square distributions

<i>df</i>	95th	99th
1	3.84	6.63
2	5.99	9.21
3	7.81	11.34
4	9.49	13.28
5	11.07	15.09
6	12.59	16.81
7	14.07	18.48
8	15.51	20.09
9	16.92	21.67
10	18.31	23.21
11	19.68	24.72
12	21.03	26.22
13	22.36	27.69
14	23.68	29.14
15	25.00	30.58
16	26.30	32.00
17	27.59	33.41
18	28.87	34.81
19	30.14	36.19
20	31.41	37.57
21	32.67	38.93
22	33.92	40.29
23	35.17	41.64
24	36.42	42.98
25	37.65	44.31
26	38.89	45.64
27	40.11	46.96
28	41.34	48.28
29	42.56	49.59
30	43.77	50.89
35	49.80	57.34
40	55.76	63.69
45	61.66	69.96
50	67.50	76.15
75	96.22	106.39
100	124.34	135.81

Table 10.4 Upper 95th percentiles of *F*-distributions

	Upper 95th percentiles of <i>F</i> -distributions									
<i>df</i> ₁	1	2	3	4	5	6	7	8	9	10
<i>df</i> ₂										
2	18.51	19.00	19.16	19.25	19.30	19.33	19.35	19.37	19.38	19.40
3	10.13	9.55	9.28	9.12	9.01	8.94	8.89	8.85	8.81	8.79
4	7.71	6.94	6.59	6.39	6.26	6.16	6.09	6.04	6.00	5.96
5	6.61	5.79	5.41	5.19	5.05	4.95	4.88	4.82	4.77	4.74
6	5.99	5.14	4.76	4.53	4.39	4.28	4.21	4.15	4.10	4.06
7	5.59	4.74	4.35	4.12	3.97	3.87	3.79	3.73	3.68	3.64
8	5.32	4.46	4.07	3.84	3.69	3.58	3.50	3.44	3.39	3.35
9	5.12	4.26	3.86	3.63	3.48	3.37	3.29	3.23	3.18	3.14
10	4.96	4.10	3.71	3.48	3.33	3.22	3.14	3.07	3.02	2.98
11	4.84	3.98	3.59	3.36	3.20	3.09	3.01	2.95	2.90	2.85
12	4.75	3.89	3.49	3.26	3.11	3.00	2.91	2.85	2.80	2.75
13	4.67	3.81	3.41	3.18	3.03	2.92	2.83	2.77	2.71	2.67
14	4.60	3.74	3.34	3.11	2.96	2.85	2.76	2.70	2.65	2.60
15	4.54	3.68	3.29	3.06	2.90	2.79	2.71	2.64	2.59	2.54
16	4.49	3.63	3.24	3.01	2.85	2.74	2.66	2.59	2.54	2.49
17	4.45	3.59	3.20	2.96	2.81	2.70	2.61	2.55	2.49	2.45
18	4.41	3.55	3.16	2.93	2.77	2.66	2.58	2.51	2.46	2.41
19	4.38	3.52	3.13	2.90	2.74	2.63	2.54	2.48	2.42	2.38
20	4.35	3.49	3.10	2.87	2.71	2.60	2.51	2.45	2.39	2.35
21	4.32	3.47	3.07	2.84	2.68	2.57	2.49	2.42	2.37	2.32
22	4.30	3.44	3.05	2.82	2.66	2.55	2.46	2.40	2.34	2.30
23	4.28	3.42	3.03	2.80	2.64	2.53	2.44	2.37	2.32	2.27
24	4.26	3.40	3.01	2.78	2.62	2.51	2.42	2.36	2.30	2.25
25	4.24	3.39	2.99	2.76	2.60	2.49	2.40	2.34	2.28	2.24
26	4.23	3.37	2.98	2.74	2.59	2.47	2.39	2.32	2.27	2.22
27	4.21	3.35	2.96	2.73	2.57	2.46	2.37	2.31	2.25	2.20
28	4.20	3.34	2.95	2.71	2.56	2.45	2.36	2.29	2.24	2.19
29	4.18	3.33	2.93	2.70	2.55	2.43	2.35	2.28	2.22	2.18
30	4.17	3.32	2.92	2.69	2.53	2.42	2.33	2.27	2.21	2.16
35	4.12	3.27	2.87	2.64	2.49	2.37	2.29	2.22	2.16	2.11
40	4.08	3.23	2.84	2.61	2.45	2.34	2.25	2.18	2.12	2.08
45	4.06	3.20	2.81	2.58	2.42	2.31	2.22	2.15	2.10	2.05
50	4.03	3.18	2.79	2.56	2.40	2.29	2.20	2.13	2.07	2.03
55	4.02	3.16	2.77	2.54	2.38	2.27	2.18	2.11	2.06	2.01
60	4.00	3.15	2.76	2.53	2.37	2.25	2.17	2.10	2.04	1.99
75	3.97	3.12	2.73	2.49	2.34	2.22	2.13	2.06	2.01	1.96
100	3.94	3.09	2.70	2.46	2.31	2.19	2.10	2.03	1.97	1.93

*df*₁ Numerator *df*

*df*₂ Denominator *df*

Table 10.5 Upper 99th percentiles of *F*-distributions

	Upper 99th percentiles of <i>F</i> -distributions									
<i>df</i> ₁	1	2	3	4	5	6	7	8	9	10
<i>df</i> ₂										
2	98.50	99.00	99.17	99.25	99.30	99.33	99.36	99.37	99.39	99.40
3	34.12	30.82	29.46	28.71	28.24	27.91	27.67	27.49	27.35	27.23
4	21.20	18.00	16.69	15.98	15.52	15.21	14.98	14.80	14.66	14.55
5	16.26	13.27	12.06	11.39	10.97	10.67	10.46	10.29	10.16	10.05
6	13.75	10.92	9.78	9.15	8.75	8.47	8.26	8.10	7.98	7.87
7	12.25	9.55	8.45	7.85	7.46	7.19	6.99	6.84	6.72	6.62
8	11.26	8.65	7.59	7.01	6.63	6.37	6.18	6.03	5.91	5.81
9	10.56	8.02	6.99	6.42	6.06	5.80	5.61	5.47	5.35	5.26
10	10.04	7.56	6.55	5.99	5.64	5.39	5.20	5.06	4.94	4.85
11	9.65	7.21	6.22	5.67	5.32	5.07	4.89	4.74	4.63	4.54
12	9.33	6.93	5.95	5.41	5.06	4.82	4.64	4.50	4.39	4.30
13	9.07	6.70	5.74	5.21	4.86	4.62	4.44	4.30	4.19	4.10
14	8.86	6.51	5.56	5.04	4.69	4.46	4.28	4.14	4.03	3.94
15	8.68	6.36	5.42	4.89	4.56	4.32	4.14	4.00	3.89	3.80
16	8.53	6.23	5.29	4.77	4.44	4.20	4.03	3.89	3.78	3.69
17	8.40	6.11	5.18	4.67	4.34	4.10	3.93	3.79	3.68	3.59
18	8.29	6.01	5.09	4.58	4.25	4.01	3.84	3.71	3.60	3.51
19	8.18	5.93	5.01	4.50	4.17	3.94	3.77	3.63	3.52	3.43
20	8.10	5.85	4.94	4.43	4.10	3.87	3.70	3.56	3.46	3.37
21	8.02	5.78	4.87	4.37	4.04	3.81	3.64	3.51	3.40	3.31
22	7.95	5.72	4.82	4.31	3.99	3.76	3.59	3.45	3.35	3.26
23	7.88	5.66	4.76	4.26	3.94	3.71	3.54	3.41	3.30	3.21
24	7.82	5.61	4.72	4.22	3.90	3.67	3.50	3.36	3.26	3.17
25	7.77	5.57	4.68	4.18	3.85	3.63	3.46	3.32	3.22	3.13
26	7.72	5.53	4.64	4.14	3.82	3.59	3.42	3.29	3.18	3.09
27	7.68	5.49	4.60	4.11	3.78	3.56	3.39	3.26	3.15	3.06
28	7.64	5.45	4.57	4.07	3.75	3.53	3.36	3.23	3.12	3.03
29	7.60	5.42	4.54	4.04	3.73	3.50	3.33	3.20	3.09	3.00
30	7.56	5.39	4.51	4.02	3.70	3.47	3.30	3.17	3.07	2.98
35	7.42	5.27	4.40	3.91	3.59	3.37	3.20	3.07	2.96	2.88
40	7.31	5.18	4.31	3.83	3.51	3.29	3.12	2.99	2.89	2.80
45	7.23	5.11	4.25	3.77	3.45	3.23	3.07	2.94	2.83	2.74
50	7.17	5.06	4.20	3.72	3.41	3.19	3.02	2.89	2.78	2.70
55	7.12	5.01	4.16	3.68	3.37	3.15	2.98	2.85	2.75	2.66
60	7.08	4.98	4.13	3.65	3.34	3.12	2.95	2.82	2.72	2.63
75	6.99	4.90	4.05	3.58	3.27	3.05	2.89	2.76	2.65	2.57
100	6.90	4.82	3.98	3.51	3.21	2.99	2.82	2.69	2.59	2.50

*df*₁ Numerator *df*
*df*₂ Denominator *df*

Table 10.6 Sample sizes for independent samples *t*-tests

Sample size per group for independent samples <i>t</i> -test (normal approximation)				
Effect size = (mean difference/common SD)	$\alpha = 0.01$ (two-sided)		$\alpha = 0.05$ (two-sided)	
	Power 0.8	Power 0.9	Power 0.8	Power 0.9
0.25	373	476	251	336
0.30	259	330	174	233
0.35	190	242	128	171
0.40	145	185	98	131
0.45	115	146	77	103
0.50	93	119	62	84
0.55	77	98	51	69
0.60	64	82	43	58
0.65	55	70	37	49
0.70	47	60	32	42
0.75	41	52	27	37
0.80	36	46	24	32
0.85	32	41	21	29
0.90	28	36	19	25
0.95	25	32	17	23
1.00	23	29	15	21
1.05	21	26	14	19
1.10	19	24	12	17
1.15	17	22	11	15
1.20	16	20	10	14
1.25	14	19	10	13

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